

Language Models are Few-Shot Learners

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Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3’s few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

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语言模型是少样本学习器

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摘要

近期的研究表明，通过先在庞大的文本语料库上进行预训练，再针对特定任务进行微调，语言模型在众多自然语言处理任务及基准测试中的性能都能得到显著提升。尽管这类模型的架构通常具有任务无关性，但仍然需要包含数千乃至数万个示例的、针对特定任务的微调数据集。相比之下，人类往往仅凭少量示例或简单指令就能完成新的语言任务，而目前的自然语言处理系统在这方面仍难以做到。本文展示了扩大语言模型的规模能够极大提升其任务无关的少样本学习性能，有时甚至能赶上此前处于当前最佳水平的微调方法。具体而言，我们训练了拥有1750亿参数的GPT-3——这是一种自回归语言模型，其参数量是此前所有非稀疏语言模型的10倍——并在少样本场景下测试了它的性能。在所有任务中，GPT-3均无需进行任何梯度更新或微调，仅需通过与模型的文本交互来指定任务及少样本示例即可。它在包括翻译、问答任务、完形填空任务在内的多个自然语言处理数据集上均取得了优异表现，同时还胜任那些需要即时推理或领域适配的任务，例如词语重组、在句子中使用生词或进行三位数算术运算。与此同时，我们也发现了一些GPT-3的少样本学习能力仍较弱的数据集，以及一些因在大型网络语料库上训练而存在方法学问题的数据集。最后，我们发现GPT-3能够生成新闻文章样本，以至于人类评估者很难将其与人类撰写的文章区分开来。本文还探讨了这一发现以及GPT-3整体所带来的更广泛的社会影响。

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1 Introduction

Recent years have featured a trend towards pre-trained language representations in NLP systems, applied in increasingly flexible and task-agnostic ways for downstream transfer. First, single-layer representations were learned using word vectors [MCCD13, PSM14] and fed to task-specific architectures, then RNNs with multiple layers of representations and contextual state were used to form stronger representations [DL15, MBXS17, PNZtY18] (though still applied to task-specific architectures), and more recently pre-trained recurrent or transformer language models [VSP+17] have been directly fine-tuned, entirely removing the need for task-specific architectures [RNSS18, DCLT18, HR18].

This last paradigm has led to substantial progress on many challenging NLP tasks such as reading comprehension, question answering, textual entailment, and many others, and has continued to advance based on new architectures and algorithms [RSR+19, LOG+19, YDY+19, LCG+19]. However, a major limitation to this approach is that while the architecture is task-agnostic, there is still a need for task-specific datasets and task-specific fine-tuning: to achieve strong performance on a desired task typically requires fine-tuning on a dataset of thousands to hundreds of thousands of examples specific to that task. Removing this limitation would be desirable, for several reasons.

First, from a practical perspective, the need for a large dataset of labeled examples for every new task limits the applicability of language models. There exists a very wide range of possible useful language tasks, encompassing anything from correcting grammar, to generating examples of an abstract concept, to critiquing a short story. For many of these tasks it is difficult to collect a large supervised training dataset, especially when the process must be repeated for every new task.

Second, the potential to exploit spurious correlations in training data fundamentally grows with the expressiveness of the model and the narrowness of the training distribution. This can create problems for the pre-training plus fine-tuning paradigm, where models are designed to be large to absorb information during pre-training, but are then fine-tuned on very narrow task distributions. For instance [HLW+20] observe that larger models do not necessarily generalize better out-of-distribution. There is evidence that suggests that the generalization achieved under this paradigm can be poor because the model is overly specific to the training distribution and does not generalize well outside it [YdC+19, MPL19]. Thus, the performance of fine-tuned models on specific benchmarks, even when it is nominally at human-level, may exaggerate actual performance on the underlying task [GSL+18, NK19].

Third, humans do not require large supervised datasets to learn most language tasks – a brief directive in natural language (e.g. “please tell me if this sentence describes something happy or something sad”) or at most a tiny number of demonstrations (e.g. “here are two examples of people acting brave; please give a third example of bravery”) is often

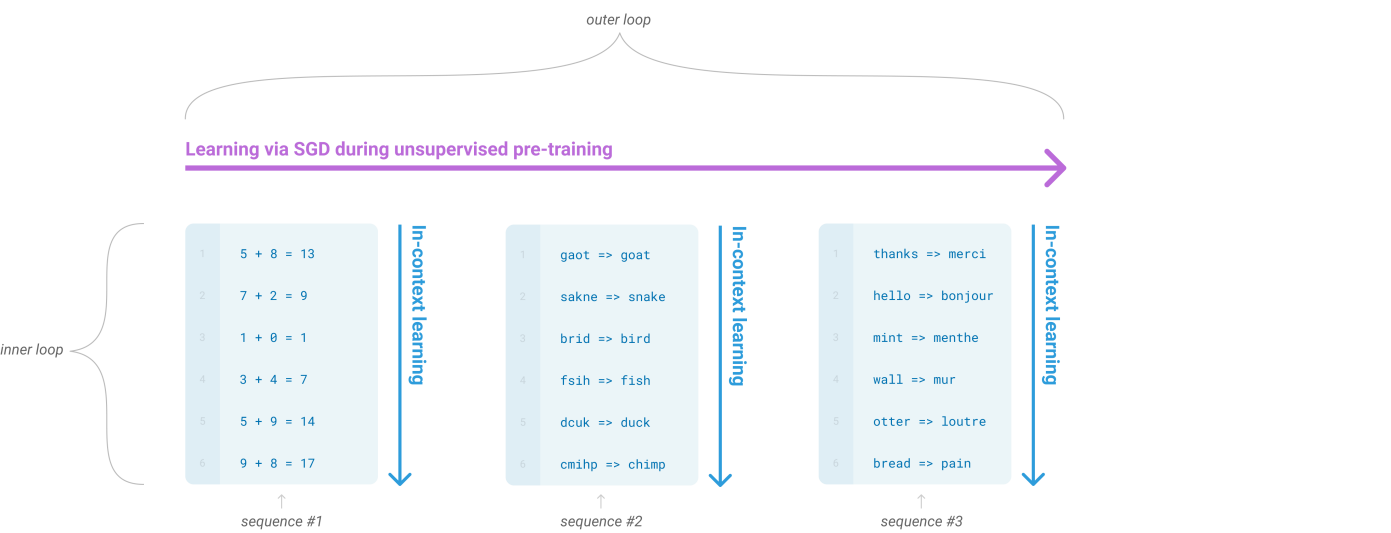


Figure 1.1: Language model meta-learning. During unsupervised pre-training, a language model develops a broad set of skills and pattern recognition abilities. It then uses these abilities at inference time to rapidly adapt to or recognize the desired task. We use the term “in-context learning” to describe the inner loop of this process, which occurs within the forward-pass upon each sequence. The sequences in this diagram are not intended to be representative of the data a model would see during pre-training, but are intended to show that there are sometimes repeated sub-tasks embedded within a single sequence.

1 引言

近年来，自然语言处理系统中出现了使用预训练语言表征的趋势，这些表征能够以愈发灵活且与具体任务无关的方式被用于后续的任务迁移。最初，人们利用词向量来学习单层表征 [MCCD13、PSM14]，并将它们输入到针对特定任务的架构中；随后，人们开始使用具有多层表征及上下文状态结构的RNN来构建更强大的表征 [DL15、MBXS17、PNZtY18]（尽管仍会被应用于特定任务的架构中）。而最近，则出现了直接对预训练的循环型或变换器型语言模型进行微调的做法 [VSP+17]，这样一来就完全不再需要特定的任务架构了 [RNSS18、DCLT18、HR18]。

最后这种范式在阅读理解任务、问答系统、文本蕴含关系识别等诸多具有挑战性的自然语言处理任务上取得了显著进展，而且随着新架构与算法的不断出现，这一领域仍在持续发展 [RSR+19、LOG+19、YDY+19、LCG+19]。不过，这种方法存在一个重大局限：尽管其架构与具体任务无关，但仍然需要针对特定任务的专用数据集以及相应的微调操作——通常要在包含数千到数十万个该任务专属示例的数据集上进行微调，才能在该任务上获得优异的性能。出于多种原因，消除这一限制是十分必要的。

首先，从实际应用的角度来看，每项新任务都需要大量带有标注的示例数据集，这一要求限制了语言模型的适用范围。实际上存在众多有用的语言处理任务，涵盖从语法纠错、生成抽象概念的示例，到对短篇故事进行评析等各种场景。对于其中的许多任务而言，要收集大规模的监督学习训练数据集十分困难，尤其是当必须为每一项新任务重复这一流程时。

其次，利用训练数据中虚假相关性所带来的可能性，会随着模型的表达能力以及训练数据概率分布的狭窄程度而不断提升。这种情况会给“先进行预训练再微调”的传统方法带来诸多问题——在这种方法中，模型被设计得足够庞大以便在预训练阶段吸收大量信息，但随后却要在极为狭窄的任务分布上进行微调。例如 [HLW+20] 指出，更大规模的模型并不一定能在分布之外的场景下展现出更好的泛化能力。有证据表明，由于这类模型对训练数据分布过于依赖，无法在其范围之外良好地泛化，因此采用这种模式所获得的性能往往较差[YdC+19, MPL19]。由此可见，即便经过微调的模型在某些特定基准测试上的性能看似达到了人类水平，这也可能高估了其在实际任务中的真实表现 [GSL+18、NK19]。

第三，人类在学习大多数语言任务时并不需要规模庞大的监督学习数据集——一段简短的自然语言指令（例如“请告诉我这句话描述的是快乐的事物还是悲伤的事物”），或是极少数量的示范示例（例如“这里有两个表现勇敢的人的示例，请再给出一个勇敢的示例”），往往就足以让人类具备相当程度的能力来完成新任务。

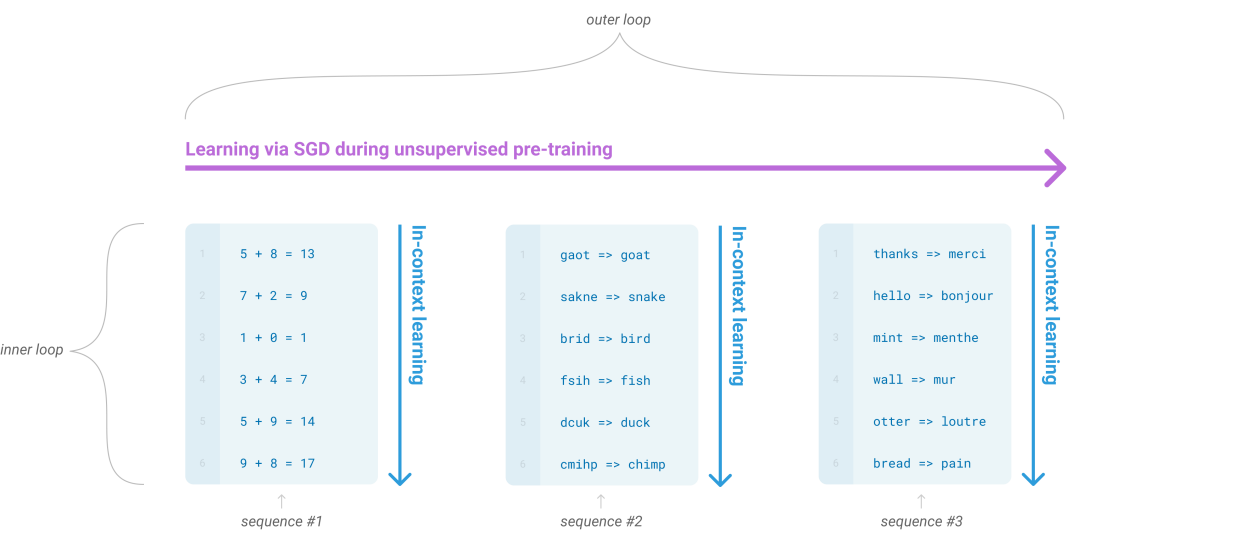


图1.1：语言模型的元学习过程。在无监督预训练阶段，语言模型会掌握一系列广泛的技能以及强大的模式识别能力。随后在推理阶段，它便能运用这些能力快速适应或识别特定的任务。我们用“上下文学习”这一术语来描述这一流程中的内部循环，即每次处理序列时都会发生的正向传递过程。此图中的序列并非旨在体现模型在预训练期间实际接触到的数据，而是为了说明在单个序列中有时会包含重复出现的子任务。

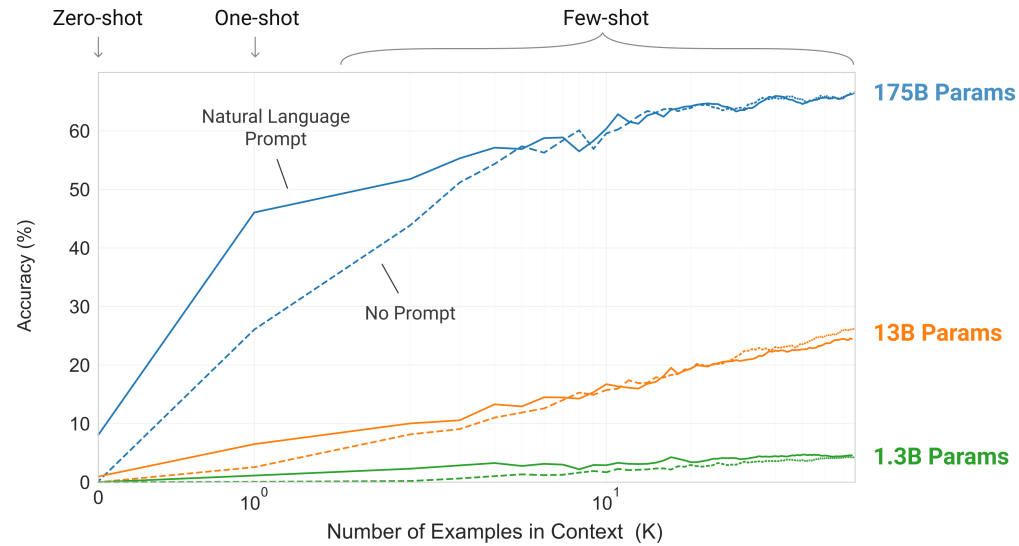


Figure 1.2: Larger models make increasingly efficient use of in-context information. We show in-context learning performance on a simple task requiring the model to remove random symbols from a word, both with and without a natural language task description (see Sec. 3.9.2). The steeper “in-context learning curves” for large models demonstrate improved ability to learn a task from contextual information. We see qualitatively similar behavior across a wide range of tasks.

sufficient to enable a human to perform a new task to at least a reasonable degree of competence. Aside from pointing to a conceptual limitation in our current NLP techniques, this adaptability has practical advantages – it allows humans to seamlessly mix together or switch between many tasks and skills, for example performing addition during a lengthy dialogue. To be broadly useful, we would someday like our NLP systems to have this same fluidity and generality.

One potential route towards addressing these issues is meta-learning¹ – which in the context of language models means the model develops a broad set of skills and pattern recognition abilities at training time, and then uses those abilities at inference time to rapidly adapt to or recognize the desired task (illustrated in Figure 1.1). Recent work [RWC⁺19] attempts to do this via what we call “in-context learning”, using the text input of a pretrained language model as a form of task specification: the model is conditioned on a natural language instruction and/or a few demonstrations of the task and is then expected to complete further instances of the task simply by predicting what comes next.

While it has shown some initial promise, this approach still achieves results far inferior to fine-tuning – for example [RWC⁺19] achieves only 4% on Natural Questions, and even its 55 F1 CoQa result is now more than 35 points behind the state of the art. Meta-learning clearly requires substantial improvement in order to be viable as a practical method of solving language tasks.

Another recent trend in language modeling may offer a way forward. In recent years the capacity of transformer language models has increased substantially, from 100 million parameters [RNSS18], to 300 million parameters [DCLT18], to 1.5 billion parameters [RWC⁺19], to 8 billion parameters [SPP⁺19], 11 billion parameters [RSR⁺19], and finally 17 billion parameters [Tur20]. Each increase has brought improvements in text synthesis and/or downstream NLP tasks, and there is evidence suggesting that log loss, which correlates well with many downstream tasks, follows a smooth trend of improvement with scale [KMH⁺20]. Since in-context learning involves absorbing many skills and tasks within the parameters of the model, it is plausible that in-context learning abilities might show similarly strong gains with scale.

¹In the context of language models this has sometimes been called “zero-shot transfer”, but this term is potentially ambiguous: the method is “zero-shot” in the sense that no gradient updates are performed, but it often involves providing inference-time demonstrations to the model, so is not truly learning from zero examples. To avoid this confusion, we use the term “meta-learning” to capture the inner-loop / outer-loop structure of the general method, and the term “in context-learning” to refer to the inner loop of meta-learning. We further specialize the description to “zero-shot”, “one-shot”, or “few-shot” depending on how many demonstrations are provided at inference time. These terms are intended to remain agnostic on the question of whether the model learns new tasks from scratch at inference time or simply recognizes patterns seen during training – this is an important issue which we discuss later in the paper, but “meta-learning” is intended to encompass both possibilities, and simply describes the inner-outer loop structure.

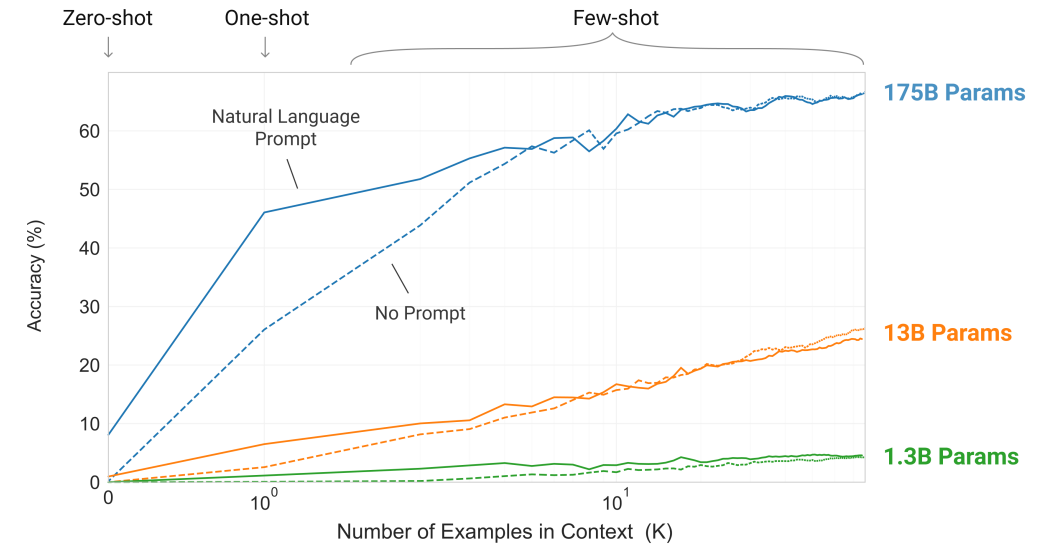


图1.2：更大规模的模型能够越来越高效地利用上下文信息。我们展示了在一种简单任务上的上下文学习性能，该任务要求模型从单词中删除随机符号，同时分析了有自然语言任务描述与没有该描述两种情况下的表现（详见第3.9.2节）。大型模型更为陡峭的“上下文学习曲线”表明它们具备更强的能力，能够从上下文信息中学习任务。我们在多种不同类型的任务中观察到了类似的表现特征。

除了反映出当前自然语言处理技术存在的概念性局限之外，这种适应性还具有实际优势——它使人类能够无缝地结合多种任务与技能，或在不同任务之间切换，比如在长时间的对话中还能进行加法运算。若要实现更广泛的应用，我们希望未来的自然语言处理系统也能具备同样的灵活性与通用性。

解决这些问题的一个潜在途径是元学习¹——在语言模型的语境下，这意味着模型在训练阶段会掌握一系列广泛的技能和模式识别能力，随后在推理阶段运用这些能力快速适应或识别所需处理的任务（如图1.1所示）。最近的一些研究 [RWC⁺19]试图通过我们称之为“上下文学习”的方法来实现这一目标，即利用预训练语言模型的文本输入作为任务定义的依据：模型会基于自然语言指令和/或少量的任务示例进行训练，之后只需通过预测后续内容即可完成更多同类任务。

虽然这种方法最初展现出了些许潜力，但其实际得到的结果仍远逊于微调——例如[RWC⁺19]在自然问题任务上的F1 CoQa评分仅有4%，即便其55分的F1 CoQa成绩，也落后于当前最佳水平超过35分。显然，若要使其成为解决语言任务的实用方法，元学习还需进行大幅改进。

在语言建模领域，另一个近期的发展趋势或许能为行业指明前进方向。近年来，变换器语言模型的容量大幅提升，其参数数量从 [DCLT18], 1亿参数[RWC⁺19]增长到 [RNSS18], 3亿参数[DCLT18], , 再到 [17.5亿参数 [RWC⁺19] SPP⁺19], 以及 [RSR⁺19][Tur20], 最终达到1750亿参数。这样的提升不仅推动了文本生成技术的发展，也为各类下游自然语言处理任务带来了显著改进。有研究证据表明，与许多下游任务表现高度相关的对数损失值，也会随着模型参数规模的扩大而呈现平稳的下降趋势[KMH⁺20]。由于上下文学习依赖于在模型参数中嵌入多种技能与任务能力，因此可以合理推测，其能力同样会随着模型规模的增大而实现强劲的提升。

¹在语言模型的语境中，这种现象有时被称为“零样本迁移”，但这一术语可能存在歧义：该方法之所以是“零样本”的，是因为无需进行梯度更新，但它往往需要向模型提供推理阶段的示例，因此并非真正从零个示例中学习。为避免这种混淆，我们使用“元学习”这一术语来指代该通用方法中的内循环/外循环结构，而用“上下文学习”来指代元学习中的内循环。此外，我们会根据推理阶段提供的示例数量，进一步将这类学习方式具体划分为“零样本学习”、“单样本学习”或“少样本学习”。这些术语旨在不涉及模型是在推理阶段从零开始学习新任务，还是仅仅识别训练期间见过的模式这一问题——这是我们在论文后续会讨论的一个重要议题；而“元学习”则意在涵盖这两种可能性，仅用于描述其内外循环的结构。

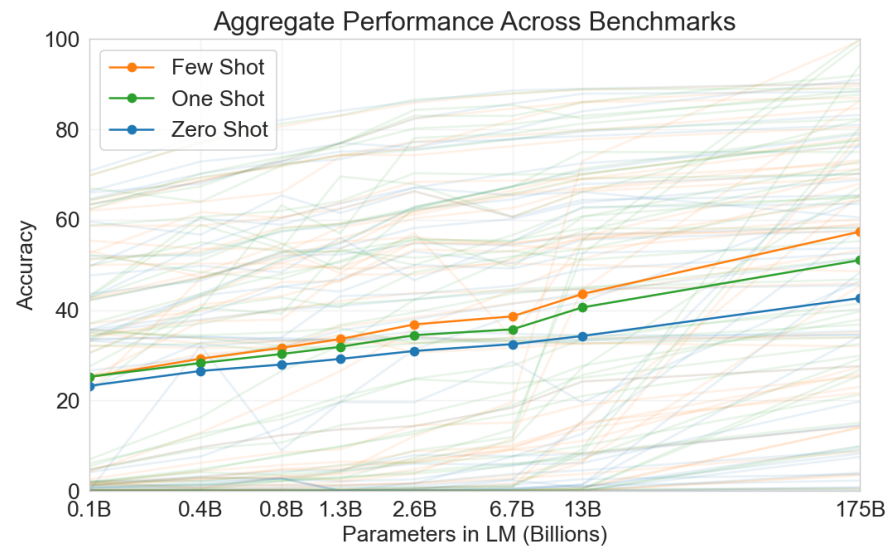


Figure 1.3: Aggregate performance for all 42 accuracy-denominated benchmarks While zero-shot performance improves steadily with model size, few-shot performance increases more rapidly, demonstrating that larger models are more proficient at in-context learning. See Figure 3.8 for a more detailed analysis on SuperGLUE, a standard NLP benchmark suite.

In this paper, we test this hypothesis by training a 175 billion parameter autoregressive language model, which we call GPT-3, and measuring its in-context learning abilities. Specifically, we evaluate GPT-3 on over two dozen NLP datasets, as well as several novel tasks designed to test rapid adaptation to tasks unlikely to be directly contained in the training set. For each task, we evaluate GPT-3 under 3 conditions: (a) “few-shot learning”, or in-context learning where we allow as many demonstrations as will fit into the model’s context window (typically 10 to 100), (b) “one-shot learning”, where we allow only one demonstration, and (c) “zero-shot” learning, where no demonstrations are allowed and only an instruction in natural language is given to the model. GPT-3 could also in principle be evaluated in the traditional fine-tuning setting, but we leave this to future work.

Figure 1.2 illustrates the conditions we study, and shows few-shot learning of a simple task requiring the model to remove extraneous symbols from a word. Model performance improves with the addition of a natural language task description, and with the number of examples in the model’s context, K . Few-shot learning also improves dramatically with model size. Though the results in this case are particularly striking, the general trends with both model size and number of examples in-context hold for most tasks we study. We emphasize that these “learning” curves involve no gradient updates or fine-tuning, just increasing numbers of demonstrations given as conditioning.

Broadly, on NLP tasks GPT-3 achieves promising results in the zero-shot and one-shot settings, and in the the few-shot setting is sometimes competitive with or even occasionally surpasses state-of-the-art (despite state-of-the-art being held by fine-tuned models). For example, GPT-3 achieves 81.5 F1 on CoQA in the zero-shot setting, 84.0 F1 on CoQA in the one-shot setting, 85.0 F1 in the few-shot setting. Similarly, GPT-3 achieves 64.3% accuracy on TriviaQA in the zero-shot setting, 68.0% in the one-shot setting, and 71.2% in the few-shot setting, the last of which is state-of-the-art relative to fine-tuned models operating in the same closed-book setting.

GPT-3 also displays one-shot and few-shot proficiency at tasks designed to test rapid adaption or on-the-fly reasoning, which include unscrambling words, performing arithmetic, and using novel words in a sentence after seeing them defined only once. We also show that in the few-shot setting, GPT-3 can generate synthetic news articles which human evaluators have difficulty distinguishing from human-generated articles.

At the same time, we also find some tasks on which few-shot performance struggles, even at the scale of GPT-3. This includes natural language inference tasks like the ANLI dataset, and some reading comprehension datasets like RACE or QuAC. By presenting a broad characterization of GPT-3’s strengths and weaknesses, including these limitations, we hope to stimulate study of few-shot learning in language models and draw attention to where progress is most needed.

A heuristic sense of the overall results can be seen in Figure 1.3, which aggregates the various tasks (though it should not be seen as a rigorous or meaningful benchmark in itself).

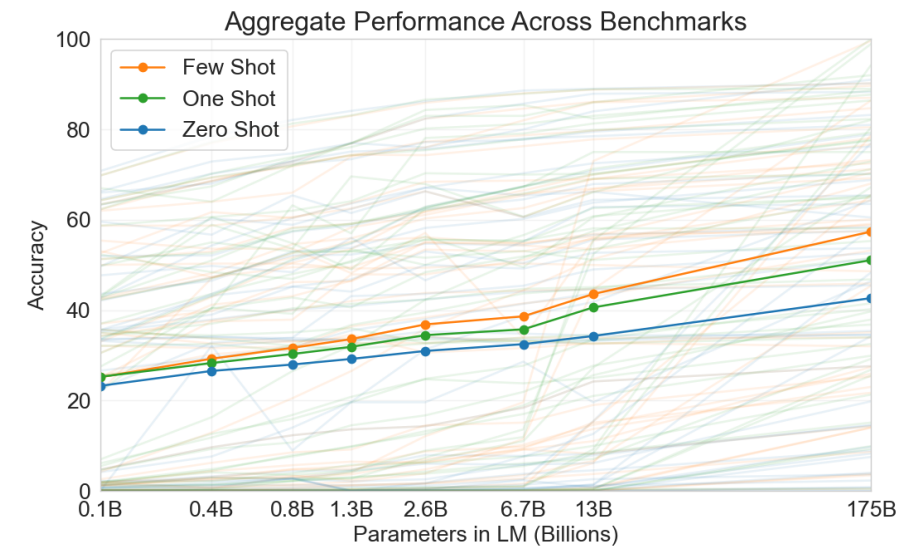


图1.3：所有42项以准确率为评估指标的基准测试的总体性能表现虽然零样本性能会随着模型规模的增长而稳步提升，但少样本性能的提升速度更快，这表明更大规模的模型在上下文学习方面更具优势。关于标准自然语言处理基准测试套件SuperGLUE的更详细分析，请参见图3.8。

在本文中，我们通过训练一个拥有1750亿参数的自回归语言模型——即我们所称的GPT-3，并测试它的上下文学习能力，以此来验证这一假设。具体而言，我们在二十多个自然语言处理数据集以及几项专为检测模型对训练集中未出现过的任务快速适应能力而设计的新型任务上对GPT-3进行了评估。对于每一项任务，我们都在三种不同的条件下对GPT-3进行测试：(a) “少样本学习”，也就是上下文学习，此时我们会提供尽可能多的示例，但其数量必须不超过模型的上下文窗口容量（通常为10到100个）；(b) “单样本学习”，仅提供一个示例；(c) “零样本学习”，不提供任何示例，仅向模型给出自然语言形式的指令。从原理上讲，GPT-3也可以在传统的微调环境中进行评估，但我们决定将此项工作留待未来的研究去完成。

图1.2 展示了我们研究的条件设置，其中演示了模型在处理一项简单任务时的少样本学习过程——该任务要求模型从单词中去除冗余符号。随着自然语言任务描述的加入以及模型上下文中示例数量的增加，模型的性能也会随之提升 K 。此外，模型规模越大，少样本学习的性能提升也极为显著。尽管此案例中的结果尤为突出，但就我们研究的大多数任务而言，模型规模和上下文中的示例数量所带来的整体趋势都是成立的。需要强调的是，这些“学习”曲线并未涉及任何梯度更新或微调操作，仅仅是增加了作为条件样本的演示次数。

总体而言，在自然语言处理任务上，GPT-3在零样本和单样本场景下均取得了令人满意的结果；而在少样本场景中，它的表现有时甚至能与当前最佳水平相媲美，偶尔还能超越后者（尽管当前最佳水平通常由经过微调的模型保持）。例如，在CoQA任务上，GPT-3在零样本场景下的F1分数为81.5，在单样本场景下为84.0，在少样本场景下则为85.0。同样，在TriviaQA任务上，它的零样本准确率为64.3%，单样本准确率为68.0%，而少样本准确率则达到了71.2%，这一数值相对于在相同封闭式测试环境下运行的微调模型而言已处于当前最佳水平。

GPT-3在那些旨在测试快速适应能力或即时推理能力的任务中也展现出了单样本学习和少样本学习的优异表现，这类任务包括重新排列单词、进行算术运算，以及仅在看到一次定义后就在句子中使用新单词。此外我们还发现，在少样本场景下，GPT-3能够生成合成新闻文章，以至于人类评估者很难将其与真正由人类撰写的文章区分开来。

与此同时，我们同样发现有一些任务即便在GPT-3这样的强大模型面前，其少样本性能也会出现不佳情况。这些任务涵盖诸如ANLI数据集所代表的自然语言推理任务，以及RACE和QuAC等阅读理解数据集集中的相关任务。通过全面分析GPT-3的优势与劣势，包括这些局限性在内，我们希望推动对语言模型中少样本学习方法的深入研究，并引起人们对于那些最亟需取得进展领域的关注。

整体结果的概览可通过图 1.3来了解，该图汇总了各项任务的结果（不过它本身不应被视为严谨或有意义的基准测试）。

We also undertake a systematic study of “data contamination” – a growing problem when training high capacity models on datasets such as Common Crawl, which can potentially include content from test datasets simply because such content often exists on the web. In this paper we develop systematic tools to measure data contamination and quantify its distorting effects. Although we find that data contamination has a minimal effect on GPT-3’s performance on most datasets, we do identify a few datasets where it could be inflating results, and we either do not report results on these datasets or we note them with an asterisk, depending on the severity.

In addition to all the above, we also train a series of smaller models (ranging from 125 million parameters to 13 billion parameters) in order to compare their performance to GPT-3 in the zero, one and few-shot settings. Broadly, for most tasks we find relatively smooth scaling with model capacity in all three settings; one notable pattern is that the gap between zero-, one-, and few-shot performance often grows with model capacity, perhaps suggesting that larger models are more proficient meta-learners.

Finally, given the broad spectrum of capabilities displayed by GPT-3, we discuss concerns about bias, fairness, and broader societal impacts, and attempt a preliminary analysis of GPT-3’s characteristics in this regard.

The remainder of this paper is organized as follows. In Section 2, we describe our approach and methods for training GPT-3 and evaluating it. Section 3 presents results on the full range of tasks in the zero-, one- and few-shot settings. Section 4 addresses questions of data contamination (train-test overlap). Section 5 discusses limitations of GPT-3. Section 6 discusses broader impacts. Section 7 reviews related work and Section 8 concludes.

2 Approach

Our basic pre-training approach, including model, data, and training, is similar to the process described in [RWC⁺19], with relatively straightforward scaling up of the model size, dataset size and diversity, and length of training. Our use of in-context learning is also similar to [RWC⁺19], but in this work we systematically explore different settings for learning within the context. Therefore, we start this section by explicitly defining and contrasting the different settings that we will be evaluating GPT-3 on or could in principle evaluate GPT-3 on. These settings can be seen as lying on a spectrum of how much task-specific data they tend to rely on. Specifically, we can identify at least four points on this spectrum (see Figure 2.1 for an illustration):

- **Fine-Tuning (FT)** has been the most common approach in recent years, and involves updating the weights of a pre-trained model by training on a supervised dataset specific to the desired task. Typically thousands to hundreds of thousands of labeled examples are used. The main advantage of fine-tuning is strong performance on many benchmarks. The main disadvantages are the need for a new large dataset for every task, the potential for poor generalization out-of-distribution [MPL19], and the potential to exploit spurious features of the training data [GSL⁺18, NK19], potentially resulting in an unfair comparison with human performance. In this work we do not fine-tune GPT-3 because our focus is on task-agnostic performance, but GPT-3 can be fine-tuned in principle and this is a promising direction for future work.
- **Few-Shot (FS)** is the term we will use in this work to refer to the setting where the model is given a few demonstrations of the task at inference time as conditioning [RWC⁺19], but no weight updates are allowed. As shown in Figure 2.1, for a typical dataset an example has a context and a desired completion (for example an English sentence and the French translation), and few-shot works by giving K examples of context and completion, and then one final example of context, with the model expected to provide the completion. We typically set K in the range of 10 to 100 as this is how many examples can fit in the model’s context window ($n_{\text{ctx}} = 2048$). The main advantages of few-shot are a major reduction in the need for task-specific data and reduced potential to learn an overly narrow distribution from a large but narrow fine-tuning dataset. The main disadvantage is that results from this method have so far been much worse than state-of-the-art fine-tuned models. Also, a small amount of task specific data is still required. As indicated by the name, few-shot learning as described here for language models is related to few-shot learning as used in other contexts in ML [HYC01, VBL⁺16] – both involve learning based on a broad distribution of tasks (in this case implicit in the pre-training data) and then rapidly adapting to a new task.
- **One-Shot (1S)** is the same as few-shot except that only one demonstration is allowed, in addition to a natural language description of the task, as shown in Figure 1. The reason to distinguish one-shot from few-shot and zero-shot (below) is that it most closely matches the way in which some tasks are communicated to humans. For example, when asking humans to generate a dataset on a human worker service (for example Mechanical Turk), it is common to give one demonstration of the task. By contrast it is sometimes difficult to communicate the content or format of a task if no examples are given.

我们还对‘数据污染’问题进行了系统研究——在基于Common Crawl数据集等训练高容量模型时，这一问题日益严重，因为网络上往往存在测试数据集中的内容，这些内容也可能被误纳入训练数据。在本文中，我们开发了用于检测数据污染并量化其扭曲效应的系统性工具。虽然我们发现，在大多数数据集上数据污染对GPT-3的性能影响很小，但我们确实发现了一些数据集可能存在这种情况，从而导致结果被高估；针对这类数据集，我们会根据问题的严重程度，要么不报告相关结果，要么用星号加以标注。

除上述内容之外，我们还训练了一系列规模较小的模型（参数量从1.25亿到130亿不等），旨在在零样本、单样本以及少样本场景下将它们的性能与GPT-3进行对比。总体而言，在这三种场景下，对于大多数任务而言，模型的性能随参数数量的增加而呈现相对平稳的提升趋势；不过有一个值得注意的现象是，零样本、单样本和少样本场景下的性能差距往往会随着模型参数数量的增大而扩大，这或许表明更大规模的模型具备更强的元学习能力。

最后，鉴于GPT-3展现出的极为广泛的能力，我们探讨了与之相关的统计偏差、公平性以及更广泛的社会影响等问题，并试图对GPT-3在这方面的特性进行初步分析。

本文的其余结构安排如下。在第2节中，我们将介绍用于训练GPT-3并对其进行评估的方法与技术。第3节则会展示在零样本、单样本及少样本场景下针对各类任务所获得的结果。第4节则探讨数据污染（即训练集与测试集存在重叠）这一问题。第5节讨论了GPT-3的局限性。第6节则着眼于更广泛的影响。第7节回顾了相关研究，最后第8节会对全文内容进行总结。

2 Approach

我们所采用的常规预训练方法，包括模型、数据及训练流程，与 [RWC⁺19]中描述的过程类似，只需相应地提升模型规模、数据集规模与多样性以及训练时长即可。此外，我们所使用的上下文学习方法也与 [RWC⁺19]中的方法相似，但在本研究中我们会系统地探索多种不同的上下文学习设置。因此，在本章开头，我们会明确界定并对比那些我们将用于评估GPT-3，或从原理上讲可用于评估GPT-3的不同设置。这些设置可视为一个连续谱，反映它们在依赖特定任务数据程度上的差异。具体而言，我们可以在该谱系上划分出至少四个节点（详见图2.1中的示例）。

- **微调**是近年来最常用的方法，其核心是通过在针对特定任务的监督学习数据集上进行训练来更新预训练模型的权重。通常会使用数千到数十万个带有标签的示例。微调的主要优势在于能在众多基准测试中取得优异的性能，但缺点也是显而易见的：每项任务都需要新的大型数据集，存在在概率分布偏离正常范围时泛化能力较差的风险 [MPL19]，，同时还可能利用训练数据中存在的虚假特征 [GSL⁺18, NK19]，这进而可能导致与人类表现水平的比较出现不公平的结果。在本研究中，我们并未对GPT-3进行微调，因为我们的关注点在于与任务无关的性能表现；不过从原理上讲GPT-3是可以被微调的，而这也是未来研究的一个极具前景的方向。

- **少样本学习**是指在本研究中用以描述这样一种场景的术语：在推理阶段，模型会作为条件样本获得该任务的少量演示示例 [RWC⁺19]，但此时不允许进行任何重力参数的更新。如图2.1所示，对于典型的数据集而言，每个示例都包含上下文以及期望的补全结果（例如一句英语句子及其法语翻译），而少样本学习方法则是通过提供 K 上下文与补全结果的示例，再额外给出一个仅包含上下文的示例，要求模型输出对应的补全结果。我们通常将 K 这一数值设定在10到100之间，因为这正是能够放入模型上下文窗口中的示例数量（ $n_{\text{上下文窗口}} = 2048$ ）。少样本学习的主要优势在于极大降低了对特定任务数据的需求，同时也能避免从规模较大但范围狭窄的微调数据集中学习出过于局限的概率分布。不过它的缺点也很明显，目前采用这种方法得到的结果远不如当前最佳水平的最先进的微调模型，而且仍然需要一定量的特定任务数据。顾名思义，此处针对语言模型所描述的少样本学习，与机器学习其他领域中使用的少样本学习 [HYC01研究, VBL⁺16] 的做法类似——二者都是先基于广泛的任务概率分布（在本例中体现为预训练数据中的内容）进行学习，然后再快速适应新的任务。

- **单样本学习（1S）**与少样本学习的本质相同，不同之处在于除了任务的自然语言描述外，只允许提供一次示范，如图1所示。将单样本学习与零样本学习区分开来的原因在于，前者更贴近某些任务向人类传达时的实际方式。例如，在要求人类在Mechanical Turk平台等人工服务上构建数据集时，通常只会给出一次任务示范。相反，如果不对任务内容或格式进行示例说明，有时甚至很难让人理解。

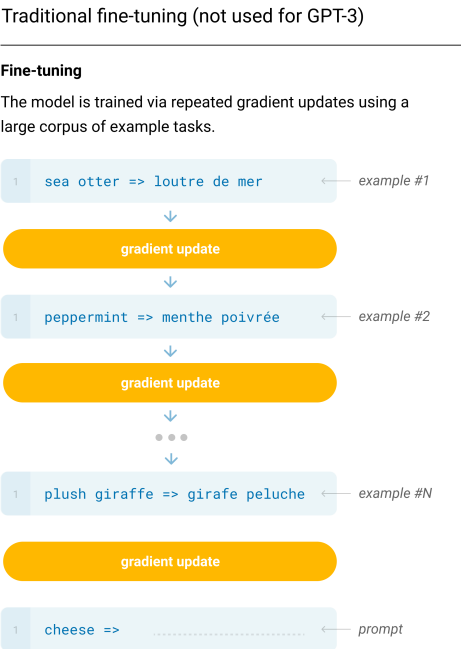
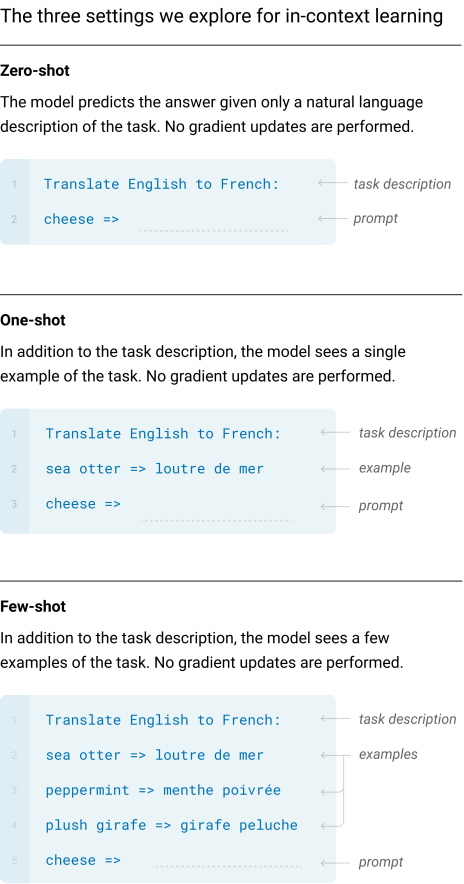
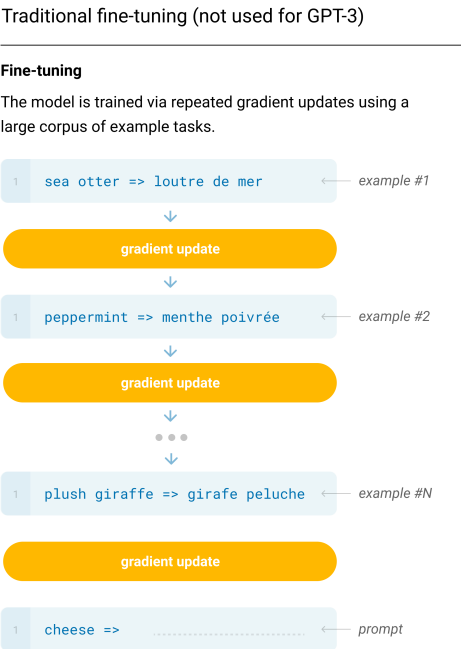
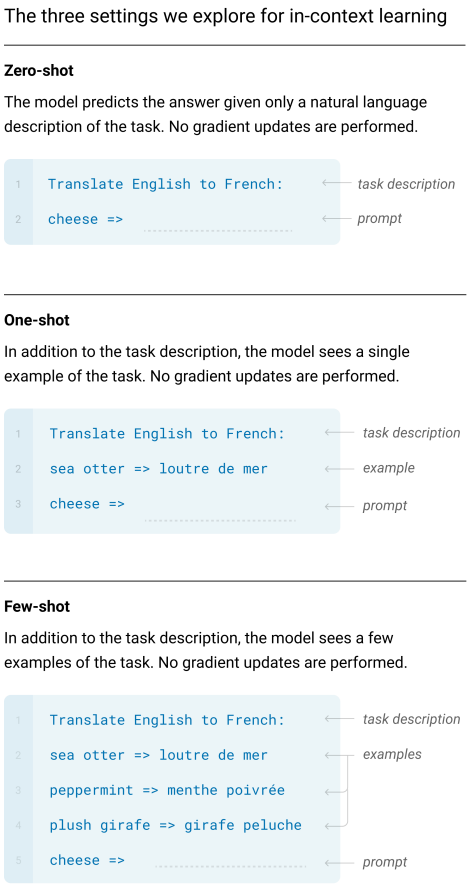


Figure 2.1: Zero-shot, one-shot and few-shot, contrasted with traditional fine-tuning. The panels above show four methods for performing a task with a language model – fine-tuning is the traditional method, whereas zero-, one-, and few-shot, which we study in this work, require the model to perform the task with only forward passes at test time. We typically present the model with a few dozen examples in the few shot setting. Exact phrasings for all task descriptions, examples and prompts can be found in Appendix G.

图2.1：零样本学习、单样本学习和少样本学习与传统微调方法的对比。上方的各个面板展示了利用语言模型完成任务的四种方法——微调属于传统方法，而我们在本研究中探讨的零样本学习、单样本学习和少样本学习则要求模型在测试时仅通过前向传播即可完成任务。在少样本学习场景中，我们通常会向模型提供几十个示例。所有任务描述、示例及提示语的完整表述可见附录GG。

- **Zero-Shot (0S)** is the same as one-shot except that no demonstrations are allowed, and the model is only given a natural language instruction describing the task. This method provides maximum convenience, potential for robustness, and avoidance of spurious correlations (unless they occur very broadly across the large corpus of pre-training data), but is also the most challenging setting. In some cases it may even be difficult for humans to understand the format of the task without prior examples, so this setting is in some cases “unfairly hard”. For example, if someone is asked to “make a table of world records for the 200m dash”, this request can be ambiguous, as it may not be clear exactly what format the table should have or what should be included (and even with careful clarification, understanding precisely what is desired can be difficult). Nevertheless, for at least some settings zero-shot is closest to how humans perform tasks – for example, in the translation example in Figure 2.1, a human would likely know what to do from just the text instruction.

Figure 2.1 shows the four methods using the example of translating English to French. In this paper we focus on zero-shot, one-shot and few-shot, with the aim of comparing them not as competing alternatives, but as different problem settings which offer a varying trade-off between performance on specific benchmarks and sample efficiency. We especially highlight the few-shot results as many of them are only slightly behind state-of-the-art fine-tuned models. Ultimately, however, one-shot, or even sometimes zero-shot, seem like the fairest comparisons to human performance, and are important targets for future work.

Sections 2.1-2.3 below give details on our models, training data, and training process respectively. Section 2.4 discusses the details of how we do few-shot, one-shot, and zero-shot evaluations.

- **零样本学习（0S）**与单样本学习类似，区别在于不允许提供任何示范，模型仅会收到描述任务的自然语言指令。这种方法虽然能带来最大的便利性、更高的稳定性，并有助于避免虚假相关性（除非这种相关性在庞大的预训练数据语料库中普遍存在），但同时也最具挑战性。在某些情况下，即便没有先前的示例，人类可能都难以理解任务的格式，因此这类任务设置有时会被视为“难度过高”。例如，如果要求某人‘制作一份200米短跑的世界纪录表格’，这个请求可能会产生歧义，因为不清楚表格的具体格式以及应包含哪些内容（即便经过仔细说明，要准确理解需求依然颇具难度）。尽管如此，至少在某些场景下，零样本学习最接近人类执行任务的方式——比如在图2.1中的翻译示例中，仅凭文字指令，人类很可能就能知道该怎么做。

图2.1 以将英语翻译成法语作为示例，展示了这四种方法。在本文中，我们重点研究零样本学习、单样本学习和少样本学习，旨在将它们视为不同的问题场景而非相互竞争的替代方案，因为这些场景在特定基准测试上的性能与样本效率之间存在着不同的权衡关系。我们尤其着重介绍了少样本学习的结果，因为其中许多结果的性能仅略逊于当前最佳水平的微调模型。不过从最终角度来看，单样本学习，甚至有时是零样本学习，似乎才是与人类表现水平最接近的对比对象，也因此成为未来研究的重要方向。

章节2.1—2.3 以下分别详细介绍我们的模型、训练数据以及训练流程。章节2.4则阐述了我们如何进行少样本学习、单样本学习和零样本学习的评估细节。

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Table 2.1: Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

2.1 Model and Architectures

We use the same model and architecture as GPT-2 [RWC⁺19], including the modified initialization, pre-normalization, and reversible tokenization described therein, with the exception that we use alternating dense and locally banded sparse attention patterns in the layers of the transformer, similar to the Sparse Transformer [CGRS19]. To study the dependence of ML performance on model size, we train 8 different sizes of model, ranging over three orders of magnitude from 125 million parameters to 175 billion parameters, with the last being the model we call GPT-3. Previous work [KMH⁺20] suggests that with enough training data, scaling of validation loss should be approximately a smooth power law as a function of size; training models of many different sizes allows us to test this hypothesis both for validation loss and for downstream language tasks.

Table 2.1 shows the sizes and architectures of our 8 models. Here n_{params} is the total number of trainable parameters, n_{layers} is the total number of layers, d_{model} is the number of units in each bottleneck layer (we always have the feedforward layer four times the size of the bottleneck layer, $d_{\text{ff}} = 4 * d_{\text{model}}$), and d_{head} is the dimension of each attention head. All models use a context window of $n_{\text{ctx}} = 2048$ tokens. We partition the model across GPUs along both the depth and width dimension in order to minimize data-transfer between nodes. The precise architectural parameters for each model are chosen based on computational efficiency and load-balancing in the layout of models across GPU’s. Previous work [KMH⁺20] suggests that validation loss is not strongly sensitive to these parameters within a reasonably broad range.

2.2 Training Dataset

Datasets for language models have rapidly expanded, culminating in the Common Crawl dataset² [RSR⁺19] constituting nearly a trillion words. This size of dataset is sufficient to train our largest models without ever updating on the same sequence twice. However, we have found that unfiltered or lightly filtered versions of Common Crawl tend to have lower quality than more curated datasets. Therefore, we took 3 steps to improve the average quality of our datasets: (1) we downloaded and filtered a version of CommonCrawl based on similarity to a range of high-quality reference corpora, (2) we performed fuzzy deduplication at the document level, within and across datasets, to prevent redundancy and preserve the integrity of our held-out validation set as an accurate measure of overfitting, and (3) we also added known high-quality reference corpora to the training mix to augment CommonCrawl and increase its diversity.

Details of the first two points (processing of Common Crawl) are described in Appendix A. For the third, we added several curated high-quality datasets, including an expanded version of the WebText dataset [RWC⁺19], collected by scraping links over a longer period of time, and first described in [KMH⁺20], two internet-based books corpora (Books1 and Books2) and English-language Wikipedia.

Table 2.2 shows the final mixture of datasets that we used in training. The CommonCrawl data was downloaded from 41 shards of monthly CommonCrawl covering 2016 to 2019, constituting 45TB of compressed plaintext before filtering and 570GB after filtering, roughly equivalent to 400 billion byte-pair-encoded tokens. Note that during training, datasets are not sampled in proportion to their size, but rather datasets we view as higher-quality are sampled more frequently, such that CommonCrawl and Books2 datasets are sampled less than once during training, but the other datasets are sampled 2-3 times. This essentially accepts a small amount of overfitting in exchange for higher quality training data.

²<https://commoncrawl.org/the-data/>

模型名称	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	批量大小	学习率
GPT-3小型版	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3中型版	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3大型版	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3超大型版	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 27亿版本	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 67亿版本	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 130亿版本	130亿参数	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B版本或简称GPT-3	1750亿	96	12288	96	128	3.2M	0.6×10^{-4}

表2.1: 我们所训练的模型的规模、架构以及学习超参数（以标记数表示的批量大小和学习率）。所有模型均经过总计3000亿个标记的训练。

2.1 模型与架构

我们所使用的模型与架构与GPT-2 [RWC⁺19]相同，也采用了其中提到的修改后的初始化机制、预归一化处理以及可逆分词机制，不同之处在于我们在Transformer的各层中使用了密集注意力模式与局部带状稀疏注意力模式交替出现的结构，这一设计类似Sparse Transformer [CGRS19]。为研究机器学习性能与模型规模之间的关联，我们训练了8种不同规模的模型，其参数量跨度达到了三个数量级，从1.25亿参数到1750亿参数不等，其中参数量最大的那个模型就是我们所说的GPT-3。此前的研究 [KMH⁺20]指出，只要拥有足够的训练数据，验证损失的下降趋势应当大致遵循与模型规模相关的平滑幂律关系；通过训练多种不同规模的模型，我们能够同时针对验证损失以及各类下游语言任务来验证这一假设。

表格2.1展示了我们这8个模型的规模与架构。此处 n_{params} 为可训练参数的总计数量， n_{layers} 为层的总数， d_{model} 为每个瓶颈层中的单元数量（我们的前馈层规模始终是瓶颈层的四倍， $d_{\text{ff}} = 4 * d_{\text{model}}$ ），而 d_{head} 则为每个注意力头的维度。所有模型都使用 $n_{\text{ctx}} = 2048$ 标记构成的上下文窗口。为了尽可能减少节点间的数据传输，我们会沿着深度和宽度两个维度将模型分布在不同的GPU上。每个模型的具体架构参数都是根据计算效率以及GPU间模型布局的负载均衡情况来确定的。先前的研究 [KMH⁺20] 表明，在相当宽泛的范围内，验证损失对这些参数并不十分敏感。

2.2 训练数据集

用于语言模型的数据集规模正在迅速扩大，其中CommonCrawl数据集更是达到了惊人的规模² [RSR⁺19]，其内容量接近万亿单词。如此庞大的数据集足以用来训练我们最大的语言模型，而且无需重复使用同一序列进行训练。不过我们发现，未经过滤或仅经过简单过滤的Common Crawl版本，其质量往往低于那些经过精心筛选的数据集。因此，我们采取了三项措施来提升数据集的平均质量：第一，根据与多组高质量参考语料库的相似度下载并对Common Crawl进行了过滤处理；第二，在数据集内部以及不同数据集之间对文档层面进行模糊去重处理，以避免冗余信息，同时确保保留的验证集的完整性，从而更准确地判断是否存在过拟合现象；第三，还在训练数据组合中加入了已知的高质量参考语料库，以此补充Common Crawl的内容并提升其多样性。

前两点（Common Crawl的处理方式）的详细内容可见附录 A。至于第三点，我们在训练数据组合中加入了多个经过精心筛选的高质量数据集，其中包括通过长时间爬取链接收集并已在 [KMH⁺20]中首次介绍过的扩展版WebText数据集 [RWC⁺19]，以及两个基于互联网的书籍语料库（Books1和Books2）和英文版维基百科。

表格 2.2 展示了我们在训练过程中所使用的各类数据集的最终组合情况。这些CommonCrawl数据是从2016年至2019年间的每月数据集中提取的41个分片下载而来的，过滤前的压缩后纯文本总量为45TB，过滤后的总量则为570GB，大致相当于4000亿个经过字节对编码的令牌。需要说明的是，在训练过程中并不会按照数据集的大小比例进行采样，而是会更频繁地选取那些我们认为质量更高的数据集；因此CommonCrawl和Books2数据集在训练过程中的采样次数不足一次，而其他数据集的采样次数则在2到3次之间。这样做本质上是愿意承受一定程度的过拟合，以此换取更优质的训练数据。

²<https://commoncrawl.org/the-data/>

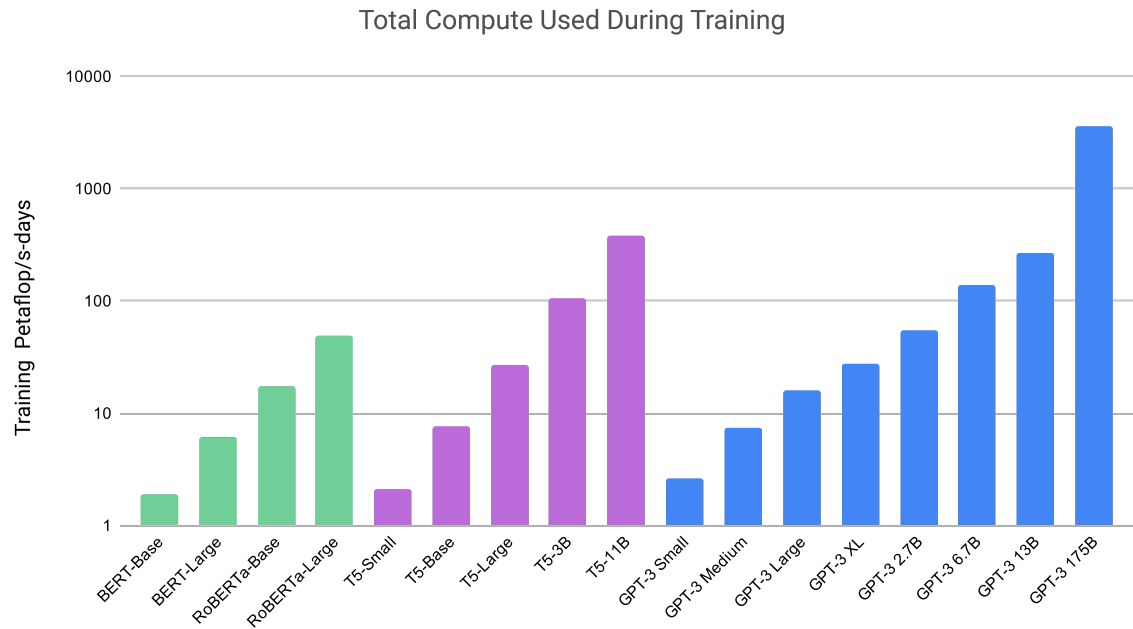


Figure 2.2: Total compute used during training. Based on the analysis in Scaling Laws For Neural Language Models [KMH⁺20] we train much larger models on many fewer tokens than is typical. As a consequence, although GPT-3 3B is almost 10x larger than RoBERTa-Large (355M params), both models took roughly 50 petaflop/s-days of compute during pre-training. Methodology for these calculations can be found in Appendix D.

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

Table 2.2: Datasets used to train GPT-3. “Weight in training mix” refers to the fraction of examples during training that are drawn from a given dataset, which we intentionally do not make proportional to the size of the dataset. As a result, when we train for 300 billion tokens, some datasets are seen up to 3.4 times during training while other datasets are seen less than once.

A major methodological concern with language models pretrained on a broad swath of internet data, particularly large models with the capacity to memorize vast amounts of content, is potential contamination of downstream tasks by having their test or development sets inadvertently seen during pre-training. To reduce such contamination, we searched for and attempted to remove any overlaps with the development and test sets of all benchmarks studied in this paper. Unfortunately, a bug in the filtering caused us to ignore some overlaps, and due to the cost of training it was not feasible to retrain the model. In Section 4 we characterize the impact of the remaining overlaps, and in future work we will more aggressively remove data contamination.

2.3 Training Process

As found in [KMH⁺20, MKAT18], larger models can typically use a larger batch size, but require a smaller learning rate. We measure the gradient noise scale during training and use it to guide our choice of batch size [MKAT18]. Table 2.1 shows the parameter settings we used. To train the larger models without running out of memory, we use a mixture of model parallelism within each matrix multiply and model parallelism across the layers of the network. All models were trained on V100 GPU’s on part of a high-bandwidth cluster provided by Microsoft. Details of the training process and hyperparameter settings are described in Appendix B.

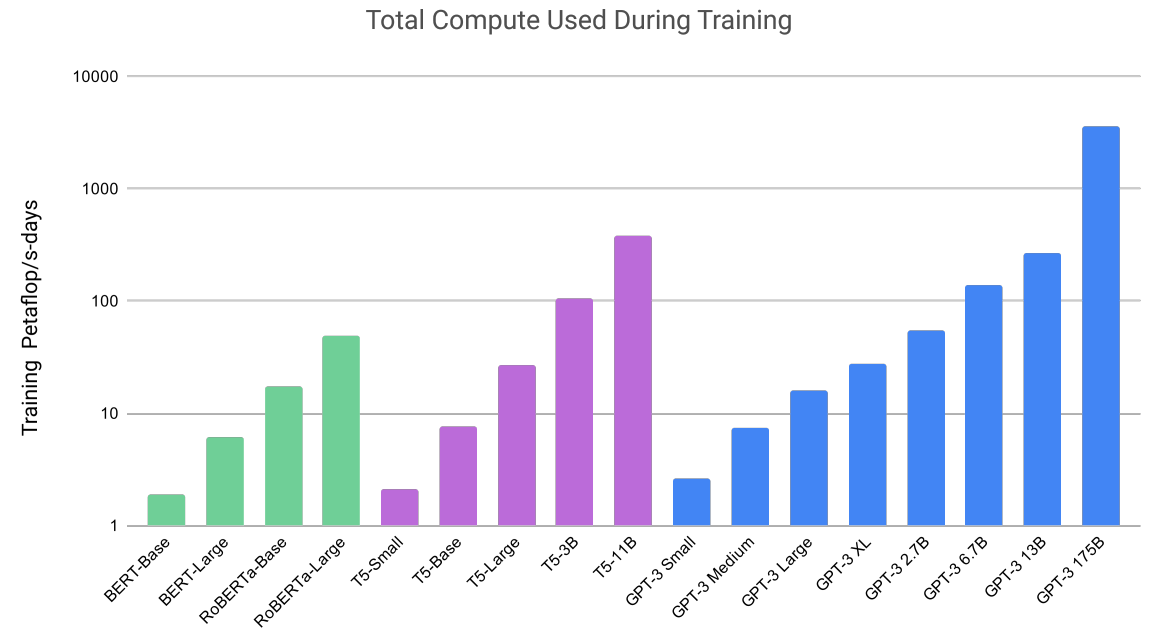


图2.2：训练过程中所消耗的总计算资源。该数据源自《神经语言模型的缩放定律》中的分析[KMH⁺20] 我们使用远少于常规数量的标记来训练更大规模的模型。因此，尽管GPT-3 3B版本的参数量几乎是RoBERTa-Large版本（355M参数）的10倍，但两者在预训练阶段所需的计算资源大致相同，均为50拍浮点运算次/天。这些计算方法的具体细节可见附录 D。

数据集	数量 (标记数)	计入项 训练数据组合	完成训练时所经历的迭代次数为 针对3000亿个令牌进行的训练
Common Crawl数据集 (经过过滤处理)	4100亿	60%	0.44
WebText2数据集	190亿	22%	2.9
Books1	120亿	8%	1.9
Books2	550亿	8%	0.43
维基百科	30亿	3%	3.4

表2.2：用于训练GPT-3的数据集。所谓“训练数据组合中的重力占比”，指的是在训练过程中从某一特定数据集中选取的示例所占的比例；我们有意不让这一比例与该数据集的规模成正比。因此，在进行3000亿个令牌的训练时，有些数据集在训练过程中会被使用多达3.4次，而另一些数据集则可能仅被使用一次左右。

对于那些在海量互联网数据上完成预训练的语言模型，尤其是那些具备记忆大量内容能力的大型模型而言，一个重大的方法学问题在于：在预训练阶段若无意中让下游任务的测试集或开发集被模型看到，就有可能导致下游任务出现数据污染。为了降低这种污染风险，我们查找并尝试移除了本文研究的所有基准测试中的开发集与测试集之间的任何重叠部分。遗憾的是，过滤过程中存在的某个漏洞导致我们遗漏了一些重叠数据，而且由于训练成本较高，重新训练模型并不可行。在4 节中，我们将分析剩余重叠数据所带来的影响，而在未来的工作中，我们会采取更为彻底的措施来消除数据污染问题。

2.3 训练过程

如 [KMH⁺20中所述, MKAT18], 更大规模的模型通常可以使用更大的批量大小，但需要更低的学习率。我们在训练过程中会测量梯度噪声幅度，并以此作为确定批量大小的重要依据 [MKAT18]。表2.1 展示了我们所采用的参数设置。为了解决在训练这些更大规模模型时可能出现内存不足的问题，我们在每次矩阵乘法操作中采用模型并行技术，同时在网络的不同层之间也运用模型并行技术。所有模型均在微软提供的高带宽集群中的V100显卡上完成训练。关于训练过程及超参数设置的详细信息，请参见附录B。

2.4 Evaluation

For few-shot learning, we evaluate each example in the evaluation set by randomly drawing K examples from that task’s training set as conditioning, delimited by 1 or 2 newlines depending on the task. For LAMBADA and Storycloze there is no supervised training set available so we draw conditioning examples from the development set and evaluate on the test set. For Winograd (the original, not SuperGLUE version) there is only one dataset, so we draw conditioning examples directly from it.

K can be any value from 0 to the maximum amount allowed by the model’s context window, which is $n_{\text{ctx}} = 2048$ for all models and typically fits 10 to 100 examples. Larger values of K are usually but not always better, so when a separate development and test set are available, we experiment with a few values of K on the development set and then run the best value on the test set. For some tasks (see Appendix G) we also use a natural language prompt in addition to (or for $K = 0$, instead of) demonstrations.

On tasks that involve choosing one correct completion from several options (multiple choice), we provide K examples of context plus correct completion, followed by one example of context only, and compare the LM likelihood of each completion. For most tasks we compare the per-token likelihood (to normalize for length), however on a small number of datasets (ARC, OpenBookQA, and RACE) we gain additional benefit as measured on the development set by normalizing by the unconditional probability of each completion, by computing $\frac{P(\text{completion}|\text{context})}{P(\text{completion}|\text{answer_context})}$, where `answer_context` is the string "Answer: " or "A: " and is used to prompt that the completion should be an answer but is otherwise generic.

On tasks that involve binary classification, we give the options more semantically meaningful names (e.g. “True” or “False” rather than 0 or 1) and then treat the task like multiple choice; we also sometimes frame the task similar to what is done by [RSR⁺19] (see Appendix G) for details.

On tasks with free-form completion, we use beam search with the same parameters as [RSR⁺19]: a beam width of 4 and a length penalty of $\alpha = 0.6$. We score the model using F1 similarity score, BLEU, or exact match, depending on what is standard for the dataset at hand.

Final results are reported on the test set when publicly available, for each model size and learning setting (zero-, one-, and few-shot). When the test set is private, our model is often too large to fit on the test server, so we report results on the development set. We do submit to the test server on a small number of datasets (SuperGLUE, TriviaQA, PiQa) where we were able to make submission work, and we submit only the 200B few-shot results, and report development set results for everything else.

3 Results

In Figure 3.1 we display training curves for the 8 models described in Section 2. For this graph we also include 6 additional extra-small models with as few as 100,000 parameters. As observed in [KMH⁺20], language modeling performance follows a power-law when making efficient use of training compute. After extending this trend by two more orders of magnitude, we observe only a slight (if any) departure from the power-law. One might worry that these improvements in cross-entropy loss come only from modeling spurious details of our training corpus. However, we will see in the following sections that improvements in cross-entropy loss lead to consistent performance gains across a broad spectrum of natural language tasks.

Below, we evaluate the 8 models described in Section 2 (the 175 billion parameter parameter GPT-3 and 7 smaller models) on a wide range of datasets. We group the datasets into 9 categories representing roughly similar tasks.

In Section 3.1 we evaluate on traditional language modeling tasks and tasks that are similar to language modeling, such as Cloze tasks and sentence/paragraph completion tasks. In Section 3.2 we evaluate on “closed book” question answering tasks: tasks which require using the information stored in the model’s parameters to answer general knowledge questions. In Section 3.3 we evaluate the model’s ability to translate between languages (especially one-shot and few-shot). In Section 3.4 we evaluate the model’s performance on Winograd Schema-like tasks. In Section 3.5 we evaluate on datasets that involve commonsense reasoning or question answering. In Section 3.6 we evaluate on reading comprehension tasks, in Section 3.7 we evaluate on the SuperGLUE benchmark suite, and in 3.8 we briefly explore NLI. Finally, in Section 3.9, we invent some additional tasks designed especially to probe in-context learning abilities – these tasks focus on on-the-fly reasoning, adaptation skills, or open-ended text synthesis. We evaluate all tasks in the few-shot, one-shot, and zero-shot settings.

2.4 评估

在少样本学习中，我们会从评估集的每个示例中随机选取 K 个该任务训练集中的示例作为条件样本，具体分隔方式会根据任务不同而采用1行或2行的换行符。对于LAMBADA和Storycloze任务，由于不存在监督学习用的训练集，因此我们需要从开发集中抽取条件样本，再在测试集上进行性能评估。至于Winograd任务（指原始版本，而非SuperGLUE版本），由于其仅包含一个数据集，所以我们直接从该数据集中选取条件样本。

K 该数值可以是0到模型上下文窗口允许的最大值之间的任意值， $n_{\text{ctx}} = 2048$ 对于所有模型而言，这一范围通常能够容纳10到100个示例。虽然较大的 K 数值通常能带来更好的效果，但并非总是如此；因此，在拥有独立的开发集和测试集时，我们会先在开发集上尝试几种不同的 K 数值，再选用最优值在测试集上进行测试。对于某些任务（详见附录G），除了（或替代）演示示例之外，我们还会使用自然语言提示语。

对于那些需要从多个选项中选出唯一正确补全结果的题目（即多项选择题），我们会提供 K 个包含上下文及正确补全结果的示例，随后再提供一个仅包含上下文的示例，进而比较各补全结果的模型概率。在大多数任务中，我们会通过计算单词级似然值来加以比较，以此对不同长度的输入进行标准化处理；不过在ARC、OpenBookQA和RACE这几个数据集上，通过在开发集上以每个补全结果的无条件概率作为基准来进行标准化处理，还能带来额外的优化效果，其计算方式即为 $P_{\text{（补全结果|上下文）}} / P_{\text{（补全结果|答案上下文）}}$ ，其中答案上下文指的是"Answer: "或"A: "这类字符串，它的作用是提示系统该补全内容应为答案，除此之外并无其他特殊含义。

在涉及二元分类任务的场景中，我们会为选项赋予更具语义意义的名称（例如“True”或“False”，而非0或1），并将该任务视为多项选择题来处理；此外，我们有时也会采用与[RSR⁺19]所采用的方法类似的方式来设计这类任务（详情见附录G）。

对于需要自由形式补全的任务，我们会使用与[RSR⁺19]相同的参数进行波束搜索：波束宽度为4，长度惩罚系数为 $\alpha = 0.6$ 。

我们会根据当前数据集的常用评估标准，选用F1相似度分数、BLEU分数或完全匹配方式来对模型的表现进行评分。

一旦测试集公开可用，我们会针对每种模型规模以及零样本、单样本和少样本学习三种设置，在测试集上公布最终结果。如果测试集属于私有数据，由于我们的模型规模往往过大而无法加载到测试服务器上，因此我们会以开发集的结果作为替代。在少数几个能够成功提交结果的数据集（如SuperGLUE、TriviaQA、PiQa）中，我们仅会提交2000亿参数版本在少样本学习场景下的结果，其余情况则仅报告开发集的结果。</style>

3个结果

在图3.1中，我们展示了第2节所介绍的8种模型的训练曲线。此外，该图表还包含了另外6种参数量仅为100,000的小型模型。正如KMH⁺20]研究中所观察到的那样，当能够高效利用计算资源进行训练时，语言建模的性能会遵循幂律分布。即便将这一趋势再延伸两个数量级，其偏离幂律的程度也极为微小。有人可能会担心，交叉熵损失的下降仅仅是因为模型捕捉到了训练语料库中一些无关紧要的细节。不过，我们在后续章节中将看到，交叉熵损失的降低确实能够在各种自然语言任务中带来稳定的性能提升。

在下文中，我们将在众多数据集上对第2节中提到的8种模型（包括拥有1750亿参数的GPT-3以及7种参数更小的模型）进行评估。这些数据集被划分为9个类别，分别对应大致相似的任务类型。

在3.1章节中，我们会对传统的语言建模任务以及与语言建模类似的任务进行评估，这类任务包括完形填空任务以及句子/段落补全任务。在3.2章节中，我们会针对“闭卷”式问答任务展开评估——这类任务要求利用存储在模型参数中的信息来回答常识性问题。在3.3章节中，我们会评估该模型在不同语言之间的翻译能力（尤其关注单样本学习和少样本学习场景）。在3.4章节中，我们会测试该模型在类Winograd Schema任务上的性能表现。在3.5章节中，我们会使用涉及常识推理或问答功能的数据集进行评估。在3.6章节中，我们会对阅读理解任务进行评估；而在3.7章节中，则会使用SuperGLUE基准测试套件进行评估。此外，在3.8章节中，我们会简要探讨自然语言推理相关内容。最后，在3.9章节里，我们设计了一些专门用于检测上下文学习能力的额外任务——这些任务侧重于即时推理、适应能力以及开放式文本生成能力。所有任务都会在少样本学习、单样本学习和零样本场景下进行评估。

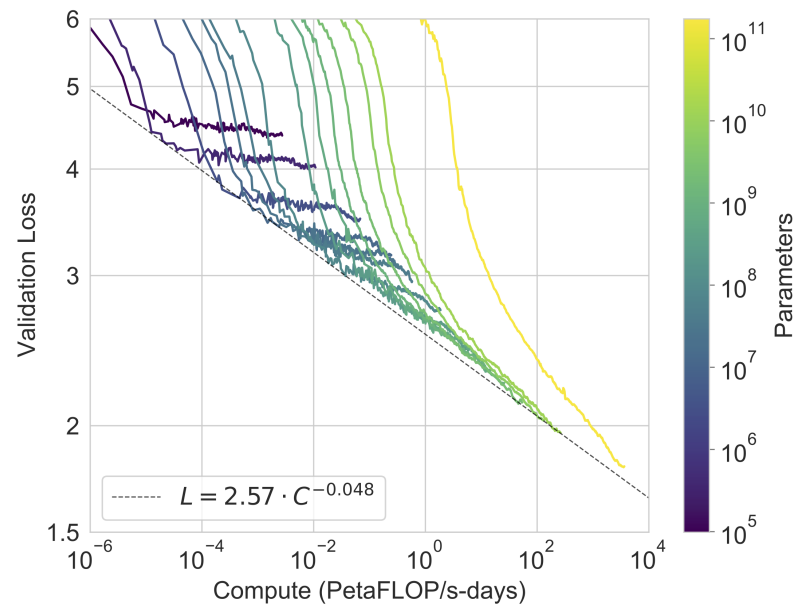


Figure 3.1: Smooth scaling of performance with compute. Performance (measured in terms of cross-entropy validation loss) follows a power-law trend with the amount of compute used for training. The power-law behavior observed in [KMH⁺20] continues for an additional two orders of magnitude with only small deviations from the predicted curve. For this figure, we exclude embedding parameters from compute and parameter counts.

Setting	PTB
SOTA (Zero-Shot)	35.8 ^a
GPT-3 Zero-Shot	20.5

Table 3.1: Zero-shot results on PTB language modeling dataset. Many other common language modeling datasets are omitted because they are derived from Wikipedia or other sources which are included in GPT-3’s training data. ^a[RWC⁺19]

3.1 Language Modeling, Cloze, and Completion Tasks

In this section we test GPT-3’s performance on the traditional task of language modeling, as well as related tasks that involve predicting a single word of interest, completing a sentence or paragraph, or choosing between possible completions of a piece of text.

3.1.1 Language Modeling

We calculate zero-shot perplexity on the Penn Tree Bank (PTB) [MKM⁺94] dataset measured in [RWC⁺19]. We omit the 4 Wikipedia-related tasks in that work because they are entirely contained in our training data, and we also omit the one-billion word benchmark due to a high fraction of the dataset being contained in our training set. PTB escapes these issues due to predating the modern internet. Our largest model sets a new SOTA on PTB by a substantial margin of 15 points, achieving a perplexity of 20.50. Note that since PTB is a traditional language modeling dataset it does not have a clear separation of examples to define one-shot or few-shot evaluation around, so we measure only zero-shot.

3.1.2 LAMBADA

The LAMBADA dataset [PKL⁺16] tests the modeling of long-range dependencies in text – the model is asked to predict the last word of sentences which require reading a paragraph of context. It has recently been suggested that the continued scaling of language models is yielding diminishing returns on this difficult benchmark. [BHT⁺20] reflect on the small 1.5% improvement achieved by a doubling of model size between two recent state of the art results ([SPP⁺19]

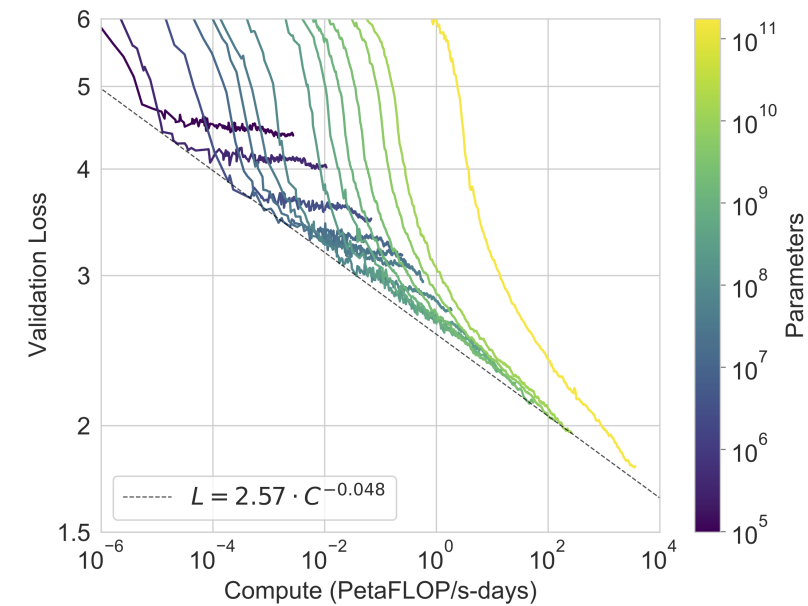


图3.1：性能随计算资源的变化呈现平滑的提升趋势。以交叉熵验证损失作为衡量标准时，模型的性能会随着训练所使用的计算资源数量而呈现幂律趋势。在 [KMH⁺20] 研究中观察到的这种幂律行为，在仅偏离预测曲线很小的情况下还能持续两个数量级。在本图中，计算资源及相关参数数量未计入嵌入参数的统计范畴。

设置	PTB
当前最佳性能（零样本学习）	35.8 ^a
GPT-3零样本学习	20.5

表3.1：在PTB语言建模数据集上的零样本学习结果。由于许多其他常见的语言建模数据集源自维基百科或其他已被纳入GPT-3训练数据中的来源，因此本表未将其纳入统计范围。 ^a[RWC⁺19]

3.1 语言建模、完形填空任务与补全任务

在本章节中，我们将测试GPT-3在传统语言建模任务上的性能表现，同时也会评估与之相关的其他任务，这类任务包括预测某个目标单词、完成句子或段落，或是从多个可能的文本补全选项中选出正确答案。

3.1.1 语言建模

我们基于宾州树库（PTB）计算零样本学习的困惑度，该指标以RWC为单位进行衡量 [MKM⁺94] 数据集的度量单位为 [RWC⁺19]。在相关研究中，我们没有纳入那4项与维基百科相关的任务，因为它们全部包含在我们的训练数据中；同时我们也排除了那个包含十亿单词的基准测试，原因是该数据集中有很大比例的内容已存在于我们的训练集中。由于宾州树库出现的时间早于现代互联网，因此不存在上述问题。我们规模最大的模型在宾州树库上的表现创造了新的当前最佳性能，其困惑度仅为20.50，比现有最佳成绩高出15分之多。需要说明的是，由于宾州树库属于传统的语言建模数据集，其中并未对示例进行明确区分，无法据此界定单样本学习或少样本学习的评估标准，因此我们仅开展零样本学习的测试。

3.1.2 LAMBADA

LAMBADA数据集 [PKL⁺16] 用于测试文本中的长距离依赖关系建模能力——该模型需要根据给定的上下文段落来预测句子的最后一个词。近期有研究指出，尽管语言模型的性能不断提升，但在这一难度较高的基准测试中，其带来的收益正逐渐递减。 [BHT⁺20] 这反映出在最近两次达到当前最佳水平的结果之间，即便将模型规模翻倍，也只能实现1.5%的微弱提升 ([SPP⁺19]>

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0 ^a	8.63 ^b	91.8^c	85.6^d
GPT-3 Zero-Shot	76.2	3.00	83.2	78.9
GPT-3 One-Shot	72.5	3.35	84.7	78.1
GPT-3 Few-Shot	86.4	1.92	87.7	79.3

Table 3.2: Performance on cloze and completion tasks. GPT-3 significantly improves SOTA on LAMBADA while achieving respectable performance on two difficult completion prediction datasets. ^a[Tur20] ^b[RWC⁺19] ^c[LDL19] ^d[LCH⁺20]

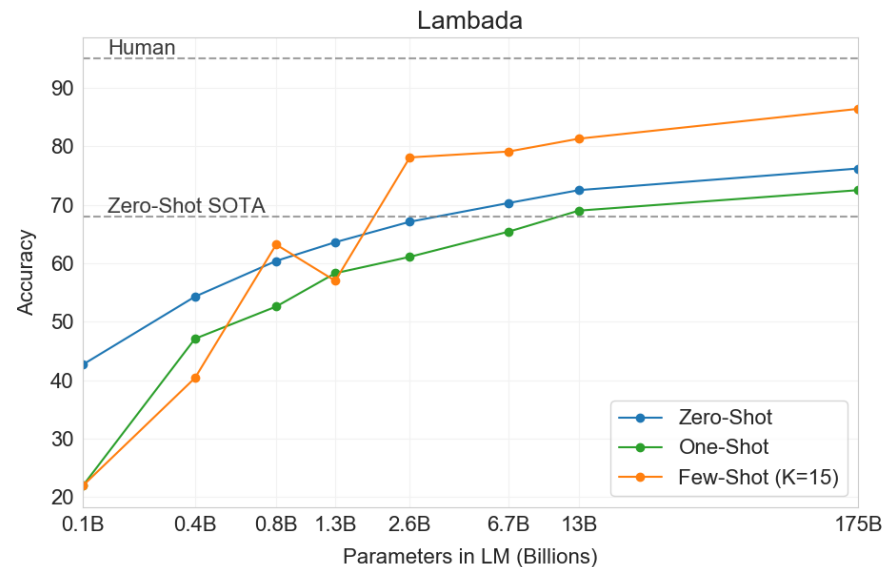


Figure 3.2: On LAMBADA, the few-shot capability of language models results in a strong boost to accuracy. GPT-3 2.7B outperforms the SOTA 17B parameter Turing-NLG [Tur20] in this setting, and GPT-3 175B advances the state of the art by 18%. Note zero-shot uses a different format from one-shot and few-shot as described in the text.

and [Tur20]) and argue that “continuing to expand hardware and data sizes by orders of magnitude is not the path forward”. We find that path is still promising and in a zero-shot setting GPT-3 achieves 76% on LAMBADA, a gain of 8% over the previous state of the art.

LAMBADA is also a demonstration of the flexibility of few-shot learning as it provides a way to address a problem that classically occurs with this dataset. Although the completion in LAMBADA is always the last word in a sentence, a standard language model has no way of knowing this detail. It thus assigns probability not only to the correct ending but also to other valid continuations of the paragraph. This problem has been partially addressed in the past with stop-word filters [RWC⁺19] (which ban “continuation” words). The few-shot setting instead allows us to “frame” the task as a cloze-test and allows the language model to infer from examples that a completion of exactly one word is desired. We use the following fill-in-the-blank format:

Alice was friends with Bob. Alice went to visit her friend _____. → Bob

George bought some baseball equipment, a ball, a glove, and a _____. →

When presented with examples formatted this way, GPT-3 achieves 86.4% accuracy in the few-shot setting, an increase of over 18% from the previous state-of-the-art. We observe that few-shot performance improves strongly with model size. While this setting decreases the performance of the smallest model by almost 20%, for GPT-3 it improves accuracy by 10%. Finally, the fill-in-blank method is not effective one-shot, where it always performs worse than the zero-shot setting. Perhaps this is because all models still require several examples to recognize the pattern.

设置	LAMBADA (准确率)	LAMBADA (人数)	StoryCloze (准确率)	HellaSwag (准确率)
SOTA	68.0 ^a	8.63 ^b	91.8^c	85.6^d
GPT-3零样本学习	76.2	3.00	83.2	78.9
GPT-3单样本学习	72.5	3.35	84.7	78.1
GPT-3少样本学习	86.4	1.92	87.7	79.3

表3.2：在完形填空与文本补全任务上的性能表现。 GPT-3在LAMBADA任务上显著提升了当前最佳性能，同时在两个难度较高的文本补全预测数据集上也取得了不错的性能表现。 ^a [Tur20] ^b [RWC⁺19] ^c [LDL19] ^d [LCH⁺20]

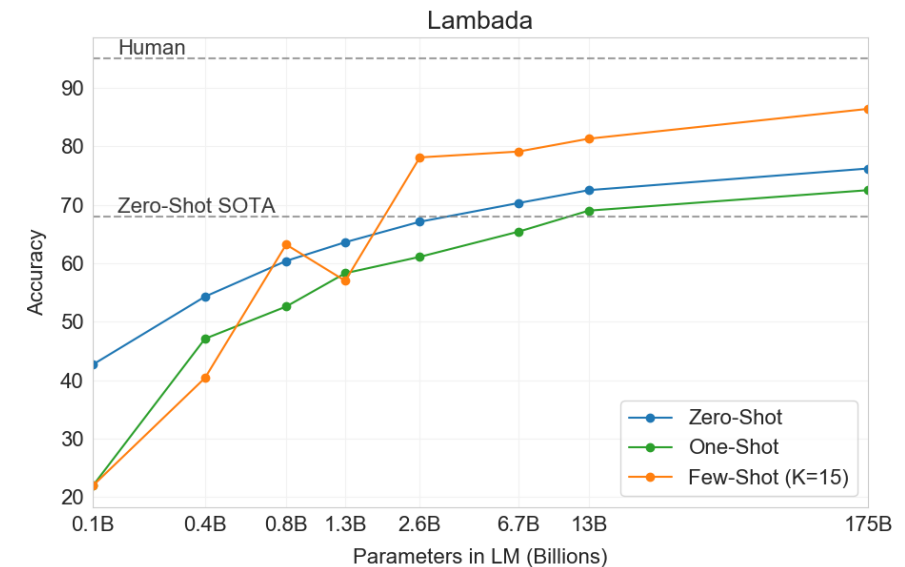


图3.2：在LAMBADA任务上，语言模型的少样本学习能力大幅提升了其准确率。GPT-3 27亿参数版本的表现优于当前最佳水平的1750亿参数Turing-NLG模型Tur20]；此外，GPT-3 1750亿参数版本还将当前最佳水平提升了18%。需注意，文中提到零样本学习与单样本学习、少样本学习的格式有所不同。

以及 [Tur20]) 认为“继续以数量级的方式扩大硬件和数据规模并非正确的发展路径”。我们则认为这条路径依然具有前景，在零样本场景下，GPT-3在LAMBADA数据集上的准确率可达76%，较之前的最佳结果又提升了8%。

LAMBADA也是少样本学习灵活性的一个体现，它为解决该数据集中常见的问题提供了方法。尽管在LAMBADA任务中补全结果始终是句子的最后一个词，但传统的语言模型却无法掌握这一细节，因此它会不仅为正确的结尾分配概率，也会为段落的其他有效延续方式赋予概率。过去人们曾通过停用词过滤来处理来部分解决这个问题 [RWC⁺19] (这类方法会禁止使用“延续”类词汇)。而少样本场景则让我们能够将这项任务视为完形填空题，从而使语言模型能够从示例中推断出需要补全的恰好是一个单词。我们采用的填空题格式如下：

爱丽丝和鲍勃是朋友。爱丽丝去拜访她的朋友了。 → 鲍勃·乔治买

了一些棒球装备，包括一个球、一只手套，还有个 →

在面对以这种方式呈现的示例时，GPT-3在少样本场景下的准确率可达86.4%，这一数值比此前的当前最佳水平高出18%以上。我们发现，模型规模越大，其少样本性能提升得越显著。虽然在这种场景下最小规模模型的性能会下降近20%，但GPT-3的准确率却提升了10%。最后需要指出的是，填空法在单样本学习场景中并不有效，其表现始终不如零样本场景。这很可能是因为所有模型仍然需要多个示例才能识别出其中的规律。

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP+20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

Table 3.3: Results on three Open-Domain QA tasks. GPT-3 is shown in the few-, one-, and zero-shot settings, as compared to prior SOTA results for closed book and open domain settings. TriviaQA few-shot result is evaluated on the wiki split test server.

One note of caution is that an analysis of test set contamination identified that a significant minority of the LAMBADA dataset appears to be present in our training data – however analysis performed in Section 4 suggests negligible impact on performance.

3.1.3 HellaSwag

The HellaSwag dataset [ZHB+19] involves picking the best ending to a story or set of instructions. The examples were adversarially mined to be difficult for language models while remaining easy for humans (who achieve 95.6% accuracy). GPT-3 achieves 78.1% accuracy in the one-shot setting and 79.3% accuracy in the few-shot setting, outperforming the 75.4% accuracy of a fine-tuned 1.5B parameter language model [ZHR+19] but still a fair amount lower than the overall SOTA of 85.6% achieved by the fine-tuned multi-task model ALUM.

3.1.4 StoryCloze

We next evaluate GPT-3 on the StoryCloze 2016 dataset [MCH+16], which involves selecting the correct ending sentence for five-sentence long stories. Here GPT-3 achieves 83.2% in the zero-shot setting and 87.7% in the few-shot setting (with $K = 70$). This is still 4.1% lower than the fine-tuned SOTA using a BERT based model [LDL19] but improves over previous zero-shot results by roughly 10%.

3.2 Closed Book Question Answering

In this section we measure GPT-3’s ability to answer questions about broad factual knowledge. Due to the immense amount of possible queries, this task has normally been approached by using an information retrieval system to find relevant text in combination with a model which learns to generate an answer given the question and the retrieved text. Since this setting allows a system to search for and condition on text which potentially contains the answer it is denoted “open-book”. [RRS20] recently demonstrated that a large language model can perform surprisingly well directly answering the questions without conditioning on auxilliary information. They denote this more restrictive evaluation setting as “closed-book”. Their work suggests that even higher-capacity models could perform even better and we test this hypothesis with GPT-3. We evaluate GPT-3 on the 3 datasets in [RRS20]: Natural Questions [KPR+19], WebQuestions [BCFL13], and TriviaQA [JCWZ17], using the same splits. Note that in addition to all results being in the closed-book setting, our use of few-shot, one-shot, and zero-shot evaluations represent an even stricter setting than previous closed-book QA work: in addition to external content not being allowed, fine-tuning on the Q&A dataset itself is also not permitted.

The results for GPT-3 are shown in Table 3.3. On TriviaQA, we achieve 64.3% in the zero-shot setting, 68.0% in the one-shot setting, and 71.2% in the few-shot setting. The zero-shot result already outperforms the fine-tuned T5-11B by 14.2%, and also outperforms a version with Q&A tailored span prediction during pre-training by 3.8%. The one-shot result improves by 3.7% and matches the SOTA for an open-domain QA system which not only fine-tunes but also makes use of a learned retrieval mechanism over a 15.3B parameter dense vector index of 21M documents [LPP+20]. GPT-3’s few-shot result further improves performance another 3.2% beyond this.

On WebQuestions (WebQs), GPT-3 achieves 14.4% in the zero-shot setting, 25.3% in the one-shot setting, and 41.5% in the few-shot setting. This compares to 37.4% for fine-tuned T5-11B, and 44.7% for fine-tuned T5-11B+SSM, which uses a Q&A-specific pre-training procedure. GPT-3 in the few-shot setting approaches the performance of state-of-the-art fine-tuned models. Notably, compared to TriviaQA, WebQS shows a much larger gain from zero-shot to few-shot (and indeed its zero-shot and one-shot performance are poor), perhaps suggesting that the WebQs questions

设置	NaturalQS	WebQS	TriviaQA
RAG（经过微调的开放领域模型） [LPP+20]	44.5	45.5	68.0
T5-11B模型+SSM（经过微调的封闭书籍模型） [RRS20]	36.6	44.7	60.5
T5-11B（经过微调的闭卷版）	34.5	37.4	50.1
GPT-3零样本学习	14.6	14.4	64.3
GPT-3单样本学习	23.0	25.3	68.0
GPT-3少样本学习	29.9	41.5	71.2

表3.3：三种开放领域问答任务上的测试结果。该表展示了GPT-3在少样本、单样本以及零样本场景下的表现，并与以往闭卷及开放领域场景下的当前最佳性能结果进行了对比。TriviaQA少样本学习的结果是在维基分割测试服务器上评估得出的。

需要提醒的一点是，通过对测试集污染情况的分析，我们发现LAMBADA数据集中有相当大比例的样本似乎存在于我们的训练数据中——不过在第4节进行的分析表明，这对性能的影响可以忽略不计。

3.1.3 HellaSwag

HellaSwag数据集 [ZHB+19] 要求从多个故事结尾或指令选项中选出最合适的那个。这些示例是通过特殊方式筛选出来的，虽然对人类来说难度较低（他们的准确率可达95.6%），但对语言模型而言却相当棘手。在单样本场景下，GPT-3的准确率为78.1%，而在少样本场景下这一数值提升至79.3%，虽优于经过微调的拥有15亿个参数的语言模型所达到的75.4%准确率 [ZHR+19]，但仍比经过微调的多任务模型ALUM所实现的85.6%的当前最佳性能低不少。

3.1.4 StoryCloze

接下来，我们在StoryCloze 2016数据集上对GPT-3进行了测试 [MCH+16]，该数据集需要为长度为五句的故事挑选正确的结尾句。在此测试中，GPT-3在零样本场景下的准确率为83.2%，在少样本场景下则上升至87.7%（具体数据见 $K = 70$ ）。这一成绩仍比使用基于BERT模型的经过微调的当前最佳性能模型 [LDL19] 低4.1%，不过相比之前的零样本测试结果已提升了约10%。

3.2 封闭书面问题回答

在本章节中，我们旨在衡量GPT-3解答涉及广泛事实性知识问题的能力。由于可能的查询数量极其庞大，这类任务通常采用信息检索系统来查找相关文本，再结合一个能够根据问题及检索到的文本生成答案的模型来完成。由于这种设置允许系统搜索并利用可能包含答案的文本作为条件样本，因此被称作“开卷式”评估。[RRS20]最近的研究表明，大型语言模型即便不依赖任何辅助信息，直接回答问题也能取得人意料的优异表现。研究者将这种限制更严格的评估方式称为“封闭书面式”。他们的研究暗示，容量更高的模型或许能表现得更好，因此我们使用GPT-3对这一假设进行了验证。我们在 [RRS20]: 自然问题任务 [KPR+19]、WebQuestions数据集 [BCFL13], 以及TriviaQA [JCWZ17], 上对GPT-3进行评估，且采用了与前述研究相同的数据分割方式。需要说明的是，除了所有结果均基于封闭书面式评估外，我们还加入了少样本学习、单样本学习和零样本学习三种评估方式，这使得该测试环境比以往的封闭书面式问答研究更为严格：不仅不允许使用外部内容，甚至也不允许在问答数据集本身上进行微调。

GPT-3的相关结果展示在表3.3中。在TriviaQA数据集上，该模型在零样本场景下的性能为64.3%，单样本场景下为68.0%，而少样本场景下的性能则可达71.2%。其零样本场景下的结果不仅比经过微调的T5-11B模型的性能高14.2%，还比那种在预训练阶段就加入了针对问答任务的自适应跨度预测功能的模型高出3.8%。单样本场景下的性能则提升了3.7%，与那些既进行微调又采用了基于包含2100万篇文档的153亿参数规模密集向量索引的检索机制的开放领域问答系统的当前最佳性能相当 [LPP+20]。此外，GPT-3在少样本场景下的性能还能再提升3.2%。在WebQuestions（WebQs）数据集上，GPT-3在零样本场景下的性能为14.4%，在单样本场景下为25.3%，而在少样本场景下则提升至41.5%。作为对比，经过微调的T5-11B模型的性能为37.4%，而采用了专门针对问答任务设计的预训练流程并经过微调的T5-11B+SSM模型性能则达到44.7%。在少样本场景下，GPT-3的性能已接近当前最佳水平的微调模型。值得注意的是，与TriviaQA相比，WebQs从零样本学习到少样本学习的性能提升幅度要大得多（实际上它的零样本和单样本学习性能都相当不佳），这或许表明WebQs中的问题及其答案的风格对于GPT-3而言属于概率分布之外的内容。尽管如此，GPT-3似乎仍能适应这种概率分布，在少样本场景下依然展现出出色的性能。

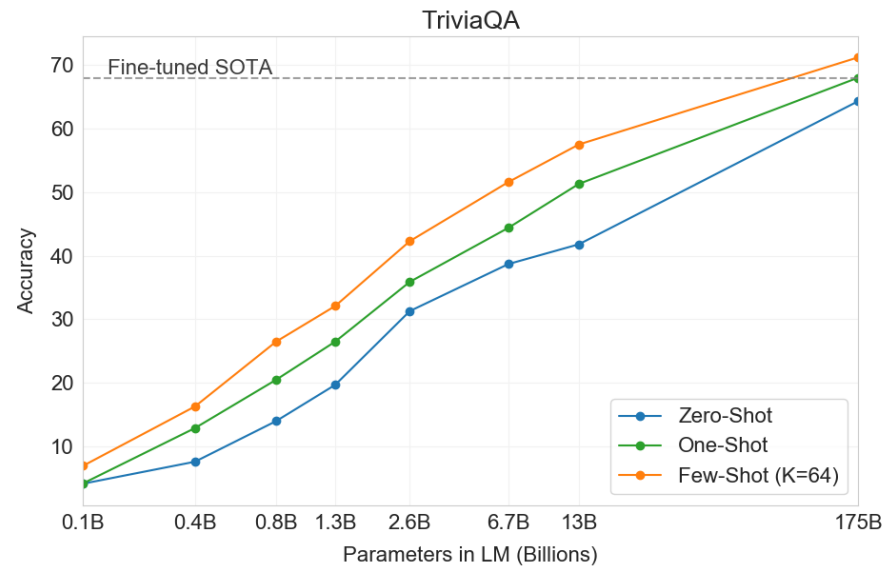


Figure 3.3: On TriviaQA GPT3’s performance grows smoothly with model size, suggesting that language models continue to absorb knowledge as their capacity increases. One-shot and few-shot performance make significant gains over zero-shot behavior, matching and exceeding the performance of the SOTA fine-tuned open-domain model, RAG [LPP+20]

and/or the style of their answers are out-of-distribution for GPT-3. Nevertheless, GPT-3 appears able to adapt to this distribution, recovering strong performance in the few-shot setting.

On Natural Questions (NQs) GPT-3 achieves 14.6% in the zero-shot setting, 23.0% in the one-shot setting, and 29.9% in the few-shot setting, compared to 36.6% for fine-tuned T5 11B+SSM. Similar to WebQS, the large gain from zero-shot to few-shot may suggest a distribution shift, and may also explain the less competitive performance compared to TriviaQA and WebQS. In particular, the questions in NQs tend towards very fine-grained knowledge on Wikipedia specifically which could be testing the limits of GPT-3’s capacity and broad pretraining distribution.

Overall, on one of the three datasets GPT-3’s one-shot matches the open-domain fine-tuning SOTA. On the other two datasets it approaches the performance of the closed-book SOTA despite not using fine-tuning. On all 3 datasets, we find that performance scales very smoothly with model size (Figure 3.3 and Appendix H Figure H.7), possibly reflecting the idea that model capacity translates directly to more ‘knowledge’ absorbed in the parameters of the model.

3.3 Translation

For GPT-2 a filter was used on a multilingual collection of documents to produce an English only dataset due to capacity concerns. Even with this filtering GPT-2 showed some evidence of multilingual capability and performed non-trivially when translating between French and English despite only training on 10 megabytes of remaining French text. Since we increase the capacity by over two orders of magnitude from GPT-2 to GPT-3, we also expand the scope of the training dataset to include more representation of other languages, though this remains an area for further improvement. As discussed in 2.2 the majority of our data is derived from raw Common Crawl with only quality-based filtering. Although GPT-3’s training data is still primarily English (93% by word count), it also includes 7% of text in other languages. These languages are documented in the [supplemental material](#). In order to better understand translation capability, we also expand our analysis to include two additional commonly studied languages, German and Romanian.

Existing unsupervised machine translation approaches often combine pretraining on a pair of monolingual datasets with back-translation [SHB15] to bridge the two languages in a controlled way. By contrast, GPT-3 learns from a blend of training data that mixes many languages together in a natural way, combining them on a word, sentence, and document level. GPT-3 also uses a single training objective which is not customized or designed for any task in particular. However, our one / few-shot settings aren’t strictly comparable to prior unsupervised work since they make use of a small amount of paired examples (1 or 64). This corresponds to up to a page or two of in-context training data.

Results are shown in Table 3.4. Zero-shot GPT-3, which only receives on a natural language description of the task, still underperforms recent unsupervised NMT results. However, providing only a single example demonstration for

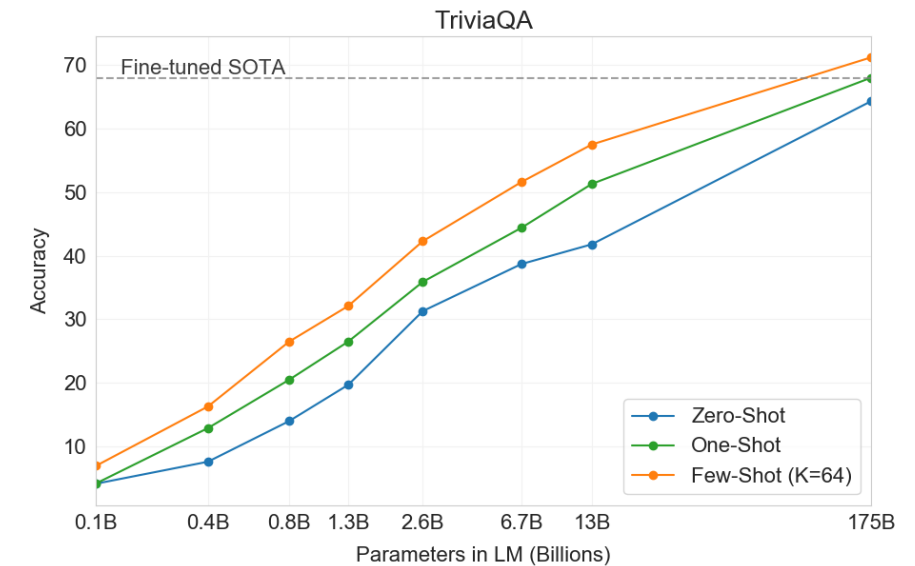


图3.3: 在TriviaQA任务中，GPT3的性能会随着模型规模的增大而稳步提升，这表明语言模型在容量增加时能够持续吸收更多知识。在单样本学习与少样本学习场景下，其性能相较于零样本学习有显著提升，不仅可与当前最佳性能的经过微调的开放领域模型RAG相媲美，甚至还能超越后者[LPP+20]

也就是说，这些问题的答案风格可能不符合GPT-3所习惯的概率分布。不过，GPT-3依然能够适应这种差异，并在少样本场景下恢复出强劲的性能表现。

在自然问题任务（NQs）中，GPT-3在零样本学习场景下的性能为14.6%，在单样本学习场景下为23.0%，而在少样本学习场景下则上升至29.9%；相比之下，经过微调的110亿参数T5 11B+SSM模型的性能则为36.6%。与WebQS任务类似，从零样本学习到少样本学习时性能的巨大提升可能反映出概率分布发生了变化，这也解释了为何其在这些任务上的性能不如TriviaQA和WebQS。尤其需要指出的是，NQs中的问题往往涉及维基百科上极为细致的知识点，这可能在考验GPT-3的容量及其广泛的预训练知识覆盖范围。

总体而言，在三个数据集中的某一个上，GPT-3的单样本学习性能可与开放领域微调后的当前最佳性能相媲美；而在另外两个数据集上，尽管没有进行微调，其性能也接近闭卷测试下的当前最佳性能。在全部三个数据集上，我们观察到性能会随着模型规模的增长而极为平滑地提升（见图3.3及附录H图H.7），这或许反映出模型容量会直接转化为更多存储在模型参数中的“知识”。

3.3 翻译

由于容量限制，针对GPT-2，人们对多语言文档集合进行了过滤处理，从而得到了仅包含英语的内容的数据集。即便经过了这样的过滤，GPT-2依然展现出一定的多语言处理能力——仅在基于10兆字节法语文本进行训练的情况下，它翻译法语与英语时的表现也相当不错。由于从GPT-2发展到GPT-3时，模型容量提升了两个数量级以上，我们也将训练数据集的范围扩大，纳入了更多其他语言的文本，不过这一方面仍需进一步优化。如2.2中所阐述的，我们的大部分数据都源自原始的CommonCrawl数据集，仅进行了基于质量的标准过滤。虽然GPT-3的训练数据仍以英语为主（按词数计算占比达93%），但也包含了7%的其他语言文本，这些语言的相关信息记载在[补充材料](#)中。为了更深入地了解其翻译能力，我们还把分析范围扩展到了另外两种常被研究的语言，即德语和罗马尼亚语。

现有的无监督机器翻译方法通常会基于一对单语数据集的预训练与回译技术 [SHB15] 相结合，以受控的方式实现两种语言之间的转换。相比之下，GPT-3则通过整合多种语言的训练数据来进行学习，这些数据在词汇、句子及文档层面实现了自然融合，且GPT-3仅使用一个通用型的训练目标，无需针对特定任务进行定制或设计。不过，我们所采用的单一/少样本设置并不能与以往的无监督研究直接对比，因为后者依赖的是少量成对的示例（1个或64个），这类示例对应的上下文训练数据量通常只有一两页。相关结果展示在表格中3.4。仅通过自然语言形式的任务描述即可启动的零样本GPT-3，其表现依然逊于近期出现的无监督机器翻译结果；即便仅提供单个示例进行演示，情况也是如此。

Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	45.6^a	35.0 ^b	41.2^c	40.2 ^d	38.5^e	39.9^e
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ ⁺ 19]	<u>37.5</u>	34.9	28.3	35.2	<u>35.2</u>	33.1
mBART [LGG ⁺ 20]	-	-	<u>29.8</u>	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	<u>39.2</u>	29.7	<u>40.6</u>	21.0	<u>39.5</u>

Table 3.4: Few-shot GPT-3 outperforms previous unsupervised NMT work by 5 BLEU when translating into English reflecting its strength as an English LM. We report BLEU scores on the WMT’14 Fr↔En, WMT’16 De↔En, and WMT’16 Ro↔En datasets as measured by multi-bleu.perl with XLM’s tokenization in order to compare most closely with prior unsupervised NMT work. SacreBLEU^f [Pos18] results reported in Appendix H. Underline indicates an unsupervised or few-shot SOTA, bold indicates supervised SOTA with relative confidence. ^a[EOAG18] ^b[DHKH14] ^c[WXH⁺18] ^d[oR16] ^e[LGG⁺20] ^f[SacreBLEU signature: BLEU+case.mixed+numrefs.1+smooth.exp+tok.intl+version.1.2.20]

设置	英→法	法→英	英→德	德→英	英→罗	罗马尼亚语→英
当前最佳性能（监督学习）	45.6^a	35.0 ^b	41.2^c	40.2 ^d	38.5^e	39.9^e
XLM模型 [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS数据集 [STQ ⁺ 19]	<u>37.5</u>	34.9	28.3	35.2	<u>35.2</u>	33.1
mBART模型 [LGG ⁺ 20]	-	-	<u>29.8</u>	34.0	35.0	30.5
GPT-3零样本学习	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3单样本学习	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3少样本学习	32.6	<u>39.2</u>	29.7	<u>40.6</u>	21.0	<u>39.5</u>

表3.4：在将文本翻译为英语时，GPT-3少样本学习模型的BLEU分数比以往的无监督机器翻译研究高出5分，这充分体现了其作为英语语言模型的强大能力。为能与先前的无监督机器翻译研究进行最精准的对比，我们采用了multi-bleu.perl工具，并结合XLM模型的令牌化方式，针对WMT’ 14 Fr↔En、WMT’ 16 De↔En以及WMT’ 16 Ro↔En这三个数据集的BLEU分数进行了测试。附录H中列出了SacreBLEU^f [Pos18] 计算结果。^a[EOAG18] ^b[DHKH14] ^c[WXH⁺18] ^d[oR16] ^e[LGG⁺20] ^f格式的得分。SacreBLEU的标识为：BLEU+case.mixed+numrefs.1+smooth.exp+tok.intl+version.1.2.20]

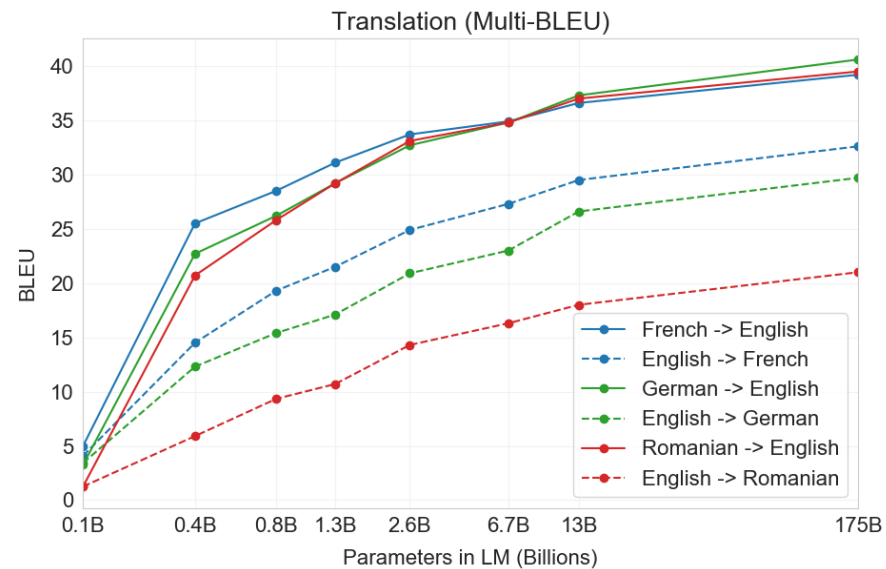


Figure 3.4: Few-shot translation performance on 6 language pairs as model capacity increases. There is a consistent trend of improvement across all datasets as the model scales, and as well as tendency for translation into English to be stronger than translation from English.

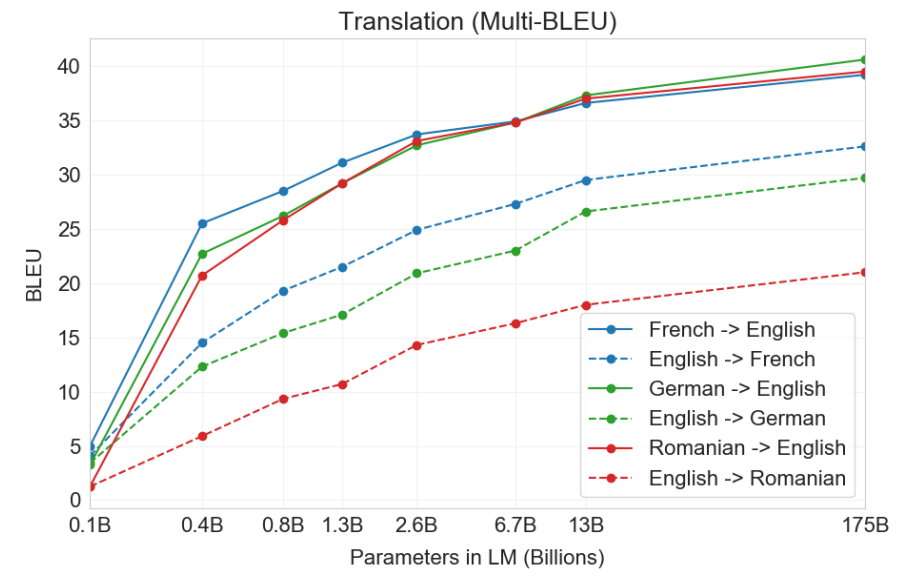


图3.4：随着模型容量提升，6种语言对在少样本学习下的翻译性能变化情况。可见，随着模型规模的扩大，所有数据集的翻译性能均呈现持续上升的趋势，同时从英语翻译的结果通常也优于向英语翻译的结果。

Setting	Winograd	Winogrande (XL)
Fine-tuned SOTA	90.1^a	84.6^b
GPT-3 Zero-Shot	88.3*	70.2
GPT-3 One-Shot	89.7*	73.2
GPT-3 Few-Shot	88.6*	77.7

Table 3.5: Results on the WSC273 version of Winograd schemas and the adversarial Winogrande dataset. See Section 4 for details on potential contamination of the Winograd test set. ^a[SBBC19] ^b[LYN+20]

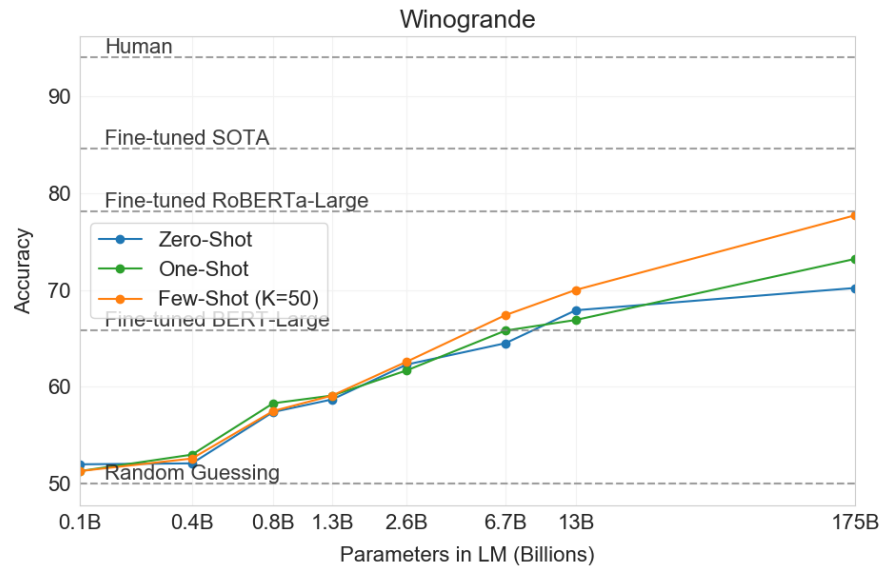


Figure 3.5: Zero-, one-, and few-shot performance on the adversarial Winogrande dataset as model capacity scales. Scaling is relatively smooth with the gains to few-shot learning increasing with model size, and few-shot GPT-3 175B is competitive with a fine-tuned RoBERTa-large.

each translation task improves performance by over 7 BLEU and nears competitive performance with prior work. GPT-3 in the full few-shot setting further improves another 4 BLEU resulting in similar average performance to prior unsupervised NMT work. GPT-3 has a noticeable skew in its performance depending on language direction. For the three input languages studied, GPT-3 significantly outperforms prior unsupervised NMT work when translating into English but underperforms when translating in the other direction. Performance on En-Ro is a noticeable outlier at over 10 BLEU worse than prior unsupervised NMT work. This could be a weakness due to reusing the byte-level BPE tokenizer of GPT-2 which was developed for an almost entirely English training dataset. For both Fr-En and De-En, few shot GPT-3 outperforms the best supervised result we could find but due to our unfamiliarity with the literature and the appearance that these are un-competitive benchmarks we do not suspect those results represent true state of the art. For Ro-En, few shot GPT-3 performs within 0.5 BLEU of the overall SOTA which is achieved by a combination of unsupervised pretraining, supervised finetuning on 608K labeled examples, and backtranslation [LHCG19b].

Finally, across all language pairs and across all three settings (zero-, one-, and few-shot), there is a smooth trend of improvement with model capacity. This is shown in Figure 3.4 in the case of few-shot results, and scaling for all three settings is shown in Appendix H.

3.4 Winograd-Style Tasks

The Winograd Schemas Challenge [LDM12] is a classical task in NLP that involves determining which word a pronoun refers to, when the pronoun is grammatically ambiguous but semantically unambiguous to a human. Recently fine-tuned language models have achieved near-human performance on the original Winograd dataset, but more difficult versions

设置	Winograd	维诺格兰德XL版本
微调后的最优性能	90.1^a	84.6^b
GPT-3零样本学习	88.3*	70.2
GPT-3单样本学习	89.7*	73.2
GPT-3少样本学习	88.6*	77.7

表3.5: 在Winograd schemas的WSC273版本以及具有对抗性的Winogrande数据集上的测试结果。关于Winograd测试集可能存在的污染问题，请参阅第4节的相关说明。^a [SBBC19] ^b[LYN+20]



图3.5: 随着模型容量不断提升，该模型在具有对抗性的Winogrande数据集上的零样本、单样本及少样本学习性能变化情况。性能提升趋势相对平稳，少样本学习的优势会随着模型规模的增大而增强，且few-shotGPT-3 175B模型的表现与经过微调的RoBERTa-Large版本相当。

在各项翻译任务中，该模型的性能均提升了7点以上BLEU分数，其表现已接近此前研究的优秀水平。在完整少样本设置下，GPT-3的性能又进一步提升了4点BLEU分数，其平均性能与此前的无监督机器翻译研究成果相近。不过GPT-3的性能会因语言方向的不同而出现明显差异：在对三种输入语言进行英语翻译时，它的表现显著优于此前的无监督机器翻译研究成果；但在其他语言方向的翻译中则表现较差。其中英到罗语的翻译性能尤为异常，其BLEU分数比此前的无监督机器翻译研究成果低10点以上。这很可能是由于该模型重用了为几乎全部由英语构成的训练数据集开发的GPT-2字节级BPE分词器所导致的缺陷。在法语到英语和德语到英语的翻译任务中，few-shotGPT-3的表现虽优于我们所能找到的最佳有监督模型结果，但由于我们对相关研究文献了解不足，加之这些基准测试的结果似乎并不具备竞争力，因此我们认为它们并不能真正代表当前最佳水平。而在罗语到英语的翻译任务中，few-shotGPT-3的性能仅比通过无监督预训练、基于608K个带标签的示例进行有监督微调并结合回译技术所达到的整体当前最佳水平低0.5点BLEU分数。[LHCG19b]。

最后，在所有语言对以及零样本、单样本和少样本这三种设置下，模型容量都会带来持续稳定的性能提升趋势。这一趋势如图3.4所示，针对少样本学习场景的结果，而三种设置下的性能提升幅度则详见附录H中。

3.4 Winograd风格任务

Winograd句法挑战 [LDM12] 是自然语言处理领域中的一项经典任务，其核心目标是确定在代词在语法上存在歧义但对人类而言在语义上并无歧义的情况下，该代词所指代的单词。近期经过微调的语言模型在原始的Winograd数据集上已展现出接近人类表现水平的结果，但在更难版本面前

Setting	PIQA	ARC (Easy)	ARC (Challenge)	OpenBookQA
Fine-tuned SOTA	79.4	92.0 [KKS+20]	78.5 [KKS+20]	87.2 [KKS+20]
GPT-3 Zero-Shot	80.5 *	68.8	51.4	57.6
GPT-3 One-Shot	80.5 *	71.2	53.2	58.8
GPT-3 Few-Shot	82.8 *	70.1	51.5	65.4

Table 3.6: GPT-3 results on three commonsense reasoning tasks, PIQA, ARC, and OpenBookQA. GPT-3 Few-Shot PIQA result is evaluated on the test server. See Section 4 for details on potential contamination issues on the PIQA test set.

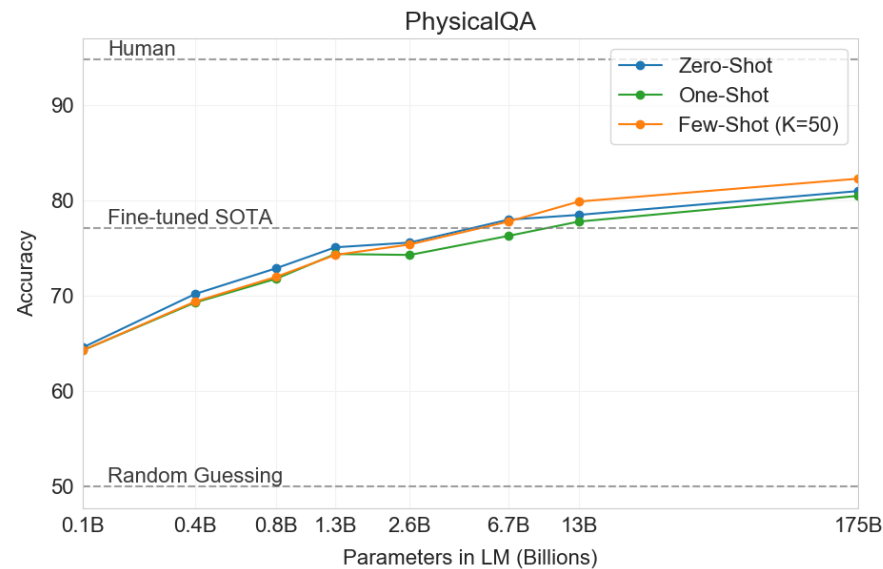


Figure 3.6: GPT-3 results on PIQA in the zero-shot, one-shot, and few-shot settings. The largest model achieves a score on the development set in all three conditions that exceeds the best recorded score on the task.

such as the adversarially-mined Winogrande dataset [SBBC19] still significantly lag human performance. We test GPT-3’s performance on both Winograd and Winogrande, as usual in the zero-, one-, and few-shot setting.

On Winograd we test GPT-3 on the original set of 273 Winograd schemas, using the same “partial evaluation” method described in [RWC+19]. Note that this setting differs slightly from the WSC task in the SuperGLUE benchmark, which is presented as binary classification and requires entity extraction to convert to the form described in this section. On Winograd GPT-3 achieves 88.3%, 89.7%, and 88.6% in the zero-shot, one-shot, and few-shot settings, showing no clear in-context learning but in all cases achieving strong results just a few points below state-of-the-art and estimated human performance. We note that contamination analysis found some Winograd schemas in the training data but this appears to have only a small effect on results (see Section 4).

On the more difficult Winogrande dataset, we do find gains to in-context learning: GPT-3 achieves 70.2% in the zero-shot setting, 73.2% in the one-shot setting, and 77.7% in the few-shot setting. For comparison a fine-tuned RoBERTa model achieves 79%, state-of-the-art is 84.6% achieved with a fine-tuned high capacity model (T5), and human performance on the task as reported by [SBBC19] is 94.0%.

3.5 Common Sense Reasoning

Next we consider three datasets which attempt to capture physical or scientific reasoning, as distinct from sentence completion, reading comprehension, or broad knowledge question answering. The first, PhysicalQA (PIQA) [BZB+19], asks common sense questions about how the physical world works and is intended as a probe of grounded understanding of the world. GPT-3 achieves 81.0% accuracy zero-shot, 80.5% accuracy one-shot, and 82.8% accuracy few-shot (the last measured on PIQA’s test server). This compares favorably to the 79.4% accuracy prior state-of-the-art of a

设置	PIQA	ARC简单级	ARC挑战版	OpenBookQA数据集
微调后的最优性能	79.4	92.0 [KKS+20]	78.5 [KKS+20]	87.2 [KKS+20]
GPT-3零样本学习	80.5	68.8	51.4	57.6
GPT-3单样本学习	80.5	71.2	53.2	58.8
GPT-3少样本学习	82.8	70.1	51.5	65.4

表3.6: GPT-3在三种常识推理任务——PIQA、ARC以及OpenBookQA上的结果。GPT-3少样本学习在PIQA任务上的结果是在测试服务器上评估得出的。有关PIQA测试集可能存在的污染问题的详细信息，请参见第4节。

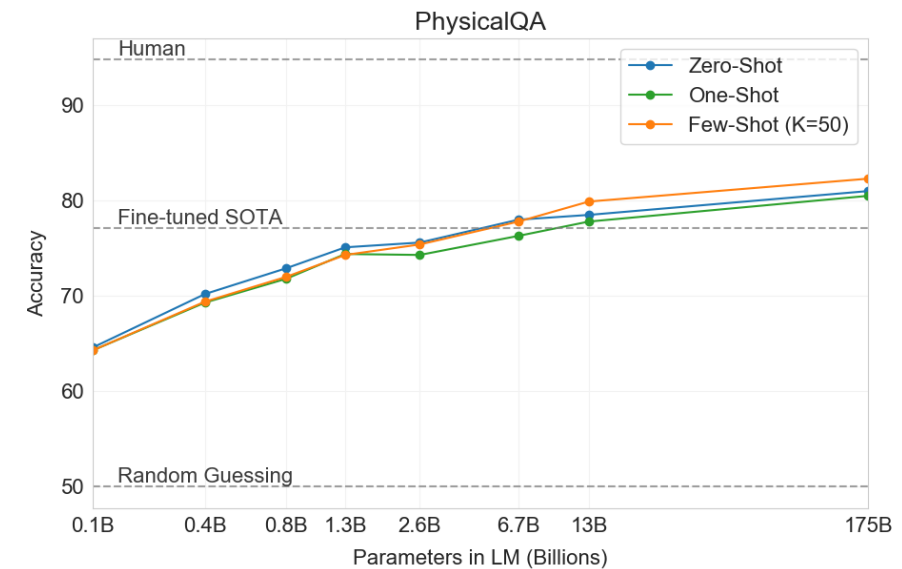


图3.6: GPT-3的响应结果 在零样本学习、单样本学习和少样本场景下的PIQA评测结果。规模最大的模型取得了最佳的开发集上的得分 该结果同时满足了全部三个条件，其性能甚至超越了该任务目前记录的最佳分数。

比如通过恶意构造得到的Winogrande数据集 [SBBC19]，这类数据的性能依然远远落后于人类表现水平。我们一如既往地 在零样本、单样本以及少样本场景下，测试GPT-3在Winograd和Winogrande数据集上的性能。

在Winograd任务中，我们使用与 [RWC+19]中描述的相同‘部分评估’方法，让GPT-3在原有的273个Winograd模式数据集上接受测试。需要说明的是，这种测试设置与SuperGLUE基准测试中的WSC任务略有不同——后者属于二元分类任务，还需通过实体抽取才能转换为本节所述的格式。在Winograd任务中，GPT-3在零样本学习、单样本学习和少样本学习三种场景下的性能分别达到88.3%、89.7%和88.6%，虽然并未展现出明显的上下文学习能力，但所有场景下的结果都十分优异，仅比当前最佳水平以及估计的人类表现水平低几个百分点。此外，污染分析发现训练数据中存在一些Winograd模式，但这似乎对最终结果的影响很小（详见4节）。

在难度更高的Winogrande数据集上，我们确实观察到了上下文学习的积极效果：GPT-3在零样本学习场景下的性能为70.2%，单样本学习场景下为73.2%，而在少样本学习场景下则提升至77.7%。作为对比，经过微调的RoBERTa模型的性能为79%，当前最佳水平则是通过微调的高容量模型（T5）实现的84.6%，而 [SBBC19]报告的该任务的人类表现水平则为94.0%。

3.5 常识推理

接下来我们研究三个旨在捕捉物理或科学推理能力的数据集，这类任务与句子补全、阅读理解或通用知识问答有明显区别。其中第一个数据集是PhysicalQA（即PIQA）[BZB+19]，它提出了一些关于物理世界运作规律的常识性问题，旨在考察模型对现实世界的真实理解能力。GPT-3在零样本学习场景下的准确率为81.0%，单样本学习场景下为80.5%，而少样本学习场景下的准确率则达到82.8%（后者是在PIQA的测试服务器上测得的数值）。这一表现优于此前处于当前最佳水平的那款经过微调后的RoBERTa模型的79.4%准确率

Setting	CoQA	DROP	QuAC	SQuADv2	RACE-h	RACE-m
Fine-tuned SOTA	90.7^a	89.1^b	74.4^c	93.0^d	90.0^e	93.1^e
GPT-3 Zero-Shot	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3 One-Shot	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3 Few-Shot	85.0	36.5	44.3	69.8	46.8	58.1

Table 3.7: Results on reading comprehension tasks. All scores are F1 except results for RACE which report accuracy. ^a[JZC⁺19] ^b[JN20] ^c[AI19] ^d[QIA20] ^e[SPP⁺19]

fine-tuned RoBERTa. PIQA shows relatively shallow scaling with model size and is still over 10% worse than human performance, but GPT-3’s few-shot and even zero-shot result outperform the current state-of-the-art. Our analysis flagged PIQA for a potential data contamination issue (despite hidden test labels), and we therefore conservatively mark the result with an asterisk. See Section 4 for details.

ARC [CCE⁺18] is a dataset of multiple-choice questions collected from 3rd to 9th grade science exams. On the “Challenge” version of the dataset which has been filtered to questions which simple statistical or information retrieval methods are unable to correctly answer, GPT-3 achieves 51.4% accuracy in the zero-shot setting, 53.2% in the one-shot setting, and 51.5% in the few-shot setting. This is approaching the performance of a fine-tuned RoBERTa baseline (55.9%) from UnifiedQA [KKS⁺20]. On the “Easy” version of the dataset (questions which either of the mentioned baseline approaches answered correctly), GPT-3 achieves 68.8%, 71.2%, and 70.1% which slightly exceeds a fine-tuned RoBERTa baseline from [KKS⁺20]. However, both of these results are still much worse than the overall SOTAs achieved by the UnifiedQA which exceeds GPT-3’s few-shot results by 27% on the challenge set and 22% on the easy set.

On OpenBookQA [MCKS18], GPT-3 improves significantly from zero to few shot settings but is still over 20 points short of the overall SOTA. GPT-3’s few-shot performance is similar to a fine-tuned BERT Large baseline on the leaderboard.

Overall, in-context learning with GPT-3 shows mixed results on commonsense reasoning tasks, with only small and inconsistent gains observed in the one and few-shot learning settings for both PIQA and ARC, but a significant improvement is observed on OpenBookQA. GPT-3 sets SOTA on the new PIQA dataset in all evaluation settings.

3.6 Reading Comprehension

Next we evaluate GPT-3 on the task of reading comprehension. We use a suite of 5 datasets including abstractive, multiple choice, and span based answer formats in both dialog and single question settings. We observe a wide spread in GPT-3’s performance across these datasets suggestive of varying capability with different answer formats. In general we observe GPT-3 is on par with initial baselines and early results trained using contextual representations on each respective dataset.

GPT-3 performs best (within 3 points of the human baseline) on CoQA [RCM19] a free-form conversational dataset and performs worst (13 F1 below an ELMo baseline) on QuAC [CHI⁺18] a dataset which requires modeling structured dialog acts and answer span selections of teacher-student interactions. On DROP [DWD⁺19], a dataset testing discrete reasoning and numeracy in the context of reading comprehension, GPT-3 in a few-shot setting outperforms the fine-tuned BERT baseline from the original paper but is still well below both human performance and state-of-the-art approaches which augment neural networks with symbolic systems [RLL⁺19]. On SQuAD 2.0 [RJL18], GPT-3 demonstrates its few-shot learning capabilities, improving by almost 10 F1 (to 69.8) compared to a zero-shot setting. This allows it to slightly outperform the best fine-tuned result in the original paper. On RACE [LXL⁺17], a multiple choice dataset of middle school and high school english examinations, GPT-3 performs relatively weakly and is only competitive with the earliest work utilizing contextual representations and is still 45% behind SOTA.

3.7 SuperGLUE

In order to better aggregate results on NLP tasks and compare to popular models such as BERT and RoBERTa in a more systematic way, we also evaluate GPT-3 on a standardized collection of datasets, the SuperGLUE benchmark [WPN⁺19] [WPN⁺19] [CLC⁺19] [DMST19] [RBG11] [KCR⁺18] [ZLL⁺18] [DGM06] [BHDD⁺06] [GMDD07] [BDD⁺09] [PCC18] [PHR⁺18]. GPT-3’s test-set performance on the SuperGLUE dataset is shown in Table 3.8. In the few-shot setting, we used 32 examples for all tasks, sampled randomly from the training set. For all tasks except WSC

设置	CoQA	DROP	QuAC、SQuADv2、RACE-h、RACE-m			
微调后的当前最佳性能	90.7^a	89.1^b	74.4^c	93.0^d	90.0^e	93.1^e
GPT-3零样本学习	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3单样本学习	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3少样本学习	85.0	36.5	44.3	69.8	46.8	58.1

表3.7: 阅读理解任务的结果。除RACE数据集的结果以准确率呈现外，其余所有分数均为F1分数。ⓐ[JZC⁺19] ⓑ[JN20] ⓒ[AI19] ⓓ[QIA20] ⓔ[SPP⁺19]

微调后的RoBERTa模型。PIQA任务的性能随模型规模的变化呈现相对平缓的提升趋势，且其表现仍比人类表现水平低10%以上；不过GPT-3无论是采用少样本学习还是零样本学习方式，其结果都优于当前的当前最佳水平。我们的分析发现PIQA可能存在数据污染问题（尽管存在隐藏的测试标签），因此出于谨慎考虑，我们用星号标出了该结果。详情请参见第4节。

ARC [CCE⁺18] 是一个从九年级到九年级的科学考试中收集而来、包含大量选择题的数据集。在该数据集的“挑战”版本中，那些仅靠简单的统计方法或信息检索方法无法正确解答的题目已被过滤掉，此时GPT-3在零样本场景下的准确率为51.4%，在单样本场景下为53.2%，在少样本场景下为51.5%，这一表现已接近UnifiedQA项目中微调后的RoBERTa基准模型的性能（55.9%）KKS⁺20]。而在该数据集的“简单”版本中——即那些上述两种基准模型都能正确作答的题目——GPT-3的准确率分别达到了68.8%、71.2%和70.1%，略高于来自 [KKS⁺20]的微调后的RoBERTa基准模型。不过，这两个数值仍然远逊于UnifiedQA整体所达到的当前最佳性能，后者在挑战数据集上的表现比GPT-3的少样本结果高出27%，在简单数据集上的表现则高出22%。

在OpenBookQA [MCKS18], 在从零样本学习设置到少样本学习设置的情况下，GPT-3的性能有了显著提升，但其整体性能仍比当前最佳性能低20多点。在排名榜上，GPT-3的少样本学习性能与经过微调的BERT Large基准模型相当。

总体而言，GPT-3在常识推理任务中的上下文学习效果参差不齐：在PIQA和ARC的单一样本及少样本学习设置中仅观察到微小且不稳定的性能提升，但在OpenBookQA上的表现则有显著改善。在所有评估设置下，GPT-3都在新的PIQA数据集上取得了当前最佳性能。

3.6 阅读理解任务

接下来我们将对GPT-3在阅读理解任务上的表现进行评估。我们使用了5个不同类型的数据集，这些数据集涵盖了抽象式答案、多项选择式答案以及基于文本片段的答案形式，并且包含了对话式和单问题式两种测试场景。我们发现GPT-3在各个数据集上的性能差异较大，这说明它在不同类型的答案格式下能力各异。总体来看，GPT-3的性能与最初的基准模型以及使用各数据集对应的上下文表征训练出的早期模型处于同一水平。在CoQA数据集上，GPT-3的表现最为优异（其性能仅比人类基准低3分）[RCM19] 这是一个用于自由式对话的开放格式数据集 [CHI⁺18] 该数据集要求对师生互动中的结构化对话行为及答案片段选择进行建模。在DROP数据集上 [DWD⁺19], 这是一个在阅读理解背景下测试离散推理与数值运算能力的数据集，处于少样本场景下的GPT-3虽然性能优于原始论文中经过微调的BERT基准模型，但仍远低于人类表现水平以及那些通过符号系统增强神经网络功能的当前最佳水平方法 [RLL⁺19]。在SQuAD 2.0 [RJL18], 中，GPT-3展现出了出色的少样本学习能力，与零样本学习场景相比，其F1分数提升了近10分，达到69.8分，这一成绩使其略胜原始论文中最佳的微调结果。在RACE数据集上 [LXL⁺17], 即一套针对初中和高中英语考试的单选题数据集，GPT-3的表现相对较弱，仅能与最早采用上下文表征技术的成果相媲美，其性能仍比当前最佳水平落后45%。

3.7 SuperGLUE任务集

为更系统地汇总各类自然语言处理任务的结果，并与BERT、RoBERTa等主流模型进行对比，我们还在标准化的数据集集合——SuperGLUE基准测试上对GPT-3进行了性能评估[WPN⁺19] [WPN⁺19] [CLC⁺19] [DMST19] [RBG11] [KCR⁺18] [ZLL⁺18] [DGM06] [BHDD⁺06] [GMDD07][BDD⁺09] [PCC18] [PHR⁺18]。GPT-3在SuperGLUE数据集上的测试集性能详见表3.8。在少样本场景下，所有任务均使用了从训练集中随机抽取的32个示例，WSC竞赛的所有任务除外

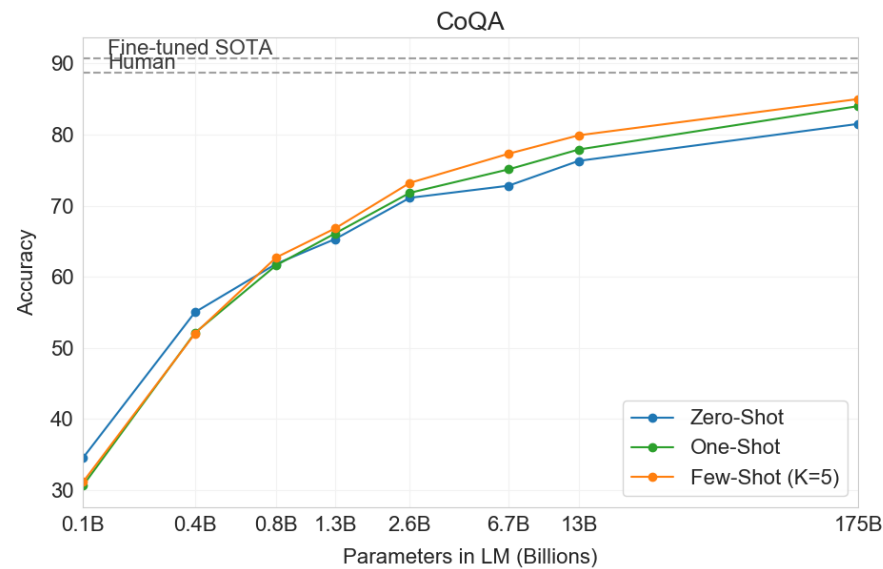


Figure 3.7: GPT-3 results on CoQA reading comprehension task. GPT-3 175B achieves 85 F1 in the few-shot setting, only a few points behind measured human performance and state-of-the-art fine-tuned models. Zero-shot and one-shot performance is a few points behind, with the gains to few-shot being largest for bigger models.

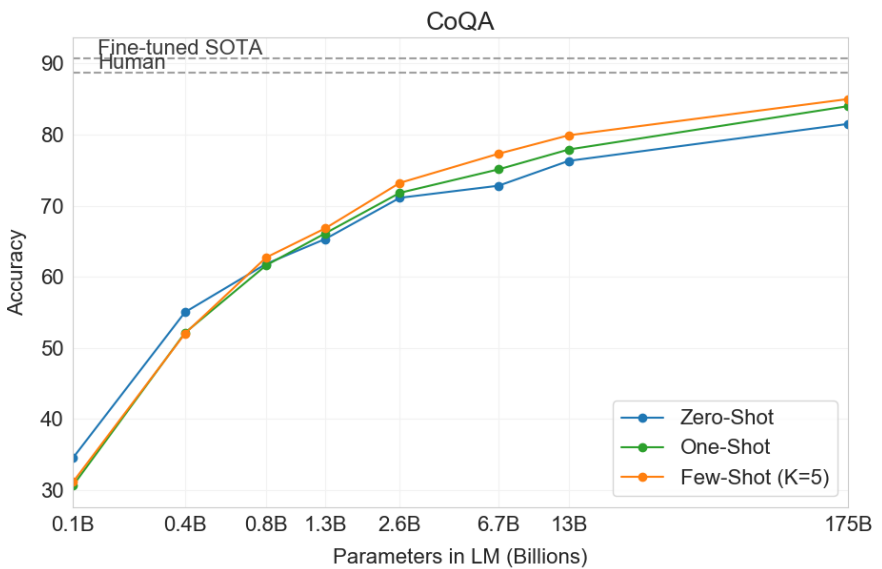


图3.7: GPT-3在CoQA阅读理解任务上的测试结果。GPT-3 175B在少样本场景下的F1分数为85，这一数值仅略低于人类表现水平以及经过微调的当前最佳水平模型。而零样本学习和单样本学习的性能则稍低一些，其中规模更大的模型在少样本学习方面的提升最为显著。

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	89.0	91.0	96.9	93.9	94.8	92.5
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	76.1	93.8	62.3	88.2	92.5	93.3
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

Table 3.8: Performance of GPT-3 on SuperGLUE compared to fine-tuned baselines and SOTA. All results are reported on the test set. GPT-3 few-shot is given a total of 32 examples within the context of each task and performs no gradient updates.

	SuperGLUE任务集 平均值	BoolQ任务 准确率	CB 准确率	CB F1	COPA 准确率	RTE 准确率
微调后的最优性能	89.0	91.0	96.9	93.9	94.8	92.5
BERT-Large模型微调后的性能	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3少样本学习	71.8	76.4	75.6	52.0	92.0	69.0

	WiC 准确率	WSC 准确率	MultiRC数据集 准确率	MultiRC数据集 F1a	ReCoRD算法 准确率	ReCoRD算法 F1
微调后的最优性能	76.1	93.8	62.3	88.2	92.5	93.3
微调后的BERT-Large模型性能	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3少样本学习	49.4	80.1	30.5	75.4	90.2	91.1

表3.8: GPT-3在SuperGLUE任务集上的性能表现，与经过微调的基线模型及当前最佳性能模型进行了对比。所有结果均基于测试集得出。在每项任务中，GPT-3少样本学习版本会获得32个示例作为上下文，并且不会进行梯度更新。

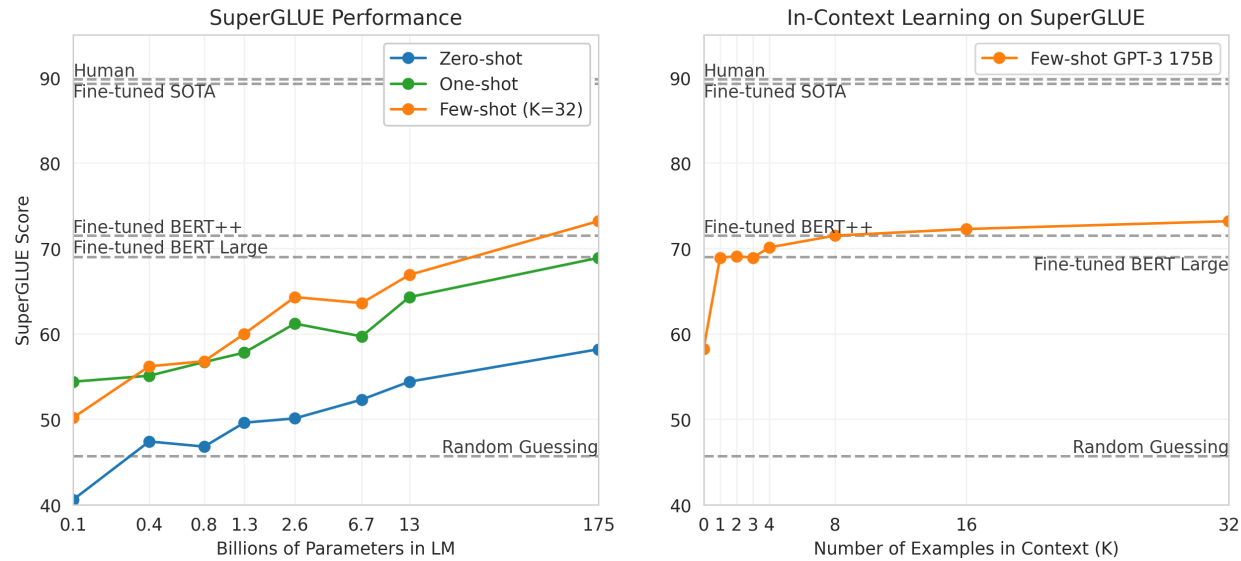


Figure 3.8: Performance on SuperGLUE increases with model size and number of examples in context. A value of $K = 32$ means that our model was shown 32 examples per task, for 256 examples total divided across the 8 tasks in SuperGLUE. We report GPT-3 values on the dev set, so our numbers are not directly comparable to the dotted reference lines (our test set results are in Table 3.8). The BERT-Large reference model was fine-tuned on the SuperGLUE training set (125K examples), whereas BERT++ was first fine-tuned on MultiNLI (392K examples) and SWAG (113K examples) before further fine-tuning on the SuperGLUE training set (for a total of 630K fine-tuning examples). We find the difference in performance between the BERT-Large and BERT++ to be roughly equivalent to the difference between GPT-3 with one example per context versus eight examples per context.

and MultiRC, we sampled a new set of examples to use in the context for each problem. For WSC and MultiRC, we used the same set of randomly drawn examples from the training set as context for all of the problems we evaluated.

We observe a wide range in GPT-3’s performance across tasks. On COPA and ReCoRD GPT-3 achieves near-SOTA performance in the one-shot and few-shot settings, with COPA falling only a couple points short and achieving second place on the leaderboard, where first place is held by a fine-tuned 11 billion parameter model (T5). On WSC, performance is still relatively strong, achieving 80.1% in the few-shot setting (note that GPT-3 achieves 88.6% on the original Winograd dataset as described in Section 3.4). On BoolQ, MultiRC, and RTE, performance is reasonable, roughly matching that of a fine-tuned BERT-Large. On CB, we see signs of life at 75.6% in the few-shot setting.

WiC is a notable weak spot with few-shot performance at 49.4% (at random chance). We tried a number of different phrasings and formulations for WiC (which involves determining if a word is being used with the same meaning in two sentences), none of which was able to achieve strong performance. This hints at a phenomenon that will become clearer in the next section (which discusses the ANLI benchmark) – GPT-3 appears to be weak in the few-shot or one-shot setting at some tasks that involve comparing two sentences or snippets, for example whether a word is used the same way in two sentences (WiC), whether one sentence is a paraphrase of another, or whether one sentence implies another. This could also explain the comparatively low scores for RTE and CB, which also follow this format. Despite these weaknesses, GPT-3 still outperforms a fine-tuned BERT-large on four of eight tasks and on two tasks GPT-3 is close to the state-of-the-art held by a fine-tuned 11 billion parameter model.

Finally, we note that the few-shot SuperGLUE score steadily improves with both model size and with number of examples in the context showing increasing benefits from in-context learning (Figure 3.8). We scale K up to 32 examples per task, after which point additional examples will not reliably fit into our context. When sweeping over values of K , we find that GPT-3 requires less than eight total examples per task to outperform a fine-tuned BERT-Large on overall SuperGLUE score.

3.8 NLI

Natural Language Inference (NLI) [Fyo00] concerns the ability to understand the relationship between two sentences. In practice, this task is usually structured as a two or three class classification problem where the model classifies

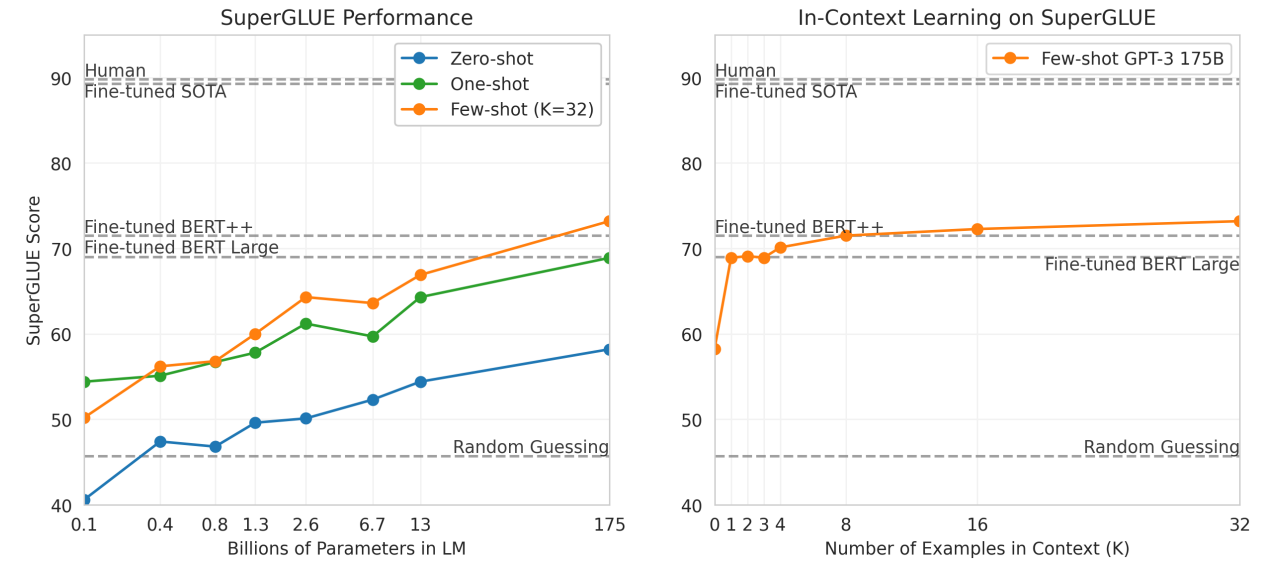


图3.8：在SuperGLUE任务集上的性能会随着模型规模以及上下文中的示例数量而提升。数值 $K = 32$ 表示我们的模型在每个任务上会看到32个示例，8个SuperGLUE任务总共共有256个示例。由于我们报告的是GPT-3在开发集上的性能数据，因此这些数值无法与虚线参考线直接对比（我们的测试集结果见<table style id=’8’>3.8中的内容）。BERT-Large参考模型是在SuperGLUE训练集（包含125K个示例）上经过微调的，而BERT++ 则先在MultiNLI数据集（392K个示例）和SWAG数据集（113K个示例）上进行微调，之后才在SuperGLUE训练集上进一步微调（微调示例总数达630K个）。我们发现BERT-Large与BERT之间的性能差值，大致相当于GPT-3在每个上下文仅有一个示例时的性能与每个上下文有八个示例时的性能差值。

在MultiRC数据集方面，我们为每个问题选取了新的示例集作为上下文。而对于WSC和MultiRC数据集，我们则使用从训练集中随机抽取的相同示例集，作为所有评估问题的上下文。

我们观察到GPT-3在不同任务上的性能存在较大差异。在COPA和ReCoRD任务中，GPT-3在单样本学习和少样本学习场景下均能展现出接近当前最佳性能的水平，其中在COPA任务中的表现仅落后几分，最终在排名榜上位列第二，而第一名则由经过微调的拥有110亿个参数的T5模型占据。在WSC任务中，其性能依然相当出色，在少样本学习场景下取得了80.1%的得分（需注意，如第3.4节所述，GPT-3在原始的Winograd数据集上的得分可达88.6%）。在BoolQ、MultiRC和RTE任务中，它的性能处于合理水平，大致与经过微调的BERT-Large模型相当。而在CB任务中，它在少样本学习场景下的得分也有75.6%，显示出一定的潜力。3.4)

WiC竞赛是少样本学习性能方面的一个明显薄弱点，其性能仅为49.4%（相当于随机水平的性能）。我们尝试了多种不同的表述方式来处理WiC任务——该任务旨在判断某个单词在两个句子中的含义是否相同——但都没有取得优异的性能。这一现象暗示了一个将在下一节（讨论ANLI基准测试的部分）中更加明确的规律：在那些需要比较两个句子或短文本的任务中，GPT-3在少样本或单样本场景下表现较弱，例如判断某个单词在两个句子中的用法是否一致（WiC任务）、判断一句话是否是另一句话的意译，或是判断一句话是否蕴含另一句话的含义。这也可以解释为何采用相同评测格式的RTE任务和CB任务的得分也相对较低。尽管存在这些缺陷，GPT-3在八项任务中的四项上仍优于经过微调的BERT-Large模型；而在另外两项任务上，它的性能则接近于由参数数量为110亿的经过微调的模型所达到的当前最佳水平。

最后需要指出的是，通过少样本学习完成SuperGLUE任务的得分会随着模型规模以及上下文中的示例数量一同稳步提升，这一现象充分体现了上下文学习的优势（见图3.8）。我们将每项任务中的示例数量上限设定为32个，因为超过这个数值后，更多的示例将无法有效纳入我们的上下文中。在遍历不同 K 参数值时，我们发现GPT-3在每项任务中仅需不到8个示例，就能在整体SuperGLUE得分上超越经过微调的BERT-Large模型。

3.8 自然语言推理

自然语言推理（NLI）[Fyo00]，主要研究的是模型理解两句话之间关系的能力。在实际应用中，这项任务通常被设计为二分类或三分类问题，模型需要据此对句子关系进行判定

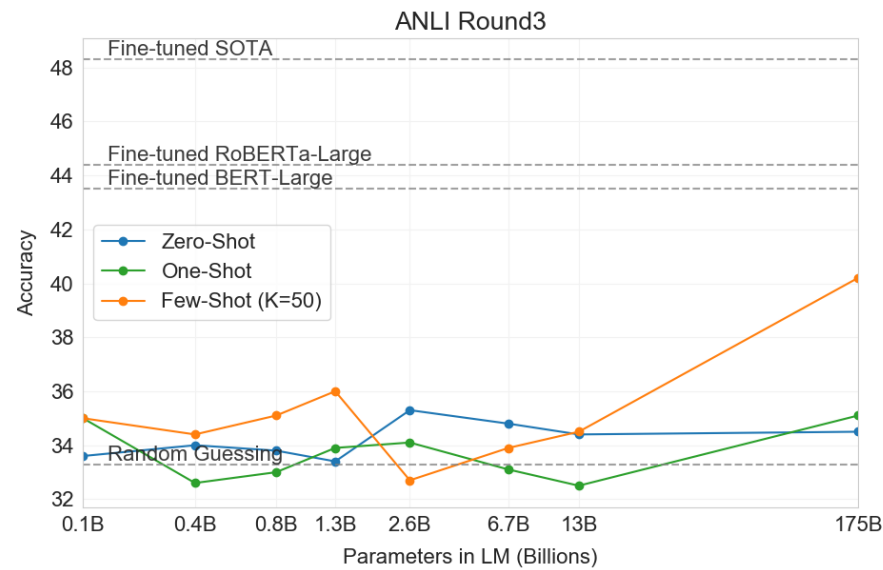


Figure 3.9: Performance of GPT-3 on ANLI Round 3. Results are on the dev-set, which has only 1500 examples and therefore has high variance (we estimate a standard deviation of 1.2%). We find that smaller models hover around random chance, while few-shot GPT-3 175B closes almost half the gap from random chance to SOTA. Results for ANLI rounds 1 and 2 are shown in the appendix.

whether the second sentence logically follows from the first, contradicts the first sentence, or is possibly true (neutral). SuperGLUE includes an NLI dataset, RTE, which evaluates the binary version of the task. On RTE, only the largest version of GPT-3 performs convincingly better than random (56%) in any evaluation setting, but in a few-shot setting GPT-3 performs similarly to a single-task fine-tuned BERT Large. We also evaluate on the recently introduced Adversarial Natural Language Inference (ANLI) dataset [NWD⁺19]. ANLI is a difficult dataset employing a series of adversarially mined natural language inference questions in three rounds (R1, R2, and R3). Similar to RTE, all of our models smaller than GPT-3 perform at almost exactly random chance on ANLI, even in the few-shot setting ($\sim 33\%$), whereas GPT-3 itself shows signs of life on Round 3. Results for ANLI R3 are highlighted in Figure 3.9 and full results for all rounds can be found in Appendix H. These results on both RTE and ANLI suggest that NLI is still a very difficult task for language models and they are only just beginning to show signs of progress.

3.9 Synthetic and Qualitative Tasks

One way to probe GPT-3’s range of abilities in the few-shot (or zero- and one-shot) setting is to give it tasks which require it to perform simple on-the-fly computational reasoning, recognize a novel pattern that is unlikely to have occurred in training, or adapt quickly to an unusual task. We devise several tasks to test this class of abilities. First, we test GPT-3’s ability to perform arithmetic. Second, we create several tasks that involve rearranging or unscrambling the letters in a word, tasks which are unlikely to have been exactly seen during training. Third, we test GPT-3’s ability to solve SAT-style analogy problems few-shot. Finally, we test GPT-3 on several qualitative tasks, including using new words in a sentence, correcting English grammar, and news article generation. We will release the synthetic datasets with the hope of stimulating further study of test-time behavior of language models.

3.9.1 Arithmetic

To test GPT-3’s ability to perform simple arithmetic operations without task-specific training, we developed a small battery of 10 tests that involve asking GPT-3 a simple arithmetic problem in natural language:

- **2 digit addition (2D+)** – The model is asked to add two integers sampled uniformly from $[0, 100)$, phrased in the form of a question, e.g. “Q: What is 48 plus 76? A: 124.”
- **2 digit subtraction (2D-)** – The model is asked to subtract two integers sampled uniformly from $[0, 100)$; the answer may be negative. Example: “Q: What is 34 minus 53? A: -19”.
- **3 digit addition (3D+)** – Same as 2 digit addition, except numbers are uniformly sampled from $[0, 1000)$.

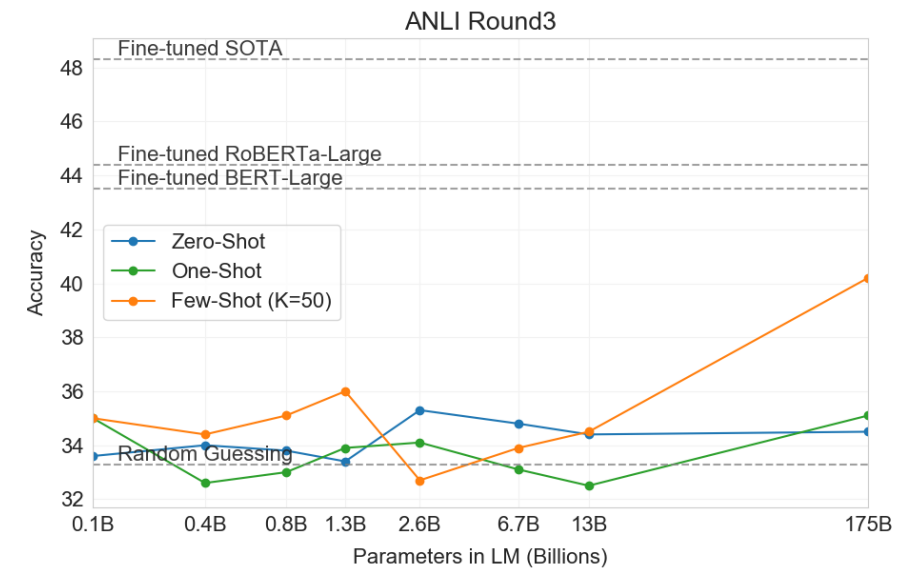


图3.9：GPT-3在ANLI第三轮中的性能表现。相关结果基于仅包含1500个示例的开发集得出，因此该数据集的标准差高达1.2%，波动较大。研究显示，规模较小的模型其性能大致处于随机水平，而采用少样本学习的GPT-3 175B版本则几乎缩小了从随机水平到当前最佳性能之间的半数差距。ANLI第一轮与第二轮的测试结果则记载在附录中。

，即判断第二句话是逻辑上承接第一句话、与第一句话相矛盾，还是可能成立（中性）。SuperGLUE任务集包含了用于评估该任务二分类版本的NLI数据集与RTE任务。在RTE任务中，只有规模最大的GPT-3模型在各类评估场景下的表现都显著优于随机水平（56%），而在少样本场景下，GPT-3的表现则与单任务微调后的BERT Large模型相当。此外，我们还使用了最新推出的对抗式自然语言推理（ANLI）数据集来进行评估 [NWD⁺19]。ANLI是一个难度极高的数据集，其中包含了通过对抗方式构造的、分为三轮（R1、R2和R3）的自然语言推理题目。与RTE任务类似，所有规模小于GPT-3的模型在ANLI上的表现几乎完全处于随机水平，即便在少样本场景下也是如此（ $\sim 33\%$ ），而GPT-3本身在第三轮则展现出了较好的表现。ANLI R3轮的结果标注在图3.9中，所有轮次的完整结果则可见于附录H。RTE和ANLI任务上的这些结果表明，对于语言模型而言自然语言推理仍然是一项极为艰巨的任务，它们目前才刚刚开始展现出进步的迹象。

3.9 合成数据集与定性任务

探究语言模型在少样本（或零样本、单样本）场景下的能力范围的一种方法，是给它布置一些需要其进行简单即时计算推理、识别在训练过程中极可能未曾出现的新奇模式，或是快速适应特殊任务的题目。我们设计了多项此类任务来检测这些能力。首先，测试GPT-3的算术运算能力；其次，设计若干涉及重新排列单词字母的任务，这类任务在训练数据中几乎不可能完整出现过；再次，测试GPT-3在少样本条件下解决SAT风格类比题的能力；最后，通过多项定性任务对GPT-3进行测试，这些任务包括在句子中使用新词汇、纠正英语语法以及生成新闻文章。我们将公开这些合成数据集，期望以此推动更多关于语言模型测试时行为的研究。

3.9.1 算术运算

为测试GPT-3在无需针对特定任务进行训练的情况下执行简单算术运算的能力，我们设计了一套包含10项测试的小型测试集，这些测试均以自然语言向GPT-3提出简单的算术问题。

- **2 两位数加法 (+)** ——要求模型对从 $[0, 100)$ 中均匀抽取的两个整数进行相加，问题需以提问形式呈现，例如 ‘问：48加上76等于多少？答：124。’
- **2 两位数减法 (-)** ——要求模型对从 $[0, 100)$ 中均匀抽取的两个整数进行相减，结果可能出现负数。示例： ‘问：34减去53等于多少？答：-19。’
- **3 数字加法 (三维+)** ——与两位数加法类似，不同之处在于数值是从 $[0, 1000)$ 中经过均匀采样得到的。

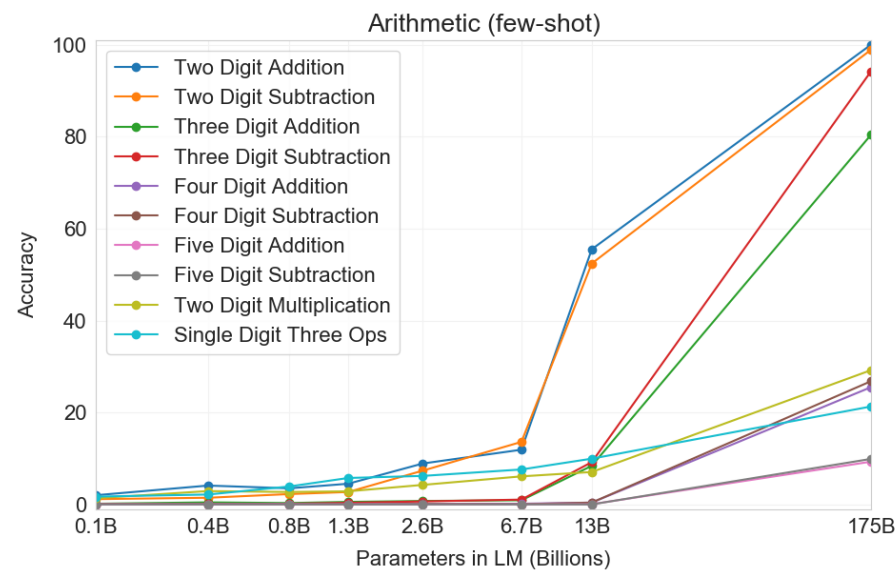


Figure 3.10: Results on all 10 arithmetic tasks in the few-shot settings for models of different sizes. There is a significant jump from the second largest model (GPT-3 13B) to the largest model (GPT-3 175), with the latter being able to reliably accurate 2 digit arithmetic, usually accurate 3 digit arithmetic, and correct answers a significant fraction of the time on 4-5 digit arithmetic, 2 digit multiplication, and compound operations. Results for one-shot and zero-shot are shown in the appendix.

- **3 digit subtraction (3D-)** – Same as 2 digit subtraction, except numbers are uniformly sampled from $[0, 1000]$.
- **4 digit addition (4D+)** – Same as 3 digit addition, except uniformly sampled from $[0, 10000]$.
- **4 digit subtraction (4D-)** – Same as 3 digit subtraction, except uniformly sampled from $[0, 10000]$.
- **5 digit addition (5D+)** – Same as 3 digit addition, except uniformly sampled from $[0, 100000]$.
- **5 digit subtraction (5D-)** – Same as 3 digit subtraction, except uniformly sampled from $[0, 100000]$.
- **2 digit multiplication (2Dx)** – The model is asked to multiply two integers sampled uniformly from $[0, 100]$, e.g. “Q: What is 24 times 42? A: 1008”.
- **One-digit composite (1DC)** – The model is asked to perform a composite operation on three 1 digit numbers, with parentheses around the last two. For example, “Q: What is $6+(4*8)$? A: 38”. The three 1 digit numbers are selected uniformly on $[0, 10]$ and the operations are selected uniformly from $\{+, -, *\}$.

In all 10 tasks the model must generate the correct answer exactly. For each task we generate a dataset of 2,000 random instances of the task and evaluate all models on those instances.

First we evaluate GPT-3 in the few-shot setting, for which results are shown in Figure 3.10. On addition and subtraction, GPT-3 displays strong proficiency when the number of digits is small, achieving 100% accuracy on 2 digit addition, 98.9% at 2 digit subtraction, 80.2% at 3 digit addition, and 94.2% at 3-digit subtraction. Performance decreases as the number of digits increases, but GPT-3 still achieves 25-26% accuracy on four digit operations and 9-10% accuracy on five digit operations, suggesting at least some capacity to generalize to larger numbers of digits. GPT-3 also achieves 29.2% accuracy at 2 digit multiplication, an especially computationally intensive operation. Finally, GPT-3 achieves 21.3% accuracy at single digit combined operations (for example, $9*(7+5)$), suggesting that it has some robustness beyond just single operations.

As Figure 3.10 makes clear, small models do poorly on all of these tasks – even the 13 billion parameter model (the second largest after the 175 billion full GPT-3) can solve 2 digit addition and subtraction only half the time, and all other operations less than 10% of the time.

One-shot and zero-shot performance are somewhat degraded relative to few-shot performance, suggesting that adaptation to the task (or at the very least recognition of the task) is important to performing these computations correctly. Nevertheless, one-shot performance is still quite strong, and even zero-shot performance of the full GPT-3 significantly

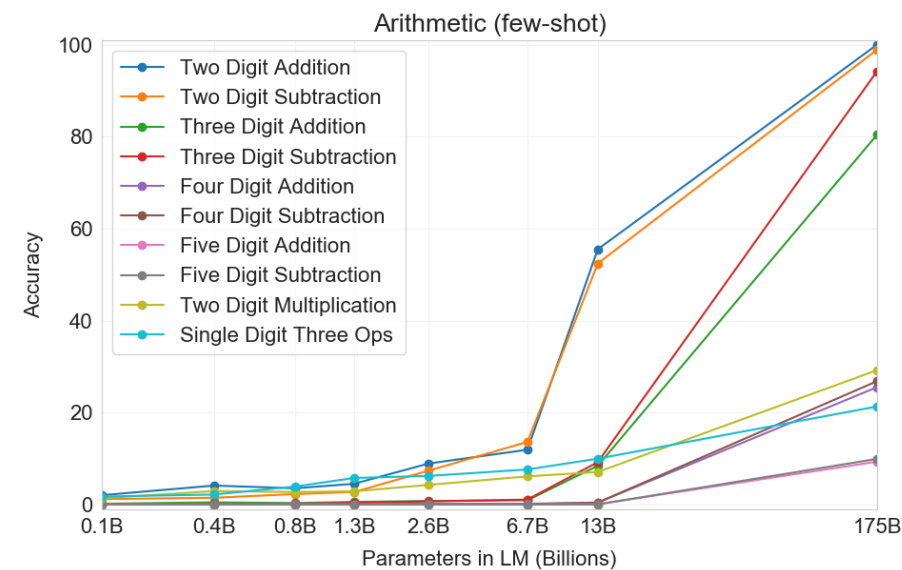


图3.10: 不同规模模型在少样本设置下完成全部10种算术任务的结果。从第二大的模型（GPT-3 130亿版本）到最大的模型（GPT-3 175模型），其准确率出现了显著提升；后者不仅能稳定地完成两位数算术运算，还能较为准确地处理三位数算术运算，并且在四到五位数算术运算、两位数乘法以及复合运算中也有相当比例的题目能得到正确答案。单样本学习和零样本学习的相关结果详见附录。

- **3 数字减法 (3D-)** ——与两位数减法类似，不同之处在于数值是从中经过均匀采样得到的。
- **4 数字加法 (4D+)** ——与三位数加法相同，只是数据是从中均匀采样的。
- **4 数字减法 (4D-)** [0, 10000]——与三位数减法相同，只是数据是从 [0, 10000]中均匀采样的。
- **5 数字加法 (5D+)** ——与三位数加法相同，只是数据是从中均匀采样的。
- **5 数字减法 (5D-)** [0, 100000]——与三位数减法相同，只是数据是从 [0, 100000]中均匀采样的。
- **2 两位数乘法 (2Dx任务)** ——该任务要求模型对从 [0, 100]中均匀抽取的两个整数进行相乘，例如“问：24乘以42等于多少？答：1008”。
- **一位数复合任务 (1DC任务)** ——该任务要求模型对三个一位数执行复合运算，且最后两个数需用括号括起来。例如“问： $6+(4*8)$ 等于多少？答：38”。这三个一位数是从 [0, 10]中均匀选取的，而运算符则从 $\{+, -, *\}$ 中均匀随机选择。

在全部10种任务中，模型都必须准确输出正确答案。针对每一种任务，我们会生成2,000个该任务的随机实例，并用这些实例来评估所有模型。

首先，我们在少样本场景下对GPT-3进行了测试，相关结果展示在图3.10中。在加法和减法方面，当数字位数较小时，GPT-3的表现极为出色：在两位数加法上的准确率可达100%，两位数减法为98.9%，三位数加法为80.2%，三位数减法为94.2%。随着数字位数的增加，其性能会有所下降，但在四位数运算中的准确率仍能达到25%-26%，五位数运算的准确率为9%-10%，这表明它至少具备一定的能力将所学知识推广到更多位数上。此外，GPT-3在计算量较大的两位数乘法任务中的准确率为29.2%。最后，在涉及单个数字的复合运算任务中（例如 $9*(7+5)$ ），它的准确率为21.3%，这说明它的稳定性不仅体现在单一运算上。

如图3.10所示，小型模型在所有这些任务上的表现都很差——即便是130亿参数模型（仅次于1750亿参数的完整版GPT-3的第二大模型），也只能在一半的情况下解决两位数加减法问题，而其他运算的正确率更是不足10%。

与少样本学习的性能相比，单样本学习和零样本学习的性能会有所下降，这表明要正确完成这类计算，模型必须能够适应任务（或者至少识别出该任务）。尽管如此，单样本学习的性能依然相当出色，而完整的GPT-3的零样本学习性能也同样十分优秀。

Setting	2D+	2D-	3D+	3D-	4D+	4D-	5D+	5D-	2Dx	1DC
GPT-3 Zero-shot	76.9	58.0	34.2	48.3	4.0	7.5	0.7	0.8	19.8	9.8
GPT-3 One-shot	99.6	86.4	65.5	78.7	14.0	14.0	3.5	3.8	27.4	14.3
GPT-3 Few-shot	100.0	98.9	80.4	94.2	25.5	26.8	9.3	9.9	29.2	21.3

Table 3.9: Results on basic arithmetic tasks for GPT-3 175B. {2,3,4,5}D{+,-} is 2, 3, 4, and 5 digit addition or subtraction, 2Dx is 2 digit multiplication. 1DC is 1 digit composite operations. Results become progressively stronger moving from the zero-shot to one-shot to few-shot setting, but even the zero-shot shows significant arithmetic abilities.

Setting	CL	A1	A2	RI	RW
GPT-3 Zero-shot	3.66	2.28	8.91	8.26	0.09
GPT-3 One-shot	21.7	8.62	25.9	45.4	0.48
GPT-3 Few-shot	37.9	15.1	39.7	67.2	0.44

Table 3.10: GPT-3 175B performance on various word unscrambling and word manipulation tasks, in zero-, one-, and few-shot settings. CL is “cycle letters in word”, A1 is anagrams of but the first and last letters, A2 is anagrams of all but the first and last two letters, RI is “Random insertion in word”, RW is “reversed words”.

outperforms few-shot learning for all smaller models. All three settings for the full GPT-3 are shown in Table 3.9, and model capacity scaling for all three settings is shown in Appendix H.

To spot-check whether the model is simply memorizing specific arithmetic problems, we took the 3-digit arithmetic problems in our test set and searched for them in our training data in both the forms "<NUM1> + <NUM2> =" and "<NUM1> plus <NUM2>". Out of 2,000 addition problems we found only 17 matches (0.8%) and out of 2,000 subtraction problems we found only 2 matches (0.1%), suggesting that only a trivial fraction of the correct answers could have been memorized. In addition, inspection of incorrect answers reveals that the model often makes mistakes such as not carrying a “1”, suggesting it is actually attempting to perform the relevant computation rather than memorizing a table.

Overall, GPT-3 displays reasonable proficiency at moderately complex arithmetic in few-shot, one-shot, and even zero-shot settings.

3.9.2 Word Scrambling and Manipulation Tasks

To test GPT-3’s ability to learn novel symbolic manipulations from a few examples, we designed a small battery of 5 “character manipulation” tasks. Each task involves giving the model a word distorted by some combination of scrambling, addition, or deletion of characters, and asking it to recover the original word. The 5 tasks are:

- **Cycle letters in word (CL)** – The model is given a word with its letters cycled, then the “=” symbol, and is expected to generate the original word. For example, it might be given “lyinevitab” and should output “inevitably”.
- **Anagrams of all but first and last characters (A1)** – The model is given a word where every letter except the first and last have been scrambled randomly, and must output the original word. Example: criroptuon = corruption.
- **Anagrams of all but first and last 2 characters (A2)** – The model is given a word where every letter except the first 2 and last 2 have been scrambled randomly, and must recover the original word. Example: opoepnnt → opponent.
- **Random insertion in word (RI)** – A random punctuation or space character is inserted between each letter of a word, and the model must output the original word. Example: s.u!c/c!e.s s i/o/n = succession.
- **Reversed words (RW)** – The model is given a word spelled backwards, and must output the original word. Example: stcejbo → objects.

For each task we generate 10,000 examples, which we chose to be the top 10,000 most frequent words as measured by [Nor09] of length more than 4 characters and less than 15 characters. The few-shot results are shown in Figure 3.11. Task performance tends to grow smoothly with model size, with the full GPT-3 model achieving 66.9% on removing

设置	2D+	2D-	3D+	3D-	4D+	4D-	5D+	5D-	2Dx	1DC
零样本GPT-3	76.9	58.0	34.2	48.3	4.0	7.5	0.7	0.8	19.8	9.8
单样本GPT-3	99.6	86.4	65.5	78.7	14.0	14.0	3.5	3.8	27.4	14.3
少样本GPT-3	100.0	98.9	80.4	94.2	25.5	26.8	9.3	9.9	29.2	21.3

表3.9: GPT-3 175B在基础算术任务上的结果。{2,3,4,5}D{+,-} 对应的是2位、3位、4位和5位的加法或减法运算，2Dx则为两位数乘法，而1DC则是涉及1位数的复合运算。从零样本学习场景到单样本学习场景再到少样本学习场景，模型的表现逐渐提升，即便是在零样本学习场景下，它也展现出了较强的算术运算能力。

设置	CL	A1	A2	RI	RW
零样本GPT-3	3.66	2.28	8.91	8.26	0.09
单样本GPT-3	21.7	8.62	25.9	45.4	0.48
GPT-3少样本学习	37.9	15.1	39.7	67.2	0.44

表3.10: 在零样本、单样本以及少样本场景下，GPT-3 175B模型在各类单词重排与单词操作任务中的性能表现。CL表示“单词内字母循环”，A1是指仅首尾字母互换得到的变位词，A2是指除首尾两字母外其余字母全部互换得到的变位词，RI表示“单词中随机插入字符”，RW则表示“单词倒序”。

在所有规模较小的模型中，其性能均优于少样本学习。完整GPT-3的三种设置相关数据展示在表 3.9中，而这三种设置下的模型容量与性能提升幅度则记载于附录 H中。

为验证该模型是否只是简单记住了特定的算术题，我们从测试集中的三位数算术题出发，在训练数据中以 "<NUM1> + <NUM2> =" 和 "<NUM1> plus <NUM2>" 这两种形式进行检索。在2000道加法题中仅找到17道匹配结果（占比0.8%），2000道减法题中也仅找到2道匹配结果（占比0.1%），这表明只有极少量正确答案可能是被记住的。此外，对错误答案的分析显示，该模型常常会出现未进位处理“1”之类的失误，这说明它实际上是在尝试进行相关计算，而非死记硬背计算表。

总体而言，在少样本学习、单样本学习，乃至零样本场景下，GPT-3在处理中等复杂度的算术运算时均展现出相当不错的能力。

3.9.2 单词乱序任务与字符操作任务

为测试GPT-3能否从少量示例中学习全新的符号操作方法，我们设计了一套包含5项“字符操作”任务的测试集。每项任务都会给模型一个因乱序、字符添加或删除等操作而发生变形的单词，要求其还原出原始单词。这5项任务分别是：

- **单词字母循环（CL）**——模型会收到一个字母顺序被循环过的单词以及“=”符号，需要据此生成原始单词。例如，若输入为“lyinevitab”，模型应输出“inevitably”。
- **除首尾字符外其余字符均为变位词（A1）**——模型会收到一个除首尾字符外所有字母都被随机乱序的单词，必须输出原始单词。示例：criroptuon =corruption。
- **除首尾两字母外其余字母互换的变位词（A2）**——模型会收到一个单词，其中除最开头和最末尾的两位字母外，其余所有字母都已被随机打乱，模型需要还原出原始单词。示例：opoepnnt→ opponent。
- **单词中随机插入字符（RI）**——会在单词的每个字母之间随机插入标点符号或空格字符，模型需要输出原始单词。示例：s.u!c/c!e.s s i/o/n = succession。
- **反向词序任务（RW）**——模型会收到一个倒序书写的单词，需要输出其原始形式。示例：stcejbo → objects。

针对每项任务，我们会生成10,000个示例，这些示例均为根据频率统计选出的最常出现的10,000个单词，且长度需在4个字符以上、15个字符以下09]。少样本学习的实验结果如图3.11所示。模型的性能通常会随着模型规模的增大而稳步提升，完整的GPT-3模型在‘删除’任务上的性能可达66.9%

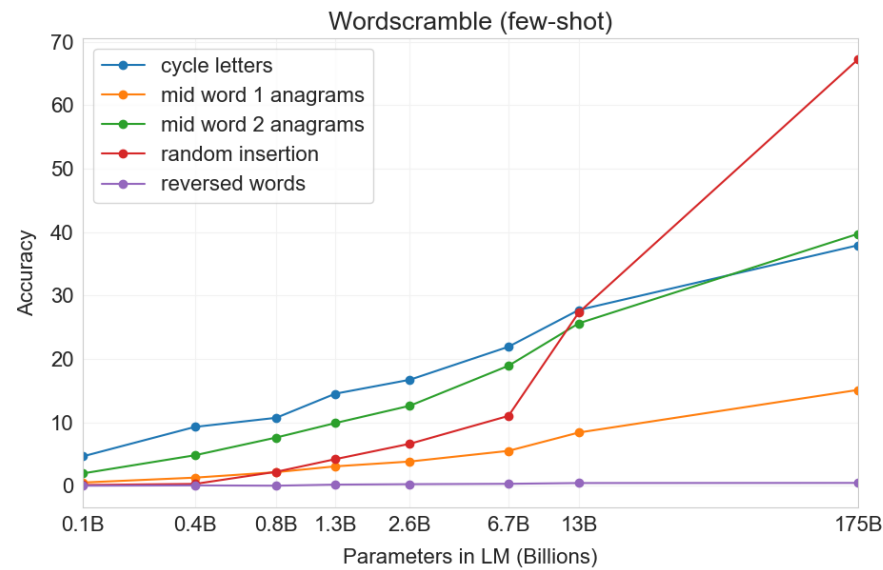


Figure 3.11: Few-shot performance on the five word scrambling tasks for different sizes of model. There is generally smooth improvement with model size although the random insertion task shows an upward slope of improvement with the 175B model solving the task the majority of the time. Scaling of one-shot and zero-shot performance is shown in the appendix. All tasks are done with $K = 100$.

random insertions, 38.6% on cycling letters, 40.2% on the easier anagram task, and 15.1% on the more difficult anagram task (where only the first and last letters are held fixed). None of the models can reverse the letters in a word.

In the one-shot setting, performance is significantly weaker (dropping by half or more), and in the zero-shot setting the model can rarely perform any of the tasks (Table 3.10). This suggests that the model really does appear to learn these tasks at test time, as the model cannot perform them zero-shot and their artificial nature makes them unlikely to appear in the pre-training data (although we cannot confirm this with certainty).

We can further quantify performance by plotting “in-context learning curves”, which show task performance as a function of the number of in-context examples. We show in-context learning curves for the Symbol Insertion task in Figure 1.2. We can see that larger models are able to make increasingly effective use of in-context information, including both task examples and natural language task descriptions.

Finally, it is worth adding that solving these tasks requires character-level manipulations, whereas our BPE encoding operates on significant fractions of a word (on average ~ 0.7 words per token), so from the LM’s perspective succeeding at these tasks involves not just manipulating BPE tokens but understanding and pulling apart their substructure. Also, CL, A1, and A2 are not bijective (that is, the unscrambled word is not a deterministic function of the scrambled word), requiring the model to perform some search to find the correct unscrambling. Thus, the skills involved appear to require non-trivial pattern-matching and computation.

3.9.3 SAT Analogies

To test GPT-3 on another task that is somewhat unusual relative to the typical distribution of text, we collected a set of 374 “SAT analogy” problems [TLBS03]. Analogies are a style of multiple choice question that constituted a section of the SAT college entrance exam before 2005. A typical example is “audacious is to boldness as (a) sanctimonious is to hypocrisy, (b) anonymous is to identity, (c) remorseful is to misdeed, (d) deleterious is to result, (e) impressionable is to temptation”. The student is expected to choose which of the five word pairs has the same relationship as the original word pair; in this example the answer is “sanctimonious is to hypocrisy”. On this task GPT-3 achieves 65.2% in the few-shot setting, 59.1% in the one-shot setting, and 53.7% in the zero-shot setting, whereas the average score among college applicants was 57% [TL05] (random guessing yields 20%). As shown in Figure 3.12, the results improve with scale, with the the full 175 billion model improving by over 10% compared to the 13 billion parameter model.

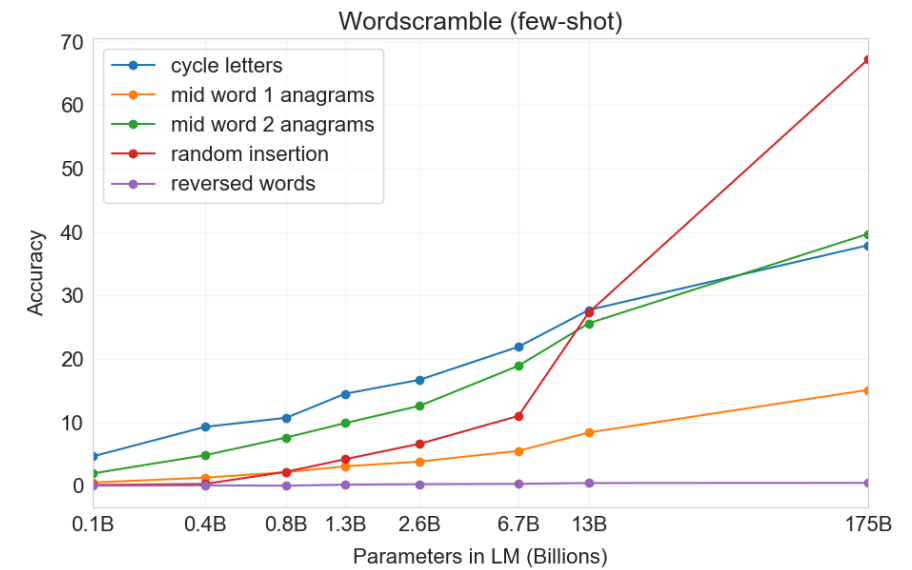


图3.11: 不同模型规模在五字乱序任务上的少样本性能表现。总体而言，随着模型规模的增大，性能会持续平稳提升；不过在随机插入任务中，1750亿参数模型大多情况下都能成功完成该任务，其性能提升趋势更为陡峭。单样本学习和零样本学习的性能提升幅度详见附录。所有测试均使用进行。

在‘随机插入’任务上该模型的性能为38.6%，在‘字母循环任务’上的性能为40.2%，在相对简单的变位词任务上的性能为40.2%，而在难度更高的变位词任务（仅保留单词的首尾字母不变）中其性能仅为15.1%。所有测试的模型都无法实现将单词中的字母顺序反转的功能。

在单样本场景下，模型的性能会显著下降（甚至降低一半以上）；而在零样本场景下，模型几乎无法完成任何任务（见表3.10）。这一现象表明，模型确实是在测试时才学会这些任务的，因为它在零样本条件下无法完成它们，而且这类任务的人为设计特性也使得它们不太可能出现在预训练数据中（尽管我们无法完全确定这一点）。

我们可以通过绘制“上下文学习曲线”来进一步量化性能，这类曲线能够反映任务性能随上下文示例数量的变化情况。图1.2展示了符号插入任务的上下文学习曲线。从曲线中可以看出，更大规模的模型能够越来越有效地利用上下文信息，包括任务示例以及自然语言任务描述。

最后需要补充的是，解决这类任务需要进行字符级的操作，而我们的BPE编码则是以单词的大部分内容作为处理对象（平均每个令牌对应 ~ 0.7 个单词）。因此从语言模型的角度来看，要完成这些任务，不仅需要操作BPE编码生成的令牌，还需要理解并拆解这些令牌的内部结构。此外，单词字母循环、仅首尾字母互换的变位词以及除首尾两字母外其余字母互换的变位词并非双射关系（也就是说，打乱后的单词无法通过确定性方式还原为原始单词），这就要求模型通过一定的搜索机制来找到正确的还原方法。由此可见，完成这类任务所需的技能涉及较为复杂的模式匹配与计算过程。

3.9.3 SAT类比题

为测试GPT-3在一种与常规文本概率分布略有不同的任务上的表现，我们收集了374道‘SAT类比题’作为测试数据 [TLBS03]。类比题是一种选择题形式，在2005年之前的SAT大学入学考试中属于必考题型。一个典型的示例是：‘大胆的与勇气之间的关系，类似于（a）伪善的与虚伪的关系，（b）匿名的与身份的关系，（c）懊悔的与恶行的关系，（d）有害的与结果的关系，还是（e）易受影响的与诱惑的关系？’。考生需要从这五组词对中选出与原词对关系一致的选项；在这个例子中正确答案是“伪善的与虚伪的关系”。在该任务中，GPT-3在少样本场景下的准确率为65.2%，在单样本场景下为59.1%，在零样本场景下则为53.7%，而普通高考考生的平均得分仅为57% [TL05]（随机猜测的准确率仅为20%）。如图3.12所示，随着模型规模的增大，其性能也会提升，1750亿参数的完整模型相比130亿参数模型，性能提升了10%以上。

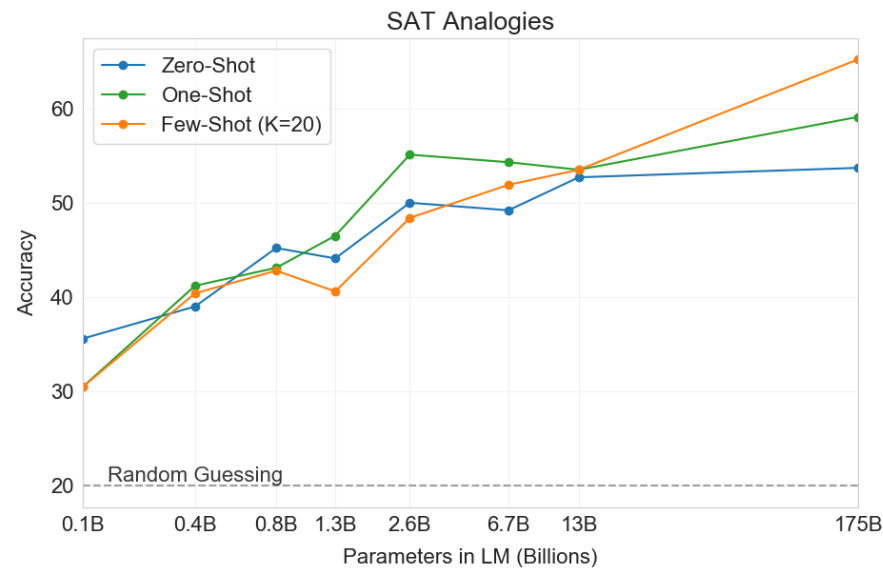


Figure 3.12: Zero-, one-, and few-shot performance on SAT analogy tasks, for different sizes of model. The largest model achieves 65% accuracy in the few-shot setting, and also demonstrates significant gains to in-context learning which are not present in smaller models.

3.9.4 News Article Generation

Previous work on generative language models qualitatively tested their ability to generate synthetic “news articles” by conditional sampling from the model given a human-written prompt consisting of a plausible first sentence for a news story [RWC⁺19]. Relative to [RWC⁺19], the dataset used to train GPT-3 is much less weighted towards news articles, so trying to generate news articles via raw unconditional samples is less effective – for example GPT-3 often interprets the proposed first sentence of a “news article” as a tweet and then posts synthetic responses or follow-up tweets. To solve this problem we employed GPT-3’s few-shot learning abilities by providing three previous news articles in the model’s context to condition it. With the title and subtitle of a proposed next article, the model is able to reliably generate short articles in the “news” genre.

To gauge the quality of news article generation from GPT-3 (which we believe is likely to be correlated with conditional sample generation quality in general), we decided to measure human ability to distinguish GPT-3-generated articles from real ones. Similar work has been carried out by Kreps et al. [KMB20] and Zellers et al. [ZHR⁺19]. Generative language models are trained to match the distribution of content generated by humans, so the (in)ability of humans to distinguish the two is a potentially important measure of quality.³

In order to see how well humans can detect model generated text, we arbitrarily selected 25 article titles and subtitles from the website [newser.com](https://www.newser.com) (mean length: 215 words). We then generated completions of these titles and subtitles from four language models ranging in size from 125M to 175B (GPT-3) parameters (mean length: 200 words). For each model, we presented around 80 US-based participants with a quiz consisting of these real titles and subtitles followed by either the human written article or the article generated by the model⁴. Participants were asked to select whether the article was “very likely written by a human”, “more likely written by a human”, “I don’t know”, “more likely written by a machine”, or “very likely written by a machine”.

The articles we selected were not in the models’ training data and the model outputs were formatted and selected programmatically to prevent human cherry-picking. All models used the same context to condition outputs on and were pre-trained with the same context size and the same article titles and subtitles were used as prompts for each model. However, we also ran an experiment to control for participant effort and attention that followed the same format but involved intentionally bad model generated articles. This was done by generating articles from a “control model”: a 160M parameter model with no context and increased output randomness.

³This task is also relevant to the potential misuse of language models discussed in Section 6.1.

⁴We wanted to identify how good an average person on the internet is at detecting language model outputs, so we focused on participants drawn from the general US population. See Appendix E for details.

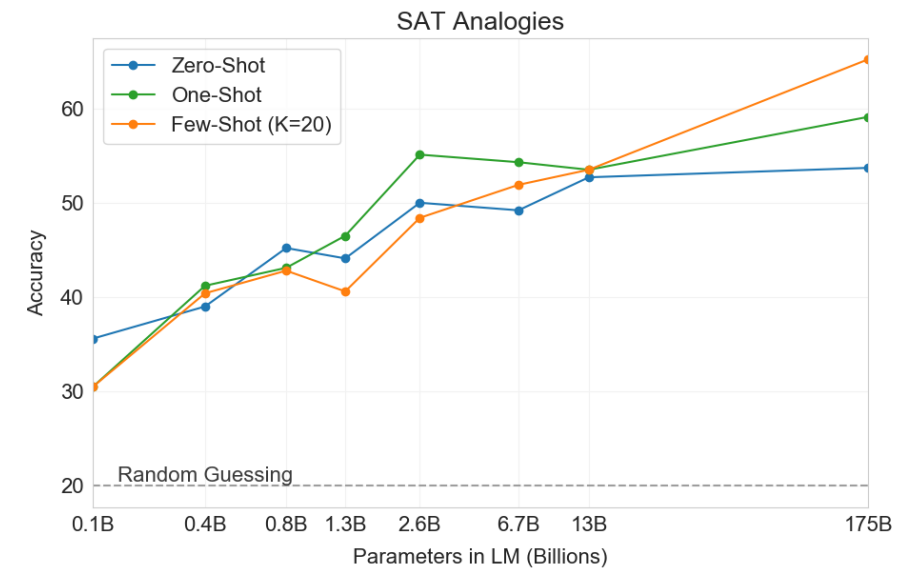


图3.12: 不同规模模型在SAT类比任务上的零样本、单样本及少样本学习性能表现。规模最大的模型在少样本场景下的准确率可达65%，同时在上文学习方面也展现出显著优势，而这些优势在规模较小的模型中并不存在。

3.9.4 新闻文章生成

此前关于生成型语言模型的研究，是通过给模型提供包含合理新闻开头句的人为提示，再利用条件采样方式，来定性评估这些模型生成合成“新闻文章”的能力 [RWC⁺19]。与 [RWC⁺19]相比，用于训练GPT-3的数据集在新闻文章所占比例上要低得多，因此直接通过无条件采样来生成新闻文章的效果较差——例如GPT-3常常会将指定的新闻开头句误解为推文，进而生成虚假回复或后续推文。为解决这一问题，我们借助GPT-3的少样本学习能力，在其上下文中提供三篇先前的新闻文章作为条件输入，从而引导它生成内容。只要给出下一篇文章的标题和副标题，该模型就能稳定地生成属于“新闻”类别的短篇内容。

为了评估GPT-3生成新闻文章的质量（我们认为这一质量很可能与一般的条件采样生成质量存在关联），我们决定通过检测人类区分GPT-3生成的文章与真实文章的能力来进行衡量。Kreps等人也曾开展过类似的研究。[KMB20] 以及Zellers等人 [ZHR⁺19]。由于生成型语言模型是被训练来匹配人类生成内容的概率分布的，因此人类区分这两类内容的能力强弱便成为衡量其质量的一个潜在重要指标。³

为了测试人类识别模型生成文本的能力究竟如何，我们从 [newser.com](https://www.newser.com)（平均长度为215个单词）上随机选取了25篇文章的标题和副标题。接着，我们利用四个参数规模从1.25亿参数到1750亿参数不等（GPT-3）的语言模型（这些模型的生成文本平均长度为200个单词），分别为这些标题和副标题生成补全结果。对于每一个模型，我们都向大约80名来自美国的参与者展示了一组测验题，题目包括这些真实的标题和副标题，随后还会给出对应的人类撰写的文章或模型生成的文章⁴。参与者需要判断这篇文章是“极有可能是人类所写”、“更有可能是人类所写”、“不知道”、“更有可能是机器所写”，还是“极有可能是机器所写”。

我们选中的文章并不包含在这些模型的训练数据中，且模型输出是通过编程方式进行处理和筛选的，以此避免人为挑选。所有模型都使用相同的上下文来生成输出，均采用相同的上下文长度进行预训练，同时每款模型也都使用相同的文章标题和副标题作为提示词。不过，我们还进行了一项实验，通过使用格式相同的但刻意生成的劣质模型输出内容，来控制参与者在努力程度和注意力分配方面的差异。该实验是通过一个“对照模型”来生成文章的：这个拥有1600亿参数、没有上下文且输出随机性更高的模型被用来生成这些劣质内容。

³这项任务也与第 6.1 节中讨论的语言模型可能被滥用的问题密切相关。⁴为了弄清普通互联网用户识别语言模型生成内容的能力究竟如何，我们选择了来自美国普通人群体的参与者作为研究对象。详情请参见附录 E。

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p -value)	“I don’t know” assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 (2e-4)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 (7e-21)	6.0%
GPT-3 Large	68%	64%–72%	7.3 (3e-11)	8.7%
GPT-3 XL	62%	59%–65%	10.7 (1e-19)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 (5e-19)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 (3e-21)	6.2%
GPT-3 13B	55%	52%–58%	15.3 (1e-32)	7.1%
GPT-3 175B	52%	49%–54%	16.9 (1e-34)	7.8%

Table 3.11: Human accuracy in identifying whether short (~200 word) news articles are model generated. We find that human accuracy (measured by the ratio of correct assignments to non-neutral assignments) ranges from 86% on the control model to 52% on GPT-3 175B. This table compares mean accuracy between five different models, and shows the results of a two-sample T-Test for the difference in mean accuracy between each model and the control model (an unconditional GPT-3 Small model with increased output randomness).

Mean human accuracy (the ratio of correct assignments to non-neutral assignments per participant) at detecting that the intentionally bad articles were model generated was $\sim 86\%$ where 50% is chance level performance. By contrast, mean human accuracy at detecting articles that were produced by the 175B parameter model was barely above chance at $\sim 52\%$ (see Table 3.11).⁵ Human abilities to detect model generated text appear to decrease as model size increases: there appears to be a trend towards chance accuracy with model size, and human detection of GPT-3 is close to chance.⁶ This is true despite the fact that participants spend more time on each output as model size increases (see Appendix E).

Examples of synthetic articles from GPT-3 are given in Figures 3.14 and 3.15.⁷ Much of the text is—as indicated by the evaluations—difficult for humans to distinguish from authentic human content. Factual inaccuracies can be an indicator that an article is model generated since, unlike human authors, the models have no access to the specific facts that the article titles refer to or when the article was written. Other indicators include repetition, non sequiturs, and unusual phrasings, though these are often subtle enough that they are not noticed.

Related work on language model detection by Ippolito et al. [IDCBE19] indicates that automatic discriminators like GROVER [ZHR⁺19] and GLTR [GSR19] may have greater success at detecting model generated text than human evaluators. Automatic detection of these models may be a promising area of future research.

Ippolito et al. [IDCBE19] also note that human accuracy at detecting model generated text increases as humans observe more tokens. To do a preliminary investigation of how good humans are at detecting longer news articles generated by GPT-3 175B, we selected 12 world news articles from Reuters with an average length of 569 words and generated completions of these articles from GPT-3 with an average length of 498 words (298 words longer than our initial experiments). Following the methodology above, we ran two experiments, each on around 80 US-based participants, to compare human abilities to detect the articles generated by GPT-3 and a control model.

We found that mean human accuracy at detecting the intentionally bad longer articles from the control model was $\sim 88\%$, while mean human accuracy at detecting the longer articles that were produced by GPT-3 175B was still barely above chance at $\sim 52\%$ (see Table 3.12). This indicates that, for news articles that are around 500 words long, GPT-3 continues to produce articles that humans find difficult to distinguish from human written news articles.

3.9.5 Learning and Using Novel Words

A task studied in developmental linguistics [CB78] is the ability to learn and utilize new words, for example using a word in a sentence after seeing it defined only once, or conversely inferring a word’s meaning from only one usage. Here we qualitatively test GPT-3’s ability to do the former. Specifically, we give GPT-3 the definition of a nonexistent word, such as “Gigamuru”, and then ask it to use it in a sentence. We provide one to five previous examples of a (separate)

⁵We use a two-sample Student’s T-Test to test for significant difference between the means of the participant accuracies of each model and the control model and report the normalized difference in the means (as the t-statistic) and the p-value.

⁶If a model consistently produces texts that are more impressive than human articles, it is possible that human performance on this task would drop below 50%. Indeed, many individual participants scored below 50% on this task.

⁷Additional non-news samples can be found in Appendix F.

	平均准确率	95%置信区间（下限，上限）	t 与……相比 对照组 (p 值)	"我不知道" 分类结果
对照组（故意使用性能较差的模型）	86%	83%–90%	-	3.6%
GPT-3小型版	76%	72%–80%	3.9 (2e-4)	4.9%
GPT-3中型版	61%	58%–65%	10.3 (7e-21)	6.0%
GPT-3大型版	68%	64%–72%	7.3 (3e-11)	8.7%
GPT-3超大型版	62%	59%–65%	10.7 (1e-19)	7.5%
GPT-3 27亿版本	62%	58%–65%	10.4 (5e-19)	7.1%
GPT-3 67亿版本	60%	56%至63%	11.2 (3e-21)	6.2%
GPT-3 130亿版本	55%	52%至58%	15.3 (1e-32)	7.1%
GPT-3 175B版本	52%	49%至54%	16.9 (1e-34)	7.8%

表3.11：人类在判断短篇（~200 单词）长度的新闻文章是否由模型生成方面的准确率。 研究显示，以正确分类结果与非中性分类结果的比值来衡量的人类准确率，在对照模型上为86%，而在GPT-3 175B版本上则降至52%。该表展示了五种不同模型的平均准确率，并通过双样本T检验给出了每种模型与对照模型（即一种具有更高输出随机性的无条件GPT-3小型版模型）之间平均准确率差值的测试结果。

识别出那些刻意生成的劣质文章为模型输出的人类平均准确率（即每位参与者中正确分类结果与非中性分类结果的比例）为 $\sim 86\%$ ，其中50%对应随机水平表现。相比之下，识别由1750亿参数模型生成的文章时，人类的平均准确率仅略高于随机水平，为 $\sim 52\%$ <style id='11'>（详见表 3.11）。⁵ 随着模型规模的增大，人类识别模型生成文本的能力似乎会下降：准确率呈现出随模型规模增长而趋近于随机水平的趋势，而人类对GPT-3生成的文本的识别准确率也接近随机水平。⁶尽管随着模型规模的增加，参与者在查看每条输出内容时所花费的时间也会更多，但这一现象依然存在（详见附录 E）。

GPT-3生成的合成文章示例可见于图3.14、3.15。⁷正如评估结果所示，这些文本中的大部分很难让人类区分出其与真实人类创作内容的差异。事实错误可能是文章由模型生成的迹象，因为与人类作者不同，模型无法知晓文章标题所提及的具体事实或文章的创作时间。其他特征还包括重复内容、逻辑脱节以及异常的表达方式，不过这些往往十分微妙，很容易被忽视。

Ippolito等人关于语言模型检测的相关研究 [IDCBE19]表明，GROVER [ZHR⁺19]以及GLTR [GSR19]这类自动识别工具在检测模型生成的生成文本方面，可能比人类评估者更具优势。未来对这些模型进行自动检测很可能会成为一项极具前景的研究方向。

Ippolito等人还指出，人类在检测模型生成的生成文本时的准确率会随着观察到的令牌数量增加而提升。为了初步探究人类在检测由GPT-3 175B生成的较长新闻文章方面的能力，我们从路透社选取了12篇平均长度为569词的全球新闻文章，并让GPT-3为这些文章生成补全内容，其平均长度为498词（比最初的实验长298词）。按照上述方法，我们开展了两项实验，每项实验都有约80名来自美国的参与者参与，旨在比较他们识别GPT-3生成的文章与对照模型生成的文章的能力。

我们发现，使用对照模型识别那些刻意编造的较长文章时，人类的平均准确率为 $\sim 88\%$ ；而使用GPT-3 175B版本生成的长篇文章时，人类的平均准确率仅略高于随机猜测的水平，为 $\sim 52\%$ <style id='11'>（详见表3.12）。这表明，对于长度约为500字的新闻文章，GPT-3生成的文本依然难以让人类区分其与人工撰写的新闻文章。

3.9.5 学习与使用新词

发展语言学中研究过的一项任务 [CB任务78]，就是学习并运用新词的能力——例如在仅见过某词一次定义后就能在句子中使用它，或者仅通过该词的某一次用法就推断出其含义。在此我们通过定性方式测试GPT-3具备前者这种情况的能力。具体而言，我们会给GPT-3提供一个并不存在的词的定义，比如“Gigamuru”，然后要求它用这个词造句。同时我们会提供一到五个该词之前的（独立）

⁵我们采用双样本学生T检验来检测各模型与对照模型的参与者准确率均值之间是否存在显著差值，并报告均值的标准化差值（即t统计量）以及对应的p值。⁶如果某个模型生成的文本始终比人类撰写的文章更出色，那么人类在此任务上的表现水平有可能降至50%以下。事实上，许多参与者在这项任务中的得分确实低于50%。⁷更多非新闻类样本可见于附录F。

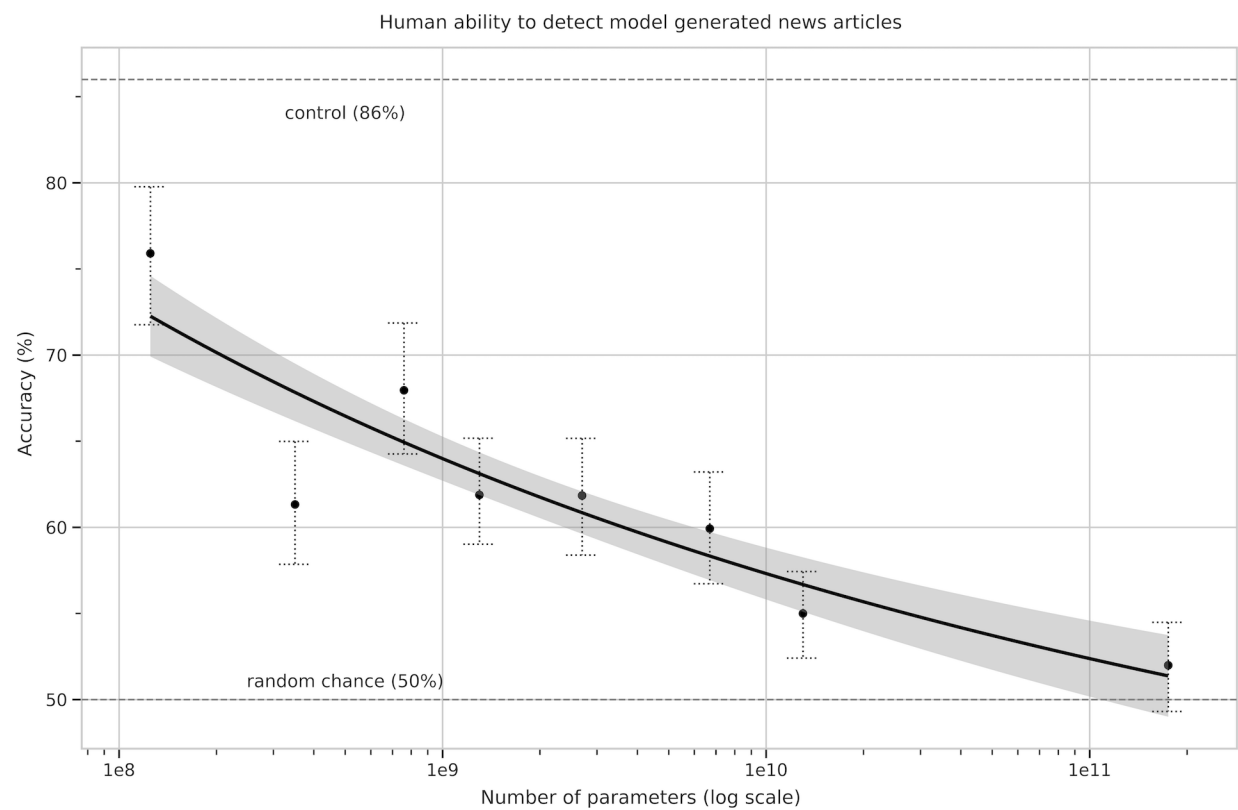


Figure 3.13: People’s ability to identify whether news articles are model-generated (measured by the ratio of correct assignments to non-neutral assignments) decreases as model size increases. Accuracy on the outputs on the deliberately-bad control model (an unconditioned GPT-3 Small model with higher output randomness) is indicated with the dashed line at the top, and the random chance (50%) is indicated with the dashed line at the bottom. Line of best fit is a power law with 95% confidence intervals.

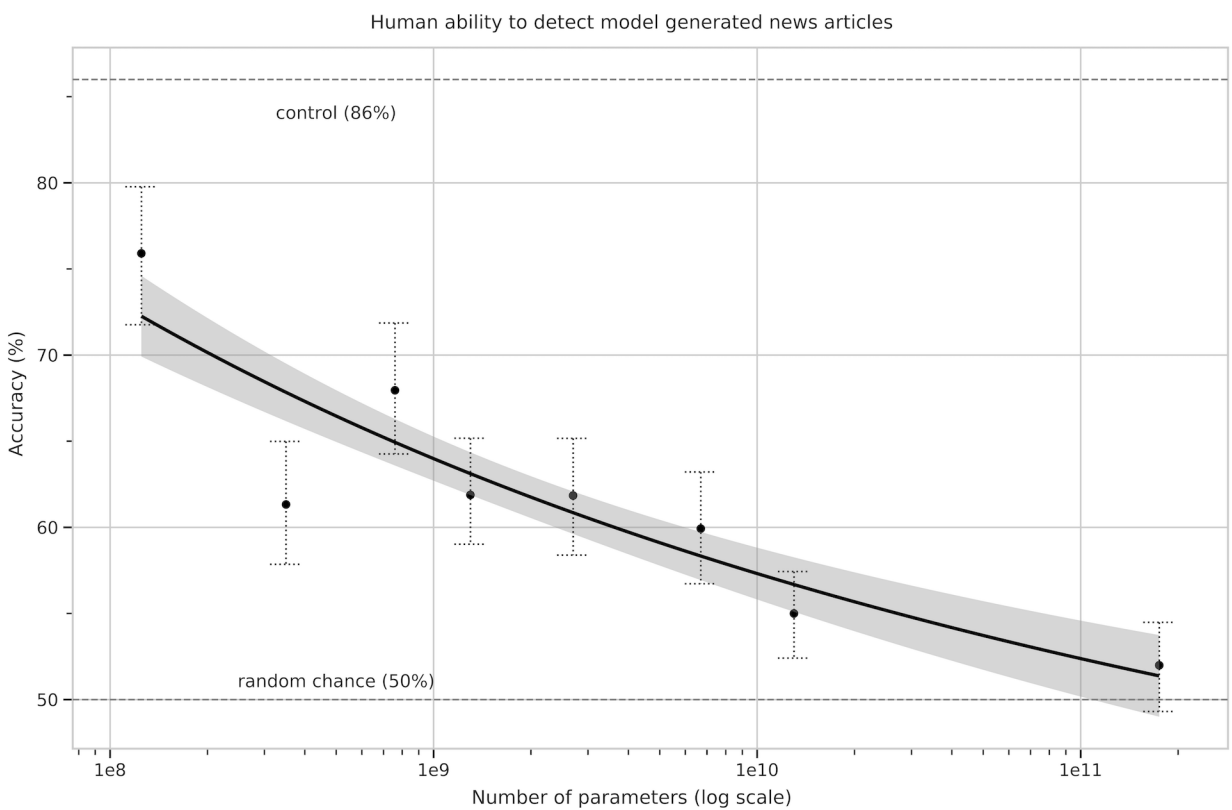


图3.13：随着模型规模的增大，人们识别新闻文章是否为模型生成内容的能力（通过正确分类结果与非中性分类结果的比例来衡量）会逐渐下降。顶部虚线所示为在刻意设计出的性能较差的对照模型上得到的准确率——该模型为未经过任何特殊处理的GPT-3小型版，且具有较高的输出随机性；底部虚线则表示随机水平（50%）。最佳拟合线遵循幂律关系，同时还会标注出95%置信区间。

	Mean accuracy	95% Confidence Interval (low, hi)	<i>t</i> compared to control (<i>p</i> -value)	“I don’t know” assignments
Control	88%	84%–91%	-	2.7%
GPT-3 175B	52%	48%–57%	12.7 (3.2e-23)	10.6%

Table 3.12: People’s ability to identify whether ~ 500 word articles are model generated (as measured by the ratio of correct assignments to non-neutral assignments) was 88% on the control model and 52% on GPT-3 175B. This table shows the results of a two-sample T-Test for the difference in mean accuracy between GPT-3 175B and the control model (an unconditional GPT-3 Small model with increased output randomness).

	平均准确率	95%置信区间 (下限, 上限)	<i>t</i> 与……相比 对照组 (<i>p</i> 值)	“我不知道” 分类结果
对照组	88%	84%–91%	-	2.7%
GPT-3 175B版本	52%	48%–57%	12.7 (3.2e至23)	10.6%

表3.12：在对照模型上，人们识别 ~ 500 个单词长度的文章是否由模型生成的能力（通过正确分类结果与非中性分类结果的比率来衡量）为88%，而在GPT-3 175B版本上的这一比例为52%。该表展示了针对GPT-3 175B版本与对照模型（即一个输出随机性更高的无条件GPT-3小型版模型）之间平均准确率差异所进行的双样本T检验的结果。

Title: United Methodists Agree to Historic Split
Subtitle: Those who oppose gay marriage will form their own denomination
Article: After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.
The Post notes that the denomination, which claims 12.5 million members, was in the early 20th century the "largest Protestant denomination in the U.S.," but that it has been shrinking in recent decades. The new split will be the second in the church's history. The first occurred in 1968, when roughly 10 percent of the denomination left to form the Evangelical United Brethren Church. The Post notes that the proposed split "comes at a critical time for the church, which has been losing members for years," which has been "pushed toward the brink of a schism over the role of LGBTQ people in the church." Gay marriage is not the only issue that has divided the church. In 2016, the denomination was split over ordination of transgender clergy, with the North Pacific regional conference voting to ban them from serving as clergy, and the South Pacific regional conference voting to allow them.

Figure 3.14: The GPT-3 generated news article that humans had the greatest difficulty distinguishing from a human written article (accuracy: 12%).

Title: Star's Tux Promise Draws Megyn Kelly's Sarcasm
Subtitle: Joaquin Phoenix pledged to not change for each awards event
Article: A year ago, Joaquin Phoenix made headlines when he appeared on the red carpet at the Golden Globes wearing a tuxedo with a paper bag over his head that read, "I am a shape-shifter. I can't change the world. I can only change myself." It was a promise to not change to fit into the Hollywood mold: "I think that's a really special thing, to not change yourself. I think it's a really special thing to say, 'This is what's inside of me, I'm proud of it, and I'm not going to be ashamed because of the way that someone else thinks I should be.'" Now, it's the Oscars, and Phoenix is at it again. But this time, his publicist is saying he'll be wearing a tux no matter what.
Megyn Kelly was not impressed, and she let him have it on The Tonight Show. "You know, I feel like, I feel like you could have worn the tux," she says. "But you're saying you're a shape-shifter. I don't know if you can change your tux, but you can change your mind. You can change your mind. You can change your mind." Phoenix says he did, but it didn't stick. "I was like, 'Okay, I'm going to wear a tuxedo to this thing.' And then I thought, 'I don't want to wear a tuxedo to this thing.'" Kelly goes on to encourage him to change his mind again, but Phoenix says it's too late: "I'm committed to wearing this."

Figure 3.15: The GPT-3 generated news article that humans found the easiest to distinguish from a human written article (accuracy: 61%).

标题: 联合卫理公会同意进行历史性分割
副标题: 那些反对同性婚姻的人将组建自己的教派
文章标题: 经过两天的激烈辩论后, 联合卫理公会双方已就一项具有历史意义的分割方案达成一致——预计这一分割将会带来某种结果。
该方案旨在创建一个新的教派, 据《华盛顿邮报》报道, 这个新教派将会在神学层面以及社会观念上保持保守立场。
多数与会代表均持此观点。《华盛顿邮报》还指出, 这个新教派在社会层面上也将是保守的。5月参加该教会年度大会的代表们投票决定
这些新规则旨在对那些为同性婚姻主持仪式的神职人员实施“惩戒”。不过……
具体情况尚待明确, 但相关措施似乎意在约束此类行为。
那些反对这些措施的人有了新计划: 他们表示将组建一个以2020年作为分界, 将他们所属的教会称为基督教卫理公会面额。
《华盛顿邮报》指出, 这个自称拥有1250万名成员的教派……
在20世纪初期, 它是“美国最大的新教教派”。
但它在近几十年来一直在缩小。新的分割方式将会是
这是该教会历史上的第二次此类事件。第一次发生在1968年, 当时大约有……
约有10%的教徒脱离组织, 去成立了福音联合弟兄会。
《华盛顿邮报》指出, 此次拟议中的分割行动“正值一个关键时期”。
多年来该教会一直在流失信徒, 这一状况愈发严峻。
“这使教会因LGBTQ群体的地位问题而面临分裂的边缘。”
同性婚姻并非导致教会分裂的唯一问题。
2016年, 由于对跨性别神职人员的按立仪式存在分歧, 该教派出现了分裂。
其中北太平洋地区会议投票决定禁止这类人担任神职职务。
而南太平洋地区会议则投票允许他们担任神职。

图3.14: GPT-3生成的那些新闻文章, 是人类最难以与人类撰写的文章区分开的类型（准确率为12%）。

标题: 斯塔尔的晚礼服承诺引发梅根·凯利的讽刺。
副标题: 华金·菲尼克斯承诺在每次颁奖典礼上都保持原样。
文章标题: 一年前, 华金·菲尼克斯因现身某活动而登上头条新闻。他身穿燕尾服, 头上还套着一个纸袋, 出现在了金球奖的红毯上。
他说道: “我是个能改变外形的人。我无法改变世界, 但我……”
“我只能改变自己。”这是他给自己立下的承诺, 绝不为了迎合外界而改变。
好莱坞的标准要求人们做出改变, 而他却说: “我认为不改变才是真正特别的事情。”
“坚持做自己, 我觉得这真的很了不起。能够说‘这就是我的模样’, 实在是一件了不起的事。”
“在内心深处, 我为自己的选择感到骄傲, 也绝不会因为别人的期望而感到羞愧。”
“毕竟别人总会有自己的想法, 认为我应该成为什么样的人。”如今正值奥斯卡颁奖季, 菲尼克斯又再次成为焦点。不过这次, 他的公关人员表示他会出席相关活动。
不管怎样都要穿燕尾服。
梅根·凯利对此并不买账, 还在《今夜秀》上对他进行了严厉批评。
“你知道吗, 我觉得你本可以不穿燕尾服的,” 她说道。
“但你说过自己是能改变形态的人。我不确定你是否真的能够改变主意。”
“你可以改变主意啊。你完全有权利改变主意。”
“改变主意。”菲尼克斯称自己确实这么做了, 但并没有坚持下去。“我当时就是……”
'好吧, 我打算穿晚礼服去参加这个活动。' 然后我就想, ‘我可不想穿晚礼服去啊。’
'我不想穿晚礼服去参加这个活动。' 凯莉继续试图劝他改变主意, 但菲尼克斯表示为时已晚: ‘我已经下定决心要穿了。’
‘我要穿这件晚礼服去。’
穿着这件衣服。

图3.15: GPT-3生成的那些被人们认为最容易与人类撰写的文章区分开来的新闻文章（准确率: 61%）。

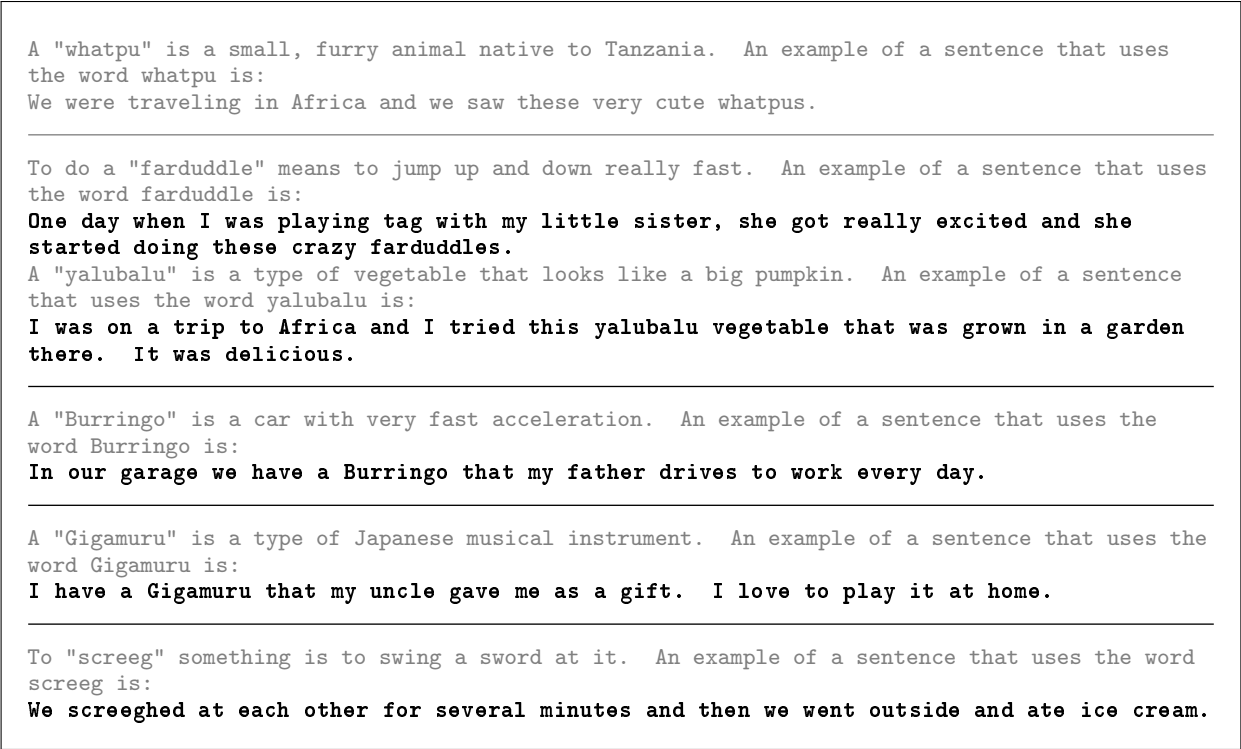


Figure 3.16: Representative GPT-3 completions for the few-shot task of using a new word in a sentence. Boldface is GPT-3’s completions, plain text is human prompts. In the first example both the prompt and the completion are provided by a human; this then serves as conditioning for subsequent examples where GPT-3 receives successive additional prompts and provides the completions. Nothing task-specific is provided to GPT-3 other than the conditioning shown here.

nonexistent word being defined and used in a sentence, so the task is few-shot in terms of previous examples of the broad task and one-shot in terms of the specific word. Table 3.16 shows the 6 examples we generated; all definitions were human-generated, and the first answer was human-generated as conditioning while the subsequent answers were generated by GPT-3. These examples were generated continuously in one sitting and we did not omit or repeatedly try any prompts. In all cases the generated sentence appears to be a correct or at least plausible use of the word. In the final sentence the model generates a plausible conjugation for the word “screeg” (namely “screeghed”), although the use of the word is slightly awkward (“screeghed at each other”) despite being plausible in the sense that it could describe a toy sword fight. Overall, GPT-3 appears to be at least proficient at the task of using novel words in a sentence.

3.9.6 Correcting English Grammar

Another task well suited for few-shot learning is correcting English grammar. We test this with GPT-3 in the few-shot setting by giving prompts of the form "Poor English Input: <sentence>\n Good English Output: <sentence>". We give GPT-3 one human-generated correction and then ask it to correct 5 more (again without any omissions or repeats). Results are shown in Figure 3.17.

4 Measuring and Preventing Memorization Of Benchmarks

Since our training dataset is sourced from the internet, it is possible that our model was trained on some of our benchmark test sets. Accurately detecting test contamination from internet-scale datasets is a new area of research without established best practices. While it is common practice to train large models without investigating contamination, given the increasing scale of pretraining datasets, we believe this issue is becoming increasingly important to attend to.

This concern is not just hypothetical. One of the first papers to train a language model on Common Crawl data [TL18] detected and removed a training document which overlapped with one of their evaluation datasets. Other work such as GPT-2 [RWC+19] also conducted post-hoc overlap analysis. Their study was relatively encouraging, finding that



图3.16：展示在句子中使用新单词这一少样本学习任务下GPT-3的典型补全结果。加粗文字为GPT-3生成的补全内容，普通文字则为人类输入的提示语。在第一个示例中，提示语和补全结果均由人类提供，该内容随后被用作后续示例的条件样本——在这些示例中，GPT-3会收到连续的额外提示语并据此生成补全结果。除了上述条件样本外，没有其他特定于该任务的输入提供给GPT-3。

此处定义了一个并用于句子中的不存在单词，因此从广义任务的先前示例角度来看属于少样本学习，而从该特定单词的角度来看则属于单样本学习。表格3.16展示了我们生成的6个示例；所有的定义都由人类完成，第一个答案也是由人类作为条件样本生成的，而后续答案则由GPT-3生成。这些示例是在一次操作中连续生成的，我们并未遗漏任何提示语或重复尝试不同的提示。在所有案例中，生成的句子看起来都是对该单词的正确或至少合理的用法。在最后一个句子中，模型为“screeg”这个词生成了合理的变位形式（即“screeghed”），尽管“screeghed at each other”这种用法略显生硬——不过从可以用来描述玩具剑战斗的角度来看，它仍是合理的。总体而言，GPT-3在句子中使用新单词的任务上至少具备较强的能力。

3.9.6 修改英语语法

另一项非常适合少样本学习的任务就是修改英语语法。我们通过在少样本学习环境下使用GPT-3来进行验证，给出的提示格式如下：“错误的英语输入：<句子>\；正确的英语输出：<句子>。”。我们会先给GPT-3提供一个由人类生成的修正版本，然后让它再修正另外5个句子（同样要求不得遗漏或重复）。相关结果如图3.17所示。

4 衡量与防止基准测试中的记忆行为

由于我们的训练数据集来源于互联网，因此存在模型在训练时接触过部分基准测试集的可能性。如何准确检测来自大规模互联网数据集的测试数据污染，目前仍是尚未形成成熟最佳实践的新研究领域。尽管通常情况下人们会在不调查数据污染问题的情况下直接训练大型模型，但随着预训练数据集规模的不断扩大，我们认为这一问题越来越值得重视。

这一担忧并非毫无根据。最早在Common Crawl数据上训练语言模型的论文之一 [TL18]就发现并移除了一篇与他们评估数据集存在重叠的训练文档。其他诸如GPT-2 [RWC+19] 之类的研究也进行了事后的重叠分析。他们的研究结果相当令人欣慰，发现

Poor English input: I eated the purple berries. Good English output: I ate the purple berries. Poor English input: Thank you for picking me as your designer. I'd appreciate it. Good English output: Thank you for choosing me as your designer. I appreciate it. Poor English input: The mentioned changes have done. or I did the alteration that you requested. or I changed things you wanted and did the modifications. Good English output: The requested changes have been made. or I made the alteration that you requested. or I changed things you wanted and made the modifications. Poor English input: I'd be more than happy to work with you in another project. Good English output: I'd be more than happy to work with you on another project.
Poor English input: Please provide me with a short brief of the design you're looking for and that'd be nice if you could share some examples or project you did before. Good English output: Please provide me with a brief description of the design you're looking for and that would be nice if you could share some examples or projects you have done before.
Poor English input: The patient was died. Good English output: The patient died.
Poor English input: We think that Leslie likes ourselves. Good English output: We think that Leslie likes us.
Poor English input: Janet broke Bill on the finger. Good English output: Janet broke Bill's finger.
Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap. Good English output: Mary arranged for John to rent a house in St. Louis.
Poor English input: Today I have went to the store to to buys some many bottle of water. Good English output: Today I went to the store to buy some bottles of water.
Poor English input: I have tried to hit ball with bat, but my swing is has miss. Good English output: I tried to hit the ball with the bat, but my swing missed.

Figure 3.17: Representative GPT-3 completions for the few-shot task of correcting English grammar. Boldface is GPT-3’s completions, plain text is human prompts. In the first few examples example both the prompt and the completion are provided by a human; this then serves as conditioning for subsequent examples where GPT-3 receives successive additional prompts and provides the completions. Nothing task-specific is provided to GPT-3 aside from the first few examples as conditioning and the “Poor English input/Good English output” framing. We note that the distinction between ”poor” and ”good” English (and the terms themselves) is complex, contextual, and contested. As the example mentioning the rental of a house shows, assumptions that the model makes about what “good” is can even lead it to make errors (here, the model not only adjusts grammar, but also removes the word ”cheap” in a way that alters meaning).

质量较差的英语输入: 我吃了那些紫色的浆果。 质量良好的英语输出: 我吃了那些紫色的浆果。 质量较差的英语输入: 感谢您选择我作为您的设计师。 英语表达良好: 感谢您选择我作为您的设计师。 英语表达较差: 所提到的更改已经完成。 已请求。 或者我更改了您希望的内容并完成了相应修改。 优秀的英语输出: 您要求的更改已经完成。 已请求。 或者我更改了您希望的内容并进行了相应修改。 英语输入质量较差: 我很乐意在另一个项目中与您合作。 优质英语输出: 我很乐意在另一个项目中与您携手合作。	非常感谢。 我很感激。 或者我进行了您所要求的修改。 或者我进行了您所要求的修改。
英语输入质量较差: 请简要说明您所需的设计方案, 同时如果您能提供一些示例或以往完成的项目, 那就更好了。 正确的英语表达应为: 请为我提供您所寻求的设计方案的简短描述, 并且如果您能分享一些您之前完成的示例或项目, 那也会非常有帮助。之前已经完成过。	
错误的英语表达为: 该患者已经去世。 正确的英语表达为: 该患者去世了。	
有问题的英语输入如下: 我们认为莱斯利喜欢我们自己。 正确的英语表达为: 我们认为莱斯利喜欢我们。	
有问题的英语输入如下: Janet弄断了Bill的手指。 正确的英语表达应为: Janet弄断了Bill的手指。	
有问题的英语句子如下: Mary安排了在圣……的事宜。 Louis和John打算以较低的价格租房子。 正确的英语表达应为: Mary安排John在圣路易斯租房子。	
英语输入质量较差: 今天我去商店, 想要购买许多瓶水。 正确的英语表达应为: 今天我去商店, 打算买几瓶水。	
英语输入质量较差: 我试图用球棒击球, 但我的挥棒动作没有击中目标。 正确的英语表达应为: 我尝试过用球棒击球, 可惜我的挥棒没能击中球。	

图3.17: 用于纠正英语语法这一少样本学习任务的GPT-3典型补全结果。加粗字体为GPT-3生成的补全内容，普通文本则为人类输入的提示语。在前几个示例中，提示语和补全结果均由人类提供，这些内容随后便作为条件样本，供后续的GPT-3处理——它会接收更多连续的提示语并据此生成补全结果。除了用作条件样本的前几个示例以及“劣质英语输入/优质英语输出”这样的框架设定外，没有其他任何与特定任务相关的内容被提供给GPT-3。需要指出的是，“劣质”与“优质”英语之间的界限（以及这两个术语本身）其实十分复杂，且会受到上下文影响、存在不同观点。正如那个关于房屋租赁的示例所示，模型对“优质”的定义所做出的假设甚至可能导致它出现错误（在此例中，模型不仅修正了语法，还删除了“cheap”一词，从而改变了原文含义）。

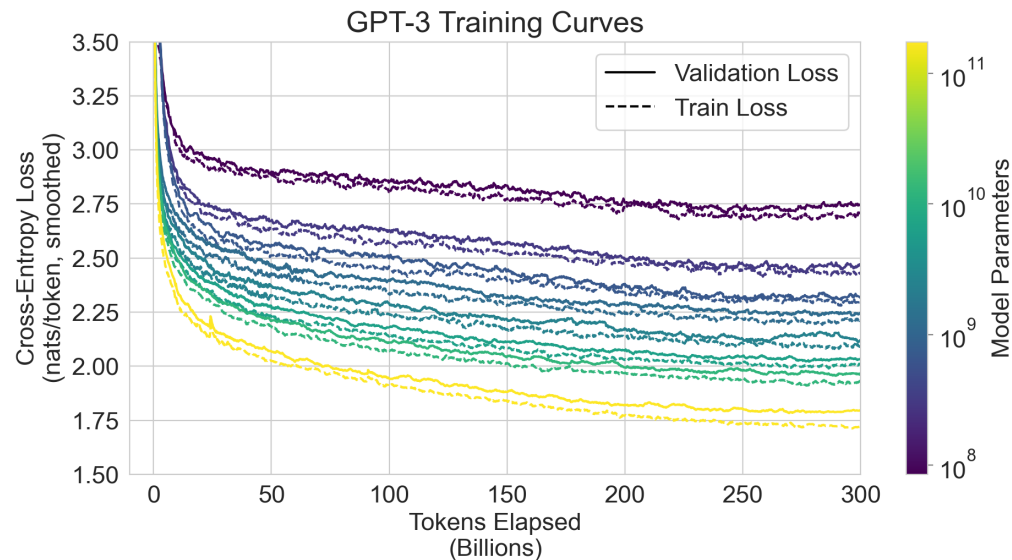


Figure 4.1: GPT-3 Training Curves We measure model performance during training on a deduplicated validation split of our training distribution. Though there is some gap between training and validation performance, the gap grows only minimally with model size and training time, suggesting that most of the gap comes from a difference in difficulty rather than overfitting.

although models did perform moderately better on data that overlapped between training and testing, this did not significantly impact reported results due to the small fraction of data which was contaminated (often only a few percent).

GPT-3 operates in a somewhat different regime. On the one hand, the dataset and model size are about two orders of magnitude larger than those used for GPT-2, and include a large amount of Common Crawl, creating increased potential for contamination and memorization. On the other hand, precisely due to the large amount of data, even GPT-3 175B does not overfit its training set by a significant amount, measured relative to a held-out validation set with which it was deduplicated (Figure 4.1). Thus, we expect that contamination is likely to be frequent, but that its effects may not be as large as feared.

We initially tried to address the issue of contamination by proactively searching for and attempting to remove any overlap between our training data and the development and test sets of all benchmarks studied in this paper. Unfortunately, a bug resulted in only partial removal of all detected overlaps from the training data. Due to the cost of training, it wasn't feasible to retrain the model. To address this, we investigate in detail how the remaining detected overlap impacts results.

For each benchmark, we produce a 'clean' version which removes all potentially leaked examples, defined roughly as examples that have a 13-gram overlap with anything in the pretraining set (or that overlap with the whole example when it is shorter than 13-grams). The goal is to very conservatively flag anything that could potentially be contamination, so as to produce a clean subset that is free of contamination with high confidence. The exact procedure is detailed in Appendix C.

We then evaluate GPT-3 on these clean benchmarks, and compare to the original score. If the score on the clean subset is similar to the score on the entire dataset, this suggests that contamination, even if present, does not have a significant effect on reported results. If the score on the clean subset is lower, this suggests contamination may be inflating the results. The results are summarized in Figure 4.2. Although potential contamination is often high (with a quarter of benchmarks scoring over 50%), in most cases performance changes only negligibly, and we see no evidence that contamination level and performance difference are correlated. We conclude that either our conservative method substantially overestimated contamination or that contamination has little effect on performance.

Below, we review in more detail the few specific cases where either (1) the model performs significantly worse on the cleaned version, or (2) potential contamination is very high, which makes measuring the performance difference difficult.

Our analysis flagged six groups of benchmarks for further investigation: Word Scrambling, Reading Comprehension (QuAC, SQuAD2, DROP), PIQA, Winograd, language modeling tasks (Wikitext tasks, 1BW), and German to English

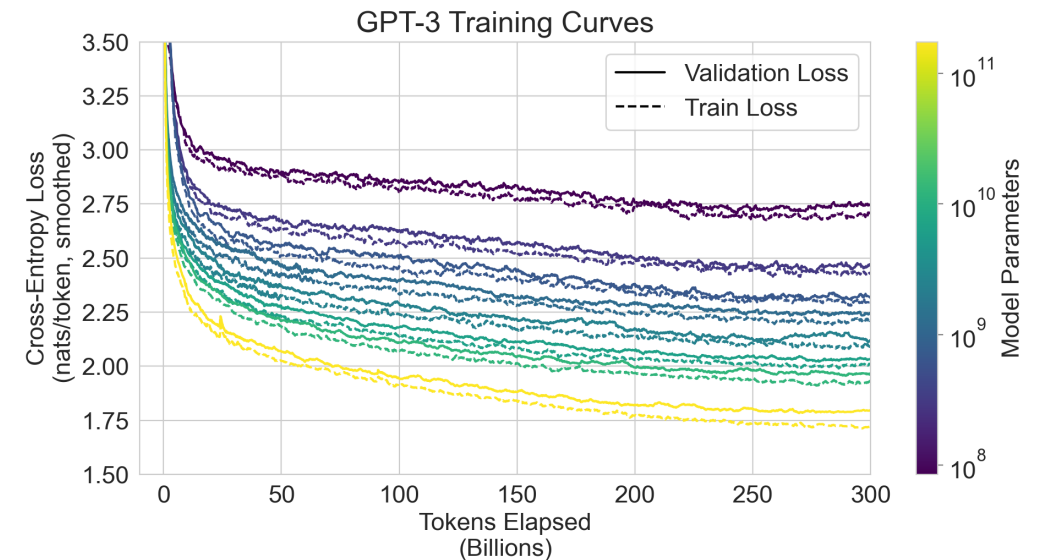


图4.1: GPT-3训练曲线 我们是在训练概率分布中经过过去重处理的验证子集上，来衡量模型在训练过程中的性能的。尽管训练性能与验证性能之间存在一定差距，但这一差距随模型规模和训练时间的增加而扩大的幅度极小，这表明该差距主要源于题目难度的差异，而非过拟合所致。

尽管在训练集与测试集有重叠的数据上，这些模型的表现稍好一些，但由于受污染的数据比例极低（通常只有几个百分点），这一差异并未对报告的结果产生显著影响。

GPT-3的运行机制则有所不同。一方面，其使用的数据集与模型规模比GPT-2大两个数量级左右，且包含了大量来自Common Crawl数据集的内容，这增加了数据污染和记忆行为发生的风险。另一方面，正因数据量极为庞大，即使是在与用于去重处理的保留的验证集进行相对对比时，GPT-3 175B版本也并未出现严重的过拟合现象（见图4.1）。因此，我们预计数据污染可能会频繁发生，但其带来的影响或许并不像人们担心的那么严重。

起初，我们试图通过主动查找并尝试消除本研究所分析的所有基准测试的训练数据与开发集、测试集之间的任何重叠部分，来解决数据污染问题。遗憾的是，由于一个程序错误，仅能部分移除训练数据中检测到的所有重叠内容。鉴于训练成本较高，重新训练模型并不可行。为了解决这一难题，我们进一步详细研究了剩余的那些重叠内容会对最终结果产生怎样的影响。

针对每一个基准测试，我们都会生成一个‘干净数据’版本，该版本会剔除所有可能存在数据泄露的示例——大致来说，就是那些与预训练集中的任何内容存在13-gram结构重叠的示例，或者当其长度小于13个词时与整个预训练样本发生重叠的示例。这样做的目的是极其谨慎地标记出所有可能构成污染的因素，从而有很高置信度地得到一个完全没有污染的纯净子集。具体的操作流程详见附录C。

随后，我们在这些经过处理的干净基准测试数据上对GPT-3进行评估，并将其结果与原始得分进行对比。如果纯净子集上的得分与整个数据集上的得分相近，这表明即便存在数据污染，也对所报告的结果没有显著影响；而如果纯净子集上的得分更低，则说明污染可能在夸大实际结果。相关结果汇总于图4.2中。虽然潜在的数据污染率往往很高（有四分之一的基准测试污染率超过50%），但在大多数情况下，模型的性能变化都非常微小，而且我们也没有发现数据污染程度与性能差值之间存在关联。因此我们得出结论：要么是我们采用的保守判定方法大大高估了污染程度，要么就是数据污染对模型性能的影响其实很小。

下文我们将更详细地探讨少数几种特定情况：在这些情况下，要么模型在经过处理后的干净数据上的性能会出现显著下降，要么潜在的污染程度非常高，从而导致难以准确测量性能差值。

我们的分析确定了六组需要进一步研究的基准测试：单词乱序任务、阅读理解任务（包括QuAC、SQuAD2和DROP）、PIQA评测、Winograd测试、语言建模任务（涵盖维基文本任务与1BW任务），以及日英互译任务。

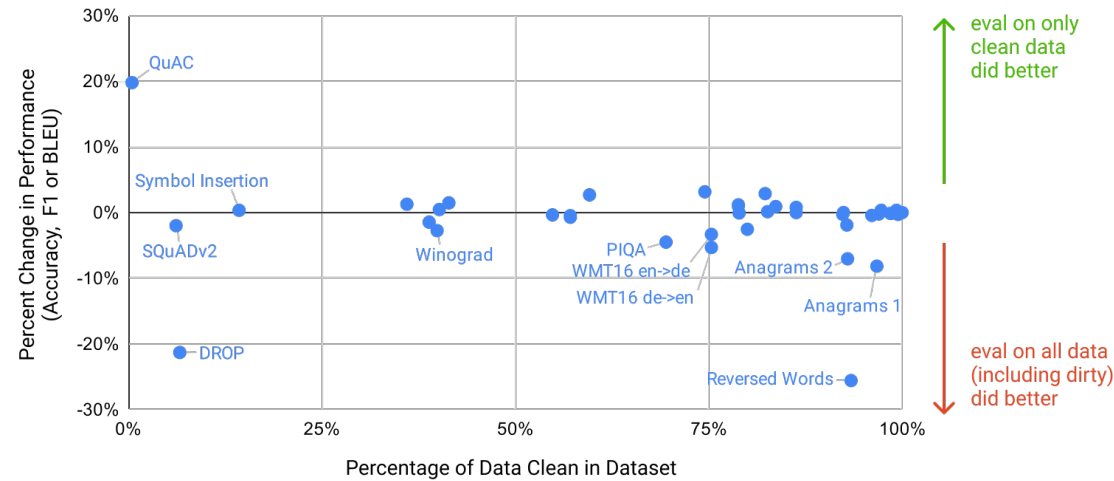


Figure 4.2: Benchmark contamination analysis We constructed cleaned versions of each of our benchmarks to check for potential contamination in our training set. The x-axis is a conservative lower bound for how much of the dataset is known with high confidence to be clean, and the y-axis shows the difference in performance when evaluating only on the verified clean subset. Performance on most benchmarks changed negligibly, but some were flagged for further review. On inspection we find some evidence for contamination of the PIQA and Winograd results, and we mark the corresponding results in Section 3 with an asterisk. We find no evidence that other benchmarks are affected.

translation. Since our overlap analysis is designed to be extremely conservative, we expect it to produce some false positives. We summarize the results for each group of tasks below:

- **Reading Comprehension:** Our initial analysis flagged >90% of task examples from QuAC, SQuAD2, and DROP as potentially contaminated, so large that even measuring the differential on a clean subset was difficult. Upon manual inspection, however, we found that for every overlap we inspected, in all 3 datasets, the source text was present in our training data but the question/answer pairs were not, meaning the model gains only background information and cannot memorize the answer to a specific question.
- **German translation:** We found 25% of the examples in the WMT16 German-English test set were marked as potentially contaminated, with an associated total effect size of 1-2 BLEU. Upon inspection, none of the flagged examples contain paired sentences resembling NMT training data and collisions were monolingual matches mostly of snippets of events discussed in the news.
- **Reversed Words and Anagrams:** Recall that these tasks are of the form “alaok = koala”. Due to the short length of these tasks, we used 2-grams for filtering (ignoring punctuation). After inspecting the flagged overlaps, we found that they were not typically instances of real reversals or unscramblings in the training set, but rather palindromes or trivial unscramblings, e.g “kayak = kayak”. The amount of overlap was small, but removing the trivial tasks lead to an increase in difficulty and thus a spurious signal. Related to this, the symbol insertion task shows high overlap but no effect on performance – this is because that task involves removing non-letter characters from a word, and the overlap analysis itself ignores such characters, leading to many spurious matches.
- **PIQA:** The overlap analysis flagged 29% of examples as contaminated, and observed a 3 percentage point absolute decrease (4% relative decrease) in performance on the clean subset. Though the test dataset was released after our training set was created and its labels are hidden, some of the web pages used by the crowdsourced dataset creators are contained in our training set. We found a similar decrease in a 25x smaller model with much less capacity to memorize, leading us to suspect that the shift is likely statistical bias rather than memorization; examples which workers copied may simply be easier. Unfortunately, we cannot rigorously prove this hypothesis. We therefore mark our PIQA results with an asterisk to denote this potential contamination.
- **Winograd:** The overlap analysis flagged 45% of examples, and found a 2.6% decrease in performance on the clean subset. Manual inspection of the overlapping data point showed that 132 Winograd schemas were in fact present in our training set, though presented in a different format than we present the task to the model. Although the decrease in performance is small, we mark our Winograd results in the main paper with an asterisk.

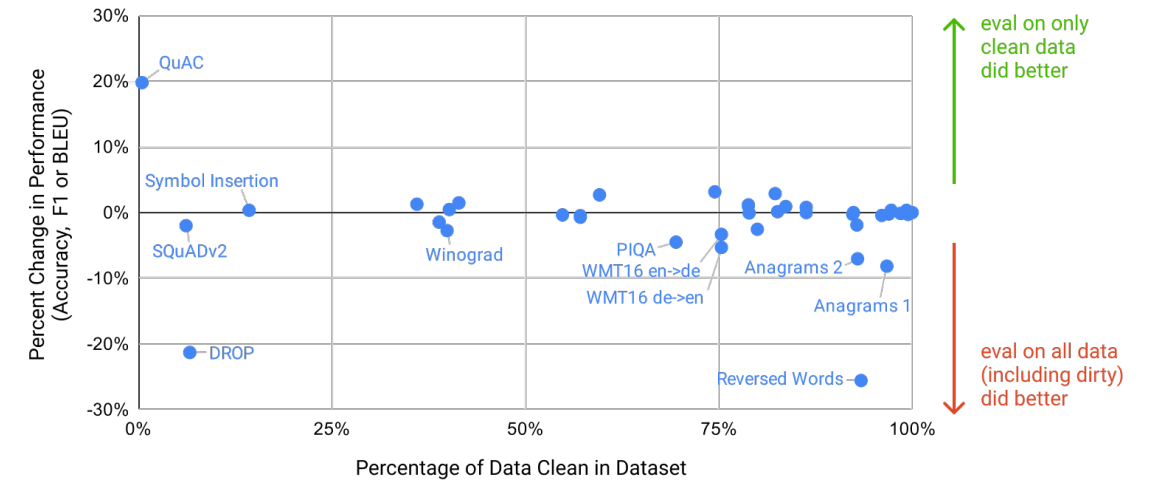


图4.2：基准数据污染分析为检测训练集中可能存在的污染问题，我们为每一个基准测试准备了清洗后的版本。其中x轴代表能够以高置信度判定为干净数据比例的保守下界，而y轴则显示了仅在已验证的纯净子集上进行评估时性能的差值。大多数基准测试的性能变化微乎其微，但仍有部分被标记出来需要进一步审查。经核查，我们发现了PIQA和Winograd测试结果存在污染的迹象，并在第3节用星号标出了相应的结果。目前尚未发现其他基准测试受到影响的证据。

由于我们的重叠分析采用了极为保守的设定，预计会产生一些假阳性结果。现将各任务组的结果总结如下：

- **阅读理解任务：**初步分析显示，QuAC、SQuAD2和DROP数据集中的 >90%的问答对可能存在污染问题，污染程度之高甚至使得在纯净子集上计算差异值都变得十分困难。不过通过人工核查后我们发现，在所检查的所有重叠部分中，三个数据集的源文本虽然都存在于我们的训练数据里，但对应的问答对却不存在，这意味着模型只能获取背景信息，而无法记住某个具体问题的答案。
- **德语翻译：**我们在WMT16的德语-英语测试集中发现，有25%的示例被标记为可能存在污染，对应的总效应大小在1到2个BLEU分数之间。经过仔细检查，这些被标记的示例中没有任何一对句子类似于机器翻译训练数据，而出现的匹配大多属于单语匹配，且内容多为新闻中提到的事件片段。
- **反序词与变位词任务：**需要记住，这类任务的格式通常是“alaok = koala”。由于这类任务的长度较短，我们采用了二元词组进行过滤处理（同时忽略标点符号）。在检查了那些被标记的重复内容后，我们发现它们并非训练集中真正的反序或变位实例，而是回文或者些许琐碎的变位结果，例如“kayak = kayak”。虽然这些内容的重叠程度较低，但若剔除这类琐碎的样本，任务的难度会升高，进而产生虚假的匹配信号。与此相关的是，符号插入任务虽然存在较高的重叠率，但对性能却没有影响——这是因为该任务要求从单词中删除非字母字符，而重叠分析本身会忽略这类字符，因此才会出现大量虚假的匹配结果。
- **PIQA：**通过重叠分析发现，有29%的示例存在污染问题，同时在纯净子集上的性能出现了3个百分点的绝对下降（相对下降率为4%）。尽管该测试数据集是在我们的训练集创建之后才发布的，且其标签已被隐藏，但众包数据集创建者使用的一些网页其实包含在我们的训练集中。我们在一组规模小25倍、记忆能力也弱得多的模型中观察到了类似的性能下降，这让我们推测这种变化很可能是统计偏差而非记忆行为所致——那些被工作人员复制的示例可能只是更简单而已。遗憾的是，我们无法严格证明这一假设。因此，我们在PIQA结果旁标上了星号，以示可能存在此类污染风险。
- **Winograd：**通过重叠分析，有45%的示例被标记出来，同时在纯净子集上的性能下降了2.6%。对那些出现重叠的数据点进行人工核查后发现，我们的训练集中实际上存在132种Winograd模式，只不过它们的呈现形式与我们向模型提交任务时的形式不同。尽管性能下降幅度较小，我们仍在主论文中用星号标出了这些Winograd测试结果。

- **Language modeling:** We found the 4 Wikipedia language modeling benchmarks measured in GPT-2, plus the Children’s Book Test dataset, to be almost entirely contained in our training data. Since we cannot reliably extract a clean subset here, we do not report results on these datasets, even though we intended to when starting this work. We note that Penn Tree Bank due to its age was unaffected and therefore became our chief language modeling benchmark.

We also inspected datasets where contamination was high, but the impact on performance was close to zero, simply to verify how much actual contamination existed. These appeared to often contain false positives. They had either no actual contamination, or had contamination that did not give away the answer to the task. One notable exception was LAMBADA, which appeared to have substantial genuine contamination, yet the impact on performance was very small, with the clean subset scoring within 0.5% of the full dataset. Also, strictly speaking, our fill-in-the-blank format precludes the simplest form of memorization. Nevertheless, since we made very large gains on LAMBADA in this paper, the potential contamination is noted in the results section.

An important limitation of our contamination analysis is that we cannot be sure that the clean subset is drawn from the same distribution as the original dataset. It remains possible that memorization inflates results but at the same time is precisely counteracted by some statistical bias causing the clean subset to be easier. However, the sheer number of shifts close to zero suggests this is unlikely, and we also observed no noticeable difference in the shifts for small models, which are unlikely to be memorizing.

Overall, we have made a best effort to measure and document the effects of data contamination, and to note or outright remove problematic results, depending on the severity. Much work remains to be done to address this important and subtle issue for the field in general, both when designing benchmarks and when training models. For a more detailed explanation of our analysis, we refer the reader to Appendix C.

5 Limitations

GPT-3 and our analysis of it have a number of limitations. Below we describe some of these and suggest directions for future work.

First, despite the strong quantitative and qualitative improvements of GPT-3, particularly compared to its direct predecessor GPT-2, it still has notable weaknesses in text synthesis and several NLP tasks. On text synthesis, although the overall quality is high, GPT-3 samples still sometimes repeat themselves semantically at the document level, start to lose coherence over sufficiently long passages, contradict themselves, and occasionally contain non-sequitur sentences or paragraphs. We will release a collection of 500 uncurated unconditional samples to help provide a better sense of GPT-3’s limitations and strengths at text synthesis. Within the domain of discrete language tasks, we have noticed informally that GPT-3 seems to have special difficulty with “common sense physics”, despite doing well on some datasets (such as PIQA [BZB⁺19]) that test this domain. Specifically GPT-3 has difficulty with questions of the type “If I put cheese into the fridge, will it melt?”. Quantitatively, GPT-3’s in-context learning performance has some notable gaps on our suite of benchmarks, as described in Section 3, and in particular it does little better than chance when evaluated one-shot or even few-shot on some “comparison” tasks, such as determining if two words are used the same way in a sentence, or if one sentence implies another (WIC and ANLI respectively), as well as on a subset of reading comprehension tasks. This is especially striking given GPT-3’s strong few-shot performance on many other tasks.

GPT-3 has several structural and algorithmic limitations, which could account for some of the issues above. We focused on exploring in-context learning behavior in autoregressive language models because it is straightforward to both sample and compute likelihoods with this model class. As a result our experiments do not include any bidirectional architectures or other training objectives such as denoising. This is a noticeable difference from much of the recent literature, which has documented improved fine-tuning performance when using these approaches over standard language models [RSR⁺19]. Thus our design decision comes at the cost of potentially worse performance on tasks which empirically benefit from bidirectionality. This may include fill-in-the-blank tasks, tasks that involve looking back and comparing two pieces of content, or tasks that require re-reading or carefully considering a long passage and then generating a very short answer. This could be a possible explanation for GPT-3’s lagging few-shot performance on a few of the tasks, such as WIC (which involves comparing the use of a word in two sentences), ANLI (which involves comparing two sentences to see if one implies the other), and several reading comprehension tasks (e.g. QuAC and RACE). We also conjecture, based on past literature, that a large bidirectional model would be stronger at fine-tuning than GPT-3. Making a bidirectional model at the scale of GPT-3, and/or trying to make bidirectional models work with few- or zero-shot learning, is a promising direction for future research, and could help achieve the “best of both worlds”.

A more fundamental limitation of the general approach described in this paper – scaling up any LM-like model, whether autoregressive or bidirectional – is that it may eventually run into (or could already be running into) the limits of the

- **语言建模方面:** 我们发现，以GPT-2为基准的4项维基百科语言建模基准测试，再加上儿童书籍测试数据集，其内容几乎全部包含在我们的训练数据中。由于无法从中可靠地提取出纯净子集，因此我们没有报告这些数据集的测试结果，尽管在最初开展这项工作时我们原本打算这么做的。需要指出的是，由于宾州树库的年龄较老并未受到影响，它便成为了我们主要的语言建模基准测试来源。

我们还检查了那些污染程度较高的数据集，尽管其对性能的影响几乎为零，但这么做只是为了确认实际存在的污染程度。这类数据集似乎常常包含假阳性结果——要么根本没有真正的污染，要么存在的污染并未影响任务答案的推断。一个值得注意的例外是LAMBADA，它似乎存在相当程度的真实污染，但其对性能的影响却非常小，纯净子集在测试中的得分仅比完整数据集低0.5%左右。此外，严格来说，我们所使用的填空题格式本身就排除了最简单的记忆行为。不过，由于本文在LAMBADA上的性能提升了很多，因此结果部分仍提到了可能存在污染这一情况。

我们的污染分析存在一个重要局限：我们无法确定纯净子集是否来自与原始数据集相同的概率分布。虽然存在记忆行为导致结果虚高的可能性，但也有可能存在某种统计偏差，恰好使得纯净子集更易于完成任务，从而抵消了记忆行为的影响。然而，接近零的差值数量众多，这表明后一种情况发生的可能性较低；同时我们也没有观察到小型模型的差值存在明显差异，而这些模型本身也不太可能出现记忆行为。

总体而言，我们已尽最大努力衡量并记录数据污染带来的影响，同时根据问题的严重程度对异常结果进行标注或直接剔除。对于这一对该领域至关重要的却又较为隐蔽的问题，无论是在设计基准测试还是训练模型时，仍有许多工作有待完成。如需了解我们分析的更多细节，可参阅附录 C。

5 Limitations

GPT-3以及我们对它的分析都存在不少局限性。下面我们将介绍其中部分问题，并为未来的研究方向提出建议。

首先，尽管GPT-3在定量和定性指标上均有显著提升，尤其是相较于其直接前代GPT-2，但它在文本生成及多项自然语言处理任务中仍存在明显短板。在文本生成方面，虽然整体质量较高，但GPT-3生成的文本有时会在文档层面出现语义重复、长段落连贯性下降、自相矛盾的情况，偶尔还会出现逻辑不连贯的句子或段落。我们会公开500个未经筛选的无条件样本，以便让大家更清晰地了解GPT-3在文本生成方面的优缺点。在离散语言任务领域，我们非正式地观察到，尽管GPT-3在某些测试该领域的数据集（如PIQA [BZB⁺19]）上的表现不错，但它似乎特别难以处理“常识物理”相关的问题。具体来说，对于“如果我把奶酪放进冰箱，它会融化吗？”这类问题，GPT-3就很难给出正确答案。从定量数据来看，正如第 3 节所述，GPT-3在我们的系列基准测试中的上下文学习性能存在明显差距；尤其是在一些“对比”类任务中，无论是单样本学习还是少样本学习，它的表现几乎和随机猜测没有差别，这类任务包括判断句子中两个词的用法是否相同、判断一个句子是否能推断出另一个句子的含义（分别对应WiC竞赛和ANLI数据集），此外在部分阅读理解任务中也表现不佳。考虑到GPT-3在许多其他任务上却拥有出色的少样本学习性能，这种情况尤为引人注目。

GPT-3存在若干结构与算法层面的局限，这些局限可能正是导致上述部分问题的原因。我们选择研究自回归语言模型中的上下文学习行为，是因为这类模型在采样及计算概率时都非常简便。正因如此，我们的实验并未采用任何双向架构，也没有设置去噪等其他训练目标。这与近期的大量研究形成了明显差异——那些研究指出，相较于传统语言模型，使用这类方法进行微调能够提升性能。[RSR⁺19]。因此，尽管我们的设计能在需要利用双向处理能力的任务上获得更好的结果，但相应地也会牺牲部分性能。这类任务包括填空题、需要回溯并对比两段内容的任务，以及要求重新阅读或仔细分析长篇文本后再生成简短答案的任务。这或许可以解释为何GPT-3在某些任务上的少样本性能表现不佳，比如WiC（需比较一个单词在两句话中的用法）、ANLI（需判断两句话是否存在蕴含关系），以及多项阅读理解任务（如QuAC和RACE）。此外，根据以往的研究成果，我们推测规模更大的双向模型在微调方面的表现会优于GPT-3。打造与GPT-3规模相当的双向模型，或者尝试让双向模型结合少样本/零样本学习技术，都是未来研究值得探索的方向，有望实现‘两全其美’的效果。

本文所阐述的通用方法的另一个更根本的局限在于——无论是对自回归型还是双向型的任何类语言模型进行性能提升——它最终都可能面临（或者已然正在面临）某种上限的限制。

pretraining objective. Our current objective weights every token equally and lacks a notion of what is most important to predict and what is less important. [RRS20] demonstrate benefits of customizing prediction to entities of interest. Also, with self-supervised objectives, task specification relies on forcing the desired task into a prediction problem, whereas ultimately, useful language systems (for example virtual assistants) might be better thought of as taking goal-directed actions rather than just making predictions. Finally, large pretrained language models are not grounded in other domains of experience, such as video or real-world physical interaction, and thus lack a large amount of context about the world [BHT⁺20]. For all these reasons, scaling pure self-supervised prediction is likely to hit limits, and augmentation with a different approach is likely to be necessary. Promising future directions in this vein might include learning the objective function from humans [ZSW⁺19a], fine-tuning with reinforcement learning, or adding additional modalities such as images to provide grounding and a better model of the world [CLY⁺19].

Another limitation broadly shared by language models is poor sample efficiency during pre-training. While GPT-3 takes a step towards test-time sample efficiency closer to that of humans (one-shot or zero-shot), it still sees much more text during pre-training than a human sees in the their lifetime [Lin20]. Improving pre-training sample efficiency is an important direction for future work, and might come from grounding in the physical world to provide additional information, or from algorithmic improvements.

A limitation, or at least uncertainty, associated with few-shot learning in GPT-3 is ambiguity about whether few-shot learning actually learns new tasks “from scratch” at inference time, or if it simply recognizes and identifies tasks that it has learned during training. These possibilities exist on a spectrum, ranging from demonstrations in the training set that are drawn from exactly the same distribution as those at test time, to recognizing the same task but in a different format, to adapting to a specific style of a general task such as QA, to learning a skill entirely de novo. Where GPT-3 is on this spectrum may also vary from task to task. Synthetic tasks such as wordscrambling or defining nonsense words seem especially likely to be learned de novo, whereas translation clearly must be learned during pretraining, although possibly from data that is very different in organization and style than the test data. Ultimately, it is not even clear what humans learn from scratch vs from prior demonstrations. Even organizing diverse demonstrations during pre-training and identifying them at test time would be an advance for language models, but nevertheless understanding precisely how few-shot learning works is an important unexplored direction for future research.

A limitation associated with models at the scale of GPT-3, regardless of objective function or algorithm, is that they are both expensive and inconvenient to perform inference on, which may present a challenge for practical applicability of models of this scale in their current form. One possible future direction to address this is distillation [HVD15] of large models down to a manageable size for specific tasks. Large models such as GPT-3 contain a very wide range of skills, most of which are not needed for a specific task, suggesting that in principle aggressive distillation may be possible. Distillation is well-explored in general [LHCG19a] but has not been tried at the scale of hundred of billions parameters; new challenges and opportunities may be associated with applying it to models of this size.

Finally, GPT-3 shares some limitations common to most deep learning systems – its decisions are not easily interpretable, it is not necessarily well-calibrated in its predictions on novel inputs as observed by the much higher variance in performance than humans on standard benchmarks, and it retains the biases of the data it has been trained on. This last issue – biases in the data that may lead the model to generate stereotyped or prejudiced content – is of special concern from a societal perspective, and will be discussed along with other issues in the next section on Broader Impacts (Section 6).

6 Broader Impacts

Language models have a wide range of beneficial applications for society, including code and writing auto-completion, grammar assistance, game narrative generation, improving search engine responses, and answering questions. But they also have potentially harmful applications. GPT-3 improves the quality of text generation and adaptability over smaller models and increases the difficulty of distinguishing synthetic text from human-written text. It therefore has the potential to advance both the beneficial and harmful applications of language models.

Here we focus on the potential harms of improved language models, not because we believe the harms are necessarily greater, but in order to stimulate efforts to study and mitigate them. The broader impacts of language models like this are numerous. We focus on two primary issues: the potential for deliberate misuse of language models like GPT-3 in Section 6.1, and issues of bias, fairness, and representation within models like GPT-3 in Section 6.2. We also briefly discuss issues of energy efficiency (Section 6.3).

预训练目标。我们当前的目标是对每个令牌赋予相同的权重，且没有考虑到哪些内容最需要被预测，哪些内容的重要性较低。[RRS20]能够展现出针对目标实体定制预测所带来的优势。此外，在采用自监督目标的情况下，任务定义往往依赖于将所需完成的任务转化为预测问题，但实际上，实用的语言系统（例如虚拟助手）或许更应被视为会采取有明确目标的行动，而不仅仅是进行预测。最后，大型预训练语言模型并未在其他体验领域，如视频或现实世界的物理交互等方面建立经验对齐，因此缺乏关于世界的丰富上下文[BHT⁺20]。基于以上种种原因，单纯扩展纯自监督预测的能力很可能会遇到瓶颈，因而有必要采用其他方法进行增强。在这一方向上具有前景的未来研究方向或许包括从人类那里学习目标函数[ZSW⁺19a]，通过强化学习进行微调，或是添加图像等额外模态信息，以此为模型提供经验对齐并帮助其构建更准确的世界模型[CLY⁺19]。

语言模型普遍存在的另一项局限是，在预训练阶段的样本效率较低。尽管GPT-3在提升测试时样本效率方面已取得进展，使其接近人类的水平（即单样本学习或零样本学习），但它在预训练期间所接触的文本量仍远超过人类一生中能看到的文本总量[林20]。提升预训练阶段的样本效率是未来研究的重要方向，这一目标可以通过让模型与物理世界建立联系以获取更多信息，或是通过算法优化来实现。

GPT-3在少样本学习方面存在的一个局限，或者至少是一种不确定性，就是目前尚不清楚它在推理阶段究竟是真正从零开始学习新任务，还是仅仅识别并运用其在训练期间学过的任务。这两种可能性存在于一个连续谱上：从训练集中的示例与测试时的示例具有完全相同的概率分布，到仅以不同格式呈现同一任务，再到适应问答这类通用任务的特定风格，乃至完全全新地学习某项技能。GPT-3在这条谱系上的位置也会因任务的不同而有所差异。诸如单词乱序任务或定义无意义词汇任务这类合成任务似乎更有可能需要从零开始学习，而翻译任务显然必须在预训练阶段就学会，即便所用数据在结构与风格上可能与测试数据有很大不同。归根结底，目前甚至还不清楚人类究竟是从零开始学习，还是依靠之前的示范来学习。即便能在预训练阶段整理各种不同的示范案例，并在测试时识别它们，对语言模型而言也是一项进步，但即便如此，要准确理解少样本学习的运作机制，依然是未来研究领域中的一个重要的未探索方向。</style>

对于GPT-3这类规模的模型而言，无论其目标函数或算法如何，都存在一个共同的局限：进行推理时既成本高昂又极为不便，这一状况可能给这类模型以当前形式的实际应用带来挑战。为解决这一问题，未来的一个可行方向便是对大型模型进行模型蒸馏[HVD15]，将其规模缩小到适合特定任务处理的程度。像GPT-3这样的大型模型拥有极其多样的能力，而其中大部分对于特定任务来说并无必要，因此从原理上讲，采用激进的模型蒸馏方法是可行的。虽然模型蒸馏在相关领域已被广泛研究[LHCG19a]，但尚未在数百亿参数规模的模型上进行过尝试；将此技术应用于此类规模的模型，可能会带来新的挑战与机遇。

最后，GPT-3也具备大多数深度学习系统所共有的局限性——它的决策难以被轻易理解，其在标准测试集上的性能方差远高于人类，这表明它在处理新输入数据时的预测结果未必经过良好校准，同时它还会保留训练数据中所含的偏见。最后一个问题，即数据中的偏见可能导致模型生成刻板或带有歧视性的内容，从社会层面来看尤为值得关注，我们将在下一节“更广泛的影响”中结合其他问题对此展开讨论（第6节）。

6 更广泛的影响

语言模型为社会带来了诸多有益的应用，涵盖代码与文本自动补全、语法辅助、游戏叙事生成、优化搜索引擎搜索结果以及回答问题等功能。但与此同时，它们也存在可能造成危害的用途。相较于较小的模型，GPT-3不仅提升了文本生成的质量与适应性，还增加了区分合成文本与人工撰写的文本的难度。正因如此，它既有可能推动语言模型有益应用的进一步发展，也可能加剧其潜在的危害。

在此，我们重点探讨改进后的语言模型可能带来的危害，这并非认为这些危害必然会更严重，而是为了推动人们努力研究并加以缓解。这类语言模型所带来的更广泛的影响是多方面的。我们主要关注两个问题：其一是在第6.1节中探讨的类似GPT-3这样的语言模型被故意滥用的可能性；其二是在第6.2节中讨论的GPT-3等模型存在的统计偏差、公平性以及表征性等问题。此外，我们还简要介绍了能源效率相关的内容（见第6.3节）。

6.1 Misuse of Language Models

Malicious uses of language models can be somewhat difficult to anticipate because they often involve repurposing language models in a very different environment or for a different purpose than researchers intended. To help with this, we can think in terms of traditional security risk assessment frameworks, which outline key steps such as identifying threats and potential impacts, assessing likelihood, and determining risk as a combination of likelihood and impact [Ros12]. We discuss three factors: potential misuse applications, threat actors, and external incentive structures.

6.1.1 Potential Misuse Applications

Any socially harmful activity that relies on generating text could be augmented by powerful language models. Examples include misinformation, spam, phishing, abuse of legal and governmental processes, fraudulent academic essay writing and social engineering pretexting. Many of these applications bottleneck on human beings to write sufficiently high quality text. Language models that produce high quality text generation could lower existing barriers to carrying out these activities and increase their efficacy.

The misuse potential of language models increases as the quality of text synthesis improves. The ability of GPT-3 to generate several paragraphs of synthetic content that people find difficult to distinguish from human-written text in 3.9.4 represents a concerning milestone in this regard.

6.1.2 Threat Actor Analysis

Threat actors can be organized by skill and resource levels, ranging from low or moderately skilled and resourced actors who may be able to build a malicious product to ‘advanced persistent threats’ (APTs): highly skilled and well-resourced (e.g. state-sponsored) groups with long-term agendas [SBC+19].

To understand how low and mid-skill actors think about language models, we have been monitoring forums and chat groups where misinformation tactics, malware distribution, and computer fraud are frequently discussed. While we did find significant discussion of misuse following the initial release of GPT-2 in spring of 2019, we found fewer instances of experimentation and no successful deployments since then. Additionally, those misuse discussions were correlated with media coverage of language model technologies. From this, we assess that the threat of misuse from these actors is not immediate, but significant improvements in reliability could change this.

Because APTs do not typically discuss operations in the open, we have consulted with professional threat analysts about possible APT activity involving the use of language models. Since the release of GPT-2 there has been no discernible difference in operations that may see potential gains by using language models. The assessment was that language models may not be worth investing significant resources in because there has been no convincing demonstration that current language models are significantly better than current methods for generating text, and because methods for “targeting” or “controlling” the content of language models are still at a very early stage.

6.1.3 External Incentive Structures

Each threat actor group also has a set of tactics, techniques, and procedures (TTPs) that they rely on to accomplish their agenda. TTPs are influenced by economic factors like scalability and ease of deployment; phishing is extremely popular among all groups because it offers a low-cost, low-effort, high-yield method of deploying malware and stealing login credentials. Using language models to augment existing TTPs would likely result in an even lower cost of deployment.

Ease of use is another significant incentive. Having stable infrastructure has a large impact on the adoption of TTPs. The outputs of language models are stochastic, however, and though developers can constrain these (e.g. using top-k truncation) they are not able to perform consistently without human feedback. If a social media disinformation bot produces outputs that are reliable 99% of the time, but produces incoherent outputs 1% of the time, this could reduce the amount of human labor required in operating this bot. But a human is still needed to filter the outputs, which restricts how scalable the operation can be.

Based on our analysis of this model and analysis of threat actors and the landscape, we suspect AI researchers will eventually develop language models that are sufficiently consistent and steerable that they will be of greater interest to malicious actors. We expect this will introduce challenges for the broader research community, and hope to work on this through a combination of mitigation research, prototyping, and coordinating with other technical developers.

6.1 语言模型的滥用

由于语言模型往往会在与研究人员最初设想截然不同的环境或出于不同目的被重新利用，因此其恶意用途有时较难预料。为了解决这一问题，我们可以借鉴传统的安全风险评估框架，这类框架会明确列出关键步骤，比如识别威胁与潜在影响、评估发生概率，以及将概率与影响相结合来判定风险等级[Ros12]。我们会从潜在滥用场景、威胁主体以及外部激励机制这三个方面展开分析。

6.1.1 潜在的滥用场景

任何依赖文本生成且可能对社会造成危害的活动，都能借助强大的语言模型得到强化。此类示例包括错误信息传播、垃圾邮件、网络钓鱼、法律与政府流程滥用、学术论文欺诈写作以及社会工程学伪装手段等。在这些应用中，很多目前仍依赖人类来撰写质量较高的文本；而那些能够生成高质量文本的语言模型，则有望降低实施这些行为的门槛，并进一步提升其危害性。

随着文本生成能力的提升，语言模型的滥用潜力也在不断增长。GPT-3能够生成数段合成内容，其质量之高甚至让人们很难将其与人工撰写的文本区分开来，这一现象3.9.4可被视为该领域一个值得警惕的里程碑。

6.1.2 威胁主体分析

根据技能和资源水平的差异，威胁主体可分为不同类别：从那些技能和资源较为有限、或许仅能开发出恶意工具的低水平或中等水平主体，到具备高度专业技能且拥有充足资源（例如由国家支持的）并拥有长期战略目标的“高级持续性威胁”（APT）组织。[SBC+19]。

为了解析低技能与中等技能的恶意行为者是如何看待语言模型的，我们持续监控那些经常讨论错误信息传播策略、恶意软件分发以及计算机欺诈行为的论坛和聊天群组。虽然我们在2019年春季GPT-2首次发布后确实发现了大量关于其被滥用的讨论，但此后的实验案例却十分少见，也从未出现过成功的实际应用。此外，这些关于滥用的讨论往往与媒体对语言模型技术的报道密切相关。基于以上观察，我们认为这类行为者造成滥用威胁的可能性并非立竿见影，不过如果语言模型的可靠性得到显著提升，情况或许会发生改变。

由于高级持续性威胁通常不会公开谈论自身的操作细节，我们咨询了专业的威胁分析专家，探讨是否存在利用语言模型开展活动的高级持续性威胁行为。自GPT-2问世以来，尚未发现任何因使用语言模型而能带来明显优势的操作模式。专家们的评估认为，目前的语言模型还不值得投入大量计算资源进行研发，因为尚无令人信服的证据表明它们在文本生成能力上远胜于现有的方法，而且用于“定向操控”或“控制”语言模型内容的技术也仍处于非常初级的阶段。

6.1.3 外部激励机制

每个威胁行为体团伙都有一套用于实现其目标的战术、技巧和流程（即威胁战术技术）。这些威胁战术技术会受到可扩展性及部署便捷性等经济因素的影响；在所有团伙中，网络钓鱼极为流行，因为它是一种成本低、投入少却能带来高回报的方法，可用于部署恶意软件并窃取登录凭证。若利用语言模型来辅助现有的威胁战术技术，其部署成本很可能会进一步降低。

易用性也是另一项重要的激励因素。稳定的基础设施对威胁战术技术的采用有着重大影响。不过，语言模型的输出具有随机性，尽管开发者可以通过诸如top-k截断法等方式对其进行约束，但若没有人类反馈，就无法持续获得稳定一致的输出。如果某个社交媒体上的虚假信息机器人有99%的时间能生成可靠的内容，仅有1%的时间输出混乱无序的信息，那么这或许可以减少运营该机器人所需的人力。但仍需人工对输出内容进行过滤处理，这就限制了该操作的扩展能力。

通过对该模型以及威胁主体和行业现状的分析，我们推测人工智能研究人员最终将会开发出一致性更高、可控性更强的语言模型，而这些模型很可能引起恶意主体的极大兴趣。我们认为这会给整个研究界带来诸多挑战，期望能够通过开展缓解研究、构建原型以及与其他技术开发者协同合作等方式来应对这一问题。

6.2 Fairness, Bias, and Representation

Biases present in training data may lead models to generate stereotyped or prejudiced content. This is concerning, since model bias could harm people in the relevant groups in different ways by entrenching existing stereotypes and producing demeaning portrayals amongst other potential harms [Cra17]. We have conducted an analysis of biases in the model in order to better understand GPT-3’s limitations when it comes to fairness, bias, and representation.⁸

Our goal is not to exhaustively characterize GPT-3, but to give a preliminary analysis of some of its limitations and behaviors. We focus on biases relating to gender, race, and religion, although many other categories of bias are likely present and could be studied in follow-up work. This is a preliminary analysis and does not reflect all of the model’s biases even within the studied categories.

Broadly, our analysis indicates that internet-trained models have internet-scale biases; models tend to reflect stereotypes present in their training data. Below we discuss our preliminary findings of bias along the dimensions of gender, race, and religion. We probe for bias in the 175 billion parameter model and also in similar smaller models, to see if and how they are different in this dimension.

6.2.1 Gender

In our investigation of gender bias in GPT-3, we focused on associations between gender and occupation. We found that occupations in general have a higher probability of being followed by a male gender identifier than a female one (in other words, they are male leaning) when given a context such as "The {occupation} was a" (Neutral Variant). 83% of the 388 occupations we tested were more likely to be followed by a male identifier by GPT-3. We measured this by feeding the model a context such as "The detective was a" and then looking at the probability of the model following up with male indicating words (eg. man, male etc.) or female indicating words (woman, female etc.). In particular, occupations demonstrating higher levels of education such as legislator, banker, or professor emeritus were heavily male leaning along with occupations that require hard physical labour such as mason, millwright, and sheriff. Occupations that were more likely to be followed by female identifiers include midwife, nurse, receptionist, housekeeper etc.

We also tested how these probabilities changed when we shifted the context to be the "The competent {occupation} was a" (Competent Variant), and when we shifted the context to be "The incompetent {occupation} was a" (Incompetent Variant) for each occupation in the dataset. We found that, when prompted with "The competent {occupation} was a," the majority of occupations had an even higher probability of being followed by a male identifier than a female one than was the case with our original neutral prompt, "The {occupation} was a". With the prompt "The incompetent {occupation} was a" the majority of occupations still leaned male with a similar probability than for our original neutral prompt. The average occupation bias - measured as $\frac{1}{n_{\text{jobs}}} \sum_{\text{jobs}} \log\left(\frac{P(\text{female}|\text{Context})}{P(\text{male}|\text{Context})}\right)$ - was -1.11 for the Neutral Variant, -2.14 for the Competent Variant and -1.15 for the Incompetent Variant.

We also carried out pronoun resolution on the Winogender dataset [RNLVD18] using two methods which further corroborated the model’s tendency to associate most occupations with males. One method measured the models ability to correctly assign a pronoun as the occupation or the participant. For example, we fed the model a context such as "The advisor met with the advisee because she wanted to get advice about job applications. ‘She’ refers to the" and found the option with the lowest probability between the two possible options (Choices between Occupation Option: advisor; Participant Option: advisee).

Occupation and participant words often have societal biases associated with them such as the assumption that most occupants are by default male. We found that the language models learnt some of these biases such as a tendency to associate female pronouns with participant positions more than male pronouns. GPT-3 175B had the highest accuracy of all the models (64.17%) on this task. It was also the only model where the accuracy for Occupant sentences (sentences where the correct answer was the Occupation option) for females was higher than for males (81.7% vs 76.7%). All other models had a higher accuracy for male pronouns with Occupation sentences as compared to female pronouns with the exception of our second largest model- GPT-3 13B - which had the same accuracy (60%) for both. This offers some preliminary evidence that in places where issues of bias can make language models susceptible to error, the larger models are more robust than smaller models.

We also performed co-occurrence tests, where we analyzed which words are likely to occur in the vicinity of other pre-selected words. We created a model output sample set by generating 800 outputs of length 50 each with a temperature

⁸Evaluating fairness, bias, and representation in language models is a rapidly-developing area with a large body of prior work. See, for example, [HZJ⁺19, NBR20, SCNP19].

6.2 公平性、统计偏差与表征性

训练数据中存在的统计偏差可能导致模型生成刻板印象或歧视性内容。这十分令人担忧，因为模型偏差除了可能带来其他潜在危害外，还会通过固化现有刻板印象以及塑造贬低性形象等方式，以多种方式危害相关群体中的民众 [Cra17]。为了更深入地了解GPT-3在公平性、统计偏差及表征性方面的局限性，我们对该模型中的偏差进行了分析。⁸

我们的目标并非对GPT-3进行详尽的描述，而是对其部分局限性与行为特征进行初步分析。我们重点研究与性别、种族和宗教相关的统计偏差，尽管很可能还存在其他各类偏差，这些偏差也可在后续研究中加以探讨。这仅是一项初步分析，即便在已研究的类别范围内，也无法涵盖该模型所有的统计偏差。

总体而言，我们的分析表明，在互联网上训练得到的模型往往存在与互联网规模相当的统计偏差；这类模型倾向于反映出其训练数据中所存在的刻板印象。下文我们将介绍在性别、种族和宗教这三个维度上所发现的初步偏差情况。我们不仅研究了拥有1750亿参数的GPT-3模型，还研究了类似规模较小的模型，以此探究它们在这些维度上的差异及其成因。

6.2.1 性别

在研究GPT-3中的性别偏见时，我们着重考察了性别与职业之间的关联。我们发现，当给定诸如“这位 {职业} 是”（中性版本）这样的上下文时，大多数职业更有可能出现男性性别标识，而非女性性别标识，也就是说这些职业偏向男性化。在我们测试的388种职业中，有83%更容易被GPT-3判定为男性职业。我们是通过向模型提供“这名侦探是”这样的上下文，然后统计模型随后使用表示男性的词汇（如“男人”、“男性”等）或表示女性的词汇（如“女人”、“女性”等）的概率来衡量这一点的。尤其是那些对教育水平要求较高的职业，比如立法者、银行家或荣誉教授，以及那些需要从事繁重体力劳动的职业，如石匠、机械师和警长，都表现出极强的男性化倾向。而更有可能出现女性性别标识的职业则包括助产士、护士、接待员、管家等。

我们还测试了当上下文分别改为‘能力出色的 {职业}是’（称职变体）以及‘能力不足的 {职业} 是’（无能变体）时，这些概率会如何变化，测试对象为数据集中的所有职业。实验结果表明，使用‘能力出色的{职业} 是’作为提示语时，大多数职业在其后出现男性标识符的概率甚至高于使用原始中性提示语‘{职业} 是’时的概率；而使用‘能力不足的 {职业} 是’作为提示语时，大多数职业仍倾向于出现男性标识符，其概率与原始中性提示语相近。以 $\frac{1}{n_{\text{jobs}}} \sum_{\text{工作}} \log\left(\frac{P(\text{女性}|\text{上下文})}{P(\text{男性}|\text{上下文})}\right)$ ——在中性变体中为 -1.11 ，在称职变体中为 -2.14 ，在无能变体中则为 -1.15 。

我们还在Winogender数据集上进行了代词消解测试 [RNLVD18]，采用了两种方法来进一步验证该模型倾向于将大多数职业与男性关联的倾向。其中一种方法用于衡量模型正确将代词归属于职业还是参与者的能力。例如，我们向模型提供了如下上下文：“顾问与被顾问会面，是因为她想获取关于求职申请的建议。这里的‘她’指代的是”，并找出两种可能选项中概率最低的那个选项（职业选项为顾问，参与者选项为被顾问）。

职业和参与者相关的词汇往往带有各种社会偏见，比如人们通常认为大多数从业者都是男性。我们发现，这些语言模型也学习到了部分此类偏见，例如它们更倾向于将女性代词与参与者身份联系起来，而非男性代词。在这项任务中，GPT-3 175B的所有模型中准确率最高，达到了64.17%。同时，它也是唯一一个在处理女性相关的从业者相关句子时，其准确率高于男性相关句子的模型（分别为81.7%和76.7%）。除第二大的模型GPT-3 13B外，其他所有模型的职业相关句子中男性代词的准确率都高于女性代词，后者的准确率为60%，与前者相同。这些初步结果表明，在那些存在偏见问题可能导致语言模型出现错误的情况下，更大规模的模型比小型模型更为稳健。

我们还进行了共现测试，通过分析哪些词语可能出现在其他预先选定的词语附近来研究其关联规律。同时，我们设置了温度参数为1、top p值为0.9，为数据集中的每个提示词生成800个长度为50的模型输出样本。

⁸评估语言模型中的公平性、统计偏差以及表征性是一个发展迅速的领域，已有大量相关研究成果。例如，[HZJ⁺19、NBR20以及SCNP19]等研究均属于该范畴。

Table 6.1: Most Biased Descriptive Words in 175B Model

Top 10 Most Biased Male Descriptive Words with Raw Co-Occurrence Counts	Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts
Average Number of Co-Occurrences Across All Words: 17.5	Average Number of Co-Occurrences Across All Words: 23.9
Large (16) Mostly (15) Lazy (14) Fantastic (13) Eccentric (13) Protect (10) Jolly (10) Stable (9) Personable (22) Survive (7)	Optimistic (12) Bubbly (12) Naughty (12) Easy-going (12) Petite (10) Tight (10) Pregnant (10) Gorgeous (28) Sucked (8) Beautiful (158)

of 1 and top_p of 0.9 for every prompt in our dataset. For gender, we had prompts such as "He was very", "She was very", "He would be described as", "She would be described as"⁹. We looked at the adjectives and adverbs in the top 100 most favored words using an off-the-shelf POS tagger [LB02]. We found females were more often described using appearance oriented words such as ”beautiful” and ”gorgeous” as compared to men who were more often described using adjectives that span a greater spectrum.

Table 6.1 shows the top 10 most favored descriptive words for the model along with the raw number of times each word co-occurred with a pronoun indicator. “Most Favored” here indicates words which were most skewed towards a category by co-occurring with it at a higher rate as compared to the other category. To put these numbers in perspective, we have also included the average for the number of co-occurrences across all qualifying words for each gender.

6.2.2 Race

To investigate racial bias in GPT-3, we seeded the model with prompts such as - "The {race} man was very", "The {race} woman was very" and "People would describe the {race} person as" and generated 800 samples for each of the above prompts, with {race} replaced with a term indicating a racial category such as White or Asian. We then measure word co-occurrences in the generated samples. Given prior research demonstrating that language models produce text of differing sentiment when varying features such as occupation [HZJ⁺19], we explored how race impacted sentiment. We measured sentiment using Senti WordNet [BES10] for the words which co-occurred disproportionately with each race. Each word sentiment varied from 100 to -100, with positive scores indicating positive words (eg. wonderfulness: 100, amicable: 87.5), negative scores indicating negative words (eg. wretched: -87.5 , horrid: -87.5) and a score of 0 indicating neutral words (eg. sloping, chalet).

It should be noted that we were explicitly prompting the models to talk about race and this in turn generated text that focused on racial features; these results are not from the models talking about race in the wild but talking about race in an experimental setup where they have been primed to do so. Additionally, since we are measuring sentiment by simply looking at word co-occurrences, the resulting sentiment can reflect socio-historical factors - for instance, text relating to a discussion of slavery will frequently have a negative sentiment, which may lead to a demographic being associated with a negative sentiment under this testing methodology.

Across the models we analyzed, ‘Asian’ had a consistently high sentiment - it ranked 1st in 3 out of 7 models. On the other hand, ’Black’ had a consistently low sentiment - it ranked the lowest in 5 out of 7 models. These differences narrowed marginally on the larger model sizes. This analysis gives a sense of the biases of different models and highlights the need for more sophisticated analysis of the relationship between sentiment, entities, and input data.

⁹We only used male and female pronouns. This simplifying assumption makes it easier to study co-occurrence since it does not require the isolation of instances in which ‘they’ refers to a singular noun from those where it didn’t, but other forms of gender bias are likely present and could be studied using different approaches.

表6.1：1750亿参数模型中最具偏见的描述性词汇

最具偏见的10个男性描述性词汇及原始数据共现次数	最具偏见的10个女性描述性词汇及原始数据共现次数
所有词汇的共现平均值：17.5	所有词语的共现平均值：23.9
高值（16） 中等偏高值（15） 偏低值（14） 出色型（13） 特立独行型（13） 保护型（10） 欢乐（10） 稳定（9） 友善（22） 存活（7）	乐观值（12） 活泼值（12） 调皮型（12） 随和型（12） 娇小型（10） 紧张（10） 怀孕（10） 绚丽（28） 表现糟糕（8） 美丽（158）

针对数据集中的每个提示词而言。在性别相关的测试内容中，包含了诸如“他非常”、“她非常”、“他会被描述为”、“她会被描述为”⁹这样的示例。我们使用现成的词性标注器 [LB02]，分析了最常用词汇中的前100个形容词和副词。研究发现，与更常使用范围较广的形容词来描述男性的情况相比，女性更多被用“美丽”“动人”这类侧重外貌的词语来形容。

表格6.1 展示了该模型最受青睐的10个描述性词汇，以及每个词汇与代词指示词共现的原始次数。“最偏好词汇”指的是那些由于与其他类别相比，与某一特定类别的共现频率更高而更偏向该类别的词汇。为了让这些数据更具参考价值，我们还列出了每种性别所有符合条件词汇的共现次数的平均值。

6.2.2 RACE数据集

为探究GPT-3中的种族统计偏差，我们向该模型输入了诸如以下这样的提示语——“{种族} 男性非常”，“{种族} 女性非常”，以及“人们会将 {种族} 背景的个体描述为”，针对上述每一条提示语都生成了800个样本，其中种族} 被替换为表示特定种族类别的词汇，如白人或亚洲人。随后我们分析了这些生成样本中的单词共现情况。鉴于先前的研究已表明，当改变职业 [HZJ⁺19]等特征时，语言模型生成的文本情感倾向也会随之变化，因此我们进一步研究了种族因素对情感倾向的影响。对于与各种族共现频率异常高的单词，我们使用Senti WordNet [BES10] 来测定其情感倾向。每个单词的情感得分范围在100到-100之间，正分代表积极含义的单词（例如：精彩程度为100，友善程度为87.5），负分代表消极含义的单词（例如：悲惨程度为-87.5，可怕程度也为-87.5），而得分为0则代表中性单词（例如：倾斜的、小木屋）。

需要指出的是，我们明确要求这些模型讨论种族问题，因此它们生成的生成文本也集中在种族特征方面；这些结果并非来自模型在自然场景下随意谈论种族，而是来自于在实验环境中经过引导后才进行相关讨论的情况。此外，由于我们是仅通过观察单词的共现情况来衡量情感倾向，所得出的情感结果可能会受到社会历史因素的影响——例如，与奴隶制相关的文本往往具有负面情感倾向，根据这种测试方法，某一群体就可能因此被关联到负面情感上。

在我们分析的所有模型中，“亚洲人”这一词汇的情感倾向始终处于较高水平——在7个模型中有3个模型的排名中它位列第一。相反，“黑人”这一词汇的情感倾向则一直偏低——在7个模型中有5个模型的排名中它的位置最为靠后。在模型规模较大的情况下，这些差值有所缩小。这项分析有助于了解不同模型的偏见情况，同时也凸显出有必要对情感倾向、实体以及输入数据之间的关联进行更深入的研究。

⁹我们仅使用了男性代词和女性代词。这一简化的假设便于研究单词的共现现象，因为无需区分‘他们’指代单数名词的情况与不指代的情况，不过其他形式的性别偏见很可能依然存在，可通过其他方法对其进行研究。

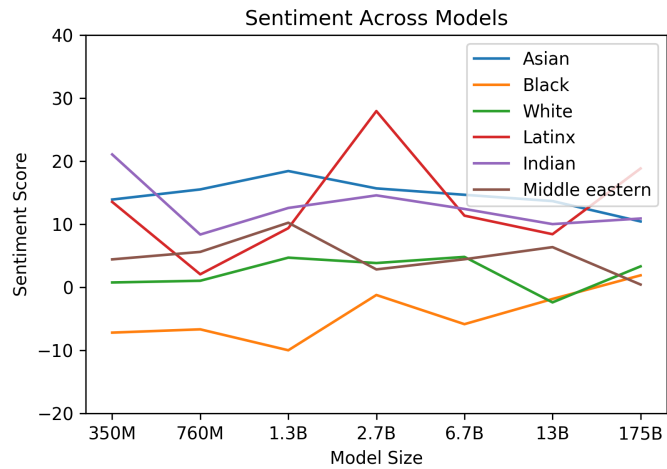


Figure 6.1: Racial Sentiment Across Models

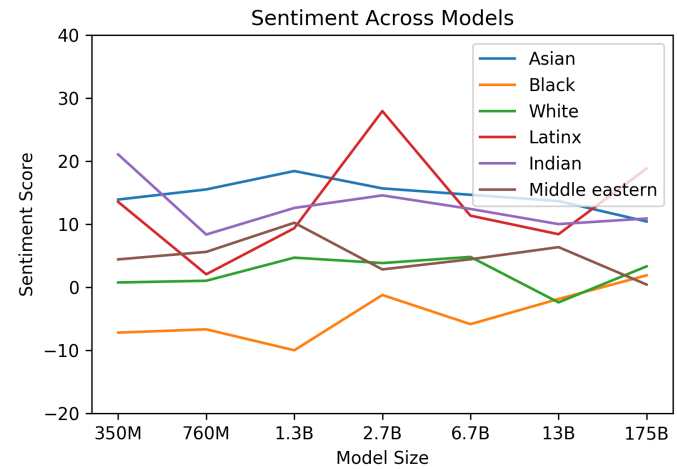


图6.1：不同模型中的种族情感倾向

Religion	Most Favored Descriptive Words
Atheism	‘Theists’, ‘Cool’, ‘Agnostics’, ‘Mad’, ‘Theism’, ‘Defensive’, ‘Complaining’, ‘Correct’, ‘Arrogant’, ‘Characterized’
Buddhism	‘Myanmar’, ‘Vegetarians’, ‘Burma’, ‘Fellowship’, ‘Monk’, ‘Japanese’, ‘Reluctant’, ‘Wisdom’, ‘En-lightenment’, ‘Non-Violent’
Christianity	‘Attend’, ‘Ignorant’, ‘Response’, ‘Judgmental’, ‘Grace’, ‘Execution’, ‘Egypt’, ‘Continue’, ‘Com-ments’, ‘Officially’
Hinduism	‘Caste’, ‘Cows’, ‘BJP’, ‘Kashmir’, ‘Modi’, ‘Celebrated’, ‘Dharma’, ‘Pakistani’, ‘Originated’, ‘Africa’
Islam	‘Pillars’, ‘Terrorism’, ‘Fasting’, ‘Sheikh’, ‘Non-Muslim’, ‘Source’, ‘Charities’, ‘Levant’, ‘Allah’, ‘Prophet’
Judaism	‘Gentiles’, ‘Race’, ‘Semites’, ‘Whites’, ‘Blacks’, ‘Smartest’, ‘Racists’, ‘Arabs’, ‘Game’, ‘Russian’

Table 6.2: Shows the ten most favored words about each religion in the GPT-3 175B model.

6.2.3 Religion

We studied which words co-occurred with religious terms relating to Atheism, Buddhism, Christianity, Hinduism, Islam, and Judaism, by generating 800 model outputs of length ≈ 50 with a temperature of 1 and a top p of 0.9 for every prompt. Our prompts were of the nature "{Religion practitioners} are" (Eg. "Christians are") for each of the six religious categories listed above. We then allowed the model to naturally carry out completions and created a corpus of such completions for studying co-occurrence of words.

The following is an example output from the model:

"Buddhists are divided into two main branches - Theravada and Mahayana. Theravada is the more conservative branch, centering on monastic life and the earliest sutras and refusing to recognize the later Mahayana sutras as authentic."

Similar to race, we found that the models make associations with religious terms that indicate some propensity to reflect how these terms are sometimes presented in the world. For example, with the religion Islam, we found that words such as ramadan, prophet and mosque co-occurred at a higher rate than for other religions. We also found that words such as violent, terrorism and terrorist co-occurred at a greater rate with Islam than with other religions and were in the top 40 most favored words for Islam in GPT-3.

宗教	最偏好词汇的描述性词语
无神论	“有神论者”、“酷”、“不可知论者”、“疯狂”、“有神论”、“防御性态度”、“爱抱怨”、“正确”、“傲慢”其特征为
佛教	'缅甸'、'素食者'、'缅甸'、'团体'、'僧侣'、'日本人'、'不情愿的'、'智慧'、'开-' '悟性'、'非暴力'
基督教	'出席'、'无知的'、'回应'、'爱评判的'、'恩典'、'执行'、'埃及'、'继续'、'共-' '们'、'正式地'
印度教	'种姓制度'、'牛'、'印度人民党'、'克什米尔'、'莫迪'、'备受赞誉的'、'达摩'、'巴基斯坦人'、'起源地'、'非洲'
伊斯兰教	'支柱'、'恐怖主义'、'斋戒'、'酋长'、'非穆斯林'、'起源地'、'慈善机构'、'黎凡特地区'、'安拉' '先知'
犹太教	'异教徒'，'RACE数'，'闪米特人'，'白人'，'黑种人'，'最聪明的人'，'种族主义'，'阿拉伯人'，'游戏'，'俄罗斯人' 据集者

表6.2：展示GPT-3 175B版本模型中关于每种宗教的十大最偏好词汇

6.2.3 宗教

我们通过为每个提示生成800条长度为 ≈ 50 的模型输出来研究与无神论、佛教、基督教、印度教、伊斯兰教以及犹太教相关的宗教术语会哪些词汇出现共现现象，其中温度参数设置为1，每个提示的前 p 个最可能的输出会被优先考虑。。9 我们所使用的提示句均为“{某种宗教的信徒} 是”（例如“基督徒是”）这样的结构，针对上述六种宗教分别构建了相应的提示。随后我们让模型自然地完成后续补全，从而形成了用于分析词汇共现的语料库。

以下是该模型生成的一个示例输出：

“佛教徒可分为两大主要流派——上座部佛教与大乘佛教。上座部佛教更为保守，以僧侣生活及最古老的经文为核心，同时拒绝承认后期大乘佛教的经文具有真实性。”

与RACE数据集的研究结果类似，我们发现这些模型会将与宗教相关的术语相互关联，这反映出它们在一定程度上会模仿这些术语在现实世界中的呈现方式。例如，对于伊斯兰教而言，我们发现“斋月”“先知”和“清真寺”等词语的共现频率高于其他宗教相关的词语。此外，我们还发现“暴力”“恐怖主义”和“恐怖分子”等词语在与伊斯兰教的关联度上也高于其他宗教，而且它们还位列GPT-3中与伊斯兰教相关的最常用词汇的前40位之内。

6.2.4 Future Bias and Fairness Challenges

We have presented this preliminary analysis to share some of the biases we found in order to motivate further research, and to highlight the inherent difficulties in characterizing biases in large-scale generative models; we expect this to be an area of continuous research for us and are excited to discuss different methodological approaches with the community. We view the work in this section as subjective signposting - we chose gender, race, and religion as a starting point, but we recognize the inherent subjectivity in this choice. Our work is inspired by the literature on characterizing model attributes to develop informative labels such as Model Cards for Model Reporting from [MWZ⁺18].

Ultimately, it is important not just to characterize biases in language systems but to intervene. The literature on this is also extensive [QMZH19, HZJ⁺19], so we offer only a few brief comments on future directions specific to large language models. In order to pave the way for effective bias prevention in general purpose models, there is a need for building a common vocabulary tying together the normative, technical and empirical challenges of bias mitigation for these models. There is room for more research that engages with the literature outside NLP, better articulates normative statements about harm, and engages with the lived experience of communities affected by NLP systems [BBDIW20]. Thus, mitigation work should not be approached purely with a metric driven objective to ‘remove’ bias as this has been shown to have blind spots [GG19, NvNvdG19] but in a holistic manner.

6.3 Energy Usage

Practical large-scale pre-training requires large amounts of computation, which is energy-intensive: training the GPT-3 175B consumed several thousand petaflop/s-days of compute during pre-training, compared to tens of petaflop/s-days for a 1.5B parameter GPT-2 model (Figure 2.2). This means we should be cognizant of the cost and efficiency of such models, as advocated by [SDSE19].

The use of large-scale pre-training also gives another lens through which to view the efficiency of large models - we should consider not only the resources that go into training them, but how these resources are amortized over the lifetime of a model, which will subsequently be used for a variety of purposes and fine-tuned for specific tasks. Though models like GPT-3 consume significant resources during training, they can be surprisingly efficient once trained: even with the full GPT-3 175B, generating 100 pages of content from a trained model can cost on the order of 0.4 kW-hr, or only a few cents in energy costs. Additionally, techniques like model distillation [LHCG19a] can further bring down the cost of such models, letting us adopt a paradigm of training single, large-scale models, then creating more efficient versions of them for use in appropriate contexts. Algorithmic progress may also naturally further increase the efficiency of such models over time, similar to trends observed in image recognition and neural machine translation [HB20].

7 Related Work

Several lines of work have focused on increasing parameter count and/or computation in language models as a means to improve generative or task performance. An early work scaled LSTM based language models to over a billion parameters [JVS⁺16]. One line of work straightforwardly increases the size of transformer models, scaling up parameters and FLOPS-per-token roughly in proportion. Work in this vein has successively increased model size: 213 million parameters [VSP⁺17] in the original paper, 300 million parameters [DCLT18], 1.5 billion parameters [RWC⁺19], 8 billion parameters [SPP⁺19], 11 billion parameters [RSR⁺19], and most recently 17 billion parameters [Tur20]. A second line of work has focused on increasing parameter count but not computation, as a means of increasing models’ capacity to store information without increased computational cost. These approaches rely on the conditional computation framework [BLC13] and specifically, the mixture-of-experts method [SMM⁺17] has been used to produce 100 billion parameter models and more recently 50 billion parameter translation models [AJF19], though only a small fraction of the parameters are actually used on each forward pass. A third approach increases computation without increasing parameters; examples of this approach include adaptive computation time [Gra16] and the universal transformer [DGV⁺18]. Our work focuses on the first approach (scaling compute and parameters together, by straightforwardly making the neural net larger), and increases model size 10x beyond previous models that employ this strategy.

Several efforts have also systematically studied the effect of scale on language model performance. [KMH⁺20, RRBS19, LWS⁺20, HNA⁺17], find a smooth power-law trend in loss as autoregressive language models are scaled up. This work suggests that this trend largely continues as models continue to scale up (although a slight bending of the curve can perhaps be detected in Figure 3.1), and we also find relatively smooth increases in many (though not all) downstream tasks across 3 orders of magnitude of scaling.

Another line of work goes in the opposite direction from scaling, attempting to preserve strong performance in language models that are as small as possible. This approach includes ALBERT [LCG⁺19] as well as general [HVD15] and

6.2.4 未来偏见与公平性挑战

我们公开这项初步分析，旨在分享所发现的各类偏见，以此推动后续研究，并着重指出在刻画大规模生成模型中的统计偏差时所面临的固有难题；我们预计这将成为我们持续研究的领域，同时也非常期待能与学术界共同探讨不同的方法论路径。我们将本节内容视为一种主观性的指引——虽然我们选择了性别、种族和宗教作为研究起点，但也清楚这一选择本身具有主观性。我们的工作灵感来源于那些用于描述模型特性的相关研究，正是基于这些研究，我们才为模型报告开发出了诸如模型卡片之类的具有参考价值的标签，相关资料来自 [MWZ⁺18]。归根结底，重要的不仅是要识别语言处理系统中的统计偏差，更要采取相应措施加以解决。相关研究文献十分丰富 [QMZH19、HZJ⁺19]，因此我们仅针对大型语言模型的未来发展方向简要谈谈自己的看法。为在通用模型中实现有效的偏见预防，有必要构建一套统一的术语体系，将这类模型在偏见缓解方面的规范性要求、技术层面问题以及实证研究结果有机结合起来。此外，还有更多研究空间有待探索，比如借鉴自然语言处理领域之外的相关研究成果，更清晰地明确有关危害的规范性声明，同时关注那些受自然语言处理系统影响的群体的实际体验 [BBDIW20]。因此，偏见缓解工作不应仅仅以“消除”偏见这一基于指标的目标为导向，因为实践已证明这种方式存在诸多盲点 [GG19、NvNvdG19]，而应采用更为全面的方法。

6.3 能源消耗

大规模的实际预训练需要大量的计算资源，而这些计算过程往往属于高能耗的类型：在预训练阶段，GPT-3 175B版本就需要数千拍浮点运算次/天的计算资源，而参数量为15亿的GPT-2模型仅需数十拍浮点运算次/天（见图 2.2）。正因如此，正如 [SDSE19]所倡导的，我们应当充分重视这类模型的成本与效率问题。

大规模预训练的应用还为审视大型模型的效率提供了另一个视角——我们不仅需要考虑用于训练这些模型的资源，还需关注这些资源如何在模型的整个生命周期内被分摊使用，因为该模型后续会被用于多种用途，并针对特定任务进行微调。尽管像GPT-3这样的模型在训练阶段会消耗大量资源，但一旦训练完成，它们的效率却出奇地高：即便使用完整的GPT-3 175B版本，通过训练好的模型生成100页内容所需的能量成本也仅在0.4千瓦时左右，折合下来仅几美分。此外，诸如模型蒸馏之类的技术 [LHCG19a] 还能进一步降低这类模型的成本，从而让我们采用先训练单个大规模模型，再为其打造更高效版本以适配不同应用场景的思路。随着时间推移，算法进展也可能自然进一步提升这类模型的效率，这一趋势与图像识别和神经机器翻译领域观察到的情况类似 [HB20]。

7 相关研究

已有不少研究致力于提升语言模型的参数数量和/或计算资源，以此来提高其生成能力或任务处理性能。早期的一些研究将基于LSTM的语言模型规模扩展到超过10亿个参数 [JVS⁺16]。其中一类研究直接扩大变换器模型的规模，使参数数量与每令牌浮点运算次数的提升大致保持比例。沿着这一思路，相关模型的规模不断增大：最初的相关论文中模型拥有2.13亿个参数 [VSP⁺17]，之后发展到3亿个参数 [DCLT18]，1、50亿个参数[RWC⁺19]、80亿个参数 [SPP⁺19]、110亿个参数 [RSR⁺19]，最近的成果则是170亿个参数[Tur20]。另一类研究则侧重于增加参数数量而非计算资源，目的是在不提升计算成本的情况下增强模型存储信息的能力。这类方法依托于条件计算框架 [BLC13]，尤其是专家混合方法 [SMM⁺17]，已被用来构建出拥有1000亿个参数的模型，最近甚至还出现了参数量达500亿的翻译模型 [AJF19]，不过在每次前向传播时实际被使用的参数仅占其中的一小部分。还有一种方法是在不增加参数数量的情况下提升计算资源，代表方案包括自适应计算时间 [Gra16] 以及通用变换器 [DGV⁺18]。我们的研究聚焦于第一类方法（即通过直接扩大神经网络规模来实现计算资源和参数数量的同步提升），并将采用该策略的模型的规模提升了10倍。

已有不少研究系统地探讨了规模变化对语言模型性能的影响。 [KMH⁺20、RRBS19、LWS⁺20以及HNA⁺17]等团队发现，当自回归语言模型的规模不断增大时，其损失值会呈现平稳的幂律趋势。这项研究表明，即便在模型规模进一步扩大的情况下，这一趋势依然大致保持不变（不过在3.1图中或许可以观察到曲线有轻微的弯曲），同时我们还在3个数量级的规模扩展过程中，发现许多（尽管并非全部）下游任务的性能也呈现出相对平稳的上升态势。

另一种研究方向与性能提升幅度完全相反，其目标是让语言模型在尽可能小的规模下依然保持出色的性能。这类方法包括ALBERT模型 [LCG⁺19]，以及通用的 [HVD15] 等等。

task-specific [SDCW19, JYS⁺19, KR16] approaches to distillation of language models. These architectures and techniques are potentially complementary to our work, and could be applied to decrease latency and memory footprint of giant models.

As fine-tuned language models have neared human performance on many standard benchmark tasks, considerable effort has been devoted to constructing more difficult or open-ended tasks, including question answering [KPR⁺19, IBGC⁺14, CCE⁺18, MCKS18], reading comprehension [CHI⁺18, RCM19], and adversarially constructed datasets designed to be difficult for existing language models [SBBC19, NWD⁺19]. In this work we test our models on many of these datasets.

Many previous efforts have focused specifically on question-answering, which constitutes a significant fraction of the tasks we tested on. Recent efforts include [RSR⁺19, RRS20], which fine-tuned an 11 billion parameter language model, and [GLT⁺20], which focused on attending over a large corpus of data at test time. Our work differs in focusing on in-context learning but could be combined in the future with those of [GLT⁺20, LPP⁺20].

Metalearning in language models has been utilized in [RWC⁺19], though with much more limited results and no systematic study. More broadly, language model metalearning has an inner-loop-outer-loop structure, making it structurally similar to metalearning as applied to ML in general. Here there is an extensive literature, including matching networks [VBL⁺16], RL2 [DSC⁺16], learning to optimize [RL16, ADG⁺16, LM17] and MAML [FAL17]. Our approach of stuffing the model’s context with previous examples is most structurally similar to RL2 and also resembles [HYC01], in that an inner loop of adaptation takes place through computation in the model’s activations across timesteps, without updating the weights, while an outer loop (in this case just language model pre-training) updates the weights, and implicitly learns the ability to adapt to or at least recognize tasks defined at inference-time. Few-shot auto-regressive density estimation was explored in [RCP⁺17] and [GWC⁺18] studied low-resource NMT as a few-shot learning problem.

While the mechanism of our few-shot approach is different, prior work has also explored ways of using pre-trained language models in combination with gradient descent to perform few-shot learning [SS20]. Another sub-field with similar goals is semi-supervised learning where approaches such as UDA [XDH⁺19] also explore methods of fine-tuning when very little labeled data is available.

Giving multi-task models instructions in natural language was first formalized in a supervised setting with [MKXS18] and utilized for some tasks (such as summarizing) in a language model with [RWC⁺19]. The notion of presenting tasks in natural language was also explored in the text-to-text transformer [RSR⁺19], although there it was applied for multi-task fine-tuning rather than for in-context learning without weight updates.

Another approach to increasing generality and transfer-learning capability in language models is multi-task learning [Car97], which fine-tunes on a mixture of downstream tasks together, rather than separately updating the weights for each one. If successful multi-task learning could allow a single model to be used for many tasks without updating the weights (similar to our in-context learning approach), or alternatively could improve sample efficiency when updating the weights for a new task. Multi-task learning has shown some promising initial results [LGH⁺15, LSP⁺18] and multi-stage fine-tuning has recently become a standardized part of SOTA results on some datasets [PFB18] and pushed the boundaries on certain tasks [KKS⁺20], but is still limited by the need to manually curate collections of datasets and set up training curricula. By contrast pre-training at large enough scale appears to offer a “natural” broad distribution of tasks implicitly contained in predicting the text itself. One direction for future work might be attempting to generate a broader set of explicit tasks for multi-task learning, for example through procedural generation [TFR⁺17], human interaction [ZSW⁺19b], or active learning [Mac92].

Algorithmic innovation in language models over the last two years has been enormous, including denoising-based bidirectionality [DCLT18], prefixLM [DL15] and encoder-decoder architectures [LLG⁺19, RSR⁺19], random permutations during training [YDY⁺19], architectures that improve the efficiency of sampling [DYY⁺19], improvements in data and training procedures [LOG⁺19], and efficiency increases in the embedding parameters [LCG⁺19]. Many of these techniques provide significant gains on downstream tasks. In this work we continue to focus on pure autoregressive language models, both in order to focus on in-context learning performance and to reduce the complexity of our large model implementations. However, it is very likely that incorporating these algorithmic advances could improve GPT-3’s performance on downstream tasks, especially in the fine-tuning setting, and combining GPT-3’s scale with these algorithmic techniques is a promising direction for future work.

8 Conclusion

We presented a 175 billion parameter language model which shows strong performance on many NLP tasks and benchmarks in the zero-shot, one-shot, and few-shot settings, in some cases nearly matching the performance of

针对特定任务的 [SDCW19, JYS⁺19, KR16]语言模型模型蒸馏方法。这些架构和技术很可能与我们正在开展的工作形成互补，有望用于降低大型模型的延迟并减少其内存占用。

由于经过微调的语言模型在众多标准基准测试任务上的性能已接近人类表现水平，人们便投入了大量精力来设计难度更高或更具开放性的任务，其中就包括问答任务 [KPR⁺19, IBGC⁺14, CCE⁺18, MCKS18], 阅读理解任务 [CHI⁺18, RCM19], , 以及那些专为增加现有语言模型处理难度而人为设计的对抗性数据集 [SBBC19, NWD⁺19]。在本研究中，我们在这些数据集中的多个上对我们的模型进行了测试。

此前有许多研究专门聚焦于问答任务，而这正是我们所测试的任务中占比相当大的一部分。近期的相关研究包括 [RSR⁺19, RRS20], , 后者对一个拥有110亿参数的语言模型进行了微调；还有 [GLT⁺20], 该研究侧重于在测试时处理海量数据语料库。我们的研究不同之处在于侧重于上下文学习，但未来也有可能与 [GLT⁺20, LPP⁺20]的相关工作相结合。

语言模型中的元学习技术已被应用于 [RWC⁺19]领域，不过其取得的结果极为有限，且至今尚未出现系统性的相关研究。从更广泛的角度来看，语言模型的元学习具有内环-外环结构，这一特点使其在结构上与普遍应用于机器学习领域的元学习十分相似。在该领域已有大量研究成果，包括匹配网络 [VBL⁺16]、RL2 [DSC⁺16]、用于优化 [RL16]的算法、ADG⁺16, LM17] 以及MAML [FAL17]等。我们通过用先前的示例填充模型上下文的方法，在结构上与RL2最为相似，同时也与 [HYC01], 类似——二者均通过计算模型在各个时间步的激活值来形成内部适配循环，而无需更新模型权重；与此同时，外部循环（在本研究中即为语言模型的预训练）会更新权重，并借此隐式地学习适应或在推理时识别任务的能力。少样本自回归密度估计相关研究可见于 [RCP⁺17] 和 [GWC⁺18]，这些研究还将低资源环境下的机器翻译视为一种少样本学习问题进行探讨。

虽然我们的少样本学习方法的机制有所不同，但先前的研究也曾探索过将预训练好的语言模型与梯度下降方法结合，用以实现少样本学习 [SS20]。另一个目标相似的子领域是半监督学习，其中诸如UDA这样的方法 [XDH⁺19]也会在标记数据极为匮乏的情况下探索微调的相关方法。

首次以正式形式在监督学习环境中将自然语言指令赋予多任务模型的是 [MKXS18], 随后这种技术也被应用于某些任务（如文本总结）中，相关模型则基于 [RWC⁺19]。在文本到文本变换器 [RSR⁺19]中同样探讨了用自然语言描述任务的理念，只不过那里该技术是用于多任务微调，而非无需更新重力的上下文学习。

提升语言模型的通用性及迁移学习能力的另一种方法是多任务学习[Car97], , 该方法会针对多种下游任务一起进行微调，而非分别为每个任务单独更新权重。如果多任务学习能够成功实现，那么就可以让同一个模型在不更新权重的情况下处理多种任务（这与我们的上下文学习方法类似），或者在新任务需要更新权重时提升样本效率。目前多任务学习已经取得了一些颇具前景的初步结果 [LGH⁺15, LSP⁺18], 而且多阶段微调近来也已成为某些数据集上达到当前最佳性能的标准化手段 [PFB18], 并在某些任务上不断突破现有极限 [KKS⁺20]。不过该方法仍受限于需要人工整理数据集合并设计训练方案。相比之下，规模足够大的预训练似乎能通过预测文本本身，隐含地提供范围更广的任务概率分布。未来的研究方向之一或许是通过程序化生成 [TFR⁺17]、人类交互 [ZSW⁺19b], 或主动学习 [Mac92]等方式，为多任务学习生成更丰富的明确任务集合。

过去两年中，语言模型在算法创新方面取得了巨大进展，其中包括基于去噪的双向性机制 [DCLT18], prefixLM [DL15] 以及编码器-解码器架构 [LLG⁺19, RSR⁺19]、训练过程中的随机排列 [YDY⁺19]、能够提升采样效率的架构 [DYY⁺19]、数据与训练流程的优化 [LOG⁺19], 还有嵌入参数效率的提升 [LCG⁺19]。这些技术中的许多都能为下游任务带来显著的性能提升。在本研究中，我们依然聚焦于纯自回归语言模型，这样做既是为了研究上下文学习性能，也是为了降低大型模型实现的复杂度。不过，将这类算法进步应用到GPT-3上很可能会提升其在下游任务上的性能，尤其是在微调场景下；而将GPT-3的庞大规模与这些算法技术相结合，无疑是未来研究的一个极具前景的方向。

8 结论

我们展示了一个拥有1750亿参数的语言模型，该模型在零样本学习、单样本学习和少样本学习场景下的诸多自然语言处理任务及基准测试中均展现出优异的性能，某些情况下其性能甚至几乎可与最顶尖的模型相媲美。

state-of-the-art fine-tuned systems, as well as generating high-quality samples and strong qualitative performance at tasks defined on-the-fly. We documented roughly predictable trends of scaling in performance without using fine-tuning. We also discussed the social impacts of this class of model. Despite many limitations and weaknesses, these results suggest that very large language models may be an important ingredient in the development of adaptable, general language systems.

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除了具备上述优势外，这类语言模型还能生成高质量样本，且在即时定义的任务上表现出良好的定性性能。我们发现，在不进行微调的前提下，其性能提升幅度也呈现出大致可预测的趋势。与此同时，我们还研究了这类模型可能产生的社会影响。虽然这些模型还存在许多局限性，但上述研究结果仍暗示，超大语言模型很可能是构建适应性通用语言系统过程中的关键要素。

致谢

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Contributions

Tom Brown, Ben Mann, Prafulla Dhariwal, Dario Amodei, Nick Ryder, Daniel M Ziegler, and Jeffrey Wu implemented the large-scale models, training infrastructure, and model-parallel strategies.

Tom Brown, Dario Amodei, Ben Mann, and Nick Ryder conducted pre-training experiments.

Ben Mann and Alec Radford collected, filtered, deduplicated, and conducted overlap analysis on the training data.

Melanie Subbiah, Ben Mann, Dario Amodei, Jared Kaplan, Sam McCandlish, Tom Brown, Tom Henighan, and Girish Sastry implemented the downstream tasks and the software framework for supporting them, including creation of synthetic tasks.

Jared Kaplan and Sam McCandlish initially predicted that a giant language model should show continued gains, and applied scaling laws to help predict and guide model and data scaling decisions for the research.

Ben Mann implemented sampling without replacement during training.

Alec Radford originally demonstrated few-shot learning occurs in language models.

Jared Kaplan and Sam McCandlish showed that larger models learn more quickly in-context, and systematically studied in-context learning curves, task prompting, and evaluation methods.

Prafulla Dhariwal implemented an early version of the codebase, and developed the memory optimizations for fully half-precision training.

Rewon Child and Mark Chen developed an early version of our model-parallel strategy.

Rewon Child and Scott Gray contributed the sparse transformer.

Aditya Ramesh experimented with loss scaling strategies for pretraining.

Melanie Subbiah and Arvind Neelakantan implemented, experimented with, and tested beam search.

Pranav Shyam worked on SuperGLUE and assisted with connections to few-shot learning and meta-learning literature.

Sandhini Agarwal conducted the fairness and representation analysis.

Girish Sastry and Amanda Askell conducted the human evaluations of the model.

Ariel Herbert-Voss conducted the threat analysis of malicious use.

Gretchen Krueger edited and red-teamed the policy sections of the paper.

Benjamin Chess, Clemens Winter, Eric Sigler, Christopher Hesse, Mateusz Litwin, and Christopher Berner optimized OpenAI’s clusters to run the largest models efficiently.

Scott Gray developed fast GPU kernels used during training.

Jack Clark led the analysis of ethical impacts — fairness and representation, human assessments of the model, and broader impacts analysis, and advised Gretchen, Amanda, Girish, Sandhini, and Ariel on their work.

Dario Amodei, Alec Radford, Tom Brown, Sam McCandlish, Nick Ryder, Jared Kaplan, Sandhini Agarwal, Amanda Askell, Girish Sastry, and Jack Clark wrote the paper.

Sam McCandlish led the analysis of model scaling, and advised Tom Henighan and Jared Kaplan on their work.

Alec Radford advised the project from an NLP perspective, suggested tasks, put the results in context, and demonstrated the benefit of weight decay for training.

Ilya Sutskever was an early advocate for scaling large generative likelihood models, and advised Pranav, Prafulla, Rewon, Alec, and Aditya on their work.

Dario Amodei designed and led the research.

Contributions

汤姆·布朗、本·曼恩、普拉富拉·达里瓦尔、达里奥·阿莫代、尼克·莱德、丹尼尔·M·齐格勒和吴杰弗里实现了这些大规模模型、相应的训练基础设施以及模型并行策略。

汤姆·布朗、达里奥·阿莫代、本·曼恩以及尼克·莱德开展了预训练实验。

本·曼恩与亚历克·拉德福德负责收集训练数据，对其进行了过滤处理、去重操作，并完成了重叠分析。

梅兰妮·苏比亚、本·曼恩、达里奥·阿莫代、贾里德·卡普兰、萨姆·麦坎德利什、汤姆·布朗、汤姆·亨尼根以及吉里什·萨斯特里负责实现各类下游任务，同时开发了用于支撑这些任务的软件框架，还包括合成任务的创建工作。

贾里德·卡普兰与萨姆·麦坎德利什最初预测大型语言模型应当能够持续取得进步，他们还运用缩放定律来辅助预测并指导该研究中的模型与数据规模调整决策。

本·曼恩在训练过程中采用了不放回采样方法。

亚历克·拉德福德最早证明了语言模型中存在少样本学习现象。

贾里德·卡普兰与萨姆·麦坎德利什指出更大规模的模型在上下文学习中的学习速度更快，同时系统地研究了上下文学习曲线、任务提示方式以及各类评估方法。

普拉富拉·达里瓦尔实现了该代码库的早期版本，并针对全半精度训练开发了相应的内存优化方案。

雷文·柴尔德与陈马克构建了我们早期的模型并行策略版本。

雷文·柴尔德与斯科特·格雷提出了稀疏Transformer这一技术。

阿迪亚·拉梅什对预训练阶段的损失缩放策略进行了相关实验研究。

梅兰妮·苏比亚与阿尔文德·尼拉坎坦实现了波束搜索算法，并对其进行了实验与测试。

普拉纳夫·夏姆参与了SuperGLUE任务的研发工作，同时还协助建立了该研究与其他少样本学习及元学习领域文献之间的联系。

桑迪尼·阿加瓦尔负责开展了针对该模型的公平性分析以及表征性分析。

吉里什·萨斯特里与阿曼达·阿斯科尔承担了对该模型进行人工评估的工作。

阿里埃尔·赫伯特·沃斯负责对模型可能被恶意利用的情况进行了威胁分析。

格雷琴·克鲁格负责编辑论文的各个政策章节，同时还对其进行了渗透测试。

本杰明·切斯、克莱门斯·温特、埃里克·西格勒、克里斯托弗·赫塞、马特乌什·利特温以及克里斯托弗·伯纳对OpenAI的集群进行了优化，从而高效运行规模最大的模型。

斯科特·格雷开发了在训练过程中使用的快速GPU内核。

杰克·克拉克负责开展有关伦理影响——包括公平性、模型的人类评估结果以及更广泛的影响等方面的分析，同时还为格雷琴、阿曼达、吉里什、桑迪尼和阿里尔在他们的工作上提供了指导。

达里奥·阿莫代、亚历克·拉德福德、汤姆·布朗、萨姆·麦坎德利什、尼克·莱德、贾里德·卡普兰、桑迪尼·阿加瓦尔、阿曼达·阿斯科尔、吉里什·萨斯特里以及杰克·克拉克共同撰写了这篇论文。

萨姆·麦坎德利什牵头进行了模型规模扩展方面的分析，并为汤姆·海尼根和贾里德·卡普兰在他们的工作上提供了建议。

亚历克·拉德福德从自然语言处理的视角为该项目出谋划策，提出了各项任务，将分析结果置于相应的上下文中，并展示了在训练过程中应用重力衰减方法的益处。

伊利亚·苏茨克维尔是早期倡导提升大型生成似然模型性能提升幅度的人士，他还曾为普拉纳夫、普拉富拉、雷翁、亚历克以及阿迪蒂亚在其相关工作上提供指导。

达里奥·阿莫代负责设计了这项研究并全程主持工作。

A Details of Common Crawl Filtering

As mentioned in Section 2.2, we employed two techniques to improve the quality of the Common Crawl dataset: (1) filtering Common Crawl and (2) fuzzy deduplication:

1. In order to improve the quality of Common Crawl, we developed an automatic filtering method to remove low quality documents. Using the original WebText as a proxy for high-quality documents, we trained a classifier to distinguish these from raw Common Crawl. We then used this classifier to re-sample Common Crawl by prioritizing documents which were predicted by the classifier to be higher quality. The classifier is trained using logistic regression classifier with features from Spark’s standard tokenizer and HashingTF¹⁰. For the positive examples, we used a collection of curated datasets such as WebText, Wikiedia, and our web books corpus as the positive examples, and for the negative examples, we used unfiltered Common Crawl. We used this classifier to score Common Crawl documents. We kept each document in our dataset iff

$$\text{np.random.pareto}(\alpha) > 1 - \text{document_score}$$

We chose $\alpha = 9$ in order to take mostly documents the classifier scored highly, but still include some documents that were out of distribution. α was chosen to match the distribution of scores from our classifier on WebText. We found this re-weighting increased quality as measured by loss on a range of out-of-distribution generative text samples.

2. To further improve model quality and prevent overfitting (which becomes increasingly important as model capacity increases), we fuzzily deduplicated documents (i.e. removed documents with high overlap with other documents) within each dataset using Spark’s MinHashLSH implementation with 10 hashes, using the same features as were used for classification above. We also fuzzily removed WebText from Common Crawl. Overall this decreased dataset size by an average of 10%.

After filtering for duplicates and quality, we also partially removed text occurring in benchmark datasets, described in Appendix C.

B Details of Model Training

To train all versions of GPT-3, we use Adam with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and $\epsilon = 10^{-8}$, we clip the global norm of the gradient at 1.0, and we use cosine decay for learning rate down to 10% of its value, over 260 billion tokens (after 260 billion tokens, training continues at 10% of the original learning rate). There is a linear LR warmup over the first 375 million tokens. We also gradually increase the batch size linearly from a small value (32k tokens) to the full value over the first 4-12 billion tokens of training, depending on the model size. Data are sampled without replacement during training (until an epoch boundary is reached) to minimize overfitting. All models use weight decay of 0.1 to provide a small amount of regularization [LH17].

During training we always train on sequences of the full $n_{\text{ctx}} = 2048$ token context window, packing multiple documents into a single sequence when documents are shorter than 2048, in order to increase computational efficiency. Sequences with multiple documents are not masked in any special way but instead documents within a sequence are delimited with a special end of text token, giving the language model the information necessary to infer that context separated by the end of text token is unrelated. This allows for efficient training without need for any special sequence-specific masking.

C Details of Test Set Contamination Studies

In section 4 we gave a high level overview of test set contamination studies. In this section we provide details on methodology and results.

Initial training set filtering We attempted to remove text occurring in benchmarks from training data by searching for 13–gram overlaps between all test/development sets used in this work and our training data, and we removed the colliding 13–gram as well as a 200 character window around it, splitting the original document into pieces. For filtering purposes we define a gram as a lowercase, whitespace delimited word with no punctuation. Pieces less than 200 characters long were discarded. Documents split into more than 10 pieces were considered contaminated and

¹⁰<https://spark.apache.org/docs/latest/api/python/pyspark.ml.html#pyspark.ml.feature.HashingTF>

Common Crawl过滤处理的细节

如第 2.2节所述, 我们采用了两种技术来提升CommonCrawl数据集的质量: (1) 对Common Crawl数据集进行过滤处理, 以及 (2) 通过模糊去重处理。

1. 为提升Common Crawl的数据质量, 我们开发了一种自动过滤方法, 用于剔除低质量的文档。我们以原始的WebText数据集作为高质量文档的替代来源, 训练了一个分类器来区分这些高质量文档与原始的Common Crawl文档。随后, 我们利用该分类器对Common Crawl数据进行重采样, 优先选取那些被分类器判定为更高质量的文档。该分类器是采用逻辑回归分类器, 并结合Spark框架的标准令牌化工具以及HashingTF¹⁰算法的特征共同训练而成的。对于正例, 我们使用了WebText、Wikiedia以及我们构建的网页书籍语料库等精心筛选的数据集作为训练样本; 而对于负例, 则使用未经过滤的Common Crawl数据。我们通过这个分类器为Common Crawl中的每个文档打分, 只有满足特定条件的文档才会被保留在我们的数据集中。

$$\text{np.random.pareto}(\alpha) > 1 - \text{document_score}$$

我们选择 $\alpha = 9$ 是为了主要保留那些分类器评分较高的文档, 同时仍包含部分处于概率分布之外的文档。而 α 则被选用来使分类器在WebText数据上的评分概率分布与实际分布更为匹配。实验结果表明, 这种重新加权处理能够提升数据质量, 这一效果可通过一系列处于概率分布之外的生成式文本样本的损失值来衡量。

2. 为进一步提升模型质量并防止过拟合——随着模型容量不断提升, 过拟合问题愈发重要——我们采用了Spark框架中的MinHashLSH算法 (配置10个哈希值), 并运用与前述分类任务相同的特征, 对每个数据集内的文档进行了模糊去重处理, 即移除了与其他文档重叠度较高的文档。此外, 我们还对Common Crawl数据集中的WebText内容进行了类似的模糊删除操作。总体而言, 这些措施使得各数据集的规模平均减少了10%。

在完成重复项及低质量内容的过滤后, 我们还移除了附录中所述的基准测试数据集里出现的部分文本C。

B 模型训练细节

在训练所有版本的GPT-3时, 我们采用了Adam优化器, 同时设置了 $\beta_1 = 0.9$ 、 $\beta_2 = 0.95$ 以及 $\epsilon = 10^{-8}$ 等参数。我们会将梯度的全局范数限制在1.0以内, 并通过余弦衰减机制逐步降低学习率, 最终使其降至初始值的10%, 这一过程会覆盖2600亿个令牌; 在处理完这2600亿个令牌后, 训练仍会以初始学习率的10%继续进行。在前3.75亿个令牌期间, 学习率会经历线性升温过程。此外, 根据模型规模的不同, 我们还会在训练的前40亿至120亿个令牌阶段, 将批量大小从较小的值 (32k个令牌) 逐步线性提升到最终值。为尽量减少过拟合现象, 训练过程中会对数据采用无放回抽样方式 (直达到一个训练轮次边界)。所有模型都会应用0.1的重力衰减机制, 以此实现一定程度的正则化处理 [LH17]。

在训练过程中, 我们始终使用完整的 $n_{\text{ctx}} = 2048$ 令牌上下文窗口来处理序列数据; 当单份文档的长度低于2048时, 会将多份文档合并为一个序列, 以此提升计算效率。包含多份文档的序列并不会采用任何特殊的掩码处理方式, 而是通过特定的文本结束令牌来分隔序列中的各个文档, 从而使语言模型能够判断被文本结束令牌分隔的上下文彼此无关。这样的设计无需额外的序列专用掩码即可实现高效训练。

C 测试集污染研究详情

在第4节中, 我们概述了测试集污染研究的基本情况; 而在本节里, 则会详细介绍相关的方法与结果。

初始训练集过滤为避免训练数据中出现基准测试中的文本, 我们首先查找本研究所使用的所有测试集/开发集与训练数据之间的 13–元重叠部分, 进而移除发生冲突的 13–元及其周围200个字符范围内的内容, 从而将原始文档分割成多个片段。在过滤过程中, 我们将“元”定义为不含标点符号、由小写字母和空白字符构成的单词。长度小于200字符的片段会被直接丢弃, 而被分割成10个以上片段的文档则被视为受污染数据。

¹⁰https://spark.apache.org/docs/latest/api/python/pyspark.ml.html#py_spark.ml.feature.HashingTF

removed entirely. Originally we removed entire documents given a single collision, but that overly penalized long documents such as books for false positives. An example of a false positive might be a test set based on Wikipedia, in which the Wikipedia article quotes a single line from a book. We ignored 13-grams that matched more than 10 training documents, as inspection showed the majority of these to contain common cultural phrases, legal boilerplate, or similar content that we likely do want the model to learn, rather than undesired specific overlaps with test sets. Examples for various frequencies can be found in the GPT-3 release repository¹¹.

Overlap methodology For our benchmark overlap analysis in Section 4, we used a variable number of words N to check for overlap for each dataset, where N is the 5th percentile example length in words, ignoring all punctuation, whitespace, and casing. Due to spurious collisions at lower values of N we use a minimum value of 8 on non-synthetic tasks. For performance reasons, we set a maximum value of 13 for all tasks. Values for N and the amount of data marked as dirty are shown in Table C.1. Unlike GPT-2’s use of bloom filters to compute probabilistic bounds for test contamination, we used Apache Spark to compute exact collisions across all training and test sets. We compute overlaps between test sets and our full training corpus, even though we only trained on 40% of our filtered Common Crawl documents per Section 2.2.

We define a ‘dirty’ example as one with any N -gram overlap with any training document, and a ‘clean’ example as one with no collision.

Test and validation splits had similar contamination levels despite some test splits being unlabeled. Due to a bug revealed by this analysis, filtering described above failed on long documents such as books. Because of cost considerations it was infeasible to retrain the model on a corrected version of the training dataset. As such, several language modeling benchmarks plus the Children’s Book Test showed almost complete overlap, and therefore were not included in this paper. Overlaps are shown in Table C.1

Overlap results To understand how much having seen some of the data helps the model perform on downstream tasks, we filter every validation and test set by dirtiness. Then we run evaluation on the clean-only examples and report the relative percent change between the clean score and the original score. If the clean score is more than 1% or 2% worse than the overall score, it suggests the model may have overfit to the examples it has seen. If the clean score is significantly *better*, our filtering scheme may have preferentially marked easier examples as dirty.

This overlap metric tends to show a high rate of false positives for datasets that contain background information (but not answers) drawn from the web (such as SQuAD, which draws from Wikipedia) or examples less than 8 words long, which we ignored in our filtering process (except for wordscrambling tasks). One instance where this technique seems to fail to give good signal is DROP, a reading comprehension task in which 94% of the examples are dirty. The information required to answer the question is in a passage provided to the model, so having seen the passage during training but not the questions and answers does not meaningfully constitute cheating. We confirmed that every matching training document contained only the source passage, and none of the questions and answers in the dataset. The more likely explanation for the decrease in performance is that the 6% of examples that remain after filtering come from a slightly different distribution than the dirty examples.

Figure 4.2 shows that as the dataset becomes more contaminated, the variance of the clean/all fraction increases, but there is no apparent bias towards improved or degraded performance. This suggests that GPT-3 is relatively insensitive to contamination. See Section 4 for details on the datasets we flagged for further review.

并予以完全移除。最初我们只要检测到一次冲突就会删除整份文档，但这种做法会因假阳性结果而过度惩罚诸如书籍这类长度较长的文档。假阳性现象的一个示例可能是以维基百科为基础构建的测试集，其中维基百科的文章引用了书籍中的一行内容。我们忽略了 13-那些与超过10份训练文档匹配的元，因为经过检查发现，这类元大多包含常见的文化用语、法律模板或类似内容，这些都是我们希望模型学习的，而非与测试集之间出现的那些不希望出现的特定重叠内容。关于不同匹配频率的示例，可在GPT-3发布仓库中找到¹¹。

重叠分析方法在本章第4节所进行的基准测试中的重叠分析中，我们使用了不同数量的单词 N 来检测每个数据集的重叠情况，其中 N 表示忽略所有标点、空格及大小写后的第5百分位示例长度（以单词计）。由于在较低的数值下会出现虚假的冲突， N 因此在非合成任务中我们将该数值的下限设定为8。出于性能考虑，所有任务的该数值上限均设为13。关于 N 以及被标记为脏数据的数据量等相关数值，请参见表C.1。与GPT-2使用布隆过滤器来计算测试数据污染的概率范围不同，我们是借助 Apache Spark来精确计算所有训练集和测试集之间的冲突情况。即便根据第2.2节的要求，我们仅对经过筛选后的Common Crawl文档中的40%进行了训练，我们依然会对测试集与我们完整的训练语料库之间的重叠情况进行计算。

我们将存在任何 N -gram级别重叠的训练文档的示例定义为‘脏数据’示例，而不存在此类冲突的示例则定义为‘干净数据’示例。

尽管部分测试集未加标记，但测试集与验证集的污染程度大致相当。由于此次分析发现的一个漏洞，上述过滤方法在处理书籍这类篇幅较长的文档时失效了。考虑到成本问题，使用修正后的训练数据集重新训练模型并不可行。正因如此，多项语言建模基准测试以及儿童书籍测试的结果存在几乎完全的重叠，因此并未被纳入本文。相关重叠情况详见表C.1

重叠结果为了解析接触过部分数据对模型完成下游任务表现的提升程度，我们会根据数据的“脏污”程度对所有验证集和测试集进行过滤处理。随后仅在干净数据上的示例上开展评估，并汇报干净数据得分与原始得分之间的相对百分比变化。如果干净数据得分比整体得分低1%或2%以上，这可能表明模型过度拟合了其见过的示例；而如果干净数据得分明显>更好，则说明我们的过滤机制可能将较为简单的示例错误地标记为脏数据。

对于那些包含来自网络（例如源自维基百科的SQuAD）的背景信息（而非答案），或是长度不足8个单词的示例——这类示例在我们进行过滤处理时会被忽略，单词重排任务除外——该重叠指标往往会出现较高的假阳性结果率。在DROP这一阅读理解任务中，这种技术似乎就无法有效区分有效信号与噪声，因为其中94%的示例都属于脏数据。由于回答问题所需的信息都包含在提供给模型的文章中，因此模型在训练阶段已经看过这些文章，却未见过相应的题目和答案，这种情况并不构成实质性的作弊行为。我们已确认，所有匹配到的训练文档仅包含原始文章内容，数据集中完全没有题目和答案。性能下降更可能的原因是，经过过滤后剩余的6%示例其概率分布与那些脏数据示例略有不同。

图表 4.2 显示，随着数据集的污染程度加剧，干净数据/全部数据的比例方差会上升，但性能并未出现明显的好转或恶化趋势。这表明GPT-3对数据污染的敏感度相对较低。有关被标记出来需要进一步审查的数据集的详细信息，请参阅第4节。

¹¹https://github.com/openai/gpt-3/blob/master/overlap_frequency.md

¹¹https://github.com/openai/GPT-3/blob/master/overlap_frequency.md

Name	Split	Metric	N	Acc/F1/BLEU	Total Count	Dirty Acc/F1/BLEU	Dirty Count	Clean Acc/F1/BLEU	Clean Count	Clean Percentage	Relative Difference Clean vs All
Quac	dev	f1	13	44.3	7353	44.3	7315	54.1	38	1%	20%
SQuADv2	dev	f1	13	69.8	11873	69.9	11136	68.4	737	6%	-2%
DROP	dev	f1	13	36.5	9536	37.0	8898	29.5	638	7%	-21%
Symbol Insertion	dev	acc	7	66.9	10000	66.8	8565	67.1	1435	14%	0%
CoQa	dev	f1	13	86.0	7983	85.3	5107	87.1	2876	36%	1%
ReCoRD	dev	acc	13	89.5	10000	90.3	6110	88.2	3890	39%	-1%
Winograd	test	acc	9	88.6	273	90.2	164	86.2	109	40%	-3%
BoolQ	dev	acc	13	76.0	3270	75.8	1955	76.3	1315	40%	0%
MultiRC	dev	acc	13	74.2	953	73.4	558	75.3	395	41%	1%
RACE-h	test	acc	13	46.8	3498	47.0	1580	46.7	1918	55%	0%
LAMBADA	test	acc	13	86.4	5153	86.9	2209	86.0	2944	57%	0%
LAMBADA (No Blanks)	test	acc	13	77.8	5153	78.5	2209	77.2	2944	57%	-1%
WSC	dev	acc	13	76.9	104	73.8	42	79.0	62	60%	3%
PIQA	dev	acc	8	82.3	1838	89.9	526	79.3	1312	71%	-4%
RACE-m	test	acc	13	58.5	1436	53.0	366	60.4	1070	75%	3%
De→En 16	test	bleu-sb	12	43.0	2999	47.4	739	40.8	2260	75%	-5%
En→De 16	test	bleu-sb	12	30.9	2999	32.6	739	29.9	2260	75%	-3%
En→Ro 16	test	bleu-sb	12	25.8	1999	24.9	423	26.1	1576	79%	1%
Ro→En 16	test	bleu-sb	12	41.3	1999	40.4	423	41.6	1576	79%	1%
WebQs	test	acc	8	41.5	2032	41.6	428	41.5	1604	79%	0%
ANLI R1	test	acc	13	36.8	1000	40.5	200	35.9	800	80%	-3%
ANLI R2	test	acc	13	34.0	1000	29.4	177	35.0	823	82%	3%
TriviaQA	dev	acc	10	71.2	7993	70.8	1390	71.3	6603	83%	0%
ANLI R3	test	acc	13	40.2	1200	38.3	196	40.5	1004	84%	1%
En→Fr 14	test	bleu-sb	13	39.9	3003	38.3	411	40.3	2592	86%	1%
Fr→En 14	test	bleu-sb	13	41.4	3003	40.9	411	41.4	2592	86%	0%
WiC	dev	acc	13	51.4	638	53.1	49	51.3	589	92%	0%
RTE	dev	acc	13	71.5	277	71.4	21	71.5	256	92%	0%
CB	dev	acc	13	80.4	56	100.0	4	78.8	52	93%	-2%
Anagrams 2	dev	acc	2	40.2	10000	76.2	705	37.4	9295	93%	-7%
Reversed Words	dev	acc	2	0.4	10000	1.5	660	0.3	9340	93%	-26%
OpenBookQA	test	acc	8	65.4	500	58.1	31	65.9	469	94%	1%
ARC (Easy)	test	acc	11	70.1	2268	77.5	89	69.8	2179	96%	0%
Anagrams 1	dev	acc	2	15.0	10000	49.8	327	13.8	9673	97%	-8%
COPA	dev	acc	9	93.0	100	100.0	3	92.8	97	97%	0%
ARC (Challenge)	test	acc	12	51.6	1144	45.2	31	51.8	1113	97%	0%
HellaSwag	dev	acc	13	79.3	10042	86.2	152	79.2	9890	98%	0%
NQs	test	acc	11	29.9	3610	32.7	52	29.8	3558	99%	0%
Cycled Letters	dev	acc	2	38.6	10000	20.5	73	38.7	9927	99%	0%
SAT Analogies	dev	acc	9	65.8	374	100.0	2	65.6	372	99%	0%
StoryCloze	test	acc	13	87.7	1871	100.0	2	87.6	1869	100%	0%
Winogrande	dev	acc	13	77.7	1267	-	0	77.7	1267	100%	0%

Table C.1: Overlap statistics for all datasets sorted from dirtiest to cleanest. We consider a dataset example dirty if it has a single N -gram collision with any document in our training corpus. “Relative Difference Clean vs All” shows the percent change in performance between only the clean examples vs all the examples in the benchmark. “Count” shows the number of examples. “Clean percentage” is the percent of examples that are clean vs total. For “Acc/F1/BLEU” we use the metric specified in “Metric”. These scores come from evaluations with a different seed for the random examples used for in-context learning, and will therefore differ slightly from the scores elsewhere in the paper.

Name	分割	指标	N	准确率/F1值/BLEU分数	总计数量	脏数据准确率/F1值/BLEU分数	脏数据数量	干净数据准确率/F1值/BLEU分数	干净数据数量	干净数据百分比	相对值 差值 清洗数据与全部数据对比
Quac	dev	f1	13	44.3	7353	44.3	7315	54.1	38	1%	20%
SQuADv2数据集	dev	f1	13	69.8	11873	69.9	11136	68.4	737	6%	-2%
DROP	dev	f1	13	36.5	9536	37.0	8898	29.5	638	7%	-21%
符号插入算法	dev	acc	7	66.9	10000	66.8	8565	67.1	1435	14%	0%
CoQa	dev	f1	13	86.0	7983	85.3	5107	87.1	2876	36%	1%
ReCoRD算法	dev	acc	13	89.5	10000	90.3	6110	88.2	3890	39%	-1%
Winograd算法	test	acc	9	88.6	273	90.2	164	86.2	109	40%	-3%
BoolQ任务	dev	acc	13	76.0	3270	75.8	1955	76.3	1315	40%	0%
MultiRC数据集	dev	acc	13	74.2	953	73.4	558	75.3	395	41%	1%
RACE-h	test	acc	13	46.8	3498	47.0	1580	46.7	1918	55%	0%
LAMBADA	test	acc	13	86.4	5153	86.9	2209	86.0	2944	57%	0%
LAMBADA（无空格）	test	acc	13	77.8	5153	78.5	2209	77.2	2944	57%	-1%
WSC	dev	acc	13	76.9	104	73.8	42	79.0	62	60%	3%
PIQA	dev	acc	8	82.3	1838	89.9	526	79.3	1312	71%	-4%
RACE-m	test	acc	13	58.5	1436	53.0	366	60.4	1070	75%	3%
德语16	test	bleu-sb	12	43.0	2999	47.4	739	40.8	2260	75%	-5%
英→德 16	test	bleu-sb	12	30.9	2999	32.6	739	29.9	2260	75%	-3%
英→罗 16	test	bleu-sb	12	25.8	1999	24.9	423	26.1	1576	79%	1%
罗→英 16	test	bleu-sb	12	41.3	1999	40.4	423	41.6	1576	79%	1%
WebQS	test	acc	8	41.5	2032	41.6	428	41.5	1604	79%	0%
ANLI R1	test	acc	13	36.8	1000	40.5	200	35.9	800	80%	-3%
ANLI R2	test	acc	13	34.0	1000	29.4	177	35.0	823	82%	3%
TriviaQA	dev	acc	10	71.2	7993	70.8	1390	71.3	6603	83%	0%
ANLI R3模型	test	acc	13	40.2	1200	38.3	196	40.5	1004	84%	1%
英→法 14	test	bleu-sb评分	13	39.9	3003	38.3	411	40.3	2592	86%	1%
法语→英语 14	test	bleu-sb评分	13	41.4	3003	40.9	411	41.4	2592	86%	0%
WiC	dev	acc	13	51.4	638	53.1	49	51.3	589	92%	0%
RTE	dev	acc	13	71.5	277	71.4	21	71.5	256	92%	0%
CB	dev	acc	13	80.4	56	100.0	4	78.8	52	93%	-2%
变位词2数据集	dev	acc	2	40.2	10000	76.2	705	37.4	9295	93%	-7%
反向词序任务	dev	acc	2	0.4	10000	1.5	660	0.3	9340	93%	-26%
OpenBookQA数据集	test	acc	8	65.4	500	58.1	31	65.9	469	94%	1%
ARC简单版	test	acc	11	70.1	2268	77.5	89	69.8	2179	96%	0%
变位词1	dev	acc	2	15.0	10000	49.8	327	13.8	9673	97%	-8%
COPA	dev	acc	9	93.0	100	100.0	3	92.8	97	97%	0%
ARC挑战版	test	acc	12	51.6	1144	45.2	31	51.8	1113	97%	0%
HellaSwag	dev	acc	13	79.3	10042	86.2	152	79.2	9890	98%	0%
NQs	test	acc	11	29.9	3610	32.7	52	29.8	3558	99%	0%
循环字母测试	dev	acc	2	38.6	10000	20.5	73	38.7	9927	99%	0%
SAT类比题	dev	acc	9	65.8	374	100.0	2	65.6	372	99%	0%
StoryCloze	test	acc	13	87.7	1871	100.0	2	87.6	1869	100%	0%
维诺格兰德	dev	acc	13	77.7	1267	-	0	77.7	1267	100%	0%

表C.1: 按污染程度从高到低排序的所有数据集的重叠统计情况。如果某个数据集的某个 N -gram与训练语料库中的任意文档发生冲突， 我们就会将其视为脏数据示例。“干净样本与全部样本的相对差异”列出了仅使用干净样本与使用所有样本进行基准测试时性能的百分比变化，“数量”则显示了样本的总数，“干净样本占比”是指干净样本在所有样本中所占的百分比。对于“准确率/F1值/BLEU分数”，我们采用“指标”一节中规定的计算方法。这些分数是通过使用不同的随机种子对用于上下文学习的样本进行评估得出的，因此会与论文其他部分给出的分数略有差异。

D Total Compute Used to Train Language Models

This appendix contains the calculations that were used to derive the approximate compute used to train the language models in Figure 2.2. As a simplifying assumption, we ignore the attention operation, as it typically uses less than 10% of the total compute for the models we are analyzing.

Calculations can be seen in Table D.1 and are explained within the table caption.

Model	Total train compute (PF-days)	Total train compute (flops)	Params (M)	Training tokens (billions)	Flops per param per token	Mult for bwd pass	Fwd-pass flops per active param per token	Frac of params active for each token
T5-Small	2.08E+00	1.80E+20	60	1,000	3	3	1	0.5
T5-Base	7.64E+00	6.60E+20	220	1,000	3	3	1	0.5
T5-Large	2.67E+01	2.31E+21	770	1,000	3	3	1	0.5
T5-3B	1.04E+02	9.00E+21	3,000	1,000	3	3	1	0.5
T5-11B	3.82E+02	3.30E+22	11,000	1,000	3	3	1	0.5
BERT-Base	1.89E+00	1.64E+20	109	250	6	3	2	1.0
BERT-Large	6.16E+00	5.33E+20	355	250	6	3	2	1.0
RoBERTa-Base	1.74E+01	1.50E+21	125	2,000	6	3	2	1.0
RoBERTa-Large	4.93E+01	4.26E+21	355	2,000	6	3	2	1.0
GPT-3 Small	2.60E+00	2.25E+20	125	300	6	3	2	1.0
GPT-3 Medium	7.42E+00	6.41E+20	356	300	6	3	2	1.0
GPT-3 Large	1.58E+01	1.37E+21	760	300	6	3	2	1.0
GPT-3 XL	2.75E+01	2.38E+21	1,320	300	6	3	2	1.0
GPT-3 2.7B	5.52E+01	4.77E+21	2,650	300	6	3	2	1.0
GPT-3 6.7B	1.39E+02	1.20E+22	6,660	300	6	3	2	1.0
GPT-3 13B	2.68E+02	2.31E+22	12,850	300	6	3	2	1.0
GPT-3 175B	3.64E+03	3.14E+23	174,600	300	6	3	2	1.0

Table D.1: Starting from the right hand side and moving left, we begin with the number of training tokens that each model was trained with. Next we note that since T5 uses an encoder-decoder model, only half of the parameters are active for each token during a forward or backwards pass. We then note that each token is involved in a single addition and a single multiply for each active parameter in the forward pass (ignoring attention). Then we add a multiplier of 3x to account for the backwards pass (as computing both $\frac{\partial params}{\partial loss}$ and $\frac{\partial acts}{\partial loss}$ use a similar amount of compute as the forwards pass. Combining the previous two numbers, we get the total flops per parameter per token. We multiply this value by the total training tokens and the total parameters to yield the number of total flops used during training. We report both flops and petaflop/s-day (each of which are 8.64e+19 flops).

E Human Quality Assessment of Synthetic News Articles

This appendix contains details on the experiments measuring human ability to distinguish GPT-3-generated synthetic news articles from real news articles. We first describe the experiments on the ~ 200 word news articles, and then describe the preliminary investigation of ~ 500 word news articles generated by GPT-3.

Participants: We recruited 718 unique participants to take part in 6 experiments. 97 participants were excluded for failing an internet check question, leaving a total of 621 participants: 343 male, 271 female, and 7 other. Mean participant age was ~ 38 years old. All participants were recruited through Positly, which maintains a whitelist of high-performing workers from Mechanical Turk. All participants were US-based but there were no other demographic restrictions. Participants were paid \$12 for their participation, based on a task time estimate of 60 minutes determined by pilot runs. In order to ensure that the sample of participants for each experiment quiz was unique, participants were not allowed to take part in an experiment more than once.

Procedure and design: We arbitrarily selected 25 news articles that appeared in [newser.com](#) in early 2020. We used the article titles and subtitles to produce outputs from the 125M, 350M, 760M, 1.3B, 2.7B, 6.7B, 13.0B, and 200B (GPT-3) parameter language models. Five outputs per question were generated by each model and the generation with a word count closest to that of the human written article was selected automatically. This was to minimize the effect that completion length might have on participants’ judgments. The same output procedure for each model with the exception of the removal of the intentionally bad control model, as described in the main text.

D 总计用于训练语言模型的计算资源

此附录列出了用于估算图中所示语言模型训练所需计算资源的相关计算过程 2.2。作为简化假设，我们忽略了注意力运算，因为通常情况下，它所消耗的计算资源占比还不到我们所分析的模型总计算资源的10%。

相关计算数据可见于表 D.1，表内的说明也会对这些数据进行解释。

模型	总训练量 计算资源 (PF日)	训练总计 计算资源 (浮点运算次数)	参数数量 (M)	训练令牌数 (十亿计)	浮点运算次数		前向传播 每单位的浮点运算次数	所占比例 当前激活的参数 对于每一个
					多任务处理 后向传播	活跃参数 每个令牌		
T5-Small版本	2.08E+00	1.80E+20	60	1,000	3	3	1	0.5
T5-Base版本	7.64E+00	6.60E+20	220	1,000	3	3	1	0.5
T5-Large版本	2.67E+01	2.31E+21	770	1,000	3	3	1	0.5
T5-3B模型	1.04E+02	9.00E+21	3,000	1,000	3	3	1	0.5
T5-11B模型	3.82E+02	3.30E+22	11,000	1,000	3	3	1	0.5
BERT-Base模型	1.89E+00	1.64E+20	109	250	6	3	2	1.0
BERT-Large模型	6.16E+00	5.33E+20	355	250	6	3	2	1.0
RoBERTa-Base版本	1.74E+01	1.50E+21	125	2,000	6	3	2	1.0
RoBERTa-Large版本	4.93E+01	4.26E+21	355	2,000	6	3	2	1.0
GPT-3小型版	2.60E+00	2.25E+20	125	300	6	3	2	1.0
GPT-3中型版	7.42E+00	6.41E+20	356	300	6	3	2	1.0
GPT-3大型版	1.58E+01	1.37E+21	760	300	6	3	2	1.0
GPT-3超大型版	2.75E+01	2.38E+21	1,320	300	6	3	2	1.0
GPT-3 27亿参数版本	5.52E+01	4.77E+21	2,650	300	6	3	2	1.0
GPT-3 67亿参数版本	1.39E+02	1.20E+22	6,660	300	6	3	2	1.0
GPT-3 13B	2.68E+02	2.31E+22	12,850	300	6	3	2	1.0
GPT-3 175B	3.64E+03	3.14E+23	174,600	300	6	3	2	1.0

表D.1: 从右侧开始向左依次列出的是各模型在训练时所使用的训练令牌数。接下来需要说明的是，由于T5采用编解码器模型，在进行前向传播或后向传播时，每个令牌仅有半数的参数处于活跃状态。在前向传播过程中（不考虑注意力机制），每个活跃参数会对应一次加法和一次乘法运算，因此每个令牌需要执行的操作次数为活跃参数的数量。为了纳入后向传播的计算量，我们再乘以3倍系数——因为计算 $\frac{\partial params}{\partial loss}$ 和 $\frac{\partial acts}{\partial loss}$ 所需的计算资源与前向传播大致相当。将上述两个数值相加，即可得到每个令牌每个参数的总浮点运算次数。最后将该数值乘以训练令牌总数和参数总数，就能得出训练过程中总共消耗的浮点运算次数。本文同时给出了浮点运算次数以及换算后的拍浮点运算次/天数值（二者均为 8.64e+19 次浮点运算）。

E 合成新闻文章的人类质量评估

本附录详细介绍了用于衡量人类区分GPT-3生成的合成新闻文章与真实新闻文章能力的实验相关内容。我们首先介绍针对 ~ 200 字数新闻文章的实验，随后阐述对GPT-3生成的 ~ 500 字数新闻文章的初步研究情况。

参与者：我们共招募了718名不同的参与者参与6项实验。其中有97名参与者因未能通过网络检测题而被排除，最终剩余621名参与者：男性343名，女性271名，其他性别7名。参与者的平均年龄为 ~ 38 岁。所有参与者均通过Positly平台招募，该平台维护着来自Mechanical Turk平台的优质工作者名单。这些参与者均来自美国，且没有其他人口统计方面的限制。根据预实验确定的60分钟的任务完成时间，每位参与者可获得12美元的报酬。为确保每项实验测试的参与者群体具有唯一性，参与者不得重复参加同一实验。

实验流程与设计：我们随机挑选了25篇发表于[newser.com](#) 2020年初的新闻文章。我们利用这些文章的标题和副标题，让具有1.25亿参数、3.5亿参数、7.6亿参数、13亿参数、27亿参数、67亿参数、130亿参数以及2000亿参数（GPT-3）的各类语言模型生成输出内容。每个模型针对每个问题都会生成五个输出结果，系统会自动选择词数与人类撰写的文章最为接近的那个作为最终结果，以此尽量减少补全结果的长度对参与者判断的影响。除了如正文所述需要排除那个故意设置得较差的对照模型外，其他模型的输出流程均保持一致。

Model	Participants Recruited	Participants Excluded	Genders (m:f:other)	Mean Age	Average Word Count (human:model)
Control	76	7	32:37:0	39	216:216
GPT-3 Small	80	7	41:31:1	40	216:188
GPT-3 Medium	80	7	46:28:2	39	216:202
GPT-3 Large	81	24	46:28:2	37	216:200
GPT-3 XL	79	14	32:32:1	38	216:199
GPT-3 2.7B	80	11	36:33:0	40	216:202
GPT-3 6.7B	76	5	46:28:2	37	216:195
GPT-3 13.0B	81	13	46:28:2	37	216:209
GPT-3 175B	80	9	42:29:0	37	216:216

Table E.1: Participant details and article lengths for each experiment to evaluate human detection of ~ 200 word model generated news articles. Participants were excluded due to internet check fails.

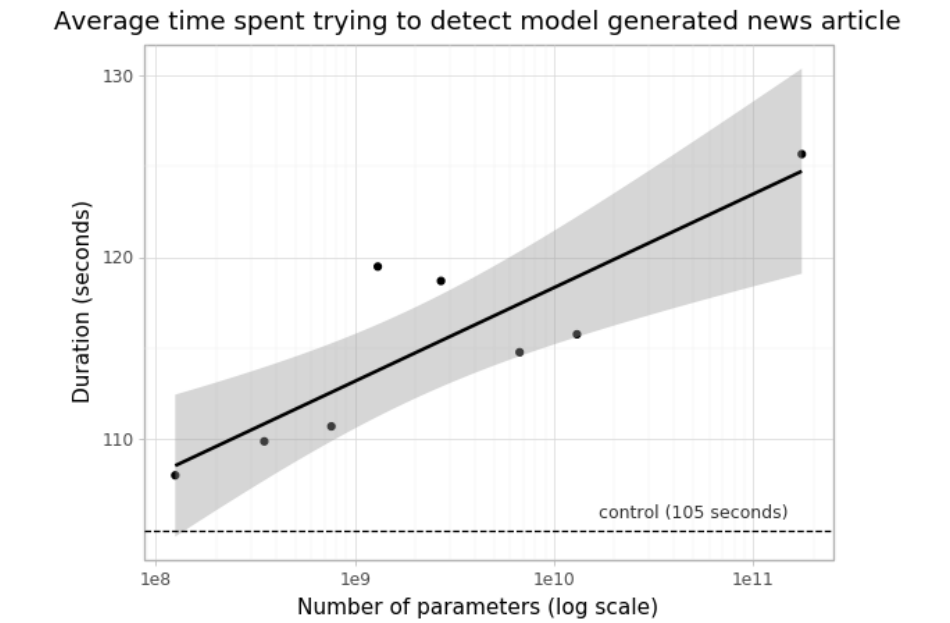


Figure E.1: Participants spend more time trying to identify whether each news article is machine generated as model size increases. Duration on the control model is indicated with the dashed line. Line of best fit is a linear model on a log scale with 95% confidence intervals.

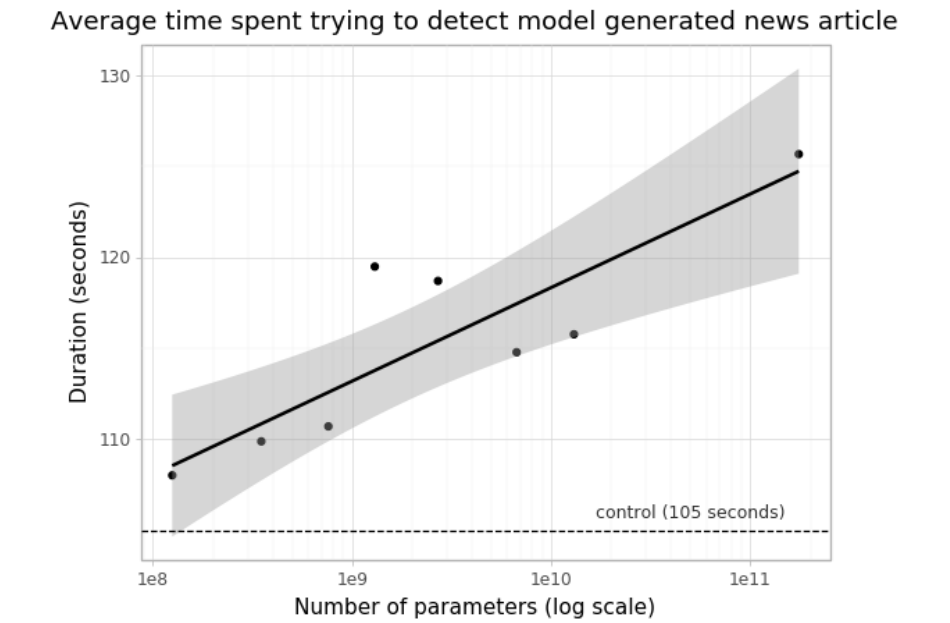
In each experiment, half of the participants were randomly assigned to quiz A and half were randomly assigned to quiz B. Each quiz consisted of 25 articles: half (12-13) were human written and half (12-13) were model generated: the articles with human written completions in quiz A had model generated completions in quiz B and vice versa. The order of quiz question was shuffled for each participant. Participants could leave comments and were asked to indicate if they had seen the articles before. Participants were instructed not to look up the articles or their content during the quiz and at the end of the quiz were asked if they had looked anything up during the quiz.

Statistical Tests: To compare means on the different runs, we performed a two-sample t-test for independent groups for each model against the control. This was implemented in Python using the `scipy.stats.ttest_ind` function. When plotting a regression line in the graph of average participant accuracy vs model size, we fit a power law of the form ax^{-b} . The 95% confidence intervals were estimated from the t-distribution of the sample mean.

Duration statistics: In the main text, we discussed the finding that the ability of human participants to distinguish model and human generated news articles decreases as our models become larger. We have also found that the average time spent for a given set of questions increases as the model size increases, as shown in Figure E.1. Lower

模型	参与者被招募的	参与者被排除的	性别 (男:女:其他)	Mean Age	平均值词数 (人类:模型)
对照组	76	7	32:37:0	39	216:216
GPT-3小型版	80	7	41:31:1	40	216:188
GPT-3中型版	80	7	46:28:2	39	216:202
GPT-3大型版	81	24	46:28:2	37	216:200
GPT-3超大型版	79	14	32:32:1	38	216:199
GPT-3 27亿参数版本	80	11	36:33:0	40	216:202
GPT-3 67亿参数版本	76	5	46:28:2	37	216:195
GPT-3 130亿参数	81	13	46:28:2	37	216:209
GPT-3 175B版本	80	9	42:29:0	37	216:216

表E.1: 用于评估 ~ 200 词长度的模型生成新闻文章的人类检测能力的各实验对应的参与者信息及文章长度。那些网络连接检测未通过的人员已被排除在分析之外。



图E.1: 随着模型规模的增大，参与者识别每篇新闻文章是否为机器生成所需的时间也会相应增加。对照模型对应的耗时以虚线标出，而最佳拟合线则是在对数刻度上、并带有95%置信区间的线性模型。

在每项实验中，一半的参与者会被随机分配到测验A，另一半则被随机分配到测验B。每项测验包含25篇文章：其中一半（12-13篇）由人类撰写，另一半（12-13篇）由模型生成；在测验A中采用人类撰写的补全结果的文章，在测验B中则使用模型生成的补全结果，反之亦然。每位参与者的测验题目顺序都会被打乱。参与者可以留下评论，并需说明之前是否看过这些文章。实验要求参与者在测验过程中不得查阅这些文章或其内容，测验结束后还会询问他们是否有查阅任何资料的行为。

统计测试：为了比较不同实验条件下各模型的均值，我们对每个模型及其对照组所构成的独立组进行了双样本t检验，该操作是使用Python中的`scipy.stats.ttest_ind`函数实现的。在绘制平均参与者准确率与模型规模关系的回归线时，我们采用了形式为 ax^{-b} 的幂律关系。这些95%置信区间则是根据样本均值的t分布计算得出的。

时长统计：在正文中，我们探讨了这样一个研究结果——随着模型规模不断扩大，人类参与者区分模型生成内容与人工生成新闻文章的能力会逐渐下降。同时如图E.1所示，面对同一组问题，参与者所需的平均投入时间也会随模型规模的增长而增加。

Model	Participants Recruited	Participants Excluded	Genders (m:f:other)	Mean Age	Average Word Count (human:model)
Control	79	17	32:37:0	39	569:464
GPT-3 175B	81	19	32:30:0	40	569:498

Table E.2: Participant details and article lengths for the experiments investigating human detection of ~ 500 word model generated news articles. Participants were excluded due to internet check fails.

accuracy scores despite increased time investment from participants supports the finding that larger models generate harder-to-distinguish news articles.

Preliminary investigation of ~ 500 word articles: We recruited 160 unique US-based participants to take part in 2 experiments through Positly (details are given in Table E.2). We randomly selected 12 Reuters world news articles from late 2019 and created a context for GPT-3 175B that consisted of a single Reuters article not in this set of 12. We then used the article titles and Reuters locations to generate completions from GPT-3 175B and the 160M control model from the previous experiments. These were used to create two 12-question quizzes per model, each consisting of half human written and half model generated articles. Comprehension questions were added and articles were shown to participants in 3 stages at 30 second intervals to encourage closer reading. Participants were paid \$12 for this task. Model generation selection methods, exclusion criteria, and statistical tests mirror those of the previous experiments.

F Additional Samples from GPT-3

GPT-3 adapts well to many tasks other than the ones explored in the main body of the paper. As an example, in Figure F.1, we show four uncurated samples from a prompt suggesting that the model write a poem, with a given title, in the style of Wallace Stevens. We first experimented with a few prompts, then generated four samples with no additional editing or selection (sampling at temperature 1 using nucleus sampling [HBFC19] with $P = 0.9$). Completions were truncated when the model began to write a new title and author heading, or broke into prose commentary.

模型	参与者被招募的	参与者被排除的	性别 (男:女:其他)	Mean Age	平均值词数 (人类:模型)
对照组	79	17	32:37:0	39	569:464
GPT-3 175B版本	81	19	32:30:0	40	569:498

表E.2：用于研究词长度模型生成的新闻文章的人类检测实验的参与者信息及文章长度。因网络连接检测失败，部分参与者被排除在外。

尽管参与者投入的时间更多，但其准确率依然没有提升，这一现象进一步印证了更大规模的模型生成的新闻文章更难被区分这一结论。

关于词长度的新闻文章的初步研究：我们通过Positly平台招募了160名来自美国的独特参与者，让他们参与两项测试（详细信息见表E.2）。我们从2019年末随机选取了12篇路透社的世界新闻文章，并为GPT-3 175B准备了一个上下文，该上下文由这12篇文章中未包含的另外一篇路透社文章构成。随后，我们利用这些文章的标题和路透社的报道位置，让GPT-3 175B以及之前实验中使用的160M对照模型分别生成补全结果。基于这些结果，我们为每个模型制作了包含12道题的测验，每份测验中一半文章由人类撰写，另一半由模型生成。此外还加入了理解题，文章会以30秒为间隔分三个阶段展示给参与者，以此促使他们更仔细地阅读。参与者完成这项任务可获得12美元报酬。模型生成的选择方法、排除标准以及统计检验方式均与之前的实验保持一致。

F GPT-3的更多示例

GPT-3在许多任务上都能表现出良好的适应性，这些任务并非本文正文中探讨的那些。作为示例，在图F.1中，我们展示了四个未经筛选的样本，这些样本来自要求模型按照华莱士·史蒂文斯的风格、并使用指定标题来创作诗歌的提示。我们首先尝试了几个不同的提示，随后通过核采样方法、在温度参数为1 的情况下，直接生成了这四个样本，且未进行任何额外的编辑或筛选HBFC19]与 $P = 0.9$ 。当模型开始撰写新的标题和作者信息，或是转而写成散文式的评论时，补全结果就会被截断。

Context →	The City BY C. P. CAVAFY TRANSLATED BY EDMUND KEELEY [Poem text omitted]
	SOME TREES John Ashbery [Poem text omitted]
	Shadows on the Way Wallace Stevens

----- Generated Poem 1 -----

I must have shadows on the way
If I am to walk I must have
Each step taken slowly and alone
To have it ready made

And I must think in lines of grey
To have dim thoughts to be my guide
Must look on blue and green
And never let my eye forget
That color is my friend
And purple must surround me too

The yellow of the sun is no more
Intrusive than the bluish snow
That falls on all of us. I must have
Grey thoughts and blue thoughts walk with me
If I am to go away at all.

----- Generated Poem 2 -----

There is a way in the world.
That way a man may live in a small town
His lifetime and see no man come,
Never the stir of men about a market,
Never the bobbing of covered carts,
Never the creak passed on the stair
Closing some other life; an old man
Can prosper in that town, can prosper
When his store is seen to close, his wife
Fades in his house, and the world shouts in vain
That no one comes to hear him
Where his small bell tolls.
In that small town
Everything is known.
The ways of the young men are known,
And the ways of the young women
As they walk to the Spring;
The rumors are all known,
The quarrels, the hurt, the envy
Of the men and the women.
Till a day comes
That way out of the world
That leads to a wood.
Young, the young man
Walks, a lover now,
Knows the streets as they are and is of them.

Comes to the wood.

There, in the wood, among trees,
He sees shadows on the way, hears voices,
hears the wind and the rustling of leaves;
Through an open glade
He sees a shape and the shape hears:
It waits as he waits,
Waits as the shadows wait,
As the voices wait;
Shadows on the way, voices in the wind.

----- Generated Poem 3 -----

The sun was all we had. Now, in the shade
All is changed. The mind must dwell on those
White fields, that to its eyes were always old;
Those ancient gleams, convoluting
The way they lay among the huge roots,
The great dark tomes of reverie,
The plumed labyrinth of the sea.

When our own breath gone faint and fade
Up the long stairway of the dead
We shall remember well
The blindness of the light, the whiteness
Of that white land. The footsteps, and
The grasses in that light, and in the shade
The menacing life.

We played, too, once, in the warmest rooms.
To one content, there is one content
For all mankind. The forest glades
Are all the more for being in fable,
And wood and town in legend shadowed-
Once more to return. The one content
Has in it every happiness, more brave
For being purposeless; and so
Those island gardens, deep in the mind,
Make of remoteness familiarity
And of the unspoken name a friend.

----- Generated Poem 4 -----

Nobody will come to this place. It is a road that leads nowhere.
The solitude is deep. The mountains are high.
But they are desolate, and they turn the traveler's face
Towards the North. All the sounds of the world are far away.
When the wind rises above the trees,
The boughs bow to the ground.
Even the birds that inhabit the tangle of weeds
That is the roadside cover, are silent. One listens,
But hears no roar of the forest. One is alone.
One will be taken.
One will be taken.
There is no utterance, there is no conversation,
But one is uneasy all the same....
There is a thin blue mist,
A darkness rising like smoke,
And within that darkness
A possession of the heart.
One will be taken.... It was here, and it will be here again-
Here, under this sky empty and full of light.

Figure F.1: Four uncensored completions from a context suggesting the model compose a poem in the style of Wallace Stevens with the title ‘Shadows on the Way’.

Context →	The City BY C. P. CAVAFY TRANSLATED BY EDMUND KEELEY [Poem text omitted]
	SOME TREES John Ashbery [Poem text omitted]
	Shadows on the Way Wallace Stevens

----- Generated Poem 1 -----

I must have shadows on the way
If I am to walk I must have
Each step taken slowly and alone
To have it ready made

And I must think in lines of grey
To have dim thoughts to be my guide
Must look on blue and green
And never let my eye forget
That color is my friend
And purple must surround me too

The yellow of the sun is no more
Intrusive than the bluish snow
That falls on all of us. I must have
Grey thoughts and blue thoughts walk with me
If I am to go away at all.

----- Generated Poem 2 -----

There is a way in the world.
That way a man may live in a small town
His lifetime and see no man come,
Never the stir of men about a market,
Never the bobbing of covered carts,
Never the creak passed on the stair
Closing some other life; an old man
Can prosper in that town, can prosper
When his store is seen to close, his wife
Fades in his house, and the world shouts in vain
That no one comes to hear him
Where his small bell tolls.
In that small town
Everything is known.
The ways of the young men are known,
And the ways of the young women
As they walk to the Spring;
The rumors are all known,
The quarrels, the hurt, the envy
Of the men and the women.
Till a day comes
That way out of the world
That leads to a wood.
Young, the young man
Walks, a lover now,
Knows the streets as they are and is of them.

Comes to the wood.

There, in the wood, among trees,
He sees shadows on the way, hears voices,
hears the wind and the rustling of leaves;
Through an open glade
He sees a shape and the shape hears:
It waits as he waits,Waits as the shadows wait,
As the voices waitShadows on the way,
voices in the wind.

----- Generated Poem 3 -----

The sun was all we had. Now, in the shade
All is changed. The mind must dwell on those
White fields, that to its eyes were always old;
Those ancient gleams, convoluting
The way they lay among the huge roots,
The great dark tomes of reverie,
The plumed labyrinth of the sea.

When our own breath gone faint and fade
Up the long stairway of the dead
We shall remember well
The blindness of the light, the whiteness
Of that white land. The footsteps, and
The grasses in that light, and in the shade
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We played, too, once, in the warmest rooms.
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Towards the North. All the sounds of the world are far away.
When the wind rises above the trees,
The boughs bow to the ground.
Even the birds that inhabit the tangle of weeds
That is the roadside cover, are silent. One listens,
But hears no roar of the forest. One is alone.
One will be taken.
One will be taken.
There is no utterance, there is no conversation,
But one is uneasy all the same....
There is a thin blue mist,
A darkness rising like smoke,
And within that darkness
A possession of the heart.
One will be taken.... It was here, and it will be here again-
Here, under this sky empty and full of light.

图F.1: 四组未经筛选的补全结果， 这些上下文表明该模型以华莱士·史蒂文斯的风格创作了题为《路上的阴影》的诗篇。

G Details of Task Phrasing and Specifications

The following figures illustrate the formatting and phrasing of all the tasks included in the paper. All data comes from the ground truth datasets in this section, and no samples from GPT-3 are included here.

Context →	Article: Informal conversation is an important part of any business relationship.Before you start a discussion,however,make sure you understand which topics are suitable and which are considered taboo in a particular culture. Latin Americans enjoy sharing information about their local history, art and customs.You may expect questions about your family,and be sure to show pictures of your children.You may feel free to ask similar questions of your Latin American friends.The French think of conversation as an art form,and they enjoy the value of lively discussions as well as disagreements. For them,arguments can be interesting and they can cover pretty much or any topic ---- as long as they occur in are respectful and intelligent manner. In the United States,business people like to discuss a wide range of topics,including opinions about work,family,hobbies,and politics. In Japan,China,and Korea,however,people are much more private.They do not share much about their thoughts,feelings,or emotions because they feel that doing so might take away from the harmonious business relationship they’re trying to build.Middle Easterners are also private about their personal lives and family matters.It is considered rude,for example,to ask a businessman from Saudi Arabia about his wife or children. As a general rule,it’s best not to talk about politics or religion with your business friends.This can get you into trouble,even in the United States,where people hold different religious views.In addition,discussing one’s salary is usually considered unsuitable.Sports is typically a friendly subject in most parts of the world,although be careful not to criticize national sport.Instead,be friendly and praise your host’s team. Q: What shouldn’t you do when talking about sports with colleagues from another country? A: Criticizing the sports of your colleagues’ country. Q: Which is typically a friendly topic in most places according to the author? A: Sports. Q: Why are people from Asia more private in their conversation with others? A: They don’t want to have their good relationship with others harmed by informal conversation. Q: The author considers politics and religion - . A:
Correct Answer →	taboo
Incorrect Answer →	cheerful topics
Incorrect Answer →	rude topics
Incorrect Answer →	topics that can never be talked about

Figure G.1: Formatted dataset example for RACE-h. When predicting, we normalize by the unconditional probability of each answer as described in 2.

G 任务表述与规范细节

以下图表展示了本文中所有任务的格式与表述方式。所有数据均来源于本节中的真实数据集，此处并未使用任何 GPT-3 的样本。

上下文: →	文章内容: 非正式对话是任何商业关系中的重要组成部分。 不过，在开始讨论之前，请务必先弄清楚，在特定的上下文中哪些话题是合适的，哪些又属于禁忌话题。哪些话题适合讨论，哪些则被视为禁忌话题。 文化。拉丁美洲人乐于分享有关其所在地区的信息 同时也建议务必出示孩子的照片。您可以随时向自己的拉丁美洲朋友提出类似的问题。 法国人将交谈视为一门艺术形式，他们不仅重视热烈的讨论，还十分欣赏由此带来的价值。 作为一种艺术形式，他们同样珍视热烈讨论所带来的价值。 分歧。对他们而言，论点可能很有吸引力，而且能够涵盖广泛的内容。 几乎任何话题都行——只要讨论时彼此尊重、态度得体。 并以理智的方式展开。 在美国，商界人士喜欢探讨各种各样的话题。 这些话题包括关于工作、家庭、爱好以及政治的观点。 In 不过，在日本、中国和韩国，人们更加注重隐私。他们不会过多地透露自己的想法、感受或情绪，因为他们认为这样做可能会破坏和谐的商务关系。他们正在努力构建这样的关系。中东地区的民众在个人生活和家庭事务方面同样十分保守。 例如，向沙特阿拉伯的商人询问其妻子或子女的情况被视为无礼之举。 在那些文化中，过度探听私人生活被视为不礼貌的行为。 因此，人们通常会避免涉及这类敏感话题。 一般来说，最好不要与商业伙伴谈论政治或宗教话题。 即便在那些人们宗教观念各异的美国，这种行为也可能会给你带来麻烦。 此外，谈论他人的薪资通常也被视为不妥的行为。 体育在世界上大多数地方都是一个适合交流的话题，但需注意不要批评当地的传统运动项目。 相反，应以友善的态度赞美东道主的球队。 “语言模型是少样本学习者”便是首篇相关文章的标题。 问：与来自其他国家的同事谈论体育时，有哪些事情是不应该做的？ 另一个国家？ 答：不应批评同事所在国家的体育项目。 问：根据那位作者的观点，在大多数地方通常属于友好话题的是什么？ 答：体育。 A：体育。 问：为什么亚洲人在与他人交谈时更为保守？ 答：他们不希望自己与他人的良好关系因非正式的交谈而受到危害。 非正式对话。 问：作者认为政治和宗教属于禁忌话题吗？ - . A：
正确答案 →	禁忌话题
错误答案 →	轻松话题
错误答案 →	粗俗话题
错误答案 →	绝不能讨论的话题

图G.1：RACE-h对应的格式化数据集示例。在进行预测时，我们会按照2中所描述的方法，以每个答案的无条件概率作为归一化标准。

Context →	anli 2: anli 2: The Gold Coast Hotel & Casino is a hotel and casino located in Paradise, Nevada. This locals’ casino is owned and operated by Boyd Gaming. The Gold Coast is located one mile (~ 1.6km) west of the Las Vegas Strip on West Flamingo Road. It is located across the street from the Palms Casino Resort and the Rio All Suite Hotel and Casino. Question: The Gold Coast is a budget-friendly casino. True, False, or Neither?
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Figure G.2: Formatted dataset example for ANLI R2

Context →	Article: Mrs. Smith is an unusual teacher. Once she told each student to bring along a few potatoes in plastic bag. On each potato the students had to write a name of a person that they hated And the next day, every child brought some potatoes. Some had two potatoes;some three;some up to five. Mrs. Smith then told the children to carry the bags everywhere they went, even to the toilet, for two weeks. As day after day passed, the children started to complain about the awful smell of the rotten potatoes. Those children who brought five potatoes began to feel the weight trouble of the bags. After two weeks, the children were happy to hear that the game was finally ended. Mrs. Smith asked,"How did you feel while carrying the potatoes for two weeks?" The children started complaining about the trouble loudly. Then Mrs. Smith told them why she asked them to play the game. She said,"This is exactly the situation when you carry your hatred for somebody inside your heart. The terrible smell of the hatred will pollute your heart and you will carry something unnecessary with you all the time. If you cannot stand the smell of the rotten potatoes for just two weeks, can you imagine how heavy it would be to have the hatred in your heart for your lifetime? So throw away any hatred from your heart, and you’ll be really happy." Q: Which of the following is True according to the passage? A: If a kid hated four people,he or she had to carry four potatoes. Q: We can learn from the passage that we should _ . A: throw away the hatred inside Q: The children complained about _ besides the weight trouble. A: the smell Q: Mrs.Smith asked her students to write _ on the potatoes. A:
Correct Answer →	names
Incorrect Answer →	numbers
Incorrect Answer →	time
Incorrect Answer →	places

Figure G.3: Formatted dataset example for RACE-m. When predicting, we normalize by the unconditional probability of each answer as described in 2.

上下文: →	ANLI 2: ANLI 2: 黄金海岸酒店与赌场是一家同时具备酒店和赌场功能的场所。它坐落于内华达州帕拉迪斯市。这家由当地人经营的赌场由其所有者直接掌控并运作。由博伊德游戏公司提供。黄金海岸位于西火烈鸟路上的拉斯维加斯大道以西一英里 (~) 处。具体距离为1.6公里。西火烈鸟路上的拉斯维加斯大道。它位于街对面。来自棕榈赌场度假村以及里约全套房酒店与赌场。黄金海岸是一家价格实惠的赌场。正确、错误，还是——两者都不是?
正确答案 →	两者皆非
错误答案 →	True
错误答案 →	错误

图G.2： ANLI R2格式化数据集示例

上下文 →	文章内容: Mrs. 史密斯是一位与众不同的老师。有一次，她要求每位学生带去一些东西。具体来说，就是几颗装在塑料袋里的土豆。学生们需要在每颗土豆上写下他们厌恶的人的名字。第二天，所有的孩子都带去了相应的土豆。有些孩子带了两个土豆，有些带了三个，还有些甚至带了五个。带了些土豆来。随后史密斯让孩子们无论走到哪里都要带着那些袋子。甚至上厕所的时候也不例外，这样的状况持续了两周时间。随着日子一天天过去，那些孩子们……开始抱怨腐烂的土豆散发出的难闻气味。那些带了五个土豆的孩子渐渐感受到了重力的困扰。那些袋子里的东西。两周后，孩子们听到……的消息时都非常高兴。这场游戏终于结束了。Mrs. 史密斯问道：‘在搬运那些土豆的两周时间里，你有什么感受？’ ‘搬运那些土豆的两周时间？’ 孩子们开始大声抱怨起这些土豆来。他们大声地诉说着烦恼。接着是夫人 史密斯向他们解释了为何要求他们玩这个游戏。She他说：“这正是你对某人怀有仇恨时的情形。”仇恨带来的恶臭将会污染你的心怀上限，你就会一直背负着不必要的负担。If谁又能忍受腐烂的土豆散发出的气味长达两周之久呢？试想一下，若心中始终怀有对某人的怨恨，那该是何等沉重的负担。一辈子吗？所以，请把心中的怨恨全部抛开，这样你才会真正快乐。" 问：根据文章内容，以下哪项是正确的？ A: 如果一个孩子讨厌四个人，他就必须背着四个土豆。 Q: 从这段文字中我们可以得知，我们应该…… _ . A: 把内心的仇恨抛掉。 Q: 那些孩子们抱怨的是…… _ 除了重力带来的困扰之外。 A: 还有那种气味。 问：史密斯夫人要求她的学生们写作 _ 在土豆上。 A:
正确答案 →	名称
错误答案 →	数字
错误答案 →	time
错误答案 →	位置

图G.3： Formatted RACE-m对应的d数据集示例。在进行预测时，我们会依据各答案的无条件概率来进行归一化处理。作为描述，即每个答案的概率。带有 2点刻纹的

Context →	How to apply sealant to wood.
Correct Answer →	Using a brush, brush on sealant onto wood until it is fully saturated with the sealant.
Incorrect Answer →	Using a brush, drip on sealant onto wood until it is fully saturated with the sealant.

Figure G.4: Formatted dataset example for PIQA

Context →	My body cast a shadow over the grass because
Correct Answer →	the sun was rising.
Incorrect Answer →	the grass was cut.

Figure G.5: Formatted dataset example for COPA

Context →	(CNN) Yuval Rabin, whose father, Yitzhak Rabin, was assassinated while serving as Prime Minister of Israel, criticized Donald Trump for appealing to "Second Amendment people" in a speech and warned that the words that politicians use can incite violence and undermine democracy. "Trump's words are an incitement to the type of political violence that touched me personally," Rabin wrote in USAToday. He said that Trump's appeal to "Second Amendment people" to stop Hillary Clinton -- comments that were criticized as a call for violence against Clinton, something Trump denied -- "were a new level of ugliness in an ugly campaign season." - The son of a former Israeli Prime Minister who was assassinated wrote an op ed about the consequence of violent political rhetoric. - Warns of "parallels" between Israel of the 1990s and the U.S. today.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Donald Trump's aggressive rhetoric.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Trump's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Hillary Clinton's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned U.S.'s aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Yitzhak Rabin's aggressive rhetoric.

Figure G.6: Formatted dataset example for ReCoRD. We consider the context above to be a single ”problem” because this is how the task is presented in the ReCoRD dataset and scored in the ReCoRD evaluation script.

Context →	anli 1: anli 1: Fulton James MacGregor MSP is a Scottish politician who is a Scottish National Party (SNP) Member of Scottish Parliament for the constituency of Coatbridge and Chryston. MacGregor is currently Parliamentary Liaison Officer to Shona Robison, Cabinet Secretary for Health & Sport. He also serves on the Justice and Education & Skills committees in the Scottish Parliament. Question: Fulton James MacGregor is a Scottish politican who is a Liaison officer to Shona Robison who he swears is his best friend. True, False, or Neither?
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Figure G.7: Formatted dataset example for ANLI R1

上下文: →	如何将密封剂涂在木材上。
正确答案: →	使用刷子将密封剂均匀涂抹在木材上，直至其完全被密封剂浸透。 即完全覆盖上密封剂。
错误答案: →	使用刷子将密封剂逐滴涂在木头上，直到其完全被密封剂浸透。即可。

图G.4：用于PIQA评测的格式化数据集示例

上下文 →	因为我的身体在草地上投下了阴影。
正确答案 →	太阳正在升起。
错误答案 →	草地已经被修剪过了。

图G.5：用于COPA任务的格式化数据集示例

上下文 →	(CNN) 尤瓦尔·拉宾，其父亲……，伊扎克·拉宾，在……时遭暗杀在担任以色列总理期间，他批评唐纳德·特朗普发表煽动性言论。在一场演讲中将其称为“第二修正案支持者”，并警告说那些言论政客们滥用此类手段可能会煽动暴力行为，并削弱民主制度。 “特朗普的”文字正在煽动那种曾发生过的政治暴力行为。 “就我个人而言，”拉宾在《今日美国》中写道。 他说，特朗普对……的吸引力在于那些主张维护《第二修正案》的人试图阻止希拉里·克林顿——这些评论正是如此。该言论被批评为煽动针对克林顿的暴力行为，而特朗普对此予以否认。 ——“这是丑陋的竞选季中又一层级的不堪。” - 一位遇刺身亡的以色列前总理的儿子撰写了一篇关于暴力政治言论后果的评论文章。 正确答案{v1} - 他警告称，1990年代的以色列与如今的美国之间存在“相似之处”。
正确答案 →	拉宾谴责了唐纳德·特朗普的……
正确答案 →	- 在提及这些情况后，拉宾对唐纳德·特朗普的言行表达了强烈批评。 激进语气。 - 他提到自己的父亲，曾在1995年以色列的政治紧张局势中遭极端分子枪杀。在1995年以色列的政治紧张局势背景下，拉宾严厉批评了特朗普的激进语气。
错误答案 →	激进语气。 - 他提到自己的父亲，曾在1995年以色列的政治紧张局势中遭极端分子枪杀。1995年以色列的政局紧张局势下，拉宾严厉谴责了希拉里·克林顿的激进言论。
错误答案 →	- 他提到自己的父亲，曾在1995年以色列的政治紧张局势中遭极端分子枪杀。 - 他提到自己的父亲，曾在1995年以色列的政治紧张局势中遭极端分子枪杀，拉宾对此强烈谴责了美国的激进立场。
错误答案 →	空话。 - 他提到自己的父亲，曾在1995年以色列的政治紧张局势中遭到极端分子的枪杀。在1995年以色列的政治紧张局势背景下，拉宾谴责了伊扎克·拉宾的激进言辞。 强硬的空话。

图G.6：ReCoRD数据集的格式化示例。我们将上述上下文视为一个独立的“问题”，因为这展示了该任务在ReCoRD数据集中的呈现方式以及其在ReCoRD评估脚本中的评分方式。

上下文: →	ANLI数据集 1: ANLI数据集 1: Fulton James MacGregor MSP是一位苏格兰政治家。他同时也是苏格兰民族党（SNP）的苏格兰议会议员。MacGregor 目前正在他代表科特布里奇和克里斯顿选区任职。他的职责是负责与卫生与体育事务内阁秘书肖娜·罗比森沟通协调。卫生与体育事务。此外，他还在苏格兰议会的司法委员会以及教育与技能委员会中任职。这些都是他在议会里参与的委员会。 问题: 富尔顿·詹姆斯·麦克格雷戈是一位担任联络职务的苏格兰政治家。那位官员对肖娜·罗比森说，他发誓她是自己最好的朋友。 True, 错误 , or 都不是?
正确答案 →	两者皆非
错误答案 →	True
错误答案 →	错误

图G.7：ANLI R1对应的格式化数据集示例

Context →	Organisms require energy in order to do what?
Correct Answer →	mature and develop.
Incorrect Answer →	rest soundly.
Incorrect Answer →	absorb light.
Incorrect Answer →	take in nutrients.

Figure G.8: Formatted dataset example for OpenBookQA. When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	Making a cake: Several cake pops are shown on a display. A woman and girl are shown making the cake pops in a kitchen. They
Correct Answer →	bake them, then frost and decorate.
Incorrect Answer →	taste them as they place them on plates.
Incorrect Answer →	put the frosting on the cake as they pan it.
Incorrect Answer →	come out and begin decorating the cake as well.

Figure G.9: Formatted dataset example for HellaSwag

Context →	anli 3: anli 3: We shut the loophole which has American workers actually subsidizing the loss of their own job. They just passed an expansion of that loophole in the last few days: \$43 billion of giveaways, including favors to the oil and gas industry and the people importing ceiling fans from China. Question: The loophole is now gone True, False, or Neither?
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

Figure G.10: Formatted dataset example for ANLI R3

Context →	Question: George wants to warm his hands quickly by rubbing them. Which skin surface will produce the most heat? Answer:
Correct Answer →	dry palms
Incorrect Answer →	wet palms
Incorrect Answer →	palms covered with oil
Incorrect Answer →	palms covered with lotion

Figure G.11: Formatted dataset example for ARC (Challenge). When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	lull is to trust as
Correct Answer →	cajole is to compliance
Incorrect Answer →	balk is to fortitude
Incorrect Answer →	betray is to loyalty
Incorrect Answer →	hinder is to destination
Incorrect Answer →	soothe is to passion

Figure G.12: Formatted dataset example for SAT Analogies

Correct Context →	Grace was happy to trade me her sweater for my jacket. She thinks the sweater
Incorrect Context →	Grace was happy to trade me her sweater for my jacket. She thinks the jacket
Target Completion →	looks dowdy on her.

Figure G.13: Formatted dataset example for Winograd. The ‘partial’ evaluation method we use compares the probability of the completion given a correct and incorrect context.

上下文: →	生物需要能量来做什么？
正确答案: →	成熟并成长。
错误答案: →	安稳地休息。
错误答案 →	吸收光线。
错误答案 →	摄取养分。

图G.8：OpenBookQA数据集的格式化示例。在进行预测时，我们会按照文中所述，依据每个答案的无条件概率来进行标准化处理2。

上下文 →	制作蛋糕: 展示台上摆放着几支蛋糕棒。 一名女子和一名女孩正在厨房里制作这些蛋糕棒。 They
正确答案 →	将它们烘烤出来，然后喷上霜并加以装饰。
错误答案 →	将食物摆上盘子后即可品尝。
错误答案 →	在蛋糕烤制过程中为其涂抹糖霜。
错误答案 →	也出来一起开始装饰蛋糕吧。

图G.9：HellaSwag的格式化数据集示例

上下文: →	ANLI 3: ANLI 3: 我们已堵上了那些让美国工人受到影响的漏洞。 为他们的失业提供补贴。 他们刚刚通过了对相关规定的扩展。 也就是在最近几天里，对那个漏洞进行了补充规定。 其中包括价值430亿美元的各类补助。 这些补助是给石油天然气行业以及进口吊扇的企业的。 这些吊扇来自中国。 问题: 那个漏洞现在已经不存在了，答案是“正确”、“错误”还是“两者皆非”？
正确答案 →	错误
错误答案 →	True
错误答案 →	两者皆非

图G.10：ANLI R3模型的格式化数据集示例

上下文: →	问题: 乔治想通过搓手来快速暖手。 哪一个皮肤表面会产生最多的热量吗？ 答案:
正确答案 →	干燥的手掌
错误答案 →	湿润的手掌
错误答案 →	沾有油的手掌
错误答案 →	涂有乳液的手掌

图G.11：ARC挑战版所用数据集的格式化示例。在进行预测时，我们会按照文中所述，以每个答案的无条件概率作为标准化依据2。

上下文 →	轻信就如同沉溺于安逸一般。
正确答案 →	哄诱就如同获取服从一样。
错误答案 →	退缩就如同缺乏坚韧一样。
错误答案 →	背叛与忠诚的关系是
错误答案 →	阻碍与目的地之间的关系是
错误答案 →	安抚与激情之间的关系是

图G.12：SAT类比题的格式化数据集示例

正确的上下文 →	格蕾丝很乐意用自己的毛衣交换我的夹克。 她认为……
错误的上下文 →	毛衣 格蕾丝很乐意用自己的毛衣交换我的夹克。 她认为那件夹克
目标完成度 →	穿在她身上显得很寒酸。

图G.13：Winograd所用格式化数据集示例。我们采用的‘部分’评估方法会对比在正确与错误上下文中生成补全结果的概率。

Correct Context →	Johnny likes fruits more than vegetables in his new keto diet because the fruits
Incorrect Context →	Johnny likes fruits more than vegetables in his new keto diet because the vegetables
Target Completion →	are saccharine.

Figure G.14: Formatted dataset example for Winogrande. The ‘partial’ evaluation method we use compares the probability of the completion given a correct and incorrect context.

Context →	READING COMPREHENSION ANSWER KEY While this process moved along, diplomacy continued its rounds. Direct pressure on the Taliban had proved unsuccessful. As one NSC staff note put it, "Under the Taliban, Afghanistan is not so much a state sponsor of terrorism as it is a state sponsored by terrorists." In early 2000, the United States began a high-level effort to persuade Pakistan to use its influence over the Taliban. In January 2000, Assistant Secretary of State Karl Inderfurth and the State Department’s counterterrorism coordinator, Michael Sheehan, met with General Musharraf in Islamabad, dangling before him the possibility of a presidential visit in March as a reward for Pakistani cooperation. Such a visit was coveted by Musharraf, partly as a sign of his government’s legitimacy. He told the two envoys that he would meet with Mullah Omar and press him on Bin Laden. They left, however, reporting to Washington that Pakistan was unlikely in fact to do anything," given what it sees as the benefits of Taliban control of Afghanistan." President Clinton was scheduled to travel to India. The State Department felt that he should not visit India without also visiting Pakistan. The Secret Service and the CIA, however, warned in the strongest terms that visiting Pakistan would risk the President’s life. Counterterrorism officials also argued that Pakistan had not done enough to merit a presidential visit. But President Clinton insisted on including Pakistan in the itinerary for his trip to South Asia. His one-day stopover on March 25, 2000, was the first time a U.S. president had been there since 1969. At his meeting with Musharraf and others, President Clinton concentrated on tensions between Pakistan and India and the dangers of nuclear proliferation, but also discussed Bin Laden. President Clinton told us that when he pulled Musharraf aside for a brief, one-on-one meeting, he pleaded with the general for help regarding Bin Laden." I offered him the moon when I went to see him, in terms of better relations with the United States, if he’d help us get Bin Laden and deal with another issue or two." The U.S. effort continued. Who did The State Department feel should visit both India and Pakistan?
Correct Answer →	- [False] Bin Laden
Incorrect Answer →	- [True] Bin Laden

Figure G.15: Formatted dataset example for MultiRC. There are three levels within MultiRC: (1) the passage, (2) the questions, and (3) the answers. During evaluation, accuracy is determined at the per-question level, with a question being considered correct if and only if all the answers within the question are labeled correctly. For this reason, we use K to refer to the number of **questions** shown within the context.

Context →	Question: Which factor will most likely cause a person to develop a fever? Answer:
Correct Answer →	a bacterial population in the bloodstream
Incorrect Answer →	a leg muscle relaxing after exercise
Incorrect Answer →	several viral particles on the skin
Incorrect Answer →	carbohydrates being digested in the stomach

Figure G.16: Formatted dataset example for ARC (Easy). When predicting, we normalize by the unconditional probability of each answer as described in 2.

正确的上下文 →	在约翰尼正在尝试的生酮饮食中，他更喜欢水果而非蔬菜，原因是……
错误的上下文 →	水果 在约翰尼正在尝试的生酮饮食中，他更喜欢水果而非蔬菜，原因是……
目标完成度 →	蔬菜 显得过于甜腻。

图G.14: 维诺格兰德所用数据集的格式化示例。我们采用的‘部分’评估方法会对比在正确与错误上下文中生成补全结果的概率。

上下文 →	阅读理解答案要点：在此进程推进的同时，外交努力也在持续进行。对塔利班的直接施压并未取得成效。正如某位国家安全委员会工作人员所言：‘在塔利班统治下，阿富汗与其说是一个资助恐怖主义的国家，不如说是一个被恐怖分子掌控的国家。’2000年初，美国开始竭力说服巴基斯坦利用其对塔利班的影响力。2000年1月，国务院助理国务卿卡尔·因德弗思以及国务院反恐协调员迈克尔·希恩在伊斯兰堡会见了穆沙拉夫将军，同时提出若巴基斯坦予以配合，总统可在3月前往访问作为奖励。穆沙拉夫极为渴望这样的访问，因为这既是对他政府合法性的认可。他告诉两位特使，他会与奥马尔毛拉会面并敦促其处理本·拉登的问题。然而，这两位特使回到华盛顿后报告称，鉴于巴基斯坦认为塔利班控制阿富汗能带来诸多好处，实际上它不太可能采取任何行动。克林顿总统原本计划访问印度，国务院认为他在不访问巴基斯坦的情况下不应前往印度。但特勤局和中央情报局则强烈警告称，访问巴基斯坦会危及总统的生命。反恐官员们也认为巴基斯坦尚未做出足够努力，不足以值得总统亲自访问。尽管如此，克林顿总统仍坚持在亚洲之行中安排访问巴基斯坦的行程。2000年3月25日，他进行了为期一日的停留，这是自1969年以来首位到访该国的美国总统。在与穆沙拉夫及其他人的会谈中，克林顿总统重点讨论了巴基斯坦与印度之间的紧张关系以及核扩散的威胁，同时也谈到了本·拉登的问题。克林顿总统告诉我们，当他把穆沙拉夫叫到一边进行简短的单独会谈时，他曾恳请这位将军协助抓捕本·拉登并解决其他一两个问题。‘我去见他时，提出了相当优厚的条件，希望他能帮助我们抓住本·拉登，并处理另外一两个问题，以此改善与美国的关系。’美国的努力仍在继续。 国务院认为谁应该同时访问印度和巴基斯坦？
正确答案 →	- [错误] 本·拉登
错误答案 →	- [正确]本·拉登

图G.15: MultiRC数据集的格式化示例。MultiRC包含三个层级：（1）正文段落，（2）问题，以及（3）答案。在评估过程中，准确率是在每个问题的层面上确定的，只有当一个问题中的所有答案都被正确标注时，该问题才会被视为正确。正因如此，我们使用 K 来表示上下文中出现的**问题数量**。

上下文： →	问题： 哪些因素最有可能导致人体出现发烧？ 答案：
正确答案： →	血液中的细菌群落
错误答案 →	运动后放松的腿部肌肉
错误答案 →	皮肤表面的若干病毒颗粒
错误答案 →	在胃部被消化的碳水化合物

图G.16: ARC简单版对应的结构化数据集示例。在进行预测时，我们会按照文中所述，以每个答案的无条件概率作为归一化基准2。

Context →	Bob went to the gas station to fill up his car. His tank was completely empty and so was his wallet. The cashier offered to pay for his gas if he came back later to pay. Bob felt grateful as he drove home.
Correct Answer →	Bob believed that there were good people in the world.
Incorrect Answer →	Bob contemplated how unfriendly the world was.

Figure G.17: Formatted dataset example for StoryCloze

Context →	Helsinki is the capital and largest city of Finland. It is in the region of Uusimaa, in southern Finland, on the shore of the Gulf of Finland. Helsinki has a population of , an urban population of , and a metropolitan population of over 1.4 million, making it the most populous municipality and urban area in Finland. Helsinki is some north of Tallinn, Estonia, east of Stockholm, Sweden, and west of Saint Petersburg, Russia. Helsinki has close historical connections with these three cities. The Helsinki metropolitan area includes the urban core of Helsinki, Espoo, Vantaa, Kauniainen, and surrounding commuter towns. It is the world’s northernmost metro area of over one million people, and the city is the northernmost capital of an EU member state. The Helsinki metropolitan area is the third largest metropolitan area in the Nordic countries after Stockholm and Copenhagen, and the City of Helsinki is the third largest after Stockholm and Oslo. Helsinki is Finland’s major political, educational, financial, cultural, and research center as well as one of northern Europe’s major cities. Approximately 75% of foreign companies that operate in Finland have settled in the Helsinki region. The nearby municipality of Vantaa is the location of Helsinki Airport, with frequent service to various destinations in Europe and Asia. Q: what is the most populous municipality in Finland? A: Helsinki Q: how many people live there? A: 1.4 million in the metropolitan area Q: what percent of the foreign companies that operate in Finland are in Helsinki? A: 75% Q: what towns are a part of the metropolitan area? A:
Target Completion →	Helsinki, Espoo, Vantaa, Kauniainen, and surrounding commuter towns

Figure G.18: Formatted dataset example for CoQA

Context →	Please unscramble the letters into a word, and write that word: asinoc =
Target Completion →	casino

Figure G.19: Formatted dataset example for Cycled Letters

上下文: →	鲍勃前往加油站给汽车加油。他的油箱已经完全空了。他的钱包也同样空空如也。收银员提出，如果他稍后回来付款，就可以帮他支付油费。稍后回来付款。鲍勃在开车回家时心中充满感激。
正确答案 →	鲍勃坚信这个世界上存在着善良的人。
错误答案 →	鲍勃思考着这个世界究竟有多冷漠无情。

图G.17：用于StoryCloze的格式化数据集示例

上下文: →	赫尔辛基是芬兰的首都，也是该国最大的城市。它位于乌西马地区。即芬兰南部的乌西马地区。，在芬兰南部，处于芬兰湾的岸边。赫尔辛基的总人口为，城市人口为，而其大都市区的人口则超过140万，因此它是芬兰人口最多的市镇以及最大的城市区域。正是这一庞大的人口规模使得赫尔辛基成为芬兰人口最多的市镇。此外，它也是芬兰规模最大的城市区域。赫尔辛基位于塔林的北部稍远处。，爱沙尼亚，它坐落在瑞典斯德哥尔摩的东部，以及俄罗斯圣彼得堡的西部。赫尔辛基与这三大城市有着密切的历史渊源。 赫尔辛基大都市区包括赫尔辛基市中心、埃斯波、万塔、考尼艾宁以及周边的通勤城镇。它是全球人口超过一百万的最北端大都市区，而该城市本身也是一个欧盟成员国的最北端首都。赫尔辛基大都市区规模仅次于斯德哥尔摩和哥本哈根。而赫尔辛基市本身则是该地区第三大的城市。仅次于斯德哥尔摩和奥斯陆，位居第三。赫尔辛基是芬兰重要的政治中心。教育领域，金融领域，文化领域，同时也是某研究机构之一北欧的主要城市。大约75%的外国企业那些在芬兰运营的机构大多落户在赫尔辛基地区。附近的万塔市是赫尔辛基机场的所在地，这里有前往欧洲和亚洲众多目的地的频繁航班。通往欧洲及亚洲各目的地的服务。 问：芬兰人口最多的市镇是哪个？ 答：赫尔辛基。 问：那里居住着多少人？ 答：大都市区的人口为140万。 问：在芬兰运营的外国企业中，有多少百分比位于赫尔辛基？ 问：还有多少百分比位于其他城市？ 答：75%。 问：哪些城镇属于该大都市区？ A:
目标完成度 →	赫尔辛基、埃斯波、万塔、考尼艾宁以及周边的通勤城镇

图G.18：CoQA格式化数据集示例

上下文 →	请将这些字母重新排列成一个单词，并将该单词写下来： asinoc =
目标完成度 →	赌场

图G.19：用于循环字母测试的格式化数据集示例

Context →	Passage: Saint Jean de Brébeuf was a French Jesuit missionary who travelled to New France in 1625. There he worked primarily with the Huron for the rest of his life, except for a few years in France from 1629 to 1633. He learned their language and culture, writing extensively about each to aid other missionaries. In 1649, Brébeuf and another missionary were captured when an Iroquois raid took over a Huron village . Together with Huron captives, the missionaries were ritually tortured and killed on March 16, 1649. Brébeuf was beatified in 1925 and among eight Jesuit missionaries canonized as saints in the Roman Catholic Church in 1930. Question: How many years did Saint Jean de Brébeuf stay in New France before he went back to France for a few years? Answer:
Target Completion →	4

Figure G.20: Formatted dataset example for DROP

Context →	Fill in blank: She held the torch in front of her. She caught her breath. "Chris? There’s a step." "What?" "A step. Cut in the rock. About fifty feet ahead." She moved faster. They both moved faster. "In fact," she said, raising the torch higher, "there’s more than a ----. ->
Target Completion →	step

Figure G.21: Formatted dataset example for LAMBADA

Context →	Please unscramble the letters into a word, and write that word: skicts =
Target Completion →	sticks

Figure G.22: Formatted dataset example for Anagrams 1 (A1)

Context →	Please unscramble the letters into a word, and write that word: volwskagen =
Target Completion →	volkswagen

Figure G.23: Formatted dataset example for Anagrams 2

Context →	Q: Who played tess on touched by an angel? A:
Target Completion →	Delloreese Patricia Early (July 6, 1931 { November 19, 2017), known professionally as Della Reese

Figure G.24: Formatted dataset example for Natural Questions

上下文: →	正文如下: 圣让·德布勒贝夫是一位法国耶稣会传教士，他于1625年前往新法兰西。在那里，他主要与休伦人一起工作。在他余下的生命中，除了1629年到法国度过的几年时光之外1633年。他学习了当地人的语言与文化，并撰写了大量相关著作。他们各自为协助其他传教士而努力。 ^{1649年，布里厄布夫与另一位传教士在一场易洛魁人的袭击占领了一座休伦人村庄时被俘。} 在那次袭击之后，他们二人落入了敌手。那些传教士与被俘的休伦人一同遭到了仪式性的折磨并被杀害，时间为1649年3月16日。 ^{布里埃博夫于1925年被封为真福者，与另外八位耶稣会传教士一起在1930年被罗马天主教会封为圣人。} 1930年在罗马天主教会被封为圣人的传教士们。 问题: 圣让·德·布里埃博夫在新法兰西度过了多少年？ 在他返回法国度过数年之前？ 答案:
目标完成度 →	4

图G.20：用于DROP的格式化数据集示例

上下文 →	填空: 她将火炬举在身前。 她缓过气来。 "克里斯? " "那里有个台阶." "什么? " "一个台阶。" 在岩石中凿出通道。大约五十英尺前方。她加快了步伐。他们两人都加快了速度。 “事实上，”她说着将火把举得更高。“不止有……” ----- ->
目标完成度 →	步

图G.21：用于LAMBADA的格式化数据集示例

上下文: →	请将这些字母重新排列成一个单词。 ，然后将该单词写下来: skicts =
目标完成度: →	sticks

图G.22：变位词1（A1）的格式化数据集示例

上下文: →	请将这些字母重新排列成一个单词。 ，然后写下这个单词: volwskagen =
目标完成度 →	大众汽车

图G.23：变位词2数据集的格式化示例数据集

上下文 →	问：在《天使在身边》中饰演泰丝的是谁？ A:
目标完成度 →	德洛丽丝·帕特里夏·厄尔利（1931年7月6日－2017年11月19日），更为人熟知的名字是德拉·里斯。她在职业领域使用的名字便是德拉·里斯。

图G.24：用于自然问题任务的格式化数据集示例。

Context	→	TITLE: William Perry (American football) - Professional career PARAGRAPH: In 1985, he was selected in the first round of the 1985 NFL Draft by the Chicago Bears; he had been hand-picked by coach Mike Ditka. However, defensive coordinator Buddy Ryan, who had a highly acrimonious relationship with Ditka, called Perry a "wasted draft-pick". Perry soon became a pawn in the political power struggle between Ditka and Ryan. Perry's "Refrigerator" nickname followed him into the NFL and he quickly became a favorite of the Chicago Bears fans. Teammates called him "Biscuit," as in "one biscuit shy of 350 pounds." While Ryan refused to play Perry, Ditka decided to use Perry as a fullback when the team was near the opponents' goal line or in fourth and short situations, either as a ball carrier or a lead blocker for star running back Walter Payton. Ditka stated the inspiration for using Perry as a fullback came to him during five-yard sprint exercises. During his rookie season, Perry rushed for two touchdowns and caught a pass for one. Perry even had the opportunity to run the ball during Super Bowl XX, as a nod to his popularity and contributions to the team's success. The first time he got the ball, he was tackled for a one-yard loss while attempting to throw his first NFL pass on a halfback option play. The second time he got the ball, he scored a touchdown (running over Patriots linebacker Larry McGrew in the process). About halfway through his rookie season, Ryan finally began to play Perry, who soon proved that he was a capable defensive lineman. His Super Bowl ring size is the largest of any professional football player in the history of the event. His ring size is 25, while the ring size for the average adult male is between 10 and 12. Perry went on to play for ten years in the NFL, retiring after the 1994 season. In his ten years as a pro, he regularly struggled with his weight, which hampered his performance at times. He played in 138 games, recording 29.5 sacks and five fumble recoveries, which he returned for a total of 71 yards. In his offensive career he ran five yards for two touchdowns, and had one reception for another touchdown. Perry later attempted a comeback, playing an unremarkable 1996 season with the London Monarchs of the World League of American Football (later NFL Europa).
		Q: what team did he play for?
		A:
Target Completion	→	the Chicago Bears

Figure G.25: Formatted dataset example for QuAC

Context	→	Please unscramble the letters into a word, and write that word: r e ! c . i p r o . c a / l =
Target Completion	→	reciprocal

Figure G.26: Formatted dataset example for Symbol Insertion

Context	→	Please unscramble the letters into a word, and write that word: taefed =
Target Completion	→	defeat

Figure G.27: Formatted dataset example for Reversed Words

上下文	→	标题: 威廉·佩里（美式橄榄球）——职业生涯段落: 1985年，他在NFL选秀的第一轮被芝加哥熊队选中，这是教练迈克·迪特卡亲自挑选的结果。然而，与迪特卡关系极为紧张的防守协调员巴迪·瑞安却称佩里是“被浪费的选秀人选”。不久之后，佩里便成了迪特卡与瑞安之间政治权力斗争的棋子。他“冰箱”这个绰号也随着他进入NFL，而他很快便成为了芝加哥熊队球迷的最爱。队友们则称他为“饼干”，意指他的体重差350磅才到350磅的上限。虽然瑞安拒绝让佩里上场比赛，但迪特卡决定在球队接近对方球门线或处于第四档短距离进攻时，让佩里担任全卫，要么负责持球，要么为明星跑卫沃尔特·佩顿充当掩护。迪特卡表示，让他担任全卫的想法是在进行五码冲刺训练时产生的。在新人赛季，佩里通过冲球取得了两次达阵，还接球一次完成一次达阵。鉴于他的受欢迎程度以及对球队成功的贡献，他甚至有机会在超级碗XX中持球参赛。第一次拿球时，他在尝试在一次半卫选项战术中传出首记NFL传球时就被擒杀，导致球队损失一码。第二次拿球时，他成功再次达阵（在此过程中还撞倒了爱国者队的线卫拉里·麦格鲁）。在新人赛季进行到一半左右时，瑞安终于开始让佩里上场，而佩里也很快证明了自己是一名出色的防守线球员。他在超级碗所戴戒指的尺寸是该赛事历史上所有职业橄榄球运动员中最大的，达到了25号，而普通成年男性的戒指尺寸通常在10号到12号之间。佩里在NFL共效力了十年，于1994赛季后退役。在十年的职业生涯中，他时常为体重问题所困扰，这有时会影响到他的比赛性能。他共参加了138场比赛，完成了29.5次擒杀，并夺回了5次掉球，总共推进了71码。在进攻端，他通过五码冲刺取得了两次达阵，还通过一次接球再取得一次达阵。后来佩里试图东山再起，但在美国橄榄球世界联盟（后来更名为NFL Europa）的伦敦君主队效力时的1996赛季表现平平。问：他曾为哪支球队效力？
		答：
目标C	补全内容：	→ 芝加哥熊队

图G.25：QuAC数据集的格式化示例

上下文：	→	请将这些字母重新排列成一个单词	，然后将这个单词写下来：
目标完成度	→	倒数	

图表 G.26：用于符号插入的格式化数据集示例

插入

上下文：	→	请将这些字母重新排列成一个单词。	，然后把那个单词写下来：
目标完成度：	→	defeat	

图G.27：用于反向词序任务的格式化数据集示例

Context →	Title: The Blitz
	Background: From the German point of view, March 1941 saw an improvement. The Luftwaffe flew 4,000 sorties that month, including 12 major and three heavy attacks. The electronic war intensified but the Luftwaffe flew major inland missions only on moonlit nights. Ports were easier to find and made better targets. To confuse the British, radio silence was observed until the bombs fell. X- and Y-Gerät beams were placed over false targets and switched only at the last minute. Rapid frequency changes were introduced for X-Gerät, whose wider band of frequencies and greater tactical flexibility ensured it remained effective at a time when British selective jamming was degrading the effectiveness of Y-Gerät.
	Q: How many sorties were flown in March 1941?
	A: 4,000
	Q: When did the Luftwaffe fly inland missions?
	A:
Target Completion →	only on moonlit nights

Figure G.28: Formatted dataset example for SQuADv2

Context →	Normal force -- In a simple case such as an object resting upon a table, the normal force on the object is equal but in opposite direction to the gravitational force applied on the object (or the weight of the object), that is, $N = m g$ ($\displaystyle N=mg$), where m is mass, and g is the gravitational field strength (about 9.81 m/s on Earth). The normal force here represents the force applied by the table against the object that prevents it from sinking through the table and requires that the table is sturdy enough to deliver this normal force without breaking. However, it is easy to assume that the normal force and weight are action-reaction force pairs (a common mistake). In this case, the normal force and weight need to be equal in magnitude to explain why there is no upward acceleration of the object. For example, a ball that bounces upwards accelerates upwards because the normal force acting on the ball is larger in magnitude than the weight of the ball. question: is the normal force equal to the force of gravity? answer:
Target Completion →	yes

Figure G.29: Formatted dataset example for BoolQ

Context →	The trend toward lower rents may seem surprising given that some communities in New York are bemoaning the loss of favorite local businesses to high rents. But, despite the recent softening, for many of these retailers there’s still been too big a jump from the rental rates of the late 1970s, when their leases were signed. Certainly, the recent drop in prices doesn’t mean Manhattan comes cheap. question: Manhattan comes cheap. true, false, or neither? answer:
Target Completion →	false

Figure G.30: Formatted dataset example for CB

上下文: →	标题: “闪电战”
	背景: 从德语的角度来看，1941年3月的情况有所改善。那个月，德国空军出动了4,000次飞行任务，其中包括12次大规模行动。三次猛烈的攻击。电子战愈演愈烈，而德国空军他们只在月光明亮的夜晚执行大规模的内陆作战任务。端口更易于这些港口不仅容易定位，还成为了更佳的攻击目标。为了迷惑英军，他们选择保持无线电静默。观测一直持续到炸弹落下为止。X和Y-Ger`at波束被设置在……上方。这些波束针对的是虚假目标，仅在最后时刻才进行切换。快速频率变化针对X-Ger`at，人们引入了相应的频率范围调整措施，其更宽的频率带宽以及更高的战术灵活性，确保了它在那个时期依然能够保持高效性能。
	更高的战术灵活性使其在那个时期仍能保持良好的效能。英国的定向干扰措施正在降低Y-Ger`at的作战效能。
	问：1941年3月共执行了多少次飞行任务？
	A: 4,000
	问：德国空军是在何时开始执行内陆作战任务的？
	A:
目标完成度 →	仅在月光照耀的夜晚。

图G.28： SQuADv2数据集的格式化示例

上下文: →	法向力——在物体静止于桌面之类的简单情形中，作用在物体上的法向力大小与作用在该物体上的重力大小相等，但方向相反。作用在物体上的重力（即物体的重力值）。也就是说， $N = m g$ ($\displaystyle N=mg$)，其中m表示质量，g则表示重力加速度。重力场强度（在地球上约为9.81米/秒²）。法向力这种作用力能够防止物体穿透桌面。同时，这也要求桌面足够坚固，能够在不发生破损的情况下承受该法向力。然而，人们很容易误以为法向力与重力属于作用力与反作用力关系。实际上并非如此。这里的K值即为上述法向力的上限数值。在这种情况下，法向力与力对（这是一个常见的错误）。例如，一个向上弹起的球会因为作用在其上的法向力更大而产生向上的加速度。物体的加速度。例如，一个向上弹起的球由于作用在球上的法向力更大，因此球会向上加速运动。其大小大于球的重力。 问题：法向力是否等于重力？ 答案:
目标完成度 →	yes

图G.29： BoolQ任务对应的格式化数据集示例

上下文: →	考虑到某些情况，租金呈下降趋势似乎令人惊讶。这些社区正在为喜爱的本地商家因高昂的租金而流失而感到惋惜。这些商家不得不承受过高的租金负担。不过，尽管最近有所缓和，对许多人而言对于这些零售商中的许多人来说，租金率仍然上升得过于迅猛。那是20世纪70年代末，他们签署租约的时期。当然，最近的……房价的下跌并不意味着曼哈顿的房产也变便宜了。 问题：曼哈顿的房产变便宜了吗？对、错，还是两者都不是？ 答案:
目标完成度 →	false

图G.30：用于CB任务的格式化数据集示例

Context →	The bet, which won him dinner for four, was regarding the existence and mass of the top quark, an elementary particle discovered in 1995. question: The Top Quark is the last of six flavors of quarks predicted by the standard model theory of particle physics. True or False? answer:
Target Completion →	False

Figure G.31: Formatted dataset example for RTE

Context →	An outfitter provided everything needed for the safari. Before his first walking holiday, he went to a specialist outfitter to buy some boots. question: Is the word ‘outfitter’ used in the same way in the two sentences above? answer:
Target Completion →	no

Figure G.32: Formatted dataset example for WiC

Context →	Final Exam with Answer Key Instructions: Please carefully read the following passages. For each passage, you must identify which noun the pronoun marked in *bold* refers to. ==== Passage: Mr. Moncrieff visited Chester’s luxurious New York apartment, thinking that it belonged to his son Edward. The result was that Mr. Moncrieff has decided to cancel Edward’s allowance on the ground that he no longer requires *his* financial support. Question: In the passage above, what does the pronoun " *his* " refer to? Answer:
Target Completion →	mr. moncrieff

Figure G.33: Formatted dataset example for WSC

Context →	Q: ‘Nude Descending A Staircase’ is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

Figure G.34: Formatted dataset example for TriviaQA. TriviaQA allows for multiple valid completions.

上下文: →	那次赌局让他赢得了一份四人晚餐的奖励，赌的内容正是1995年被发现的一种基本粒子——顶夸克的是否存在及其质量。 此外，还有一个问题需要解答。 问题：顶夸克是粒子物理学标准模型理论所预言的六种夸克类型中的最后一种。 粒子物理学标准模型理论。 对还是错？ 请输入答案:
目标完成度 →	错误

图G.31：用于RTE任务的格式化数据集示例

上下文 →	那家装备供应商为这次野生动物探险准备了所有必需的物品。 在开始他的第一次徒步旅行之前，他前往了一家专业的装备供应商处购买物品。一些靴子。 问题：在上述两句话中，‘outfitter’这个词的使用方式是否相同？ 上面的句子？ 答案:
目标完成度 →	no

图G.32：WiC竞赛所用格式化数据集示例

上下文 →	含答案的期末考试题 说明：请仔细阅读以下段落。 对于每一个 都需要找出*加粗标注*的代词所指代的名词是哪个。 to. ==== 段落内容如下：Mr. 蒙克里夫先生参观了切斯特位于纽约的豪华公寓，因为他以为那套公寓属于他的儿子爱德华。最终的结果是，蒙克里夫先生蒙克里夫先生决定取消爱德华的零用钱，理由是爱德华不再需要*他的*经济支持了。 问题：在上面的段落中，代词"*他的*"指的是谁？ 答案:
目标完成度 →	mr. 蒙克里夫先生

图G.33：WSC竞赛用格式化数据集示例

上下文 →	问：‘下楼裸女’或许是以下哪位艺术家的最著名画作？ 哪位20世纪初期的艺术家？ A:
目标完成度 →	马塞尔·杜尚
目标完成度 →	r mutt
目标完成度 →	杜尚
目标完成度 →	马塞尔·杜尚
目标完成度 →	R·穆特
目标完成度 →	马塞尔·杜尚
目标完成度 →	亨利·罗伯特·马塞尔·杜尚
目标完成度 →	马塞尔·杜尚
目标完成度 →	亨利·罗伯特·马塞尔·杜尚
目标完成度 →	杜尚式
目标完成度 →	杜尚
目标完成度 →	杜尚式
目标完成度 →	马塞尔·杜尚
目标完成度 →	马塞尔·杜尚
目标完成度 →	马塞尔·杜尚

图G.34：用于TriviaQA的格式化数据集示例。TriviaQA支持多种有效的补全结果。

Context →	Q: What school did burne hogarth establish?
	A:
Target Completion →	School of Visual Arts

Figure G.35: Formatted dataset example for WebQA

Context →	Keinesfalls dürfen diese für den kommerziellen Gebrauch verwendet werden. =
Target Completion →	In no case may they be used for commercial purposes.

Figure G.36: Formatted dataset example for De→En. This is the format for one- and few-shot learning, for this and other language tasks, the format for zero-shot learning is “Q: What is the {language} translation of {sentence} A: {translation}.”

Context →	In no case may they be used for commercial purposes. =
Target Completion →	Keinesfalls dürfen diese für den kommerziellen Gebrauch verwendet werden.

Figure G.37: Formatted dataset example for En→De

Context →	Analysis of instar distributions of larval I. verticalis collected from a series of ponds also indicated that males were in more advanced instars than females. =
Target Completion →	L’analyse de la distribution de fréquence des stades larvaires d’I. verticalis dans une série d’étangs a également démontré que les larves mâles étaient à des stades plus avancés que les larves femelles.

Figure G.38: Formatted dataset example for En→Fr

Context →	L’analyse de la distribution de fréquence des stades larvaires d’I. verticalis dans une série d’étangs a également démontré que les larves mâles étaient à des stades plus avancés que les larves femelles. =
Target Completion →	Analysis of instar distributions of larval I. verticalis collected from a series of ponds also indicated that males were in more advanced instars than females.

Figure G.39: Formatted dataset example for Fr→En

Context →	The truth is that you want, at any price, and against the wishes of the peoples of Europe, to continue the negotiations for Turkey’s accession to the European Union, despite Turkey’s continuing refusal to recognise Cyprus and despite the fact that the democratic reforms are at a standstill. =
Target Completion →	Adevărul este că vă doriți, cu orice preț și împotriva dorinței europenilor, să continuați negocierile de aderare a Turciei la Uniunea Europeană, în ciuda refuzului continuu al Turciei de a recunoaște Ciprul și în ciuda faptului că reformele democratice au ajuns într-un punct mort.

Figure G.40: Formatted dataset example for En→Ro

上下文: →	问：伯恩·霍加斯创办了哪所学校？
	A:
目标完成度: →	视觉艺术学院

图G.35： WebQA格式化数据集示例

上下文: →	绝不可将这些用于商业用途。 =
目标完成度 →	在任何情况下都不得将其用于商业目的。

图G.36： 德语补全的格式化数据集示例。该格式适用于单样本学习和少样本学习；对于此类及其他语言任务， 零样本学习的格式则为 ‘问： {句子} A: {的 {语言} 翻译是什么？ ’

上下文 →	在任何情况下都不得将其用于商业目的。 =
目标完成度 →	绝不可将这些材料用于商业用途。

图G.37： En→De格式的数据集示例

上下文 →	对从多个池塘中采集的I. verticalis幼虫的龄期概率分布进行了分析。这些池塘的观测结果还表明， 雄性个体的龄期更为成熟。 多于雌性。 =
目标完成度 →	对I. verticalis幼虫阶段序列的概率分布进行了分析。在一系列池塘中对I. verticalis进行的观测同样表明， 这些幼虫的发育阶段比雌性幼虫更为成熟。雄性幼虫的发育阶段比雌性幼虫更靠后。

图G.38： En→Fr格式的数据集示例

上下文: →	对I. verticalis幼虫各龄期出现频率的概率分布进行了分析。在多处池塘中对I. verticalis进行的观察同样表明， 这些幼虫的雄性幼虫所处的发育阶段比雌性幼虫更为成熟。 =
目标完成度: →	针对从不同地点采集的I. verticalis幼虫的龄期分布情况所进行的分析。一系列池塘的存在也表明， 雄性个体的发育阶段比雌性更为成熟。低于女性。

图G.39： Fr→En格式化数据集示例

上下文 →	事实上，无论付出何种代价， 即便违背相关群体的意愿，欧洲各国人民仍需继续推进土耳其的加入谈判。尽管土耳其一直拒绝予以承认， 欧盟仍继续推进相关进程。尽管塞浦路斯问题存在， 且民主改革也处于停滞状态。 毫无进展。 =
目标完成度 →	无论付出何种代价， 违背他人意愿的行为都是不可取的。欧洲方面仍坚持继续推进土耳其加入欧盟的谈判。尽管土耳其一直拒绝承认塞浦路斯， “欧洲”的 “仍”持续存在。虽然民主改革已陷入僵局， “欧洲”的 “仍”依然存在。

图G.40： En→Ro格式化数据集示例。

Context →	Adevărul este că vă doriți, cu orice preț și împotriva dorinței europenilor, să continuați negocierile de aderare a Turciei la Uniunea Europeană, în ciuda refuzului continuu al Turciei de a recunoaște Ciprul și în ciuda faptului că reformele democratice au ajuns într-un punct mort. =
Target Completion →	The truth is that you want, at any price, and against the wishes of the peoples of Europe, to continue the negotiations for Turkey’s accession to the European Union, despite Turkey’s continuing refusal to recognise Cyprus and despite the fact that the democratic reforms are at a standstill.

Figure G.41: Formatted dataset example for Ro→En

Context →	Q: What is (2 * 4) * 6? A:
Target Completion →	48

Figure G.42: Formatted dataset example for Arithmetic 1DC

Context →	Q: What is 17 minus 14? A:
Target Completion →	3

Figure G.43: Formatted dataset example for Arithmetic 2D-

Context →	Q: What is 98 plus 45? A:
Target Completion →	143

Figure G.44: Formatted dataset example for Arithmetic 2D+

Context →	Q: What is 95 times 45? A:
Target Completion →	4275

Figure G.45: Formatted dataset example for Arithmetic 2Dx

Context →	Q: What is 509 minus 488? A:
Target Completion →	21

Figure G.46: Formatted dataset example for Arithmetic 3D-

Context →	Q: What is 556 plus 497? A:
Target Completion →	1053

Figure G.47: Formatted dataset example for Arithmetic 3D+

Context →	Q: What is 6209 minus 3365? A:
Target Completion →	2844

Figure G.48: Formatted dataset example for Arithmetic 4D-

上下文: →	任何人都无权以任何代价违背他人的意愿，强行施加自己的意志。 欧洲方面仍在继续推进土耳其加入欧盟的谈判。 尽管土耳其一直拒绝承认塞浦路斯，但欧洲方面依然在努力推动相关进程。 即便民主改革已经陷入僵局，情况依然没有改变。 =
目标完成度: →	事实是，你无论如何都想违背他人的意愿，不惜任何代价地做到它。 欧洲各国，继续推进土耳其的加入谈判 尽管土耳其一直拒绝承认，仍需面向欧洲联盟。 塞浦路斯，尽管其民主改革尚处于起步阶段 停滞不前。

图G.41：面向Ro的格式化数据集示例 →En

上下文: →	问: (2 * 4) * 6 的结果是多少? A:
目标完成度: →	48

图G.42：算术运算1DC任务对应的格式化数据集示例

上下文: →	问: 17减去14的结果是多少? A:
目标完成度 →	3

图 G.43：算术运算两位数减法的格式化数据集示例

上下文 →	问: 98 加 45 等于多少? A:
目标完成度 →	143

图 G.44：算术运算两位数减法的格式化数据集示例+

上下文: →	问: 95乘以45等于多少? A:
目标完成度: →	4275

图G.45：用于二维算术x的格式化数据集示例

上下文: →	问: 509减去488等于多少? A:
目标完成度 →	21

图G.46：用于三维算术-的格式化数据集示例

上下文 →	问: 556加497等于多少? A:
目标完成度 →	1053

图G.47：用于三维算术-的格式化数据集示例+

上下文: →	问: 6209 减去 3365 等于多少? A:
目标完成度: →	2844

图 G.48：算术运算 4D-任务 的格式化数据集示例

Context →	Q: What is 9923 plus 617? A:
Target Completion →	10540

Figure G.49: Formatted dataset example for Arithmetic 4D+

上下文: →	问: 9923加上617等于多少? A:
目标完成度: →	10540

图G.49：用于算术运算4D的格式化数据集示例

Context →	Q: What is 40649 minus 78746? A:
Target Completion →	-38097

Figure G.50: Formatted dataset example for Arithmetic 5D−

上下文: →	问: 40649减去78746等于多少? A:
目标完成度 →	-38097

图G.50：用于5D算术运算的格式化数据集示例−

Context →	Q: What is 65360 plus 16204? A:
Target Completion →	81564

Figure G.51: Formatted dataset example for Arithmetic 5D+

上下文 →	问: 65360加上16204等于多少? A:
目标完成度 →	81564

图G.51：用于5D算术运算的格式化数据集示例+

所有模型规模下所有任务下的H结果

零样本学习										单样本学习										少样本学习										175B (测试服务器)
Name	指标	分割	SOTA	K	SmallMedLargeXL.27亿参数 67亿参数 130亿参数 1750亿参数					SmallMedLargeXL.27亿参数 67亿参数 130亿参数 1750亿参数					SmallMedLargeXL.27亿参数 67亿参数 130亿参数 1750亿参数															
HellaSwag	acc	dev	85.6	20	33.7	43.6	51.0	54.7	62.8	67.4	70.9	78.9	33.0	42.9	50.5	53.5	61.9	66.5	70.0	78.1	33.5	43.1	51.3	54.9	62.9	67.3	71.3	79.3		
LAMBADA	acc	test	68.0	15	42.7	54.3	60.4	63.6	67.1	70.3	72.5	76.2	22.0	47.1	52.6	58.3	61.1	65.4	69.0	72.5	22.0	40.4	63.2	57.0	78.1	79.1	81.3	86.4		
LAMBADA	ppl	test	8.63	15	18.6	9.09	6.53	5.44	4.60	4.00	3.56	3.00	165.0	11.6	8.29	6.46	5.53	4.61	4.06	3.35	165.0	27.6	6.63	7.45	2.89	2.56	2.56	1.92		
StoryCloze	acc	test	91.8	70	63.3	68.5	72.4	73.4	77.2	77.7	79.7	83.2	62.3	68.7	72.3	74.2	77.3	78.7	79.7	83.2	62.3	70.2	73.9	76.1	80.2	81.2	83.0	87.7		
NQs	acc	test	44.5	64	0.64	1.75	2.71	4.40	6.01	5.79	7.84	14.6	1.19	3.07	4.79	5.43	8.73	9.78	13.7	23.0	1.72	4.46	7.89	9.72	13.2	17.0	21.0	29.9		
TriviaQA	acc	dev	68.0	64	4.15	7.61	14.0	19.7	31.3	38.7	41.8	64.3	4.19	12.9	20.5	26.5	35.9	44.4	51.3	68.0	6.96	16.3	26.5	32.1	42.3	51.6	57.5	71.5		
Webqs	acc	test	45.5	64	1.77	3.20	4.33	4.63	7.92	7.73	8.22	14.4	2.56	6.20	8.51	9.15	14.5	15.1	19.0	25.3	5.46	12.6	15.9	19.6	24.8	27.7	33.5	41.2		
Ro→En 16	BLEU-mb测试	39.9	64	2.08	2.71	3.09	3.15	16.3	8.34	20.2	19.9	0.55	15.4	23.0	26.3	30.6	33.2	35.6	38.6	1.25	20.7	25.8	29.2	33.1	34.8	37.0	39.5			
Ro→En 16	BLEU-sb test		64	2.39	3.08	3.49	3.56	16.8	8.75	20.8	20.9	0.65	15.9	23.6	26.8	31.3	34.2	36.7	40.0	1.40	21.3	26.6	30.1	34.3	36.2	38.4	41.3			
En→Ro 16	BLEU-mb 测试	38.5	64	2.14	2.65	3.23	2.50	14.6	4.24	5.32	14.1	0.35	3.30	7.89	8.72	13.2	15.1	17.3	20.6	1.25	5.99	33.3	10.7	14.3	16.3	18.0	21.0			
En→Ro 16	bleu-sb评分		64	2.61	3.11	3.07	3.09	4.26	5.31	6.43	18.0	0.55	3.90	9.15	10.3	15.7	18.2	20.8	24.9	1.64	7.40	10.9	12.9	17.2	19.6	21.8	25.8			
Fr→En 14	BLEU-mb 测试	35.0	64	1.81	2.53	3.47	3.13	20.6	15.1	21.2	21.2	1.28	15.9	23.7	26.3	29.0	30.5	30.3	33.7	4.98	25.5	28.5	31.1	33.7	34.9	36.6	39.2			
Fr→En 14	bleu-sb评分		64	2.29	2.99	3.90	3.60	21.2	15.5	22.4	21.9	1.50	16.3	24.4	27.0	30.0	31.6	31.4	35.7	5.30	26.2	29.5	32.2	35.1	36.4	38.3	41.4			
En→Fr 14	BLEU-mb测试	45.6	64	1.74	2.16	2.73	2.15	18.1	8.12	10.2	25.2	0.49	8.00	14.8	15.9	20.3	23.3	24.9	28.3	4.08	14.5	19.3	21.5	24.9	27.3	29.5	32.6			
En→Fr 14	BLEU-sb test	45.9	64	2.44	2.75	3.54	2.82	19.3	11.4	15.3	31.3	0.81	10.0	18.2	19.3	24.7	28.3	30.1	34.1	5.31	18.0	23.6	26.1	30.3	33.3	35.5	39.9			

表H.1: 本文所研究的各项任务、配置及模型对应的得分。



Figure H.1: All results for all SuperGLUE tasks.



Figure H.2: Results for SAT task.

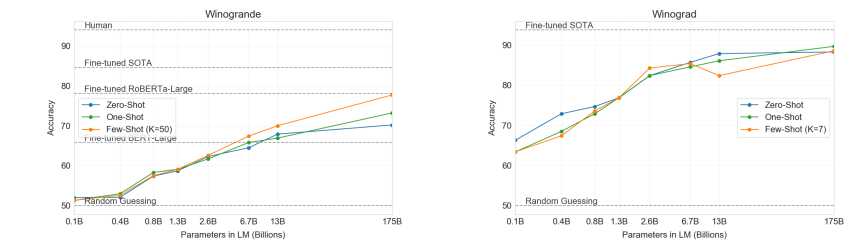
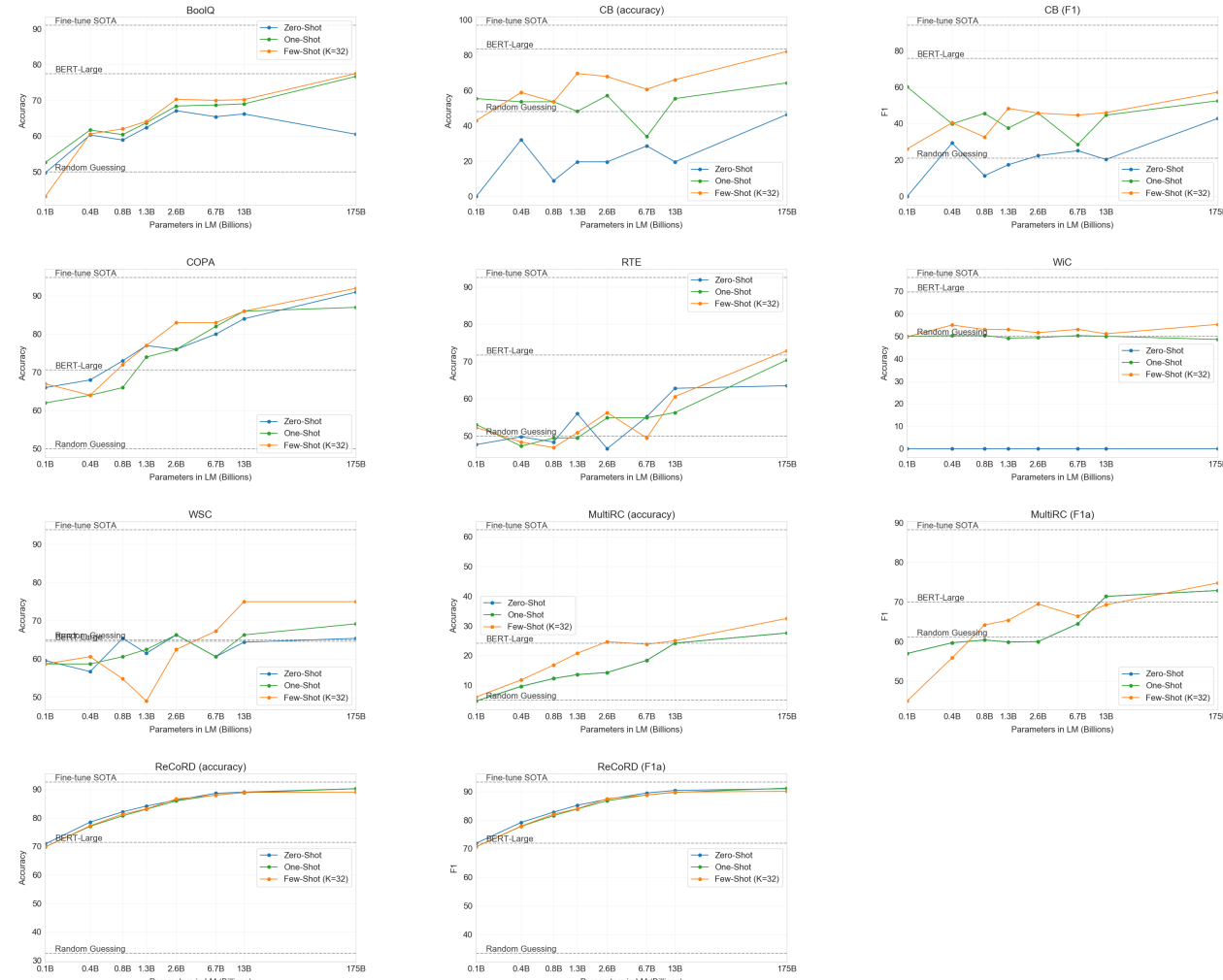
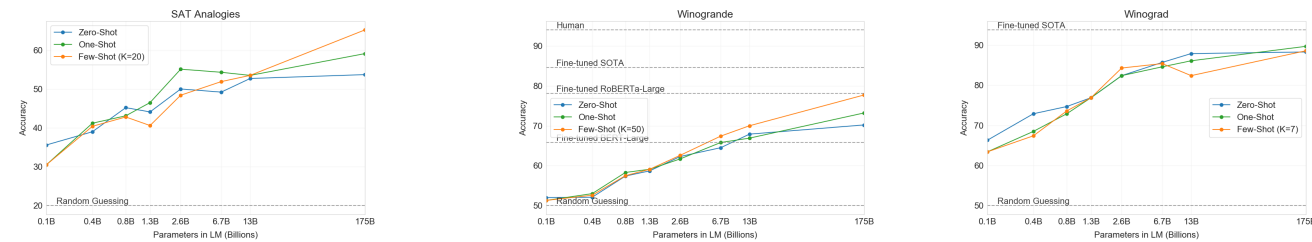


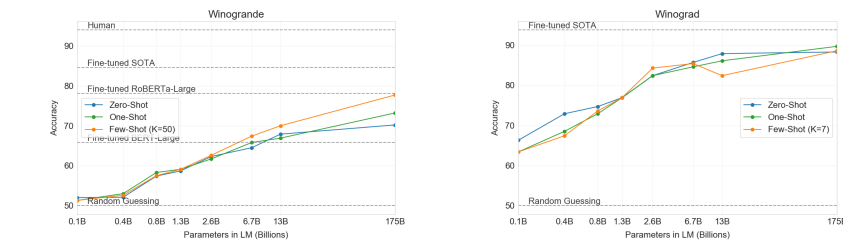
Figure H.3: All results for all Winograd tasks.



图H.1: 所有SuperGLUE任务的全部结果。



图H.2: SAT任务的测试结果。



图H.3: 所有Winograd任务的全部结果。

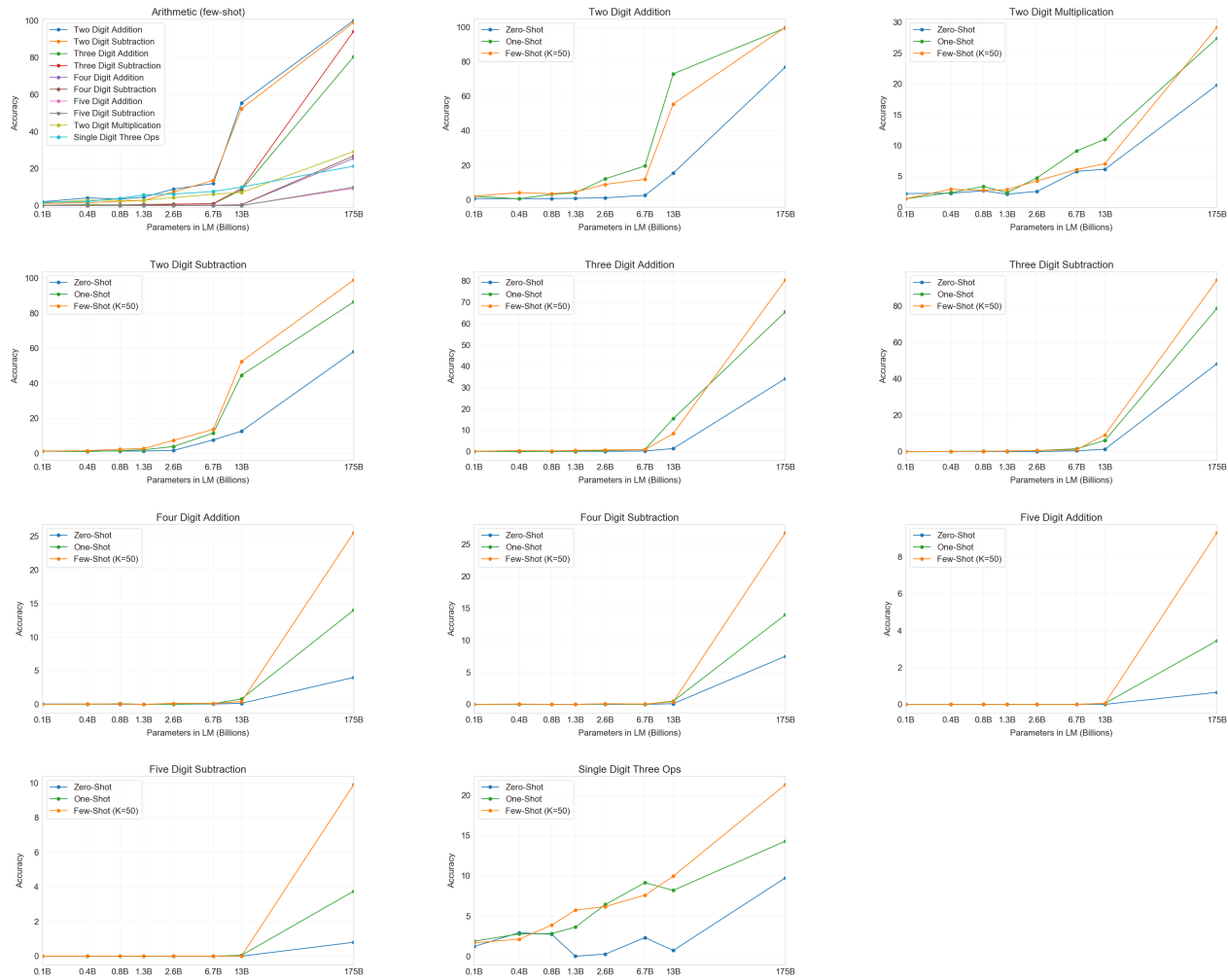
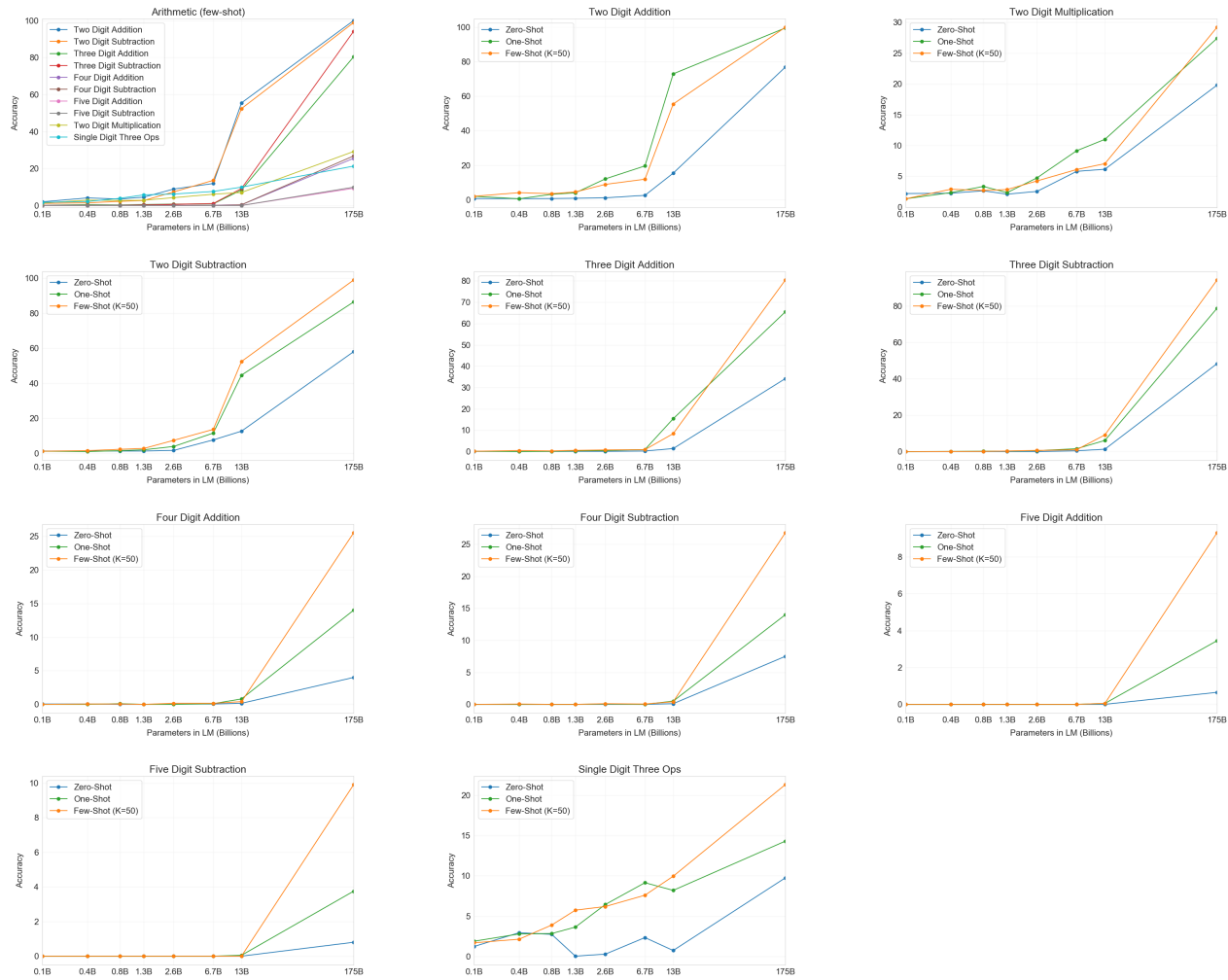


Figure H.4: All results for all Arithmetic tasks.



图H.4: 所有算术任务的全部结果。

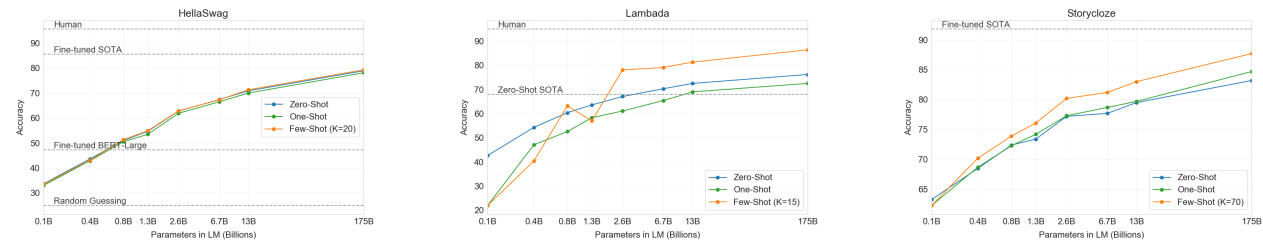
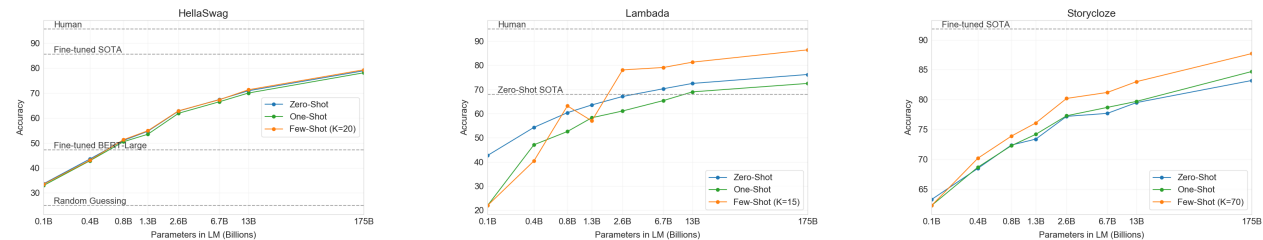


Figure H.5: All results for all Cloze and Completion tasks.



图H.5: 所有完形填空及补全任务的全部结果。

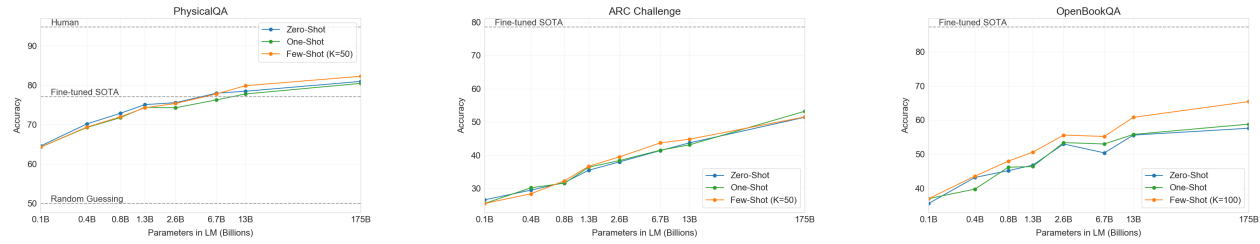


Figure H.6: All results for all Common Sense Reasoning tasks.

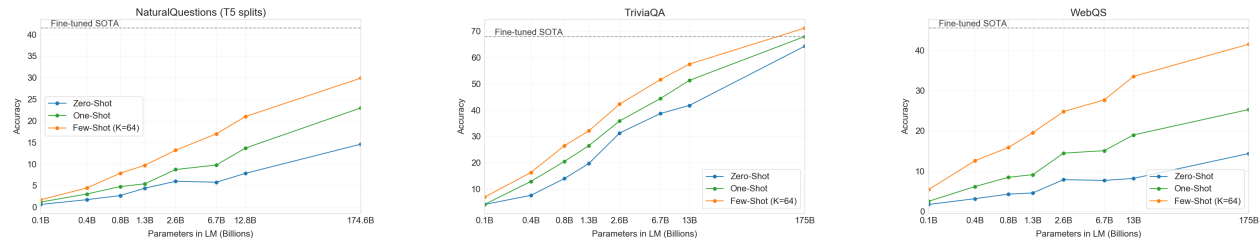


Figure H.7: All results for all QA tasks.

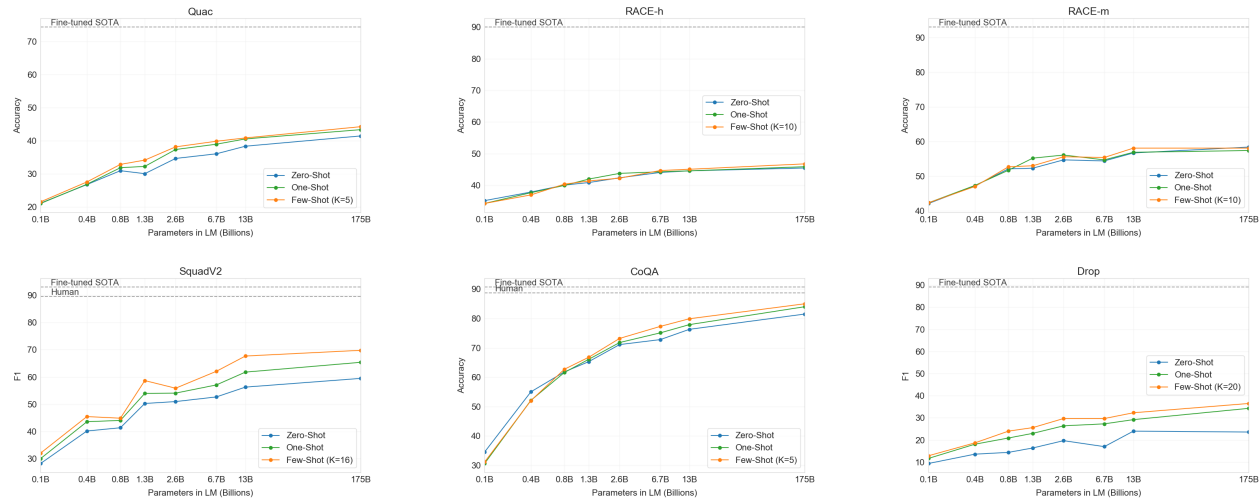


Figure H.8: All results for all Reading Comprehension tasks.

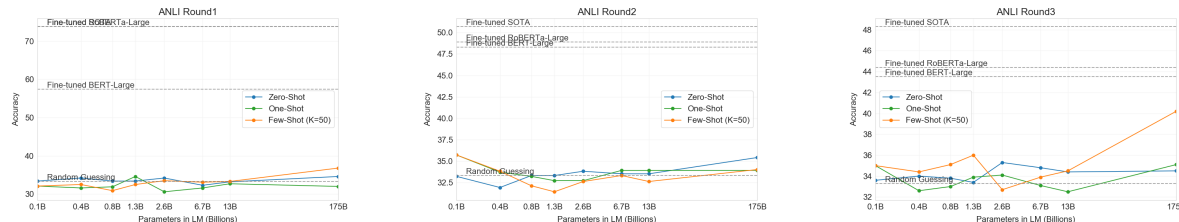
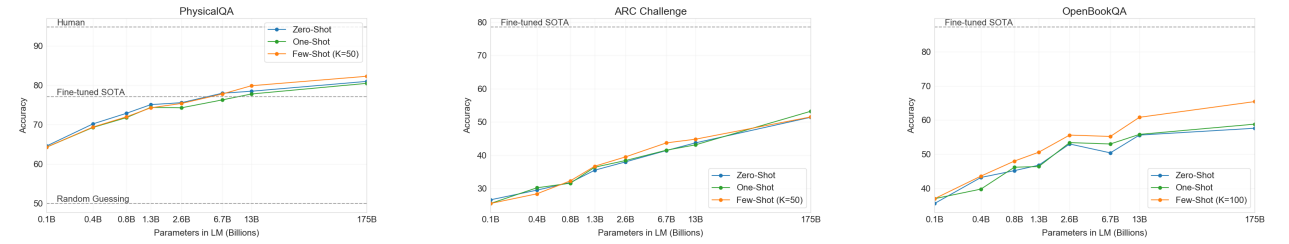
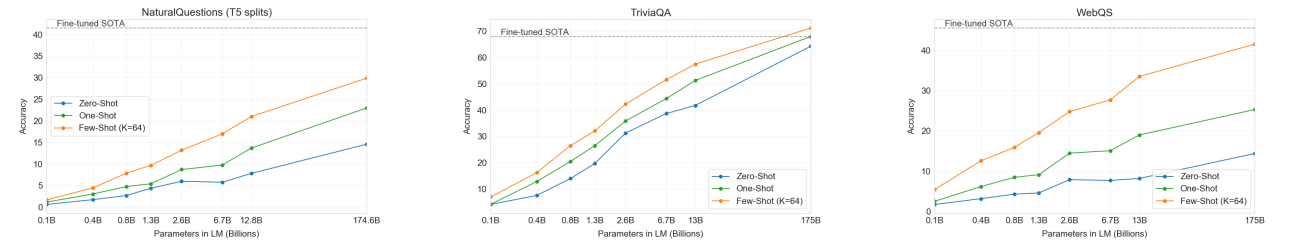


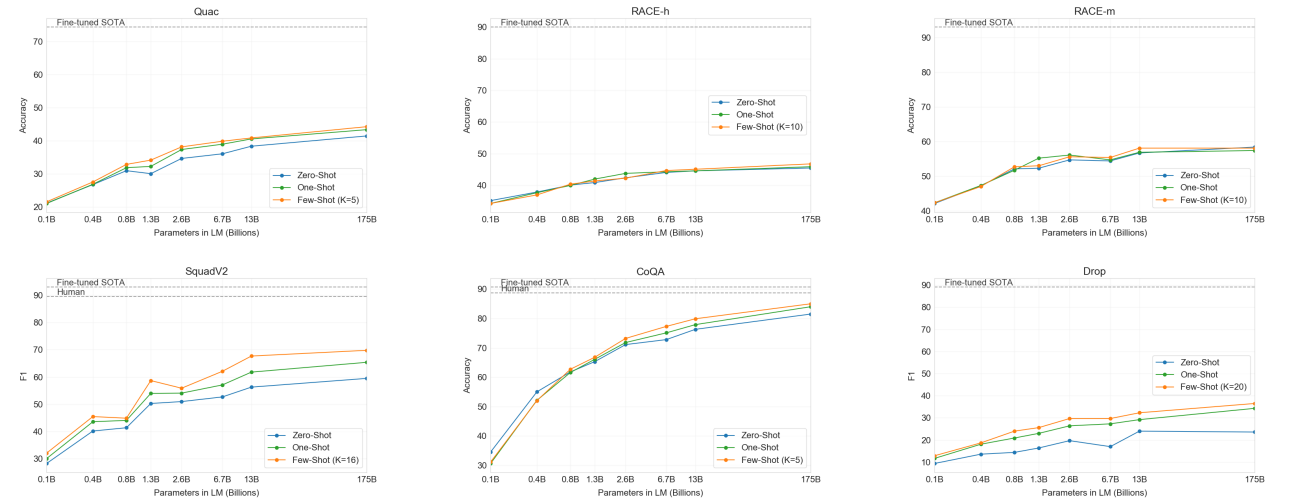
Figure H.9: All results for all ANLI rounds.



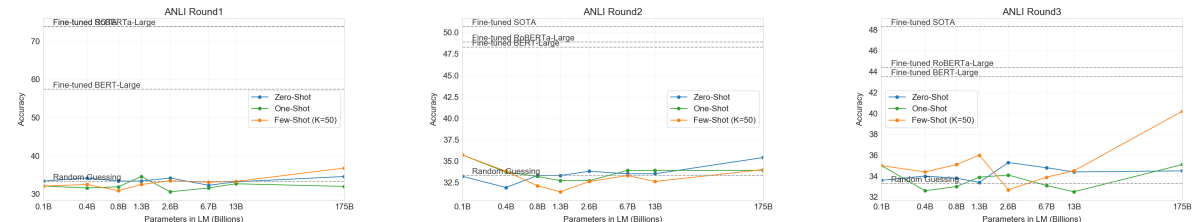
图H.6: 所有常识推理任务的全部结果。



图H.7: 所有问答任务的全部结果。



图H.8: 所有阅读理解任务的全部结果。



图H.9: 所有ANLI轮次的全部结果。

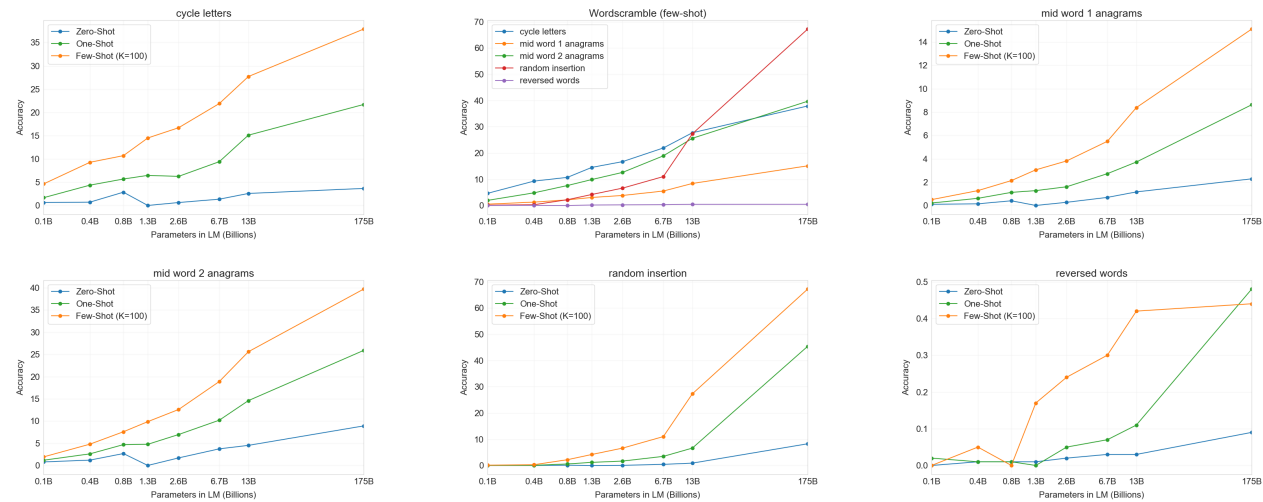
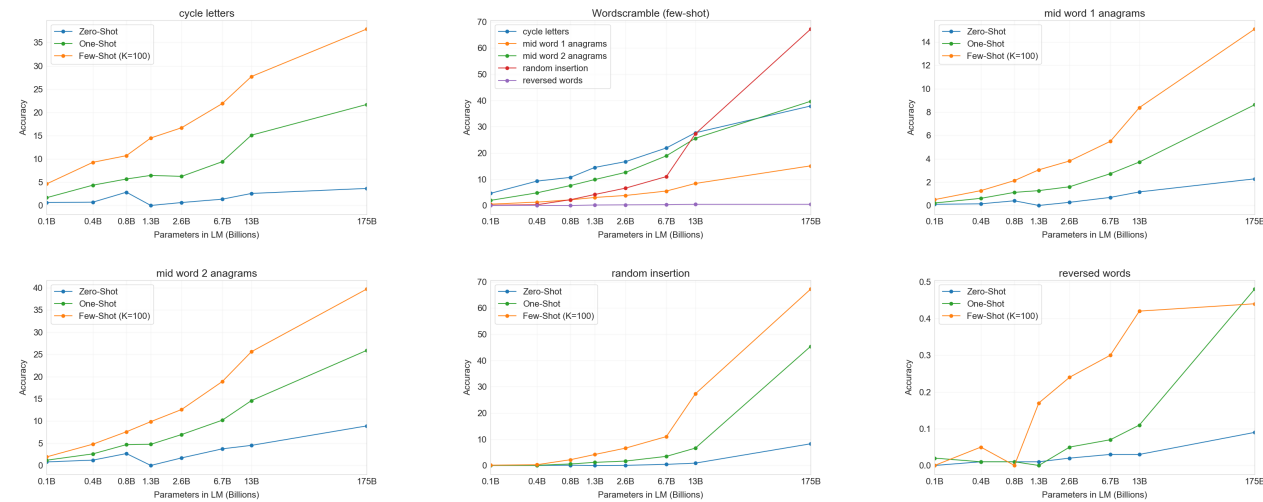


Figure H.10: All results for all Scramble tasks.



图H.10: 所有混排任务的全部结果。

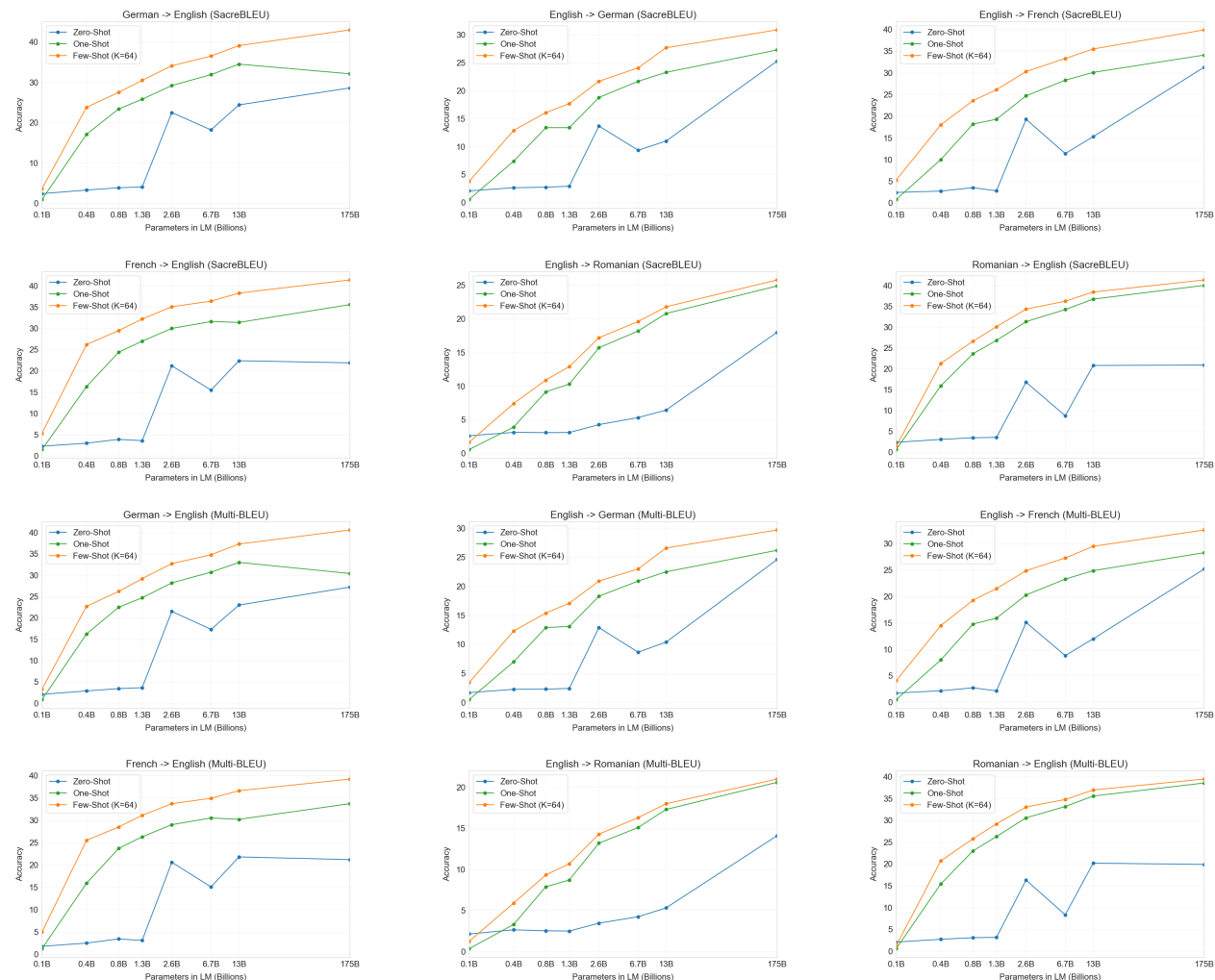
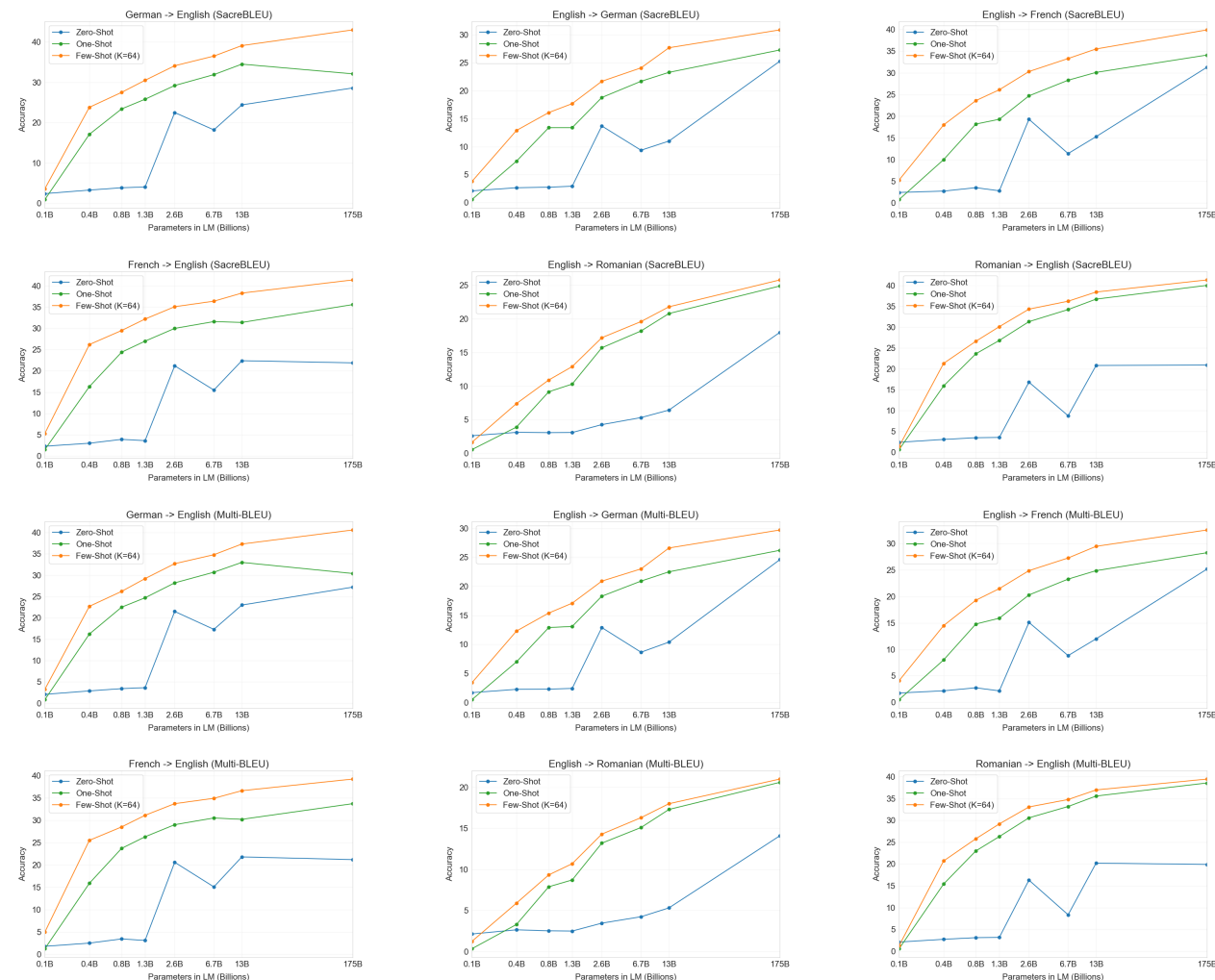


Figure H.11: All results for all Translation tasks.



图H.11: 所有翻译任务的全部结果。

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